EDNS 491 & 492
Capstone Design I & II

Student Course Text

The Design Process, Project Management, Professional Skills,
for Senior Design

Colorado School of Mines

Interdisciplinary Engineering Design

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A comprehensive reference covering the engineering design process, project milestones, hybrid project management, professional communication, client engagement, fabrication resources, safety, and professional ethics for the two-semester capstone design experience.

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How to Use This Document

This reference document accompanies the EDNS 491 and EDNS 492 course syllabi and the Canvas course site. It is organized to support you across the entire two-semester capstone design experience, from your first client meeting through the Design Showcase poster session.

Three documents work together:

1. **This Course Text**: deep technical and process content organized in five parts. Consult it when you need the methodology, design procedures, project management frameworks, communication guidance, technical reference, or safety protocols. Start with the "What Should I Be Reading Right Now?" triage guide on the next page.
2. **The Course Syllabus**: assignment details, grading weights, policies, CLOs, and faculty contact information. The syllabus is the authoritative source for deadlines and grading.
3. **Canvas**: assignment rubrics, submission portals, scheduling, lecture materials, professional development modules, purchasing forms, and the Labriola InnoHub enrollment site.

Part I: The Design Process (Chapters 1–6) covers the complete engineering design lifecycle from problem definition through verification and validation. These chapters support your work in chronological order as your project progresses through design reviews.

Part II: Major Project Milestones (Chapters 7–10) provides detailed guidance for the Statement of Work, PDR, CDR, and FDR, with deliverable checklists, evaluation criteria, and common failure patterns at each gate.

Part III: Project Management and Operations (Chapters 11–16) covers hybrid project management, fabrication spaces and resources, team dynamics, budgeting, purchasing procedures, and travel planning. These chapters are practical operations guides that you will reference repeatedly throughout both semesters.

Part IV: Professional Skills (Chapters 17–20) covers client engagement, written and oral communication, and career preparation. These skills distinguish capstone from prior coursework and directly affect your individual performance evaluations.

Part V: Safety, Compliance, and Professional Practice (Chapters 21–22) covers EHS requirements, professional ethics, PE licensure, and course policies.

Appendices include a glossary, quick reference tables, "Why Capstone Projects Fail" (12 common patterns), checklists, and references. ABET outcome mapping and the NCEES Rules of Professional Conduct are covered in Chapter 22.

Reading by Role

- **Scrum Master / Project Manager**: Chapters 11 (PM), 13 (Team Dynamics), 7 (SOW), 8–10 (Milestones)
- **Chief Financial Officer**: Chapters 14 (Budgeting), 15 (Purchasing), 16 (Travel)
- **Technical Lead**: Chapters 3 (Requirements), 4 (Concept Gen), 6 (V&V), 9 (CDR)

- **Communication Lead**: Chapters 17 (Client), 18 (Written), 19 (Oral), 13 (Team)
- **Fabrication Lead**: Chapter 12 (Fab Spaces), 21 (EHS), Front Matter (Safety)

Project Tracks

This document supports all three project tracks. Where guidance differs by track, look for **Build Track**, **Paper Track**, and **Competition Track** labels. All tracks are held to equivalent standards of engineering rigor. The Paper Track is not a reduced-effort alternative; it substitutes physical fabrication with analytical depth, simulation fidelity, and documentation completeness. The Competition Track does not reduce requirements; it reorganizes their timing around an external schedule.

What Should I Be Reading Right Now?

If you are...	Your problem	Go to
Just assigned to a project	Need to understand the course structure	Ch. 1
Preparing for your client kickoff	Need to know what to ask and how to prepare	Ch. 2
Writing the Statement of Work	Need SOW structure and scope agreement	Ch. 7
Writing requirements	Need SHALL statements and SDRD format	Ch. 3
Generating design concepts	Need brainstorming and selection methods	Ch. 4
Evaluating sustainability impacts	Need LEED, ENVISION, DFE, ESJ tools	Ch. 5
Writing the ethics essay	Need argumentative essay guidance	Ch. 5, Ch. 22
Planning your test approach	Need V&V methods, TAID, VCRM	Ch. 6
Preparing for PDR	Need PDR deliverables and common mistakes	Ch. 8
Preparing for CDR	Need CDR deliverables and eng. calc package	Ch. 9
Preparing for FDR	Need FDR deliverables and showcase prep	Ch. 10
Setting up your sprint process	Need Scrum framework for capstone	Ch. 11
Using InnoHub shops or 3D printers	Need Labriola policies and procedures	Ch. 12
Dealing with a team conflict	Need conflict resolution and PIP process	Ch. 13
Creating or updating the budget	Need budget planning and tracking	Ch. 14
Ordering parts or materials	Need purchasing workflow and forms	Ch. 15
Planning team travel	Need travel policy and budget template	Ch. 16
Meeting with your client	Need client communication best practices	Ch. 17
Writing a design report	Need report structure and writing tips	Ch. 18
Giving a design review presentation	Need presentation and Q&A guidance	Ch. 19
Preparing for Career Day	Need resume and interview tips	Ch. 20
Need EHS or safety plan guidance	Need hazard assessment or lab safety	Ch. 21
Need PE licensure or ethics info	Need FE/PE path or NCEES rules	Ch. 22

SDI Sprint-to-Chapter Mapping

Sprint	Weeks	Primary Activities	Chapters
0	1–2	Project Expo, team formation, client kickoff meeting	Ch. 1, 2
1	3–4	Project Plan, Scrum training, product backlog	Ch. 7, 11, 13
2	5–6	Concept generation, trade study, Career Day	Ch. 4, 20
3	7–8	Concept development, client approval of approach	Ch. 3, 4, 17
4	9–10	Functional prototypes / site plans, ethics assignment	Ch. 5, 12, 22
5	11–13	PDR preparation, mock PDR, PDR execution	Ch. 6, 8, 18, 19
6	14–15	Design refinement, next steps, SDII handoff	Ch. 8, 10

SDII Sprint-to-Chapter Mapping

Sprint	Weeks	Primary Activities	Chapters
7	1	Re-engagement with client, goal setting, safety plan	Ch. 2, 11, 21
8	2–5	CDR report, engineering calc package, CDR execution	Ch. 6, 9, 14, 15
9–10	6–8	Build/test/iterate, Broader Impacts essay	Ch. 5, 6, 12
11	9–11	System integration, V&V execution	Ch. 6
12–13	12–14	FDR preparation, final testing, documentation	Ch. 10, 18
14	15–16	FDR, Showcase, digital archive, final checkout	Ch. 10, 19

Laboratory and Workshop Safety

This chapter establishes mandatory safety practices for all EDNS 491/492 lab, shop, and project work. Capstone projects span electrical, mechanical, chemical, and computational domains. The hazards you encounter depend on your project, but the practices below apply universally.

EHS Training Requirement

Both SDI and SDII require completion of an EHS Safety Training module on Canvas. In SDII, you must also complete a **Hazard Assessment and Safety Plan** specific to your project by Week 2. This document identifies the hazards associated with your build, the controls you will implement, and the PPE required. Your PA must review and approve the safety plan before fabrication begins. See Chapter 21 for the full EHS framework.

Electrical Hazards

Electrical shock severity depends on the current flowing through the body, which is determined by voltage and the body's resistance. Body resistance varies widely: approximately 100 kΩ for dry, intact skin but as low as 1 kΩ hand-to-hand with wet or broken skin. **No voltage should be treated as unconditionally safe.** A source as low as 24 VDC can deliver dangerous current under the right conditions (wet skin, cuts, conductive jewelry, medical implants). IEC 61010 designates voltages below 30 VDC / 25 VAC RMS as "extra-low voltage," but this classification assumes dry-skin contact with intact insulation, not that the source is harmless.

The safest practice is always to de-energize before working. When live contact is unavoidable: below 48 V, use insulated tools and the one-hand rule (keep one hand behind your back to prevent hand-to-hand current path). Above 48 V, including all AC mains: de-energize and lock out/tag out (LOTO) before touching any conductor. **Never design your own AC-DC converter.** Always use a pre-certified (UL/CE listed) wall adapter.

Machine Shop Safety

Before using any machine in the Labriola InnoHub shops (Metal Shop, Wood Shop, Composites Shop), you must complete the machine-specific training on the Labriola InnoHub Canvas site. No drop-ins are accepted. CNC training requires prior completion of the corresponding manual machine training. Extended machine use (>4 hours) for mills, lathes, welders, wood lathe, and table saw requires manager approval. Training prioritization: **safety training always takes precedence over project work**. If a TA needs a machine or work area for training, you will be asked to return at a different time.

3D Printing Safety

FDM printers produce fumes, especially with ABS and specialty filaments. Work in ventilated areas. SLA resin is a skin sensitizer and eye irritant: wear nitrile gloves and safety glasses when handling uncured parts. UV exposure from curing stations can cause eye damage; do not look directly at UV LEDs. Review the InnoHub 3D printing policies in Chapter 12 for print limits, material restrictions, and overnight print approvals.

Lithium Battery Safety

Lithium-ion and lithium-polymer cells present fire and explosion risks if mishandled. Never short-circuit, overcharge, over-discharge, puncture, crush, or overheat cells. Always use a Battery Management System (BMS) for charging and discharge protection. If a cell swells, hisses, or smokes: clear the area, place in a fire-safe container (metal bucket with sand), and notify lab staff immediately. Never charge lithium cells unattended. Always charge in a fireproof LiPo safety bag.

Soldering Safety

Soldering irons operate at 300–400 °C. Return the iron to its stand when not actively soldering. Use a fume extractor or work in a ventilated area. Wash hands after soldering, especially if using leaded solder.

Chemical and Coatings Safety

Adhesives, solvents, paints, and coatings may be flammable, toxic, or sensitizing. Check the Safety Data Sheet (SDS) before use. Spray painting must use the **designated paint booth** in the InnoHub (complete the paint booth pre-quiz on Canvas first). Never spray paint in open lab areas.

Compressed Gas and Pneumatic Systems

Compressed gas cylinders must be secured upright and chained to a wall or cart. Never direct compressed air at a person. Pneumatic systems store energy; relieve all pressure before disconnecting fittings.

Personal Protective Equipment (PPE)

Minimum PPE varies by activity: safety glasses for all shop work, hearing protection near loud machinery, closed-toe shoes in all shop spaces, gloves for chemical handling (not for rotating machinery). Long hair must be tied back; remove dangling jewelry and loose clothing before operating any rotating equipment.

Emergency Procedures

Know the location of the nearest fire extinguisher, first aid kit, eyewash station, and emergency exit *before* beginning any work. In an emergency, call 911. For non-emergency EHS concerns,

contact the InnoHub Operations Manager or Capstone Design TAs.

Part I

The Design Process

Chapter 1

Engineering Design as a Professional Practice

1.1 What Makes Capstone Different

Capstone Design is the bridge between the classroom and professional practice. In previous courses, you solved well-defined problems with known solutions, working individually or in pairs, with a single instructor who provided both the problem and the evaluation criteria. Capstone inverts every one of those conditions. The problem is open-ended, ambiguous, and defined by a real client who may not be an engineer. The solution does not exist yet. You work on an interdisciplinary team where no single person has all the required skills. Your advisors guide but do not prescribe. And the evaluation criteria are not points on a rubric; they are whether your solution actually works, meets the client's needs, and can be justified through engineering analysis.

This shift requires a fundamentally different set of skills: the ability to define the problem before solving it, to manage ambiguity and incomplete information, to negotiate scope with a client, to coordinate technical work across team members, to make engineering judgments under uncertainty, and to communicate results to audiences with varying levels of technical sophistication. These are the skills employers mean when they say they want "engineering judgment," and they are the skills this course develops.

Table 1.1 summarizes the key differences between traditional coursework and capstone.

Table 1.1: Traditional coursework compared to capstone design.

Dimension	**Traditional Coursework**	**Capstone Design**
Problem definition	Given by instructor	Defined by team with client input
Solution	Known; one correct answer	Unknown; multiple valid approaches
Scope	Fixed, well-bounded	Negotiated, evolving
Team size	Individual or pairs	4–6 interdisciplinary members
Timeline	Days to weeks	Two full semesters
Evaluation	Points on a rubric	Client satisfaction, engineering evidence, peer assessment
Stakeholders	One instructor	Client, PA, TA, CF, end users, public
Resources	Provided	Budgeted, procured, managed
Failure consequence	Lost points	Real impact on client, team, and schedule

1.2 The Engineering Design Process

The engineering design process is iterative, not linear. While we present it as a sequence of phases (problem definition → requirements → concept generation → detailed design → build → test), in practice you will revisit earlier phases as you learn more. A test result at CDR may force you to revise a requirement from PDR. A client conversation in Sprint 4 may redefine the problem statement you wrote in Sprint 0. This iteration is normal, expected, and healthy.

The key is to make these iterations *visible and documented*. Every requirement change should be captured in the SDRD with a change rationale. Every design decision should be recorded in the team's project repository with the alternatives considered, the criteria used, and the evidence that supported the choice. The difference between professional iteration and aimless wandering is **traceability**: can you explain what changed, when it changed, why it changed, and what evidence drove the change?

Figure 1.1 illustrates the iterative nature of the process. Notice that the arrows are bidirectional between adjacent phases; feedback from later phases routinely triggers refinement of earlier phases. The outer loop (from testing back to problem definition) represents the rare but important case where testing reveals that the fundamental problem was misunderstood.

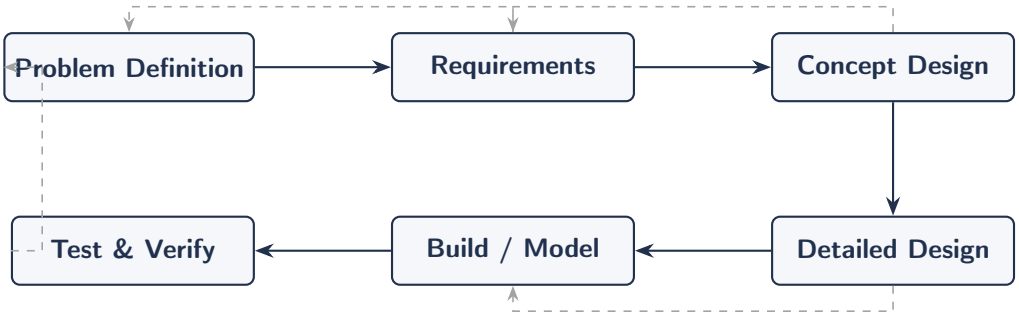

Figure 1.1: The iterative engineering design process. Solid arrows show the primary flow; dashed arrows show feedback loops where later phases trigger refinement of earlier phases.

1.3 The V-Model

The V-Model [2] is the overarching framework for capstone. It connects the process of defining what a system should do (left side) with the process of verifying that it actually does it (right side), with the build phase at the bottom of the V. Figure 1.2 illustrates this structure.

The three design reviews map directly to positions on the V-Model:

- **PDR** (upper left): problem definition, requirements, preliminary architecture. The question at PDR is "*Should* this design work?" You are demonstrating that the problem is understood, the requirements are sound, and the selected concept is feasible.
- **CDR** (lower left to bottom): detailed design, engineering analysis, build planning. The question at CDR is "*Will* this design work?" You are demonstrating that the design is complete enough to build, every component is specified, and analysis confirms feasibility.
- **FDR** (right side): verification results, validation, documentation. The question at FDR is "How *well did* it work?" You are demonstrating test results, comparing performance to requirements, and documenting lessons learned.

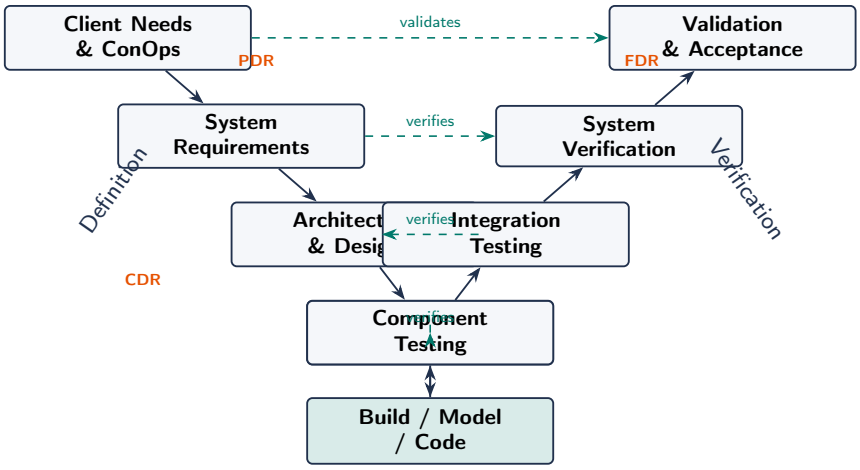

Figure 1.2: The V-Model for capstone design. The left side defines requirements and design; the right side verifies compliance. Dashed horizontal lines show the traceability between definition and verification at each level. Design reviews (PDR, CDR, FDR) are positioned at their corresponding V-Model phases.

If you remember only one thing from this chapter, remember those three questions. They tell you what every deliverable is trying to demonstrate.

1.4 Design Thinking and Systems Engineering

Capstone uses two complementary frameworks. **Design thinking** emphasizes empathy with users, divergent ideation, rapid prototyping, and iterative refinement. It is excellent for understanding needs and generating creative solutions. **Systems engineering** emphasizes requirements traceability, interface control, verification planning, and configuration management. It is excellent for ensuring that creative solutions actually work and can be proven to work.

In practice, design thinking drives the early sprints (understanding the problem, generating concepts, getting client feedback) while systems engineering drives the later sprints (detailed design, analysis, build, and verification). The transition is not abrupt; both mindsets operate throughout the project, but the emphasis shifts as the project matures from exploration to execution.

When to Use Which Framework

Design thinking when you are asking "What should we build?"; user interviews, brainstorming, concept sketches, low-fidelity prototypes, client feedback loops.
Systems engineering when you are asking "How do we prove it works?"; requirements traceability, interface control documents, test procedures, configuration management, verification matrices.
Both are always present. The question is which one is leading at any given moment.

1.5 Stakeholder Identification and Management

Every project has stakeholders beyond the client: end users (who operate the system daily), maintainers (who repair it), regulators (who must approve it), adjacent system owners (whose

systems interact with yours), the public (who may be affected by its operation or failure), and your own team (who must build and document it within the semester).

Identifying stakeholders early prevents late-stage surprises that can derail a project. "The fire marshal says your installation violates egress requirements" or "the maintenance technician cannot access the filter without removing the motor" are real examples from past capstone projects that could have been avoided with early stakeholder mapping.

Create a stakeholder map in Sprint 0 and revisit it at each design review. For each stakeholder, document: who they are, what they care about (their primary concern), how they interact with the system, and what their approval or acceptance means for your project.

Stakeholders You Might Miss

Think beyond the obvious. Who delivers supplies to the site? Who cleans the space where the system operates? Who inherits the project after your team graduates? Who trains new users? Who disposes of the system at end-of-life? Every one of these stakeholders has needs that affect your design.

1.6 The Entrepreneurial Mindset

The KEEN framework emphasizes three C's that are directly relevant to capstone:

Curiosity: investigate the problem space broadly before converging on a solution. Talk to users. Visit the site. Read the standards. Search the patent literature. The best solutions come from deeply understanding the problem, not from rushing to the first idea.

Connections: link your technical knowledge to stakeholder needs and market realities. A technically elegant solution that costs ten times the client's budget or requires skills the end user does not have is not a good solution, no matter how clever the engineering.

Creating Value: design solutions that deliver measurable benefit to someone. Value is not abstract; it is quantifiable. How much time does your solution save? How much does it reduce cost? How much does it improve safety, reliability, or user experience? If you cannot quantify the value, you cannot justify the design.

Capstone is the ultimate expression of the entrepreneurial mindset: you are solving a real problem for a real client, with a real budget and a real deadline, and the value you create is tangible.

1.7 Who's Who in the Course

Understanding the roles in capstone prevents miscommunication and misplaced expectations. Figure 1.3 shows the relationship between the key roles.

Course Faculty (CF) lead lectures, develop professional development modules, and determine final course grades based on information from PAs, clients, and Feedback Loop peer evaluations. CF set the curriculum, ensure grading consistency across sections, and may apply correction factors if grading varies significantly between PA sections.

Project Advisors (PA) are the faculty who guide your team directly. Each PA advises two or more teams. They attend your sprint reviews, grade your deliverables, assess your individual

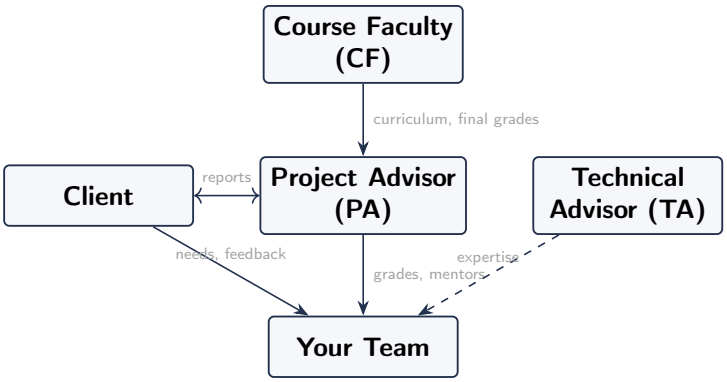

Figure 1.3: Roles and relationships in capstone design. Solid arrows indicate formal reporting/grading relationships; dashed arrows indicate advisory relationships. The team interacts with all four roles throughout the project.

performance, and serve as your primary point of contact for all project-related questions. PAs are practicing or retired engineers and faculty members. They bring real-world engineering experience and professional judgment. Treat your PA meetings as you would a project review with a supervisor in industry: come prepared with an agenda, show progress, and ask specific questions.

Clients are the external sponsors who define the problem. They may be companies, on-campus organizations, governmental agencies, nonprofits, community organizations, or private individuals. The client is the "Product Owner" in Scrum terminology (Chapter 11): they own the Product Backlog and prioritize what the system should do. Your relationship with the client is professional, not academic. They are not grading you; they are depending on you.

Technical Advisors (TA) provide domain expertise that your team may lack. TAs are *not assigned*; your team must identify, approach, and invite them. Faculty across all departments at Mines are available as TAs, and you are strongly encouraged to seek out domain experts relevant to your project. TAs attend design reviews if invited and review major deliverables. Building a relationship with a TA is a professional skill; approach them as you would a consultant in industry.

1.8 Project Tracks

All capstone projects are classified into one of three tracks at the start of SDI. Track classification is proposed by the team, confirmed by the PA, and approved by the client and CF. The classification is documented in the Statement of Work (Chapter 7) and cannot be changed after PDR without written approval from the Capstone Program Director.

Build Track: the project produces a **physical artifact**; a functioning prototype, fabricated assembly, installed system, or constructed element. The SDI proof-of-concept must demonstrate at least one physical function. The SDII outcome must be a fully tested prototype or installation verified against client requirements. Most ME, EE, and some DE/CE projects fall into this category.

Paper Track: the project produces a **non-physical deliverable**; an analytical model, simulation package, site/civil plan set, complete drawing and specification package, policy or systems engineering document, or equivalent knowledge product. Physical construction is outside scope. The SDI proof-of-concept must demonstrate clear analytical or computational goal attainment

(e.g., a validated sub-model or converging simulation). The SDII outcome must be a complete, verified technical package that a qualified engineer could use to implement the design. Most CE, EnvE, and some DE projects fall into this category.

Competition Track: the project is driven by an **external competition event** (SAE Collegiate Design Series, AIAA Design/Build/Fly, ASCE Steel Bridge, Formula SAE, EcoCAR, etc.) with a fixed, externally imposed competition schedule. Competition Track teams complete all requirements of the Build or Paper Track, as appropriate, with milestone dates adjusted to align with the competition calendar. The competition event serves as an additional public evidence point, but **competition outcomes (placement, awards) are not used in grading**. Engineering process and deliverable quality are assessed on the same standards as the corresponding base track.

Track Rigor Equivalence

All three tracks are held to equivalent standards of engineering rigor and professional quality. The Paper Track is not a reduced-effort alternative; it substitutes physical fabrication and testing with *analytical depth, simulation fidelity, and documentation completeness*. Paper Track teams are expected to produce deliverables of sufficient technical depth that a qualified engineer could use them directly to implement or fabricate the design. The Competition Track does not reduce requirements; it reorganizes their timing around an external schedule.

1.9 The Individual Design Log

You are expected to maintain a design log (a bound notebook, dedicated binder, or digital notebook (OneNote, Notion, etc.)) for the entire two-semester sequence. Record your concepts, engineering calculations, research notes, meeting notes, and design decisions with dates.

The design log serves three critical purposes:

1. **Individual contribution evidence**: during Feedback Loop peer evaluations, the design log provides contemporaneous evidence of what you personally contributed. A team member who claims to have done substantial analysis work but has an empty design log will find that claim difficult to support.
2. **Knowledge capture**: calculations, decisions, and design rationale that are not recorded are lost. When you need to explain a design choice at FDR that was made in Sprint 2, the design log is your memory.
3. **Legal and IP record**: in any intellectual property context, a contemporaneous design log with dated entries is a legal document. This may seem distant from your current concerns, but establishing the habit now will serve you throughout your career.

Your PA or client may ask to see your design log at any time, and it may become part of your project deliverables depending on project requirements.

1.10 The Project Repository

Beyond individual design logs, your team must maintain a shared electronic repository containing all notes, documents, informal and formal deliverables, and project materials. The repository is the team's institutional memory: it consolidates individual research into a shared, organized collection that any team member (or your PA, or the client) can access.

Recommended platform: Microsoft Teams group for Senior Design. Mines uses Microsoft 365 and Teams extensively, and creating a Team provides shared file storage, channels for discussion, and integration with other Microsoft tools.

Required organization: at minimum, include folders for Status Reports, Project Management (Sprint Reviews and Retrospectives), Resources and References, Analysis and Calculations, Deliverables, Pictures and Graphics, a PA Folder (shared with your PA), a Client Folder (shared with your client), and project-specific folders as needed.

Defer to your PA and client for preference of platform, format, and additional requirements. Some clients use Google Workspace; others require SharePoint. Adapt to their environment.

1.10.1 The Digital Design Archive

In SDII, the repository culminates in the **Digital Design Archive**: a final, organized submission of all project files. This includes CAD files (SolidWorks, AutoCAD), simulation source files, test results and raw data, code repositories, prototype photographs and videos, all course deliverables, cost/budget spreadsheets, and the project schedule. The archive is submitted during Final Checkout and ensures that the project knowledge persists after your team graduates.

1.11 The Two-Semester Structure

Capstone Design spans two consecutive semesters with distinct phases:

SDI (EDNS 491) focuses on **definition and design**. You progress from project assignment through problem definition, requirements engineering, concept generation and selection, proof-of-concept prototyping, and Preliminary Design Review. SDI is intellectually front-loaded: the quality of your problem definition, requirements, and concept selection in SDI determines the quality of your build in SDII. Sprints 0–6 map to the left side of the V-Model.

SDII (EDNS 492) focuses on **execution and verification**. You re-engage with the client, complete the Critical Design Review, build or model your design, test it against requirements, present at the Final Design Review, and exhibit at the Design Showcase. SDII is execution-loaded: the quality of your CDR analysis, your fabrication discipline, and your testing rigor determine the project outcome. Sprints 7–14 map to the bottom and right side of the V-Model.

The transition between semesters is a critical vulnerability. During the break, momentum is lost, team dynamics shift (members may study abroad, drop the course, or disengage), and clients may have new priorities. The SDII Sprint 7 Re-engagement process (Chapter 2) and the Hazard Assessment and Safety Plan (Chapter 6) are specifically designed to bridge this gap.

The SDI Trap: "We'll Figure It Out in SDII"

The most common SDI failure is deferring hard decisions to SDII. Teams that arrive at PDR with vague requirements, a single unanalyzed concept, and a plan to "figure out the details next semester" consistently produce weak SDII outcomes. The design work you skip in SDI does not disappear; it reappears in SDII as redesign work that competes with fabrication and testing for the same limited schedule. Do the hard work in SDI; execute in SDII.

1.12 Common First-Semester Mistakes

Having taught capstone for multiple semesters, certain patterns recur predictably. Awareness of these patterns can help you avoid them:

Solution-first thinking: jumping to a solution before understanding the problem. You cannot select the right sensor if you have not defined the measurement requirement. You cannot design the enclosure if you have not characterized the operating environment. Resist the urge to start building until the problem is defined.

Vague requirements: "the system shall be reliable" is not a requirement. "The system shall operate continuously for $\geq$720 hours (30 days) without requiring user intervention or maintenance" is. Every vague requirement becomes a fight at FDR when someone asks "how do you know this requirement is met?" and you have no answer.

Single-concept design: generating one concept and defending it instead of generating many concepts and selecting the best. The decision matrix only works if there are real alternatives to compare. Three variations of the same idea are not three concepts.

Ignoring the client: designing what the team thinks is cool instead of what the client needs. Your technical interests are valuable, but the project belongs to the client. If the client needs a simple, robust solution and you deliver a complex, fragile one because it was more technically interesting, you have failed the project.

Procrastinating procurement: every Build Track project has components with lead times longer than one sprint. Identify them in Sprint 1 and order them early. Waiting until Sprint 4 to discover that your specialty motor has a 6-week lead time is a self-inflicted schedule crisis.

Documentation debt: leaving all writing until the week before PDR. The PDR report is not a one-week deliverable; it is the cumulative documentation of 10 weeks of engineering work. If you write as you go (updating the SDRD after each requirements discussion, documenting concept sketches when you create them, recording meeting minutes within 48 hours), the PDR report assembles itself.

Ignoring advisor feedback: your PA has seen dozens of capstone projects. When they raise a concern, take it seriously. The feedback may seem premature ("you have not thought about testability yet and we are only in Sprint 2"), but it is almost always prescient.

1.13 Practice Questions

1. Your client says "I want a device that monitors soil moisture and waters my garden automatically." Identify at least five stakeholders beyond the client. For each, describe what they care about and how it might affect your design.

2. Explain the difference between the left side and right side of the V-Model using a specific example (e.g., a temperature control system). What artifacts are produced on each side, and how does each design review (PDR, CDR, FDR) map to the V?

3. Your team is assigned a Paper Track project involving site grading for a parking lot. A teammate says "Paper Track is easier because we don't have to build anything." Write a response explaining why this is incorrect, with specific examples of what Paper Track rigor requires.

4. Describe a situation where design thinking and systems engineering would recommend different next steps. How do you decide which framework to follow?
5. You are starting a project for a brewery that wants automated grain silo monitoring. Create a stakeholder map with at least six stakeholders, their primary concerns, and one design implication for each.

Chapter 2

Understanding Your Client and Defining the Problem

2.1 The Client Relationship

Your client is not your instructor. They are a professional (or organization) who has a real problem and has engaged Mines to help solve it. The relationship is professional: you represent yourself, this program, and Colorado School of Mines. Respond to emails within 24 hours. Show up on time. Come prepared. Deliver what you promise. Dress appropriately for client meetings (business casual is the minimum). Address clients by their professional title unless they invite you to use first names.

At the same time, the client is not your boss in the way a future employer will be. Your PA and the CF set the academic requirements. When client expectations and course requirements diverge (which happens), your PA is your guide. But in all client-facing interactions, conduct yourself as you would with a paying customer in industry.

The relationship has natural tensions. The client wants maximum scope for minimum cost. You want a manageable scope that allows thorough engineering within the semester. The PA wants to see evidence of engineering rigor and professional growth. Navigating these tensions is a professional skill, and learning to do it well is one of the most valuable outcomes of capstone.

2.2 The Project Expo and Team Formation

SDI begins with the **Project Expo** (Week 1), where available projects are presented by clients, faculty sponsors, or CF. You will learn about each project's scope, client, and technical domain. After the Expo, you complete a **Project Selection Survey** indicating your preferences and relevant skills. Teams of 4–6 students are formed by the CF based on preferences, skills, disciplinary mix, and course logistics.

By the end of Week 2, you will be assigned to a project, a PA, and a recitation section. Note: the recitation location and instructor shown on Trailhead may take several weeks to update after team assignments. Your PA will communicate your actual meeting location.

Choosing a Project Wisely

Pick a project based on technical interest and learning opportunity, not just ease. The projects that challenge you the most produce the strongest resumes. If you have relevant technical skills (e.g., PCB design, FEA, programming), mention them in the survey; they help the CF build

balanced teams. If you have a strong preference for a specific project, explain why in the survey comments.

2.3 The Client Kickoff Meeting

The kickoff meeting (Sprint 0, SDI Weeks 1–2) is your first formal interaction with the client. It sets the tone for the entire project. Figure 2.1 shows the recommended workflow.

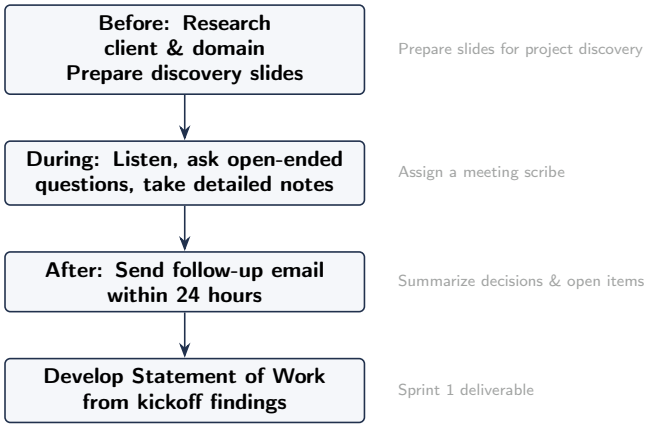

Figure 2.1: Client kickoff meeting workflow. Each phase has specific actions and deliverables.

Before the meeting: research the client's organization, industry, and the problem domain. Prepare a slide deck for project discovery and discussion (this is a Sprint 0 deliverable). Identify what you know, what you do not know, and what you need to learn from the client. Prepare at least 10 open-ended questions. Assign team roles for the meeting: facilitator (drives the agenda), scribe (captures notes and decisions), and timekeeper.

During the meeting: listen more than you talk. The meeting is for learning, not for impressing the client with your ideas. Ask open-ended questions that reveal the real problem:

- "Walk me through how this system is used today."
- "What does success look like to you at the end of this project?"
- "What constraints are absolutely non-negotiable?"
- "What have you tried before? What worked and what didn't?"
- "Who else interacts with this system on a daily basis?"
- "What would happen if we could only deliver 80% of what you want?"

After the meeting: send a follow-up email within 24 hours summarizing key decisions, open questions, and next steps. This email is critical; it serves as a de facto meeting record and gives the client an opportunity to correct misunderstandings while the conversation is still fresh. Copy your PA on this email.

The Mom Test for Client Interviews

Never ask "Would this feature be useful?"; people say yes to be polite. Instead ask about *past behavior*: "When did you last face this problem? What did you do? Why didn't that work?" Past actions predict future needs far better than hypothetical opinions. This principle,

from Rob Fitzpatrick's *The Mom Test* [9], applies to every client conversation throughout the project.

2.4 The Re-engagement Email (SDII)

At the start of SDII (Sprint 7, Week 1), your first task is reconnecting with the client after the semester break. The client has been thinking about other things for a month; you need to bring them back up to speed and re-establish the working relationship.

The Re-engagement Email should contain:

1. Reintroduce the team and note any roster changes.
2. Summarize the current state of the design (what was accomplished in SDI).
3. Outline proposed goals for SDII (aligned with the project completion plan).
4. Request a meeting to review the Product Backlog and confirm priorities for the build/analysis phase.
5. Attach the Group Reflection and Action Plan from the SDII Sprint 7 kickoff workshop.

The re-engagement meeting is also the time to revisit the Product Backlog (Chapter 11). Priorities may have shifted over the break; the client may have new information, new constraints, or revised expectations. Update the backlog before Sprint 8 planning.

2.5 Needs Elicitation

The client tells you what they *want*. Your job is to figure out what they *need*. These are not always the same. A client who says "I want a touchscreen interface" may actually need "quick access to real-time data without leaving the work area"; which might be better served by a heads-up display, voice interface, or even a simple LED indicator panel.

2.5.1 Interviews

Structured conversations with the client and end users. Prepare questions in advance organized by topic. Record the conversation (with permission) or assign a dedicated scribe. After each interview, synthesize the findings into a needs statement: "The operator needs to check system status without interrupting the production process." A need is not a solution; it is a gap between the current state and the desired state.

2.5.2 Observation

Watch how the current system or process is used in its actual environment. Observation reveals unstated needs that interviews miss. When an operator says "the system works fine" but you observe them taping a sensor cable to the frame to prevent it from catching on passing carts, you have discovered a cable management need the operator never thought to mention.

2.5.3 Contextual Inquiry

Combine observation with real-time interviewing. See Section 2.12 for a detailed treatment of contextual inquiry methods and what they reveal.

2.5.4 AI-Assisted Persona Development

When direct stakeholder access is limited (the end user is in a remote facility, the population is too large to interview comprehensively, the project involves future users who do not yet exist, or cultural or language barriers limit direct engagement), AI tools can supplement—never replace—traditional needs elicitation.

How to use AI for persona development: describe the problem domain, the operating environment, and the stakeholder type to the AI. Ask it to role-play as different stakeholder personas: the primary user, the maintenance technician, the safety officer, the purchasing manager, the regulatory inspector. Conduct the same interview questions you would use with real stakeholders. The AI will surface considerations you may not have thought of: accessibility needs, maintenance constraints, regulatory requirements, cultural preferences.

How to document AI-generated personas: for each persona, record: persona name and description, key quotes from the AI interaction, identified needs and pain points, implications for requirements, and a clear disclosure that this persona was AI-generated. Always cross-validate AI-generated needs against real stakeholder data when possible.

Limitations: AI personas reflect training data biases, may miss context-specific constraints, and cannot replace the insight that comes from observing real users in their actual environment. Use AI personas to broaden your thinking, not to shortcut your research. Always disclose AI-generated content per the course GenAI policy (Chapter 22).

WE AI Persona Example

Prompt: "You are a maintenance technician at a wastewater treatment plant in rural Colorado. Our team is designing an automated chemical dosing system. What are your biggest concerns about a new system being installed in your plant?"
Key needs surfaced: easy access for calibration without confined space entry, compatibility with existing SCADA system, alarm notification to personal cell phone (plant is not continuously staffed), resistance to freeze-thaw cycles, clear labeling of all chemical connections to prevent cross-contamination.
Several of these needs would not emerge from a specification-focused engineering discussion but are critical for operational success.

2.6 The Five-Element Problem Statement

Before you jump to solutions, articulate the problem clearly. A vague problem statement leads to a vague design. A precise problem statement constrains the solution space and focuses the team's effort. A good problem statement contains five elements:

1. **WHO** experiences the problem (the specific user or stakeholder, not "people" or "society").
2. **WHAT** is the measurable gap between the current state and the desired state.
3. **WHERE** does the problem occur (physical location, operational context, environmental conditions).
4. **WHY IT MATTERS** (quantified consequences: cost, safety incidents, time lost, environmental impact).
5. **CONSTRAINTS** that limit the solution space (budget, schedule, regulations, physical space, available infrastructure).

If your problem statement implies only one possible solution, you have jumped into the solution space too early. A good problem statement lets someone propose a fundamentally different approach and still be addressing the same problem.

WE Good vs. Bad Problem Statements

Bad: "We need to build a soil moisture monitoring system for a garden." (This is a solution, not a problem.)
Good: "Home gardeners in Colorado's Front Range (WHO) lose 30–50% of their water to over-irrigation because they have no way to assess soil moisture below the surface (WHAT) in their residential garden beds (WHERE). Over-irrigation wastes an estimated 15,000 gallons per household per summer season, contributing to municipal water stress during drought conditions (WHY). The solution must operate on battery power for an entire growing season, cost less than $150 in materials, and withstand freeze-thaw cycles from October through April (CONSTRAINTS)."

2.7 Concept of Operations (ConOps)

The ConOps [32] describes how the system will be used from the user's perspective. It is a narrative, not a specification. Walk through the primary use cases in plain language that the client can review and approve: who operates the system, in what sequence, under what conditions, what inputs they provide, what outputs they expect, and what happens when something goes wrong.

The ConOps bridges the gap between the client's needs and the engineering requirements. A requirement like "the system shall display soil moisture data updated every 15 minutes" does not tell you how the user accesses that data, what they do with it, or what happens if the sensor fails. The ConOps does.

A good ConOps includes:

- **Normal operation**: step-by-step walkthrough of the primary use case.
- **Startup and shutdown**: how the system is powered on, initialized, and safely powered off.
- **Maintenance**: routine maintenance tasks and their frequency.
- **Degraded operation**: what happens when a sensor fails, power is interrupted, or communication is lost.
- **Emergency**: what happens in a safety-critical failure mode and how the user responds.

2.8 Competitive Benchmarking and Prior Art

Before defining your solution, understand what already exists. Competitive benchmarking answers three questions: Who else has tried to solve this problem? What did they build? Where are the opportunities to do better?

Product survey: identify 3–5 competing products or existing solutions. For each, document: name, manufacturer, features, strengths, weaknesses, price point, and user reviews. Synthesize: What features are universally present (table stakes that your design must match)? What features are missing or poorly executed (your opportunity)? What do users complain about most (your value proposition)?

Patent landscape: search Google Patents (https://patents.google.com) for relevant utility patents. Document: patent number, filing date (utility patents expire after 20 years; design patents last 15 years from grant), abstract, and key claims. Expired patents contain detailed engineering you can freely use. Active patents define design boundaries you must avoid. This is not a legal opinion; it is an engineering awareness exercise that reveals prior art and protected approaches.

Standards and regulations: identify applicable codes early. UL/CSA for electrical safety, FCC Part 15 for radiated emissions, IP ratings for environmental protection, building codes for civil projects, EPA regulations for environmental projects, food-contact regulations if applicable. Standards become non-functional requirements in your SDRD. Discovering a regulatory requirement at CDR instead of PDR causes expensive, schedule-breaking redesign.

2.9 Working with Different Client Types

Not all clients are the same, and your approach should adapt to the client type:

External industry clients are professionals with specific, often well-defined problems. They understand their domain deeply but may not speak engineering language. They expect professional communication, regular updates, and deliverables that they can use directly. Their time is valuable; come to every meeting prepared with an agenda and leave with documented action items.

Mines faculty clients are engineering professors who sponsor projects related to their research or teaching needs. They understand the engineering process and can provide detailed technical guidance. However, their expectations for rigor may exceed what a typical external client demands. Faculty clients often care as much about your learning process as the project outcome.

Government and nonprofit clients may have complex approval processes, strict regulatory requirements, and stakeholder structures that are larger than a single point of contact. Budget constraints may be tighter, and the definition of "success" may include equity, accessibility, and community impact dimensions that private-sector clients do not emphasize. Allow extra time for approvals and document everything.

Competition organizers (SAE, ASCE, AIAA, etc.) are not traditional clients but function as the rule-making authority. Competition rules are non-negotiable requirements. The "client meeting" is replaced by the rulebook and the tech inspection at the event. Read the rules carefully and completely before Sprint 1; missed rules surface as failures at tech inspection.

2.10 Documenting Client Interactions

Every client interaction should produce a written record. At minimum:

- **Meeting minutes**: who attended, what was discussed, what decisions were made, and what action items were assigned (with owners and due dates). Distribute within 48 hours. Copy your PA.
- **Email summaries**: after phone calls or video meetings, send a follow-up email summarizing key decisions. This gives the client an opportunity to correct misunderstandings while the conversation is fresh.
- **Decision log**: maintain a running log of all client decisions; feature approvals, scope changes, priority changes, requirement modifications. This log becomes invaluable when, three months later, someone asks "why did we design it this way?"

Documentation protects both you and the client. If a disagreement arises about what was agreed upon, the meeting minutes and decision log are the authoritative record.

2.11 The Statement of Work

The Statement of Work (SOW) is the formal agreement between your team, the PA, and the client on what will be done, by when, and within what constraints. It is the foundation document for the entire project. Chapter 7 covers the SOW structure and content in detail. The SOW is developed during Sprint 1, incorporates findings from the kickoff meeting and needs elicitation, and is submitted to the PA and client for approval. It documents your track classification (Build, Paper, or Competition), team roles and responsibilities, project scope, deliverables, schedule, budget, and assumptions.

2.12 Contextual Inquiry: Observing Users in Their Environment

The richest source of requirements is watching people work in the actual environment where your system will be deployed. Contextual inquiry goes beyond interviews:

What to observe: how do current users perform the task your system will address? What tools do they use? What workarounds have they developed? Where do they make errors? What takes the most time? What frustrates them? What do they do that they do not mention in interviews (because it is so habitual they do not think of it)?

How to conduct: spend 1–2 hours observing the user performing their normal workflow. Take notes (timestamps, actions, observations). Ask clarifying questions only when the observation is ambiguous: "I noticed you checked the gauge twice before adjusting the valve (why?)" Do not suggest improvements during observation; your job is to understand, not to redesign.

What it reveals: contextual inquiry consistently reveals requirements that interviews miss. Users adapt to existing system limitations so thoroughly that they forget to mention them. The maintenance technician who carries a flashlight because the control panel is in a dark corner will not mention "good lighting" as a requirement) but you will notice it.

Capstone application: not all projects allow site visits, but when they do, contextual inquiry is the highest-value use of your time. For projects where site access is limited, ask the client to record a video of their current process. Analyze the video with the same observational rigor.

2.13 Elaborating the Five-Element Problem Statement

The Five-Element Problem Statement introduced in Section 2.5 is worth revisiting at this stage. As you refine your problem statement through client feedback, expand each element with the specificity that comes from deeper understanding:

Sharpen the user: move from "farmers" to "Colorado wheat farmers with <500 acres." The more precisely you define the user, the more precisely you can constrain the solution.

Frame needs, not solutions: "need real-time soil moisture data" not "need a moisture sensor." The need defines the problem; the solution comes later in concept generation (Chapter 4).

Add operational context: "during the growing season (May–September) in semi-arid conditions with limited water rights and intermittent cellular connectivity." Context constrains the solution space in ways the client may not explicitly state.

Quantify consequences: "Crop yields decrease by 15–25% due to over- or under-irrigation, costing the farmer $10,000–25,000 annually." Quantified consequences justify the project investment.

Identify the gap in existing solutions: "Commercial irrigation controllers cost $5,000+ and require reliable cellular service, pricing out small farmers and failing in areas with poor coverage." The gap defines your design target and competitive opportunity.

A complete Five-Element Problem Statement typically runs 3–5 sentences and provides enough context for any engineer to understand why the project matters, who it serves, and what success looks like. Draft it during Sprint 0 and refine it through client feedback in Sprint 1.

2.14 Distinguishing Needs from Wants

During client discussions, carefully distinguish between what the client *needs* (essential for the system to deliver value) and what they *want* (desirable but not essential):

Needs are problems that must be solved for the system to be useful. "I need to know the tank level without climbing the ladder" is a need; without it, the system has no value. Needs become Must-Have requirements.

Wants are features that enhance the system but are not essential. "I'd like the data displayed on my phone" is a want; the system is useful even with a local display. Wants become Should-Have or Could-Have requirements.

Unstated needs are the most dangerous. These are things the client assumes are obvious and does not mention: "Of course it needs to work in the rain (it's an outdoor system." Unstated needs are discovered through contextual inquiry (observing the operating environment), through systematic questioning ("What environmental conditions will the system encounter?"), and through experience (your PA has seen these gaps before).

The "5 Whys" technique: when the client states a want, ask "Why?" five times to uncover the underlying need. Client: "I want a touchscreen display." Why? "So operators can see the data easily." Why is that important? "Because the current gauge is hard to read from across the room." Why? "Because the numbers are small and the gauge is behind a pipe." The real need is not a touchscreen; it is a display that is **readable from 3 meters away in an obstructed sightline**. This need could be met by a large LED display, a remote indicator, or a touchscreen) the touchscreen is one solution among several.

Student Guide Cross-Reference

For the sprint-level checklist of kickoff deliverables and client meeting preparation, see the **Student Guide**, SDI Sprint 0 and Sprint 7 (SDII re-engagement) sections.

2.15 Practice Questions

1. Your client is a local brewery that wants to automate grain silo level monitoring. Write a Five-Element Problem Statement for this project. Ensure that your problem statement does

not imply a specific solution.

2. You are starting SDII after a semester break. Draft a Re-engagement Email to your client that covers the five required elements described in this chapter. Assume the team completed a functional proof-of-concept in SDI.

3. Your competitive benchmarking reveals three existing products that partially solve your client's problem. One product has 4.5 stars on Amazon with 2,000 reviews but costs $800; another has 2.5 stars but costs $150; the third is an academic research prototype with no commercial availability. How do you use this information in your problem statement, requirements, and concept generation?

4. A teammate says "We don't need to do a ConOps; we already know what the system does." Write a response explaining why the ConOps is still necessary, with a concrete example of what it reveals that a requirements list does not.

5. You are conducting a contextual inquiry at a wastewater treatment plant. Describe three specific observations you might make that would reveal unstated needs the operator would not have mentioned in an interview.

Chapter 3

Requirements Engineering

3.1 Why Requirements Matter

Requirements are the contract between the problem and the solution. They are the single most important set of documents in the entire project. A requirement that says "the system shall be fast" is useless; it cannot be tested, and two people will interpret "fast" differently. A requirement that says "the system shall complete a measurement cycle in $\leq$2.0 seconds under normal operating conditions" is testable, unambiguous, and traceable to a client need ("I need results quickly on the production floor").

Poorly written requirements are the single largest source of capstone project failure. If your requirements are vague, your design has no anchor. If they are wrong, your design solves the wrong problem. If they are missing, your design has gaps that surface at FDR when it is too late to fix them. If they are untraceable, you cannot prove that your design addresses the client's actual needs.

The time you invest in writing clear, testable, traceable requirements during Sprint 1–2 will save you ten times that effort during build and test. Every hour spent refining requirements saves 10–100 hours of rework downstream [36]. This is not a platitude; it is an empirically observed pattern across decades of engineering project data.

3.2 The Anatomy of a Good Requirement

Every requirement has four essential components:

1. **Unique identifier**: a short, persistent label that allows traceability through the entire project. Convention: FR-001 (functional), PR-001 (performance), NF-001 (non-functional), IF-001 (interface), SF-001 (safety). The identifier never changes, even if the requirement text is revised.
2. **SHALL statement**: the requirement itself, using "shall" for mandatory requirements and "should" for desirable goals. "The system **shall** measure temperature with an accuracy of $\pm$1.0 °C over the range 0–200 °C." Every word matters; every number must be justified.
3. **Rationale**: why this requirement exists and what client need it traces to. "This requirement ensures that the system meets the client's stated need for laboratory-grade temperature measurement (Need N-003)."
4. **Verification method**: how you will prove compliance; Test (T), Analysis (A), Inspection (I), or Demonstration (D) [2]. Assign the method when you write the requirement.

Words That Kill Requirements

Avoid these words in SHALL statements: "adequate," "appropriate," "approximately," "as needed," "best," "effective," "efficient," "fast," "flexible," "good," "high," "improved," "low," "maximize," "minimize," "optimal," "reasonable," "reliable," "robust," "safe," "significant," "sufficient," "suitable," "timely," "user-friendly." Every one of these is subjective and untestable. Replace them with numbers, units, and specific conditions.

Figure 3.1 shows how requirements flow from client needs through system requirements to design specifications.

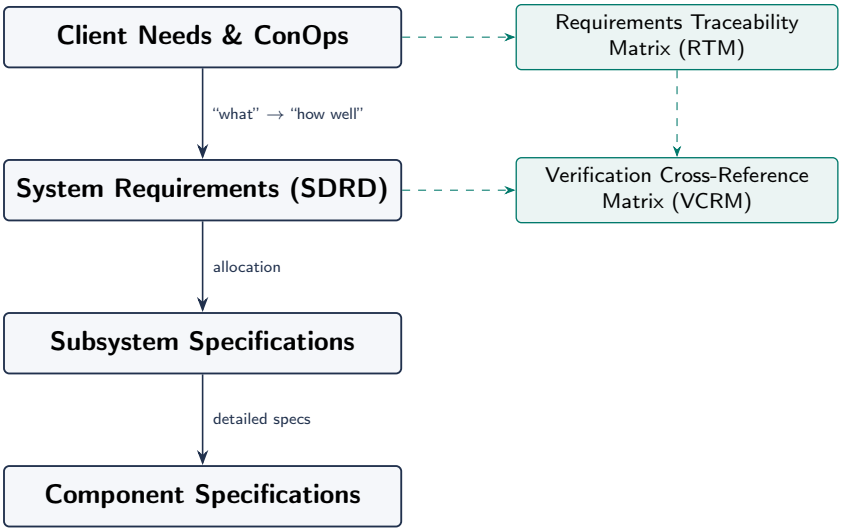

Figure 3.1: Requirements hierarchy and traceability. Client needs flow down through system requirements to component specifications. The RTM and VCRM provide bidirectional traceability and verification tracking at each level.

3.3 Requirement Priority: MoSCoW

Not all requirements are equally critical. Classify each requirement using the MoSCoW framework [10]:

Must Have (M): the system fails without this capability. It is a hard constraint that cannot be negotiated away. If a Must requirement is not met, the project has not succeeded. These become SHALL statements. Example: "The system shall shut off the heater if the temperature exceeds 200 °C." (Safety-critical; non-negotiable.)

Should Have (S): important to the client but the system still functions without it. If schedule pressure forces trade-offs, Should requirements are deferred before Must requirements. These become SHOULD statements. Example: "The system should log temperature data to an SD card at 1-second intervals." (Valuable but not safety-critical.)

Could Have (C): nice-to-have features that add value if time and budget permit. Example: "The system could display a temperature trend graph on a color LCD." (User convenience; not essential to core function.)

Won't Have (W): explicitly excluded from this version. Documenting what you will *not* do is as important as documenting what you will do; it prevents scope creep by making the boundary visible. Example: "The system will not include wireless connectivity in this version."

MoSCoW prevents scope creep by forcing the team and client to agree on what is essential *before* design begins. When schedule pressure mounts (and it will), the MoSCoW classification tells you which features to protect and which to defer. If everything is classified as "Must," either the scope is too small or the team has not been honest about priorities.

3.4 Types of Requirements

Functional requirements describe *what* the system does. Each functional requirement maps to a system function. Example: "The system shall display the current soil moisture level on an LCD screen" (FR-001). "The system shall actuate the irrigation valve when soil moisture drops below the configured threshold" (FR-002).

Performance requirements describe *how well* the system performs its functions. These are quantitative and testable. Example: "The system shall measure soil moisture with an accuracy of ±3% volumetric water content over the range 0–60%" (PR-001). "The system shall respond to a threshold crossing by actuating the valve within 5.0 seconds" (PR-002).

Non-functional requirements describe constraints on the design that are not directly related to specific functions. Categories include:

- **Environmental**: operating temperature range, IP rating, humidity tolerance, UV exposure, vibration, chemical resistance.
- **Regulatory**: UL/CSA listing, FCC compliance, building codes, EPA regulations, ADA accessibility.
- **Safety**: maximum surface temperature, emergency shutoff, overcurrent protection, mechanical guarding, fire resistance.
- **Maintainability**: mean time to repair (MTTR), tool-free access to serviceable components, standardized connectors.
- **Sustainability**: material recyclability, energy efficiency, design for disassembly (see Chapter 5).
- **Interface**: communication protocols (I2C, SPI, UART, CAN, Ethernet), connector types, mounting dimensions, power supply compatibility.
- **Usability**: setup time, training requirements, error message clarity, physical accessibility.

3.5 The System Design Requirements Document (SDRD)

The SDRD is the single authoritative document that captures all system-level requirements [33]. It is the foundational engineering document for the entire project. Standard structure:

1. **Introduction**: system overview, scope, purpose, applicable documents, glossary of terms.
2. **System Description**: ConOps summary, system boundary diagram, external interfaces, operating environment.
3. **Requirements**: organized by category (functional, performance, non-functional). Each requirement includes: unique ID, SHALL/SHOULD statement, rationale, parent traceability (which need it addresses), verification method (T/A/I/D), and MoSCoW priority.

4. **Traceability Matrix**: forward (need → requirement) and backward (requirement → need).
5. **Verification Plan Summary**: method assignments and planned test phases.

The SDRD is a **living document**. It is baselined at PDR (the first official version that the team, PA, and client agree represents the design intent) and updated under configuration management through CDR and FDR. Every change after baseline requires: a description of the change, the reason for the change, the impact on other requirements and design elements, and approval from the PA (and client, if the change affects scope or performance).

Start Your RTM on Day 1

Maintain the Requirements Traceability Matrix from the first client meeting. A simple spreadsheet with columns for Need ID, Need Description, Requirement ID, Requirement Text, Verification Method, Status, and Notes is sufficient. Update it at every team meeting and after every design decision. By FDR, it becomes the single artifact that tells the complete project story. Teams that skip this step spend hours reconstructing traceability for the final report (and the reconstructed version is always less accurate than a continuously maintained one.

3.6 Requirements Traceability

Traceability ensures that every requirement has a **parent** (the client need it addresses) and a **child** (the design element or test that addresses it). Bidirectional traceability means you can answer two questions at any time:

- **Forward**: "For each client need, which requirements address it?" (No need is left unaddressed.)
- **Backward**: "For each requirement, which need justifies it?" (No requirement exists without justification) this catches "gold plating," where the team adds features nobody asked for.)

The RTM is a table linking every requirement to its parent need, its verification method, and its current status. Open the RTM at every team meeting. By FDR, every entry should show: Verified (with evidence reference), Failed (with disposition), Waived (with client approval), or Deferred (with rationale and plan).

3.7 The Verification Cross-Reference Matrix (VCRM)

The VCRM [34] extends the RTM by adding test procedure references and results. For each requirement, the VCRM records:

- Requirement ID and text
- Verification method (T, A, I, or D)
- Test procedure ID (for requirements verified by Test)
- Planned verification phase (PDR, CDR, or FDR)
- Actual result: Open, Verified (with evidence reference), Failed (with disposition), Waived (with approval)

The VCRM evolves from a **plan** at PDR (all entries Open, methods assigned) to a **partially complete record** at CDR (analysis verifications complete, some test results available) to a **final scorecard** at FDR (all entries dispositioned). A VCRM with mostly "Open" entries at FDR indicates insufficient verification effort.

3.8 Engineering Standards

Identify applicable engineering standards early in the requirements process. Standards become non-functional requirements that are often non-negotiable. Common categories for capstone projects:

Domain	Common Standards
Electrical safety	UL/CSA, IEC 61010 (lab equipment), NEC (wiring)
EMC	FCC Part 15 Class B (residential), IEC 61000 (EU)
Environmental protection	IP ratings (IEC 60529), NEMA enclosure ratings
Structural	AISC (steel), ACI 318 (concrete), ASCE 7 (loads)
Civil/Environmental	local building codes, EPA regulations, stormwater permits
Materials compliance	RoHS, REACH, California Prop 65
Accessibility	ADA (Americans with Disabilities Act)
Pressure vessels	ASME BPVC

Discovering a regulatory requirement at CDR rather than PDR is one of the most expensive mistakes in capstone. A single missed standard can invalidate an entire design approach and force a restart with weeks remaining in the semester.

3.9 Track-Specific Requirements

Build Track: requirements emphasize physical performance (force, speed, accuracy, temperature range), environmental survivability (IP rating, vibration, thermal shock), safety (emergency shutoff, fusing, mechanical guards), and interface compatibility (connectors, protocols, mounting). Verification is primarily by Test and Demonstration.

Paper Track: requirements emphasize analytical fidelity (mesh convergence criteria, numerical error bounds, model assumptions documented), validation criteria (comparison to experimental data or published benchmarks with stated acceptance thresholds), documentation completeness (sufficient detail for a qualified engineer to implement the design), and simulation performance (convergence time, computational resource requirements). Verification is primarily by Analysis and Inspection.

Competition Track: requirements include all Build or Paper requirements *plus* competition rules as mandatory constraints. Competition rules are non-negotiable SHALL requirements that take precedence over team preferences. Schedule requirements must account for the competition calendar with margin for delays. Registration deadlines, tech inspection criteria, and safety requirements from the competition rulebook become formal requirements in your SDRD.

3.10 Requirements-Architecture Iteration

Requirements and architecture co-evolve. Initial requirements drive a candidate architecture. That architecture reveals implementation constraints (weight, power budget, thermal limits, cost) that feed back as new requirements or modified performance targets. This iteration is normal and healthy, but it must be documented. Each iteration produces a revised SDRD version with change rationale.

Convergence test: when two consecutive iterations change fewer than 5% of requirements, the design is stable enough to baseline at PDR. If requirements are still changing rapidly at PDR, the

design is not ready for review.

3.11 Requirements Change Management After Baselining

The SDRD is baselined at PDR; meaning the team, PA, and client agree that this version represents the design intent. After baselining, *every change must be formally documented*. A change request includes:

1. **Change description**: exactly what is being changed (old text → new text).
2. **Rationale**: why the change is needed (test results, client request, feasibility issue, cost constraint).
3. **Impact assessment**: which other requirements, design elements, test procedures, and schedule milestones are affected by this change.
4. **Approval**: PA signature. If the change affects scope or performance, client approval is also required.

Maintain a change log in the SDRD. At FDR, the change log tells the story of how the requirements evolved from baseline to final state.

Teams that skip change management arrive at FDR with a design that does not match their requirements document. This mismatch makes the VCRM meaningless (you are verifying against outdated requirements) and undermines the credibility of the entire project documentation.

3.12 Common Requirements Writing Mistakes

These mistakes appear in capstone SDRD documents every semester. Avoid them:

Solution masquerading as a requirement: "The system shall use an Arduino Mega 2560" is a design decision, not a requirement. The requirement might be "the system shall provide at least 8 analog input channels and 20 digital I/O pins with 5 V logic levels." Multiple MCUs could meet this requirement; selecting the Arduino is a design choice that should be justified in the architecture section, not hardcoded in the SDRD.

Untestable requirement: "The system shall be user-friendly." How do you test this? With what acceptance criteria? A testable alternative: "A first-time user shall be able to complete the primary measurement procedure without assistance in under 5 minutes, as verified by timed demonstration with 3 naive users."

Compound requirement: "The system shall measure temperature with $\pm 1\,^{\circ}\mathrm{C}$ accuracy and display the value on an LCD and log it to an SD card." This is three requirements bundled into one. If the temperature measurement works but the SD card logging fails, is this requirement met or failed? Split compound requirements into individual, independently verifiable statements.

Orphan requirement: a requirement with no parent need. If you cannot trace a requirement back to a specific client need, it is either gold-plating (adding features nobody asked for) or an implicit need that should be explicitly captured in your needs analysis.

Missing non-functional requirements: teams focus on functional and performance requirements and forget environmental constraints (operating temperature, IP rating), power constraints (maximum power consumption, battery life), maintainability requirements (access for filter replacement, calibration interval), and interface requirements (connector types, communication pro-

tocols). These omissions surface at CDR or FDR as "I didn't think of that" moments that require expensive redesign.

Requirement inflation: specifying tighter performance than the client needs. If the client needs ±5 °C and you specify ±0.5 °C, you have increased cost, complexity, and risk by an order of magnitude for no benefit. Write requirements that meet the need, not requirements that showcase your ambition.

3.13 Requirements Review Checklist

Before submitting your SDRD for PDR, verify every requirement against this checklist:

- **Unique ID**: does it have a persistent identifier (FR-001, PR-001, NF-001)?
- **SHALL/SHOULD**: is the obligation level clear? Must-have = SHALL; Should-have = SHOULD.
- **Single requirement**: does it state exactly one verifiable condition? If "and" appears, consider splitting.
- **Testable**: can you write a test procedure (or analysis, inspection, demonstration) with specific pass/fail criteria?
- **Quantitative**: does it include numbers and units where applicable?
- **No ambiguous words**: does it avoid "adequate," "appropriate," "fast," "reliable," "user-friendly," and other subjective terms?
- **Traceable upward**: does it trace to a specific client need?
- **Traceable downward**: is a TAID verification method assigned?
- **MoSCoW assigned**: is it classified as Must, Should, Could, or Won't?
- **Rationale documented**: does the rationale explain why this requirement exists?
- **Not a design solution**: does it specify *what* the system must do, not *how* to do it?

If any requirement fails this checklist, revise it before PDR. A SDRD full of vague, untestable, untraced requirements is the strongest predictor of a struggling project.

The Requirements Peer Review

Before PDR, have your PA or another team review your SDRD. Fresh eyes catch ambiguities that the authoring team has become blind to. Ask the reviewer: "Can you write a test for this requirement based solely on its text?" If they cannot, the requirement needs revision.

3.14 Integrating Engineering Standards into Requirements

CLO 3 requires you to identify appropriate engineered systems standards and incorporate them into the design. Standards appear in the SDRD as non-functional requirements:

Identifying applicable standards: start with the project domain. Electrical products: IEC 61010 (safety), IEC 61000 (EMC), UL 508A (industrial control panels). Mechanical products: ASME Y14.5 (GD&T), ASME B31.1 (power piping), ASTM material standards. Software: IEC 62304 (medical device software), DO-178C (aviation software). Environmental: EPA regulations, ISO 14001. Construction: ASCE 7 (structural loads), IBC (building code), ADA (accessibility).

Incorporating standards into requirements: do not simply list the standard in an appendix. Extract the specific clauses that apply to your design and write them as testable requirements. Example: instead of "The system shall comply with IEC 61010," write "The system shall withstand a 1500 VAC hipot test between all mains-connected conductors and the enclosure for 60 seconds without breakdown, per IEC 61010 Section 6.3." The former is unverifiable; the latter has a specific test procedure.

Standards as design constraints: some standards constrain the design space rather than specifying performance. "All food-contact surfaces shall be 304 stainless steel or FDA-approved polymer, per FDA 21 CFR 177" eliminates certain materials from consideration. Capture these as constraints in the SDRD.

Where to find standards: Mines Library provides access to many standards through the ASTM Compass and IHS Standards databases. IEEE standards are available through IEEE Xplore. ASME, SAE, and ASCE standards may require purchase; consult your PA about department copies. For capstone, referencing the relevant standard and extracting the applicable requirements is sufficient; you do not need to purchase every standard cited.

3.15 Building the Traceability Matrix

The traceability matrix links every requirement to its origin (client need) and its verification (TAID method and test procedure). Construct it incrementally:

Forward traceability (needs → requirements): for each client need identified during elicitation, show which requirement(s) address it. A need that has no corresponding requirement is an unmet need that should be either addressed (add a requirement) or explicitly documented as out of scope. A requirement that traces to no need is gold-plating and should be questioned.

Backward traceability (requirements → verification): for each requirement, show how it will be verified. The VCRM provides this mapping. A requirement with no verification method will arrive at FDR without evidence of compliance (this is a documentation failure that undermines the entire verification effort.

The allocation matrix (requirements → architecture): show which subsystem or component implements each requirement. This mapping reveals: requirements that are not allocated to any subsystem (orphans) who is responsible?), subsystems that implement too many requirements (complexity risk; single point of failure), and requirements that span multiple subsystems (interface requirements that need ICDs).

Present the traceability matrix as a table in the PDR report. It need not be a separate document; it can be columns in the SDRD itself: Need ID | Requirement ID | SHALL Statement | TAID | Allocated Subsystem | Status.

3.16 Requirements Elicitation Techniques Beyond Interviews

While client interviews are the primary elicitation method, supplement them with:

Questionnaires and surveys: useful when the client represents a larger organization and you need input from multiple stakeholders who cannot all attend meetings. Design questions that are specific and closed-ended ("On a scale of 1–5, how important is wireless connectivity?") to enable

quantitative comparison across respondents. Follow up with interviews for the highest-priority responses.

Document analysis: review existing documentation; previous designs, user manuals, maintenance logs, incident reports, regulatory filings, and competitor product specifications. These documents reveal requirements that stakeholders forget to mention because they take them for granted.

Competitive benchmarking: identify 3–5 existing products or systems that address similar needs. Create a feature comparison matrix. Identify: what do all competitors do (baseline expectations), what do the best competitors do (stretch targets), and what does no competitor do (potential differentiators). This analysis grounds your requirements in market reality.

Use case analysis: write detailed use cases ("The operator walks up to the system, presses the START button, waits for the green LED, reads the display value, and records it in the log") that trace the complete user interaction. Each step in the use case generates interface requirements, performance requirements (how fast does the display update?), and environmental requirements (is the display readable in direct sunlight?).

3.17 Practice Questions

1. Your client says "The system needs to be waterproof." This is not a testable requirement. Rewrite it as a proper SHALL statement with a specific IP rating, test method, and rationale. Then write the corresponding VCRM entry.
2. Create a mini-SDRD for a battery-powered environmental monitoring station deployed in a Colorado mountain meadow. Write at least three functional requirements, three performance requirements, and three non-functional requirements. Assign a verification method (TAID) and MoSCoW priority to each.
3. Your team has 30 requirements in the SDRD. Your PA says the design is too ambitious for one semester. Using MoSCoW, categorize the requirements and identify which Could/Won't requirements to defer. What criteria do you use, and how do you communicate the reduced scope to the client?
4. Write the forward and backward traceability entries for the following chain: Client Need N-005 ("operator must know current tank level without leaving the control room") → Requirement FR-012 ("the system shall transmit tank level data to the control room display via Ethernet") → Test Procedure TP-012 ("verify data transmission from tank to display").
5. Your requirements include "The system shall be safe." Rewrite this as three specific, testable non-functional requirements covering electrical safety, mechanical safety, and thermal safety.

Chapter 4

Concept Generation, Selection, and Early Design

4.1 Divergent Before Convergent

The most common mistake in concept generation is converging too quickly on a single idea. The first idea that comes to mind is rarely the best; it is merely the most obvious. Effective concept generation requires a deliberate **divergent phase** (generate many ideas without judgment) followed by a **convergent phase** (evaluate and select systematically). Resist the temptation to skip the divergent phase because you think you already know the answer.

Research in design methodology consistently shows that teams generating more concepts produce better final designs, even when the early concepts are impractical. The process of exploring impractical ideas exposes design constraints and triggers creative connections that would not emerge from a single-concept approach. Generate at least 15–20 concepts before evaluating any of them. The PDR requirement is to present a minimum of three distinct, developed concepts with graphical communication.

4.2 Ideation Techniques

4.2.1 Brainstorming

The classic group ideation method [26]. Rules: (1) defer all judgment (no criticizing ideas during generation, (2) encourage wild ideas) they trigger useful associations, (3) build on others' ideas ("yes, and…" not "no, but…"), (4) go for quantity over quality. A productive brainstorming session generates 30–50 ideas in 20 minutes. Most will be impractical; that is the point. The few gems are worth the pile of duds.

Tips for effective sessions: set a timer (time pressure increases output), designate a facilitator who enforces the rules, use sticky notes or a whiteboard so ideas are visible to everyone, and do not allow laptops during brainstorming (they invite distraction and reduce participation).

4.2.2 SCAMPER

SCAMPER [27] is a structured creativity prompt that forces you to transform existing solutions. For each existing solution or concept, systematically ask:

- **S**ubstitute: can you replace a component, material, or process?
- **C**ombine: can you merge two functions, systems, or ideas?

- **A**dapt: can you borrow a solution from a different domain or industry?
- **M**odify / Magnify / Minimize: can you change the size, shape, or intensity?
- **P**ut to another use: can you use this in a different context?
- **E**liminate: can you remove a component or step without losing function?
- **R**earrange / Reverse: can you change the sequence, orientation, or direction?

SCAMPER is particularly useful when you have one decent concept but need alternatives for the decision matrix. Apply each letter to your existing concept and see what emerges. Apply SCAMPER to your top 2–3 concepts from initial brainstorming to generate 10–20 variations.

4.2.3 Morphological Chart

The morphological chart [28] is the most systematic ideation technique. Decompose the system into independent functions (rows) and list multiple solution approaches for each function (columns), as shown in Table 4.1. Then systematically combine solutions across functions to generate novel system-level concepts.

Table 4.1: Example morphological chart for a soil moisture monitoring system.

Function	**Option A**	**Option B**	**Option C**	**Option D**
Sense moisture	Capacitive probe	Resistive probe	TDR sensor	Tensiometer
Power source	Solar + battery	AA batteries	USB wired	Energy harvesting
Data transmission	WiFi (ESP32)	LoRa radio	Bluetooth	Wired (RS-485)
User display	Phone app	LCD on unit	LED indicators	Web dashboard
Enclosure	IP67 molded	3D-printed PLA	NEMA 4X box	Epoxy potted

A chart with 5 functions and 4 options per function yields $4^5 = 1{,}024$ possible combinations. Not all are viable, but the chart reveals combinations that no single brainstorming session would produce. Select 3–5 promising combinations for further development.

4.2.4 Biomimicry

Biomimicry [29] studies how nature solves analogous problems. Natural systems have been optimized by billions of years of evolution for efficiency, resilience, and adaptability. Ask: "How does nature handle this function?" Examples: gecko feet for reversible adhesion, shark skin for drag reduction, termite mounds for passive ventilation, bone structure for lightweight high-strength materials, spider silk for tensile strength, elephant ears for large-surface-area heat exchangers, Namibian beetle for fog harvesting through surface wettability gradients, and woodpecker skull for layered energy absorption (note: recent research [Van Wassenbergh et al., 2022, *Current Biology*] has challenged this interpretation, finding the skull is rigid and does not absorb shock; always verify biomimicry claims against current literature). The AskNature database (`https://asknature.org`) catalogs biological strategies organized by function. Biomimicry is particularly useful for structural, thermal, fluid, and materials challenges, and often produces concepts that are lighter, more efficient, and more elegant than conventional engineering solutions.

4.3 Graphical Communication of Concepts

Every concept must be communicated visually. A verbal description of a concept is insufficient; your PA, client, and teammates need to *see* how it works. For each concept, produce:

- An **annotated sketch** showing the major components, their arrangement, and how they interact. Label key features, approximate dimensions, and critical interfaces.
- A **block diagram** showing the functional architecture: inputs, processing elements, outputs, and signal/energy/material flows between them.
- Optionally, a **CAD model** if the concept is sufficiently developed. CAD is not required at this stage; clear hand sketches with proper annotations are fully acceptable for concept selection.

The PDR deliverable requires graphical communication of at least three distinct concepts. "Distinct" means fundamentally different approaches, not minor variations of the same idea. Three versions of the same concept with different sensors are not three concepts; they are one concept with a sensor trade study.

4.4 Concept Screening: Pugh Matrix

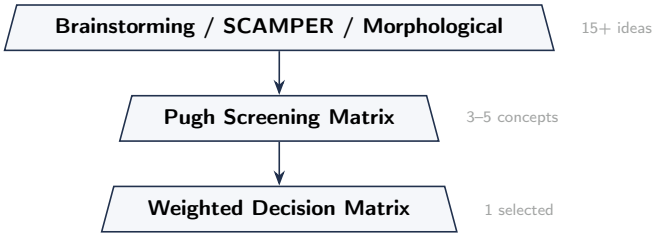

Figure 4.1: The concept selection funnel. Divergent thinking generates many ideas; systematic screening and scoring converge to a single, justified selection.

The concept selection process (Figure 4.1) is deliberately structured as a funnel: wide at the top (many ideas) and narrow at the bottom (one justified selection). The width of thinking at the top determines the quality of the selection at the bottom. Teams that start with 3 concepts and select one have not explored the design space adequately; teams that start with 15+ and screen to 3–5 before scoring have genuine confidence in their selection.

The Pugh matrix [3] is a quick, qualitative screening tool for reducing a large set of concepts to a manageable shortlist. Steps:

1. Select one concept as the **datum** (baseline for comparison). Choose a middling concept, not the one you think is best.
2. Define evaluation criteria from your requirements (performance, cost, manufacturability, schedule risk, maintainability, safety, sustainability).
3. Rate each concept against the datum on each criterion: better (+), same (S), or worse (−).
4. Sum the pluses and minuses for each concept.
5. Concepts with the most pluses advance to detailed evaluation. Concepts with many minuses are eliminated unless they have a single outstanding advantage that might be combinable with another concept.

The Pugh matrix is fast and rough. It does not produce a final answer; it reduces 10 concepts to 3–4 for the weighted decision matrix.

4.5 Concept Selection: Weighted Decision Matrix

The weighted decision matrix is the formal, quantitative selection tool presented at PDR. It provides a transparent, defensible basis for the design direction.

1. **Define criteria** from your requirements: technical performance, cost, manufacturability, schedule risk, maintainability, sustainability, safety, client preference. Each criterion must be independently assessable.
2. **Assign weights** to each criterion based on relative importance. Weights should sum to 1.0 (or 100%). Weights are derived from MoSCoW priorities and client input. Must-have requirements get higher weights than Should-have.
3. **Rate each concept** on each criterion using a consistent scale (1–5 or 1–10). Anchor the scale: 1 = does not meet the minimum, 3 = meets the requirement, 5 = exceeds the requirement significantly. Rate based on evidence (analysis, prototyping, data), not gut feeling.
4. **Calculate scores**: multiply each rating by the criterion weight and sum across criteria. The concept with the highest weighted score is the recommended selection.
5. **Sensitivity analysis**: vary the weights by ±10–20% and check whether the ranking changes. If a small change in one weight flips the winner, the decision is sensitive to that criterion and needs more data or discussion before committing.

The Decision Matrix Is Not a Black Box

The decision matrix is a tool for organizing your engineering judgment, not a substitute for it. If the matrix recommends a concept that your team's collective experience says is problematic, examine the criteria and weights; the matrix is revealing *where* the disagreement lies. This enables a focused technical discussion rather than an opinion battle. Present the matrix at PDR, explain the sensitivity analysis, and be prepared to defend the selection with engineering arguments.

4.6 Client Approval of the Selected Approach

Before proceeding to detailed design, present your selected concept (or top 2–3 with a recommendation) to the client for approval. This occurs in SDI Sprint 3 (Weeks 7–8). The client reviews the trade study, asks questions, and either approves the direction or requests modifications.

Client approval is not a formality; it is a **scope commitment**. Once the client approves a concept, subsequent changes to the fundamental approach trigger a scope change that must be documented in the SOW with the reason, the impact on schedule and budget, and re-approval from the PA and client. This is why the trade study must be thorough: changing direction after Sprint 3 costs dearly.

4.7 Proof-of-Concept Prototyping

A proof-of-concept (PoC) is not a finished product or even a complete prototype. It is a targeted demonstration that the **riskiest aspect** of your design is feasible. Focus the PoC on retiring the highest-risk technical uncertainty identified in your risk register. Three types:

Looks-like prototype: demonstrates the physical form, user interface, or ergonomics. Often built from 3D prints, foam core, or cardboard. It does not function; it helps the client and team

evaluate the physical design before committing to expensive fabrication.

Works-like prototype: demonstrates the critical function using bench-level hardware that may not resemble the final form. Example: a breadboard circuit that reads the correct temperature and displays it on a serial monitor, built before the PCB is designed or the enclosure is fabricated. The goal is proving the *function*, not the *form*.

Integration prototype: demonstrates that two or more subsystems work together across an interface. Example: the sensor subsystem communicating with the MCU via I2C and displaying data on the LCD. This proves that the interface definition (protocol, voltage levels, timing) is correct.

Choose the PoC type that retires the most risk. For Build Track projects, the SDI PoC must demonstrate at least one physical function. For Paper Track projects, the PoC must demonstrate clear analytical or computational goal attainment (e.g., a validated sub-model, a converging simulation, or a model that reproduces known benchmark results).

4.8 Risk Identification and Management

Risk management is not pessimism; it is preparation. Identify risks early so you can mitigate them before they become schedule-breaking crises.

4.8.1 Risk Matrix

Plot each identified risk on a Likelihood (1–5) vs. Impact (1–5) grid. Risks in the upper-right quadrant (high likelihood, high impact) require active mitigation before PDR. Risks in the lower-left (low likelihood, low impact) can be monitored with less urgency. The risk matrix is a visual communication tool; present it at every design review.

4.8.2 Risk Register

A living document listing each risk with: unique ID, description, likelihood rating, impact rating, risk score (L×I), mitigation strategy, responsible team member, status (Open, Mitigated, Closed), and notes. Review the register at every sprint retrospective. Add new risks as they emerge; close risks that have been mitigated.

4.8.3 FMEA: Failure Modes and Effects Analysis

FMEA [15] is a systematic risk analysis technique. For each component or function in your design: how can it fail (failure mode)? What happens to the system if it fails (effect)? Three ratings:

- **Severity (S)**: 1–10. How bad is the effect? (1 = negligible, 10 = safety hazard without warning.)
- **Occurrence (O)**: 1–10. How likely is this failure mode? (1 = extremely unlikely, 10 = near-certain.)
- **Detection (D)**: 1–10. How likely is the failure to be detected *before* it causes harm? (1 = always detected, 10 = undetectable.)

Risk Priority Number: RPN $= S \times O \times D$. Maximum RPN = 1000. Prioritize mitigation for the highest RPNs. In capstone, a simplified FMEA covering the top 10–15 failure modes is expected. Focus on: safety-critical failures (fire, shock, injury), mission-critical failures (system

cannot perform its primary function), and high-probability failures (connectors, batteries, moving parts, firmware bugs).

4.8.4 Hazard Analysis

For safety-critical projects, a formal Hazard Analysis identifies each hazard, assesses severity and probability, and defines controls (engineering controls, administrative controls, PPE). The Hazard Analysis feeds directly into the SDII Hazard Assessment and Safety Plan (Chapter 6).

4.9 Architecture Definition

Once the concept is selected and approved, define the system architecture: how the system is organized into subsystems and how they interact.

Functional decomposition: break the system into functions using verb-noun pairs ("measure temperature," "transmit data," "actuate valve," "display status"). Functions are stable; they describe *what* the system does regardless of *how* it is implemented. A function tree with 3–4 levels of decomposition is typical for capstone projects.

Allocation matrix: map each leaf-level function to a physical module (subsystem, PCB, software module, mechanical assembly). This mapping reveals integration complexity: modules that implement many functions are high-risk single points of failure. Modules that share functions with other modules require careful interface definition.

Interface Control Documents (ICDs): for each interface between subsystems, define what crosses the boundary (signals, power, data, fluid, mechanical force), in what format (voltage level, protocol, data rate, connector type), and with what constraints (maximum latency, impedance, temperature rating). Clear ICDs are what enable parallel development; team members can work on their subsystems independently if the interfaces are well-defined.

4.10 Documenting the Trade Study

The trade study (concept generation, screening, selection, and sensitivity analysis) is a PDR deliverable. It must be presented as a narrative engineering argument, not just a table of numbers. The trade study section of your PDR report should include:

1. **Criteria definition**: where did the evaluation criteria come from? Trace each criterion to a requirement or client need. Justify the weights: why is cost weighted 25% and performance 35%? The weights reflect the client's priorities and the project's constraints.
2. **Concept descriptions**: for each concept, provide a clear graphical representation (sketch, block diagram, or CAD rendering), a text description of how it works, a summary of its key advantages and disadvantages, and an honest assessment of its technical risk.
3. **Screening results**: present the Pugh matrix showing which concepts advanced and which were eliminated. Explain why eliminated concepts failed.
4. **Selection results**: present the weighted decision matrix with scores. Show the arithmetic. Highlight the winning concept and its margin of victory.
5. **Sensitivity analysis**: show which weight variations change the outcome. Interpret the results: "The selection is robust to ±20% changes in all criteria except manufacturability; if manufacturability weight exceeds 30%, Concept B becomes preferred. Since our client has emphasized

ease of manufacture, we will specifically address manufacturability risk in the detailed design phase."

6. **Recommendation**: state your recommendation clearly and acknowledge the residual risks of the selected concept.

The Trade Study Is Not a Formality

Reviewers at PDR can tell when the trade study was performed retroactively to justify a decision already made. Signs: criteria that perfectly align with the pre-selected concept, no sensitivity analysis, and competitors rated suspiciously low on subjective criteria. A genuine trade study may surprise you; the best concept is not always the one you initially favored.

4.11 Prototyping Materials and Methods

Different prototyping objectives call for different materials and methods:

Cardboard and foam core: fast, cheap, excellent for form-factor mockups, user interface layouts, and spatial planning. Build a full-scale cardboard model of your enclosure before committing to CAD. It reveals ergonomic issues and spatial conflicts that are invisible on screen. Time: hours. Cost: nearly zero.

3D-printed PLA: the default prototyping material. Good dimensional accuracy (± 0.3 mm), adequate strength for low-stress applications, and available for free from InnoHub (1,000 g per team). Limitations: brittle under impact, poor heat resistance (>55 °C deformation), poor UV resistance for outdoor use.

3D-printed PETG: better chemical resistance and heat tolerance than PLA. Slightly more flexible (less brittle). Good for functional parts that will see moderate stress or outdoor exposure. Available at cost from InnoHub.

3D-printed TPU: flexible filament for gaskets, grips, vibration isolators, and compliant mechanisms. Requires a direct-drive extruder; not all InnoHub printers support it.

Laser-cut acrylic or wood: excellent for flat parts, panels, enclosure faces, and structural frames. Fast turnaround. Dimensional accuracy better than FDM (± 0.1 mm in-plane). Acrylic is transparent; useful for visual access to internals.

CNC-machined aluminum: for structural parts requiring high strength, tight tolerances (± 0.05 mm), or metal-to-metal interfaces. More expensive and slower than 3D printing. Requires InnoHub training.

Breadboard circuits: for all initial electronics prototyping. Never solder a circuit that has not been verified on a breadboard first (unless you enjoy desoldering). Breadboard limitations: high parasitic capacitance, unreliable connections at high frequency (>1 MHz), and maximum current $\sim$1 A per bus.

The general progression: cardboard → 3D-printed mockup → breadboard circuit → functional prototype → custom PCB + machined/printed enclosure. Each stage retires a specific type of risk: form factor → spatial fit → electrical function → integrated function → final implementation.

4.12 Advanced Ideation Techniques

Beyond traditional brainstorming and the morphological chart, these techniques help generate concepts that break out of conventional thinking:

Patent search: before committing to a concept, search the US Patent Office (patents.google.com) for prior art. This serves two purposes: (1) it reveals existing solutions that may inspire your design or provide proven sub-systems, and (2) it identifies potential IP conflicts that could constrain your design freedom. In capstone, patent infringement is not a legal concern (educational use), but understanding the patent landscape demonstrates professional awareness.

Reverse engineering: if a similar product exists, obtain one and disassemble it. Document: what materials are used, how components are fastened, what manufacturing processes were employed, what design decisions were made and why (infer from the physical evidence). This is not copying; it is learning from existing solutions to inform your own. Document your reverse engineering findings in the concept generation section of the PDR report.

4.13 From Concept to Architecture: The Decomposition Process

Once a concept is selected, decompose it into an implementable architecture:

Functional decomposition: break the system function into sub-functions. A "temperature controller" decomposes into: sense temperature, compare to setpoint, compute control output, drive heater, display status, log data, and alarm on fault. Each sub-function maps to a physical implementation.

Physical decomposition: map each sub-function to a hardware or software module. "Sense temperature" maps to: thermocouple + amplifier + ADC. "Compute control output" maps to: PID algorithm in ESP32 firmware. "Drive heater" maps to: MOSFET + PWM output.

Interface definition: at every boundary between modules, define the signal, its format, and its constraints. Between "sense temperature" and "compute control output": the interface is a digital value in degrees Celsius, updated at 1 Hz, with a range of 0–500 °C and resolution of 0.25 °C. These interface specifications become interface requirements in the SDRD and ICDs in the CDR documentation.

4.14 Intellectual Property Awareness in Concept Generation

During concept generation, be aware of intellectual property (IP) considerations:

Patent awareness: searching patents.google.com reveals existing solutions and their claims. If your concept closely resembles a patented design, you have three options: (1) design around the patent claims (modify your concept to avoid the specific claimed elements), (2) use the patented concept (in capstone, educational use is generally protected, but discuss with your PA if the client intends to commercialize), or (3) contact the patent holder for licensing (rare in capstone but relevant for commercialization-track projects).

Client-owned IP: some clients require that all IP generated during the project belongs to the client. Others allow shared ownership or assign IP to the university. The IP agreement should be documented before the project begins (typically handled by the Capstone Program through the

SOW or a separate IP agreement). Understand the terms: can you include the design in your portfolio? Can you publish the results? Can you use the techniques you developed in future work?

Open-source components: many capstone projects use open-source hardware (Arduino, ESP32 ecosystem) and software (libraries, frameworks). Understand the license terms: MIT and BSD licenses allow commercial use with attribution. GPL licenses require derivative works to also be open-source. If the client intends to commercialize, GPL components may create licensing complications.

Documenting inventive contributions: if your design includes a novel approach that could be patentable, document it in your engineering notebook with: the date of conception, a clear description, the names of all contributors, and a witness signature. This establishes priority. Discuss potential patents with your PA and the Mines Office of Technology Transfer.

Student Guide Cross-Reference

For the sprint-level deliverable checklists and common mistakes during concept generation and selection, see the **Student Guide**, SDI Sprint 2 and Sprint 3 sections.

4.15 Practice Questions

1. Create a morphological chart for a portable water quality testing station with at least 5 functions and 3 solutions per function. How many total combinations exist? Select three promising combinations and explain why each might work.
2. Your weighted decision matrix selects Concept A, but when you increase the weight of "manufacturability" by 15%, Concept B wins. What does this tell you, and what should you do before committing to a direction?
3. Your team has selected a concept and the client has approved it in Sprint 3. Two weeks later, a teammate discovers a patent that may be infringed. What are your options, and how do you communicate this to the client and PA?
4. Create a simplified FMEA (5 failure modes minimum) for a solar-powered irrigation controller. Identify the highest-RPN failure mode and propose a mitigation strategy that would reduce the RPN by at least 50%.
5. Explain the difference between a "looks-like," "works-like," and "integration" prototype. For a project involving a robotic arm that picks up objects and sorts them by color, which type of PoC would retire the most technical risk, and why?

Chapter 5

Sustainability, Equity, and Societal Impact Assessment

5.1 Why This Chapter Exists

Engineering does not happen in a vacuum. Every design decision has consequences beyond the technical specification: materials are extracted from the earth, energy is consumed in manufacturing, products are used by people with different abilities and economic circumstances, and eventually everything becomes waste. This chapter equips you to evaluate those consequences systematically using industry-standard tools and to integrate sustainability and equity into your requirements rather than treating them as afterthoughts.

CLO 5 requires you to evaluate your design using a project-appropriate tool (LEED, ENVISION, DFE, or the Engineering for Social Justice checklist). CLO 15 requires you to identify and analyze the ethical, environmental, societal, and economic impacts of your engineering intervention. This chapter covers both.

These are not abstract exercises. In professional practice, sustainability assessments are increasingly required by clients, regulators, and investors. The ability to evaluate environmental and social impacts alongside technical performance is a core engineering competency, not an optional add-on.

5.2 The Triple Bottom Line

The Triple Bottom Line (TBL) framework [30], illustrated in Figure 5.1, evaluates sustainability across three dimensions that must be balanced simultaneously:

People (social equity): who benefits from this design? Who bears the costs or risks? Are workers safe during manufacturing and maintenance? Can users of all abilities access the system? Does the design reinforce or reduce existing inequities?

Planet (environmental stewardship): what natural resources does the design consume? What emissions, waste, or habitat disruption does it produce? Can materials be recycled at end-of-life? Is the design energy-efficient during the use phase (which typically dominates lifecycle energy for consumer and industrial products)?

Profit (economic viability): can the design be manufactured, operated, and maintained at a cost the market or client can sustain? What is the total lifecycle cost (not just the build cost)? Does the design create economic value that justifies its resource consumption?

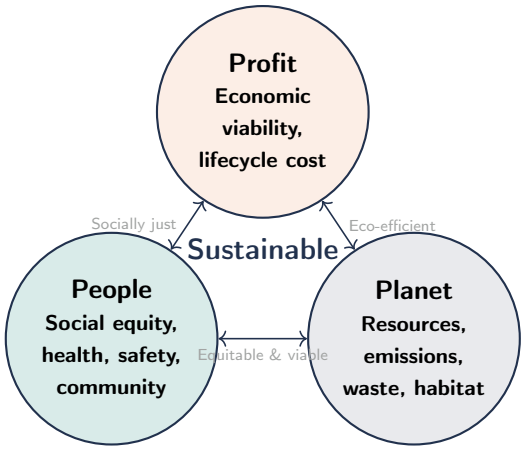

Figure 5.1: The Triple Bottom Line (TBL). Sustainable engineering designs balance social equity, environmental stewardship, and economic viability. A design that maximizes one dimension while ignoring the others is not sustainable.

A design that maximizes profit while dumping toxic waste is not sustainable. A design that minimizes environmental impact at ten times the client's budget will never be implemented and therefore has zero impact. Sustainability requires balancing all three dimensions within the constraints of your project.

5.3 Lifecycle Thinking

Every product has a lifecycle: raw material extraction → material processing → manufacturing → distribution → use phase → end-of-life (disposal, recycling, or reuse). **Cradle-to-grave** thinking accounts for impacts from extraction through disposal. **Cradle-to-cradle** thinking [31] goes further, designing products so that end-of-life materials become inputs for new products, creating a closed loop.

At capstone scale, a full quantitative Lifecycle Assessment (LCA) using tools like openLCA or SimaPro is typically beyond scope and budget. However, you can and should perform a **simplified lifecycle assessment**:

1. Identify the dominant lifecycle phase for your product category. For most consumer and industrial electronics, the use phase dominates energy consumption (the product uses more energy over its lifetime than was consumed in manufacturing). For construction materials and infrastructure, the materials phase dominates embodied energy and carbon.
2. Estimate the environmental impact of that dominant phase using published data (embodied energy databases, utility emission factors, material impact factors).
3. Design to reduce the impact of the dominant phase. If the use phase dominates, focus on energy efficiency. If the materials phase dominates, focus on material selection and material reduction.

The 80/20 Rule of Sustainability

In most products, 80% of the environmental impact comes from 20% of the design decisions. Find the dominant contributor; usually the energy source, the primary structural material, or the single most resource-intensive component; and focus your sustainability effort there. Optimizing the color of the enclosure for recyclability while ignoring a 500 W continuous power draw misses the point.

5.4 Sustainability Evaluation Tools

5.4.1 LEED (Leadership in Energy and Environmental Design)

LEED [17] is the most widely used green building rating system in the world. It evaluates buildings and infrastructure across categories: Sustainable Sites, Water Efficiency, Energy and Atmosphere, Materials and Resources, Indoor Environmental Quality, and Innovation. Projects earn credits in each category; total credits determine the certification level (Certified, Silver, Gold, Platinum).

Even if your project is not a building, the LEED categories provide a structured framework for evaluating environmental impact. Use LEED when your project involves building systems, HVAC, lighting, water management, or site development. For capstone, apply the relevant LEED categories as a checklist: does your design address water efficiency? Energy consumption? Material selection? Indoor air quality?

5.4.2 ENVISION

ENVISION [18] is the sustainability rating system for civil infrastructure projects: roads, bridges, water treatment systems, energy systems, transit, and public spaces. It evaluates projects across five categories: Quality of Life (purpose, community, well-being), Leadership (collaboration, planning, economy), Resource Allocation (materials, energy, water), Natural World (siting, conservation, ecology), and Climate and Resilience (emissions, resilience).

Use ENVISION for civil and environmental engineering projects. It is particularly relevant for CE and EnvE capstone teams working on infrastructure, water systems, and site development.

5.4.3 Design for Environment (DFE)

DFE is a design philosophy that systematically considers environmental impacts as an integral part of the engineering design process. Key DFE strategies:

- **Minimize material variety**: fewer material types simplifies recycling. A product with 15 different plastics is nearly impossible to recycle economically; one with 2–3 types is viable.
- **Select lower-impact materials**: prefer recycled content, renewable materials, and materials with lower embodied energy. Aluminum has 8× the embodied energy of steel per kg but is infinitely recyclable.
- **Design for disassembly (DfD)**: use reversible fasteners (screws, snap-fits) instead of permanent bonds (adhesives, welding, overmolding). The requirement "the system shall be fully disassembled into its base material categories using only standard hand tools in under 5 minutes" is testable by demonstration at FDR.
- **Minimize packaging**: reduce material, use recycled content, design for reusable shipping.

- **Design for energy efficiency**: during the use phase, which is typically the dominant lifecycle phase for powered products.
- **Avoid hazardous materials**: eliminate or minimize lead, mercury, cadmium, hexavalent chromium, PBBs, and PBDEs (these are restricted by RoHS in the EU and many other jurisdictions).

DFE is applicable to all project tracks and all engineering disciplines. Even Paper Track projects should address DFE in their design package recommendations.

5.4.4 Engineering for Social Justice (ESJ) Checklist

The ESJ framework asks questions that traditional engineering analysis often overlooks:

- **Who benefits** from this design? Are the benefits distributed equitably, or do they accrue primarily to those who already have advantages?
- **Who bears the costs or risks?** Are the environmental, health, or safety burdens imposed on communities that did not have a voice in the design decision?
- **Who was not consulted?** Were all affected stakeholders engaged in the needs elicitation process? Were voices systematically excluded by language barriers, geographic isolation, or power dynamics?
- **Accessibility**: can people with different physical abilities, cognitive abilities, and technology access use this system? Does the design comply with ADA requirements where applicable?
- **Affordability**: can the intended users actually afford this solution? A technically excellent irrigation controller that costs $2,000 is useless to a community garden with a $200 budget.
- **Cultural appropriateness**: does the design respect local customs, practices, and preferences? A solution designed in Golden, Colorado may not transfer directly to a community in rural Guatemala.

These are not abstract philosophical questions. They are practical design considerations that affect whether your engineering solution actually solves the problem for the people who need it most. A water treatment system designed without consulting the community it serves may be technically excellent and socially useless.

Application in capstone: ESJ is particularly relevant for: water treatment systems deployed in underserved communities, agricultural technology for small farms, assistive devices for people with disabilities, infrastructure improvements in low-income areas, and environmental monitoring in communities disproportionately affected by pollution.

Avoiding the "savior" mindset: engineering teams sometimes approach community-focused projects with the assumption that they know what the community needs. ESJ pushes back: the community is the expert on its own needs. Your role is to bring technical skills to problems defined by the community, not to impose solutions from the outside.

Practical actions: (1) include community members as stakeholders in the needs elicitation process (Chapter 2), (2) design for the actual operating context (available infrastructure, local skills, cultural practices), not for an idealized context, (3) consider the total cost of ownership (not just purchase price but installation, training, maintenance, and disposal), (4) plan for technology transfer (the community must be able to operate and maintain the system without your ongoing involvement).

5.4.5 Choosing the Right CLO 5 Tool

The following table summarizes when to use each sustainability assessment tool. Select the tool that best matches your project's domain; you are not required to use all four.

Tool	Best For	Evaluation Categories	Output Format
LEED	Buildings, HVAC, lighting, water, site development	Sustainable Sites, Water Efficiency, Energy & Atmosphere, Materials & Resources, Indoor Environmental Quality, Innovation	Credit-based rating (Certified through Platinum)
ENVISION	Civil infrastructure: roads, bridges, water treatment, transit, public spaces	Quality of Life, Leadership, Resource Allocation, Natural World, Climate & Resilience	Credit-based rating (Verified through Platinum)
DFE	All product-scale projects (ME, EE, DE); any project with a physical deliverable	Material selection, disassembly, energy efficiency, hazardous materials, packaging, end-of-life	Qualitative checklist with testable requirements
ESJ	Projects serving underserved communities or with significant equity dimensions	Access, affordability, cultural appropriateness, community engagement, power dynamics	Qualitative checklist with stakeholder input

Many projects benefit from combining tools: a building project might use LEED for environmental performance and ESJ for community impact. A consumer product might use DFE for material and energy decisions and ESJ for accessibility and affordability.

5.5 Integrating Sustainability into the SDRD

Sustainability requirements belong in your SDRD as non-functional requirements, not in a separate "sustainability" appendix that gets ignored during design. Examples of testable sustainability requirements:

- "The system shall be fully disassembled into base material categories using only standard hand tools in under 5 minutes." (TAID: Demonstration)
- "The system shall consume $\leq$50 W average power during normal operation." (TAID: Test)
- "All structural materials shall be RoHS-compliant." (TAID: Inspection of material certifications)
- "The system shall use $\leq$ 2 gallons of water per irrigation cycle." (TAID: Test)
- "The enclosure shall be manufactured from $\geq$50% post-consumer recycled content." (TAID: Inspection of supplier certification)

By embedding sustainability in the requirements, it becomes part of the normal verification process and cannot be overlooked.

5.6 Design for End-of-Life

How will your system be decommissioned? Every product eventually reaches end-of-life, and the designer's choices determine whether that end-of-life is responsible or irresponsible. Consider:

- **Disassembly**: can the system be taken apart for recycling, or is it a monolithic block of mixed materials? (DfD, discussed above.)
- **Hazardous materials**: are there lithium batteries, solvents, lead, or other materials requiring special disposal? Who is responsible for disposal, and what regulatory requirements apply?
- **Component reuse**: can any subsystems (motors, sensors, controllers, enclosures) be reused in other applications or returned to the supply chain?
- **Documentation**: the O&M manual (delivered at FDR) should include end-of-life instructions; how to safely decommission the system, what materials require special handling, and where to recycle or dispose of each component.

5.7 Simplified LCA Methodology for Capstone

A full quantitative Lifecycle Assessment requires specialized software (openLCA, SimaPro, GaBi) and access to lifecycle inventory databases that are typically beyond capstone scope. However, you can perform a meaningful **simplified LCA** that demonstrates lifecycle thinking and satisfies CLO 5:

Step 1: Define the system boundary. What is included? Typically for capstone: raw materials, manufacturing/fabrication, use phase, and end-of-life. Exclude transportation (unless your project involves logistics) and installation (unless your project is an installed system).

Step 2: Identify the dominant lifecycle phase. Research published LCA data for products similar to yours:

- **Electronics and consumer products**: the use phase typically dominates energy consumption. A device that consumes 50 W continuously for 5 years uses ∼2,200 kWh; far more energy than was consumed in manufacturing the device. Focus on energy efficiency.
- **Construction materials and infrastructure**: the materials phase dominates embodied energy and carbon. Concrete produces ∼0.13 kg CO_2/kg; steel produces ∼1.8 kg CO_2/kg; aluminum produces ∼8.1 kg CO_2/kg (values from the ICE Database; verify against current data for your materials). Focus on material selection and material reduction.
- **Vehicles and mobile systems**: the use phase dominates (fuel consumption). Focus on weight reduction and powertrain efficiency.
- **Short-lived consumer goods**: the materials and end-of-life phases dominate. Focus on recyclability and material selection.

Step 3: Estimate the dominant impact. Use published data to estimate the primary environmental metric for your dominant phase. For energy consumption: estimate annual kWh and multiply by the local grid emission factor (Colorado: ∼0.6 kg CO_2/kWh). For material impact: estimate mass of each material and multiply by the embodied energy or CO_2 factor from published databases (e.g., Granta Edupack, ICE Database).

Step 4: Identify design levers. What design changes would reduce the dominant impact? Can you reduce power consumption by 30% with a more efficient algorithm or a sleep mode? Can you reduce material mass by 20% with topology optimization or material substitution? Can you

increase product lifetime from 3 years to 10 years, spreading the embodied energy over a longer useful period?

Step 5: Document. Present your simplified LCA in the PDR and FDR reports. Show the system boundary, the dominant phase, the estimated impact, the design levers you exercised, and the resulting improvement. Even a rough estimate demonstrates that you considered environmental impact systematically.

5.8 Environmental and Regulatory Compliance

Beyond voluntary sustainability assessment, your design may need to comply with mandatory environmental regulations:

RoHS [44] (Restriction of Hazardous Substances): restricts lead, mercury, cadmium, hexavalent chromium, PBBs, and PBDEs in electrical and electronic equipment sold in the EU. Most modern electronic components are RoHS-compliant; verify by checking datasheets.

REACH (Registration, Evaluation, Authorization, and Restriction of Chemicals): EU regulation requiring registration of chemical substances. Relevant if your project uses specialty chemicals, coatings, or adhesives.

WEEE (Waste Electrical and Electronic Equipment): EU directive requiring manufacturers to provide for collection and recycling of electronic products at end-of-life. Relevant for understanding your product's full lifecycle responsibility.

EPA regulations: applicable to projects involving water treatment, air emissions, hazardous waste, or environmental monitoring. State and local regulations may be more restrictive than federal.

California Proposition 65: requires warning labels for products containing chemicals known to cause cancer or reproductive harm. Relevant if your client is in California or sells products there.

Identify applicable regulations during requirements engineering (Chapter 3) and incorporate them as non-functional requirements in the SDRD. Discovering a regulatory compliance issue at CDR or FDR is expensive and may require fundamental design changes.

5.9 The Broader Impacts Essay

In SDII, you will write an argumentative essay (1000–1500 words) analyzing the broader impacts of your specific capstone project on society, the environment, or ethics. This is not a generic essay about "engineering and society." It must connect directly to *your* project's hardware, software, or system design decisions.

The essay follows a standard argumentative structure:

1. **Introduction and Thesis**: hook the reader with a compelling opening. State your central claim clearly in the first paragraph. The thesis must be arguable; someone could reasonably disagree with your position.
2. **Background**: briefly explain the technical context of your specific project (not a generic overview of the technology domain). The reader should understand what your system does and how it works in 2–3 paragraphs.

3. **Proponent Arguments**: present the arguments supporting your thesis, buttressed by literature, data, and specific examples from your project. This is the core of the essay.
4. **Opposing Viewpoints**: present the strongest counterarguments. You must take a stand; this is an argumentative essay, not a balanced report. Present the opposition honestly, then refute it.
5. **Rebuttal**: address the opposing viewpoints with evidence and reasoning that supports your thesis.
6. **Conclusion**: restate your thesis in light of the arguments presented. Discuss implications for future design decisions or policy.

Common Essay Mistakes

The most common issues in student essays are:

1. The essay does not follow argumentative format; reads like a book report or informative summary.
2. Generic or "robotic" tone that lacks a distinct human voice or specific connection to the project, suggesting over-reliance on generative AI.
3. No opposing viewpoints presented or challenged.
4. Sources not cited, claims not supported by evidence, or references that do not exist ("hallucinated" AI citations; this is an academic integrity violation).
5. Spelling, grammar, and typographical errors suggesting lack of proofreading.
6. The essay lacks a thesis or central argument stated in one of the first sentences.

The best essays are provocative, well-researched, and deeply connected to the specific engineering decisions of the project.

5.10 Case Studies

Consider how sustainability and equity have played out in real engineering decisions:

Flint, Michigan water crisis: a cost-cutting decision to change water sources without adequate corrosion control exposed a predominantly low-income, majority-Black community to lead contamination. Engineers who raised concerns were overruled. The ESJ framework would have flagged the equity dimensions before the switch.

Solar panel end-of-life: first-generation solar panels approaching end-of-life contain cadmium, lead, and other hazardous materials. Panels designed without DfD are expensive and dangerous to recycle. Newer designs use DfD principles and less-hazardous materials.

Fast fashion supply chains: products engineered for lowest cost shift environmental and labor burdens to communities with less regulatory protection. A lifecycle perspective reveals costs hidden from the consumer but borne by factory workers and local ecosystems.

These are not hypothetical. Your capstone project (however small in scale) participates in the same system of material flows, energy consumption, and human impacts. The sustainability tools in this chapter help you evaluate and improve your contribution to that system.

The Broader Impacts Essay (SDII)

The Broader Impacts Essay is a 1000–1500 word argumentative essay on societal impacts connected to your project. The essay structure and common mistakes are covered in Section 5.10 of this chapter. For academic integrity policies related to the essay, see Chapter 22. The sustainability assessment tools in this chapter provide the analytical foundation for the essay.

5.11 Equity Impact Assessment

Even projects that do not directly serve underserved communities can have equity implications. Assess:

Access: who can use the system? Does it require technical skills, physical abilities, or financial resources that exclude certain populations? Can it be made more accessible without compromising performance?

Data and privacy: does the system collect personal data? Who owns the data? How is it protected? Could the data be used in ways that disadvantage certain groups?

Environmental justice: does the system's production, operation, or disposal disproportionately affect marginalized communities? (Example: battery production creates mining impacts in specific regions; electronic waste is often exported to developing countries.)

Economic impact: does the system create or eliminate jobs? If it automates a task currently performed by workers, what happens to those workers? Is the economic benefit distributed equitably?

Document your equity assessment in the sustainability section of the PDR and FDR reports. Even if the assessment concludes that equity impacts are minimal, demonstrating that you considered the question is important for CLO 5 compliance and professional development.

5.12 Design for Recyclability and Circular Economy

Beyond end-of-life considerations, design your system to participate in the circular economy:

Material labeling: mark all plastic parts with the resin identification code (the recycling triangle with a number). This helps recyclers sort materials correctly. PLA: Code 7 (Other). PETG: although technically in the PET family, PETG contaminates standard PET (Code 1) recycling streams; dispose of as Code 7 (Other) or check local recycling guidelines. ABS: Code 7 (Other). Acrylic: Code 7 (Other).

Minimize material variety: a product made from 3 materials is easier to recycle than one made from 12 materials. Where functionally equivalent, choose the same material for multiple components. An enclosure made entirely from PLA with brass inserts is more recyclable than one made from PLA, acrylic, aluminum, and nylon.

Avoid permanent bonding where possible: screws and snap-fits allow disassembly; adhesives and welds make disassembly difficult or impossible. Design for mechanical fastening as the default; use adhesives only when the joint requires it (sealing, vibration damping, thin material bonding).

Battery accessibility: if the system contains a rechargeable battery, design the enclosure so the

battery can be removed and replaced by the user. Batteries degrade over the system's lifetime and are hazardous waste; they must be separable from the rest of the product for proper disposal.

Modular electronics: design the PCB for component-level repair rather than board-level replacement. Use socketed ICs where practical (DIP sockets for through-hole, or SMD sockets for BGA/QFP) so failed components can be replaced without discarding the entire board.

WE Simplified LCA: Battery-Powered Environmental Monitor

System: ESP32-based temperature and humidity monitor deployed in a greenhouse. Battery powered (3.7 V, 6000 mAh LiPo), PLA enclosure, solar charging panel. 3-year deployment.
Dominant phase: materials (solar panel and battery represent >80% of embodied energy).
Design lever: extending deployment life from 3 to 10 years reduces annualized embodied energy by 70%. Alternatively, a smaller or higher-efficiency solar panel reduces the dominant material contribution.
End-of-life: LiPo battery requires certified electronics recycler. PLA enclosure is commercially compostable. Solar panel recycled through e-waste program.
This rough analysis took 30 minutes and revealed that panel selection and deployment lifetime are the most impactful design decisions; insight that a purely performance-focused analysis would miss.

Student Guide Cross-Reference

For the sprint-level checklist covering the Ethics Assignment and sustainability deliverables, see the **Student Guide**, SDI Sprint 4 section. For the Broader Impacts Essay, see SDII Sprints 9–11.

5.13 Practice Questions

1. Your team is designing a solar-powered irrigation controller for a community garden in a low-income neighborhood. Using the ESJ checklist, identify at least four equity considerations that should influence your design decisions. For each, propose a specific design response.

2. Write three sustainability-focused non-functional requirements for a battery-powered environmental monitoring station deployed in a Colorado mountain meadow. Make each requirement testable (specify the TAID method).

3. A teammate argues that sustainability analysis is unnecessary because "we're just building a prototype, not a production product." Construct a counterargument with at least three specific points.

4. Compare LEED and ENVISION: for what types of projects is each most appropriate? Could a single project benefit from applying elements of both? Give an example.

5. Your project uses a lithium-polymer battery. Evaluate the end-of-life implications using DFE principles. What design decisions can you make now (during CDR) that will make responsible end-of-life management easier?

Chapter 6

Verification, Validation, and Test Planning

6.1 Verification vs. Validation

These terms [34, 2] are frequently confused, even by practicing engineers. The distinction matters because they answer fundamentally different questions and require different evidence.

Verification answers "Did we build the system right?" (does the system meet its specified requirements as written in the SDRD? Verification compares the system's measured performance to the requirements document. It is an engineering activity performed by the design team.

Validation answers "Did we build the right system?") does the system satisfy the client's actual needs in its intended operating environment? Validation compares the system's behavior to the client's expectations and the ConOps. It requires the client's involvement and often occurs in the real operating context.

A system can pass all verification tests and still fail validation. Example: your temperature controller meets every requirement in the SDRD (verification passes), but the client discovers that the mounting location you assumed is inaccessible in their facility, making the system unusable (validation fails). The requirements were correct; the ConOps was incomplete. Verification is necessary but not sufficient; validation closes the loop between what was built and what was needed.

Figure 6.1 illustrates how verification and validation activities map to the project timeline.

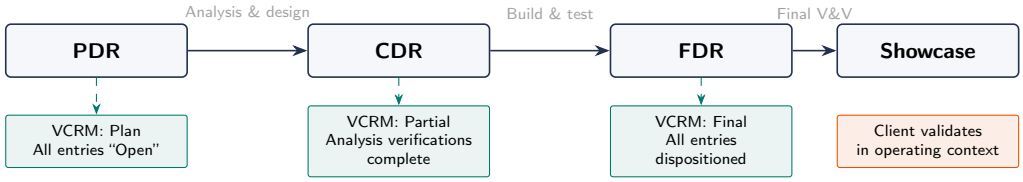

Figure 6.1: VCRM lifecycle through the project milestones. The VCRM evolves from a plan at PDR to a partially complete record at CDR to a final scorecard at FDR. Validation occurs when the client assesses the system in its intended context.

6.2 The Four Verification Methods (TAID)

Not every requirement is verified by running a physical test. The four methods [2] provide a spectrum of rigor and cost:

Test (T): measure performance under controlled conditions against quantitative pass/fail criteria. Test is the gold standard (it produces hard data) but it is also the most resource-intensive method. Example: "Apply a 5.0 V reference signal to the input. Measure the output with a calibrated multimeter. PASS if the output is 3.30 ± 0.05 V. FAIL if outside this range." Use Test for critical performance requirements where analysis alone is insufficient.

Analysis (A): verify by calculation, simulation, or modeling. Example: "Verify structural safety factor $\geq$2.0 by FEA under maximum load case per ASCE 7." Analysis is appropriate when physical testing is impractical (extreme conditions, destructive tests on expensive hardware), when the physics is well-understood and the model has been validated, or when the requirement is a design margin that can be verified by calculation. Analysis should be documented in the Engineering Calculation Package (CDR deliverable).

Inspection (I): verify by visual or dimensional examination without operating the system. Example: "Verify all structural materials are RoHS-compliant by inspecting material certifications and supplier datasheets." Also used for dimensional tolerances (measure with calipers), workmanship standards (visual inspection of solder joints), and label compliance (verify safety labels are present and legible). Inspection is the least expensive method but provides the least information about actual performance.

Demonstration (D): verify by operating the system through its use cases without quantitative measurement. Example: "Demonstrate that the system can be fully disassembled using only standard hand tools in under 5 minutes. Time the procedure." Demonstration is appropriate for functional and operational requirements where the pass/fail criterion is binary (it works or it does not) or where quantitative measurement is impractical.

Assign a verification method to each requirement *when you write it*. If you cannot assign a TAID method, the requirement is probably not testable and needs to be rewritten with specific, measurable acceptance criteria.

TAID Method Selection Guide

Method	Use When...	Evidence
Test (T)	Quantitative performance must be measured	Data, plots, statistical analysis
Analysis (A)	Physical test is impractical or insufficient alone	Calculations, simulation output
Inspection (I)	Compliance is verified by examination	Photos, datasheets, measurements
Demonstration (D)	Function is binary: works or does not	Video, witness signatures, time logs

6.3 Writing Test Procedures

A test procedure is a step-by-step recipe that allows someone who did not design the system to verify a requirement independently [2]. Every test procedure should include:

Header: test procedure ID (TP-001), associated requirement IDs (FR-003, PR-007), date written, author, revision number.

Purpose: which requirement(s) this test verifies, stated explicitly by ID.

Equipment: all instruments, fixtures, power supplies, calibration standards, and software needed, with model numbers and calibration status where relevant. If a specific instrument is required (e.g., a calibrated Pt100 RTD reference), state its accuracy and calibration date.

Setup: how to configure the system and test environment, including initial conditions. Diagrams are strongly encouraged; a wiring diagram showing how the test instruments connect to the DUT (device under test) prevents setup errors.

Preconditions: what must be true before the test begins. Examples: "System has been powered on for ≥15 minutes to reach thermal equilibrium." "Ambient temperature is $22 \pm 3\,°C$." "Firmware version is v2.1.3 or later."

Steps: numbered, imperative instructions. Each step should produce an observable or measurable result. Be specific: "Set the bench supply to 12.0 V, 500 mA limit" not "Turn on the power supply." Number every step so you can reference them in the results.

Acceptance criteria: the quantitative or qualitative pass/fail threshold for each measured parameter. "PASS if the measured temperature is within $\pm 1.0\,°C$ of the reference at all 5 test points. FAIL if any single test point deviation exceeds $1.0\,°C$." The acceptance criteria must trace directly to the requirement.

Safety: any hazards associated with the test and the precautions required. "High voltage present at test point TP3. Do not touch exposed terminals. Use insulated probes."

Data recording: what to record (raw measurements, calculated values, environmental conditions), in what format (table, datasheet, electronic log), and where to store it (project repository, lab notebook). Raw data files should be saved in the test data directory of the project repository with a naming convention: `TP-PR-004_run01_2026-04-10.csv`.

Results: actual measured values, calculated error, and pass/fail determination. Sign and date.

The Test Procedure Is a Legal Document

In professional practice, test procedures and their results are legal evidence of design compliance. They may be reviewed by regulators, auditors, or opposing counsel in litigation. Write test procedures as if someone with no knowledge of your project will read them five years from now and need to understand exactly what was tested, how, and what the results mean. In capstone, this standard of clarity is what distinguishes professional engineering documentation from lab reports.

6.4 The Engineering Calculation Package

In SDII, the CDR deliverables include an **Engineering Calculation Package**: a formal, documented set of analyses that demonstrate design feasibility. This package may include:

- Hand calculations (structural stress, thermal analysis, circuit analysis, fluid flow, motor sizing)
- Simulation results (FEA, CFD, electromagnetic, circuit simulation)
- Modeling outputs (system dynamics, control system response, energy balance)

Every calculation must: (1) state its purpose (which requirement or design question it addresses), (2) list all assumptions explicitly, (3) show the complete derivation or methodology, (4) present results with appropriate units and significant figures, (5) draw a conclusion ("the safety factor is 3.2, which exceeds the required 2.0; the design is structurally adequate"), and (6) cite its sources (equations from textbooks, material properties from datasheets, load cases from standards).

The Engineering Calculation Package is not a collection of MATLAB screenshots. It is a narrative engineering document that walks the reader through the analysis logic, demonstrates that the team understands the physics, and provides sufficient detail for an independent engineer to reproduce the results.

6.5 VCRM Management Through the Lifecycle

The VCRM evolves through three distinct phases:

At PDR: the VCRM is a **plan**. Every requirement has an assigned verification method (TAID) and a planned verification phase (PDR, CDR, or FDR), but no results yet. Most entries show "Open." The VCRM at PDR demonstrates that you have thought about how you will prove compliance for every requirement.

At CDR: the VCRM is **partially complete**. Analysis-based verifications (A) should be complete, with references to the Engineering Calculation Package. Some test results (T) may be available from subsystem testing. Inspection-based verifications (I) for component selection should be complete. The VCRM shows a mix of "Verified (by Analysis)," "Verified (by Inspection)," "In Progress," and "Open."

At FDR: the VCRM is the **final scorecard**. Every entry must show a disposition: "Verified" (with evidence reference: test report number, calculation page, inspection record), "Failed" (with disposition: redesign, retest, waiver, or risk acceptance), "Waived" (with client approval and rationale), or "Deferred" (with plan for future completion). A VCRM with mostly "Open" entries at FDR indicates insufficient verification effort and will significantly affect the project grade.

6.6 Track-Specific V&V

Build Track: verification is primarily by Test and Demonstration. Physical prototypes are tested against requirements under controlled conditions. Environmental testing (temperature cycling, humidity, vibration) may be required depending on the operating environment. Acceptance testing with the client validates the system in its intended context. Document all test results with data, photos, and video.

Paper Track: verification is primarily by Analysis and Inspection. Models must be **verified** (mesh convergence studies showing results are grid-independent, code verification against analytical solutions or published benchmarks, sensitivity analysis showing results are robust to input uncertainty) and **validated** (comparison to experimental data or published results with stated acceptance thresholds). Documentation must be inspected for completeness: could a qualified engineer implement the design from this package alone, without contacting the team?

Competition Track: tech inspection at the competition event serves as an additional verification point for safety and compliance requirements. However, passing or failing tech inspection does not automatically determine your course grade. Your VCRM and course deliverables must

independently establish design feasibility and engineering rigor. A team whose prototype fails tech inspection *and* whose deliverables do not independently demonstrate engineering quality will score poorly; a team that fails tech inspection due to a last-minute mechanical issue but has excellent documentation and analysis may still demonstrate strong engineering.

6.7 Hazard Assessment and Safety Plan

In SDII (Week 2), each team must complete a project-specific **Hazard Assessment and Safety Plan**. This document identifies:

- All hazards associated with your build, test, and demonstration activities
- The risk level of each hazard (likelihood × severity matrix)
- Engineering controls (guards, interlocks, fusing, ventilation, thermal cutoffs)
- Administrative controls (procedures, training requirements, signage, buddy system)
- Required PPE for each activity
- Emergency response procedures specific to your project's hazards

The PA must review and approve the safety plan before fabrication begins. The plan is a living document; update it as the project evolves and new hazards are identified. See Chapter 21 for the complete EHS framework.

6.8 Designing for Testability

Testability is a design property, not an afterthought. A system that is difficult to test is difficult to verify, and difficulty in verification translates directly to risk at FDR. Consider testability during the architecture phase (Sprint 3–4), not during the test phase (Sprint 11–13):

Test points: include physical test points (solder pads, header pins, or banana jack connections) on every critical signal path. Being able to measure the sensor output, the amplifier output, the ADC input, and the MCU reading separately (without cutting wires or probing tiny SMD pads) saves hours of debugging.

Modular interfaces: design subsystem interfaces with connectors that allow each subsystem to be tested independently. If the sensor module plugs into the main board via a header, you can test the sensor module on a separate test fixture with known inputs. If it is hard-wired, you must test it in the full system context, which makes fault isolation much harder.

Built-in diagnostics: add firmware-level diagnostics that report ADC values, state machine state, communication status, and loop timing over the serial port. These diagnostic outputs turn the MCU into a test instrument. You can leave them active during development and disable (or reduce) them for the final demonstration.

Calibration access: if your system requires calibration (sensor offset, gain adjustment, PID tuning), provide a user-accessible method for performing the calibration without opening the enclosure or modifying firmware. Potentiometers, DIP switches, calibration menus, or configuration files on an SD card are common approaches.

Environmental simulation: if your system must operate in an environment you cannot easily replicate (extreme heat, high humidity, vibration), design the test setup to simulate the critical environmental condition. A heat gun and thermocouple can simulate thermal stress; a function

generator driving a speaker can simulate vibration. Even crude environmental simulation is better than no environmental testing.

6.9 Statistical Confidence and Sample Size

A single test at a single condition does not prove a requirement is met. It proves that the system worked *once*, under *those* conditions. To claim with confidence that a performance requirement is met, you need:

Multiple measurements: repeat each measurement at least 3–5 times and report the mean and standard deviation. If the standard deviation is large relative to the requirement margin, you need more repetitions.

Multiple conditions: test at the extremes of the operating range, not just at the nominal point. If the requirement specifies ±1.0 °C accuracy over 0–50 °C, test at 0 °C, 25 °C, and 50 °C at minimum. A system that meets the accuracy requirement at 25 °C but fails at 0 °C does not meet the requirement.

Acceptance criterion with margin: if the requirement is ±1.0 °C and your measurement uncertainty is ±0.2 °C, your measured error must be ≤0.8 °C to claim compliance with confidence. The measurement uncertainty is subtracted from the requirement margin.

For capstone, a full statistical treatment (confidence intervals, hypothesis testing) is appreciated but not always required. At minimum: report mean, standard deviation, number of samples, and the conditions under which measurements were taken. If the result is close to the acceptance threshold (within 2 standard deviations), flag it as a marginal pass and discuss the implications.

6.10 Common Testing Mistakes

1. **Testing only the happy path**: verifying that the system works under ideal conditions but not testing edge cases, fault conditions, or environmental extremes. The system that works perfectly on the bench may fail in the field.
2. **No baseline measurement**: testing without a calibrated reference. If your thermometer reads 22.5 °C, what is the actual temperature? Without a traceable reference, you are measuring precision but not accuracy.
3. **Confirmation bias**: subconsciously discarding data points that do not support the desired result. Report all data, including outliers. If an outlier exists, investigate the cause rather than deleting the data point.
4. **Testing too late**: leaving all testing until the last two weeks of the semester. When a test reveals a failure with two weeks remaining, there is no time for redesign. Start subsystem testing at CDR and system testing as early as Sprint 10.
5. **Incomplete test documentation**: running a test but not recording the setup, conditions, and raw data. At FDR, you need to present the test results; if you did not document them, the test effectively did not happen.
6. **Skipping regression testing**: after fixing a bug or modifying the design, re-run all previously passing tests to ensure the fix did not break something else. Regression testing catches unintended consequences.

6.11 Failure Disposition

Not every requirement will be met. When a verification activity produces a FAIL result, you have four options:

1. **Redesign and retest**: fix the root cause of the failure and repeat the verification. This is the ideal outcome but requires sufficient schedule margin.
2. **Retest only**: if the failure was due to a test setup error, instrumentation error, or procedural mistake (not a design deficiency), correct the test and repeat. Document what went wrong in the test.
3. **Waiver**: the client formally agrees that the requirement is no longer necessary or that the current level of non-compliance is acceptable. Document the waiver with the client's signature and the rationale.
4. **Risk acceptance**: the client accepts the non-compliance with full knowledge of the technical risk, operational impact, and potential consequences. Document the risk assessment and the client's informed acceptance.

All failure dispositions must be documented in the VCRM, discussed at FDR, and included in the final design report. Hiding a failed test or omitting unfavorable results is an engineering integrity violation and will be treated as such.

Failures Are Data, Not Disasters

A well-documented failure that is honestly reported and intelligently dispositioned demonstrates more engineering maturity than a perfect verification record that nobody believes. Your PA and client will respect honest reporting of problems far more than suspiciously flawless results. The question at FDR is not "did everything work perfectly?" but "do you understand your system's performance, including its limitations?"

6.12 FMEA: Failure Mode and Effects Analysis

For FMEA fundamentals, see Chapter 4. This section covers the capstone-specific application of FMEA [15, 16] to verification and test planning.

FMEA is a systematic technique for identifying potential failure modes, assessing their risk, and prioritizing mitigation actions. The methodology is covered in depth in MEGN 301; here is the capstone application.

For each component or function in your system, identify:

Failure mode: how could this component fail? Examples: sensor gives incorrect reading, motor does not start, solder joint breaks, software hangs in infinite loop, battery overheats.

Effect: what happens to the system when this failure occurs? Does the system shut down? Does it produce incorrect output? Does it create a safety hazard? Does the user notice?

Severity (S): rate the worst-case effect on a 1–10 scale. 1 = no effect. 5 = performance degradation. 8 = system inoperable. 10 = safety hazard to user or public.

Occurrence (O): how likely is this failure? 1 = extremely unlikely. 5 = occasional. 10 = almost certain. Base this on: component reliability data (datasheets), your engineering judgment, and results from testing.

Detection (D): how likely is the failure to be detected before it reaches the user? 1 = certain detection (built-in diagnostic catches it). 5 = moderate (user might notice symptoms). 10 = undetectable (failure is silent and hidden).

Risk Priority Number (RPN): RPN = $S \times O \times D$. Range: 1–1000. Prioritize mitigation for the highest RPNs. There is no universal "acceptable" RPN threshold; focus on relative ranking and on any item with $S \geq 8$ (safety-critical) regardless of RPN.

Mitigation: for high-RPN items, design a specific mitigation that reduces S, O, or D. After mitigation, re-score to verify the RPN has decreased. Document the before and after scores.

Present the FMEA as a table in the PDR and CDR reports, updated at each review. The PDR FMEA identifies potential failures in the proposed concept. The CDR FMEA reflects the detailed design with specific mitigations incorporated. The FDR FMEA shows the final state with test results confirming that mitigations are effective.

Student Guide Cross-Reference

For the sprint-level checklist of VCRM deliverables and testing milestones, see the **Student Guide**, SDI Sprint 5 (VCRM plan) and SDII Sprints 9–13 (V&V execution) sections.

6.13 Practice Questions

1. Write a complete test procedure (header, purpose, equipment, setup, preconditions, steps, acceptance criteria, safety, data recording) for verifying the requirement: "The system shall measure ambient temperature with an accuracy of ±0.5 °C over the range 0–50 °C (PR-003)."
2. Explain how a system can pass all verification tests but fail validation. Give a concrete capstone-relevant example involving a specific type of project.
3. Your FDR VCRM shows 18 requirements Verified, 2 Failed, and 1 Waived. One failure is a "Should" requirement (data logging rate); the other is a "Must" requirement (emergency shutoff response time exceeds specification by 0.3 seconds). How should each be dispositioned, and what do you present at FDR?
4. A Paper Track team says "We don't need test procedures because we don't have hardware to test." Explain what verification and validation activities Paper Track projects do require, with specific examples.
5. Your subsystem test shows that the sensor accuracy is ±1.2 °C, but the requirement specifies ±1.0 °C. The system is otherwise complete and the Showcase is in two weeks. Walk through the decision process: what information do you need, who do you consult, and what are the possible dispositions?

Part II

Major Project Milestones

Chapter 7

The Statement of Work and Project Plan

7.1 The Statement of Work

For the sprint-level checklist of SOW deliverables and deadlines, see the **Student Guide, Sprint 1 section**. This chapter provides the methodology; the Student Guide provides the week-by-week action items.

The Statement of Work (SOW) is the foundational agreement between your team, the PA, and the client. It defines what will be done, by when, within what constraints, and by whom. The SOW is developed during SDI Sprint 1 (Weeks 3–4) and submitted as a formal deliverable.

The SOW is not a formality. It is the document that prevents scope creep, resolves misunderstandings about deliverables, and protects the team when the client asks for something that was never agreed upon. In professional practice, the SOW is legally binding. In capstone, it is academically binding: your PA evaluates your project against what you committed to deliver.

7.1.1 SOW Structure

A complete SOW contains the following sections:

1. **Project Title and Team**: project name, team number, all members with assigned roles (Scrum Master, CFO, Technical Lead, Communication Lead, Fabrication Lead), PA name and contact, client name and contact, Technical Advisor if identified.
2. **Track Classification**: Build, Paper, or Competition with justification. Determines PoC expectations and final deliverable type.
3. **Problem Statement**: Five-Element Problem Statement from Chapter 2.
4. **Scope**: bounded description of what is included and what is **explicitly excluded**. The "Out of Scope" section is as important as the "In Scope" section. List features the client mentioned that will not be addressed with a rationale for each exclusion.
5. **Deliverables**: tangible outputs per semester. SDI: SOW, SDRD, PDR report and presentation, PoC. SDII: CDR report and presentation, FDR report and presentation, tested prototype or technical package, O&M documentation, Showcase poster.
6. **Schedule**: milestones mapped to sprints with dates. Include client meetings, design reviews, procurement deadlines, fabrication milestones, test milestones, and Showcase.
7. **Budget Estimate**: major categories (materials, components, services, travel, contingency) with funding source.

8. **Team Roles and Responsibilities**: who is responsible for each major activity with specific duties.
9. **Assumptions and Constraints**: what you assume to be true (client resources, facility access, technology availability) and what bounds the design (budget, regulations, physical space).
10. **Risk Summary**: top 3–5 risks with likelihood, impact, and preliminary mitigation.

7.1.2 Track Classification

Track classification is documented in the SOW and approved by the PA, client, and CF. It **cannot be changed after PDR** without written approval from the Capstone Program Director.

7.1.3 Scope Change Management

When the client requests a change to the agreed scope (and they will), follow this process:

1. **Document the request**: write down exactly what is being asked. Confirm understanding with the client.
2. **Assess impact**: how does this affect schedule, budget, requirements, and risk?
3. **Present the trade-off**: show the client what must be deferred or dropped to accommodate the change.
4. **Get explicit approval**: client and PA must both agree. Document the decision in the SOW.

Without this discipline, small changes accumulate until the project bears no resemblance to the original agreement.

The SOW Is a Scope Control Document

When a client asks for additional features, the SOW is your reference: "That feature is not in the current scope. We can add it, but here is the impact on schedule and budget. Do you want to proceed?" Without a baselined SOW, every client request becomes an obligation and scope creep consumes the semester.

7.2 The Initial Project Plan

Developed during Sprint 1 (Weeks 3–4), the Project Plan expands the SOW into an operational document:

- **Product Backlog**: all known work items, prioritized by client value (MoSCoW [10]).
- **First Sprint Backlog**: items selected for Sprint 1 with owners and effort estimates.
- **Team norms**: meeting schedule, communication tools, response time expectations, file conventions, Team Contract.
- **Risk register**: expanded from SOW with mitigations assigned to specific owners.

The Plan is emailed to the client and uploaded to Canvas. Reviewed and updated at every retrospective.

7.3 Project Goals Setting

At each semester's start, set SMART goals [38] with PA and client agreement: Specific, Measurable, Achievable, Relevant, Time-bound. Examples:

- "Demonstrate a working temperature control loop with ±2 °C accuracy by PDR" (SDI)
- "Complete a validated FEA model under ASCE 7 load cases by CDR" (SDII)
- "Achieve ≥85% of VCRM requirements verified as PASS by FDR" (SDII)

At semester's end, the **Project Goal Attainment** document evaluates performance. Teams not accomplishing goals will not receive an A.

7.4 Scope Management Throughout the Project

Scope management is the discipline of controlling what is and is not included in the project. It is one of the most important project management skills and one of the hardest to maintain under the pressure of client enthusiasm and team ambition.

7.4.1 The Scope Triangle

Every project is constrained by three interdependent variables [8]: **scope** (what the system does), **schedule** (when it is delivered), and **budget** (what it costs). Changing one variable affects the others. If the client adds a feature (increasing scope), either the schedule extends, the budget increases, or both. If the schedule compresses (e.g., a competition date moves earlier), either scope decreases or budget increases (e.g., expedited shipping).

Your SOW fixes the scope-schedule-budget triangle at a specific equilibrium. Every scope change request must be evaluated against this triangle: what is the impact on schedule and budget? If the impact is acceptable to all parties (team, PA, client), proceed with the change under documented change control. If not, the change is deferred to future work.

7.4.2 Writing Effective Deliverable Descriptions

Each deliverable in the SOW must be described with enough specificity that all parties agree on what "done" looks like. Bad: "Provide a final report." Good: "Deliver a Final Design Report (FDRt) containing: executive summary, complete VCRM with all requirements dispositioned, as-designed vs. as-built comparison, test data and analysis, O&M documentation, lessons learned, and future work recommendations. Format per the Capstone Design Program Format Guide. Submitted to PA and client no later than [date]."

The deliverable description is the acceptance criterion. At FDR, if the client says "I expected more detail in the O&M section," you can point to the SOW: "The O&M documentation was specified as [description]. If you need additional detail, we can discuss a scope extension."

7.4.3 Common SOW Mistakes

1. **Scope too broad**: "Design and build a complete autonomous vehicle." No capstone team can build a complete autonomous vehicle in two semesters. Narrow the scope to a specific subsystem or capability.
2. **Scope too vague**: "Improve the client's process." Improve how? By what metric? To what target? Vague scope leads to vague deliverables and ambiguous evaluation.
3. **Missing "Out of Scope"**: not explicitly stating what is excluded. The client assumes features are included unless they are explicitly excluded.
4. **No budget contingency**: budgeting exactly to the penny with no reserve for unexpected costs.

5. **Unrealistic schedule**: promising deliverables faster than the sprint structure allows.
6. **Missing risk acknowledgment**: not identifying that the project has known technical risks that could affect scope.

7.4.4 SOW as a Professional Development Tool

Writing a SOW is a career skill. In consulting, the SOW (or proposal) is the document that wins (or loses) the contract. In industry, the SOW defines the work breakdown for your team. In government contracting, the SOW is a legally binding document with financial consequences for non-compliance. The SOW you write in capstone is practice for every project you will manage in your career.

7.5 The SOW Review Process

The SOW should be reviewed by all stakeholders before baselining:

Team review: every member reads the complete SOW and verifies that their role, responsibilities, and the scope description are accurate. Each member signs or acknowledges the SOW.

PA review: the PA checks for: realistic scope (is this achievable in two semesters?), adequate budget (does the estimate cover all foreseeable costs?), appropriate track classification (does the track match the project deliverables?), identified risks (are the top risks realistic and mitigable?), and completeness (are all required sections present?).

Client review: the client verifies that the scope, deliverables, and schedule match their expectations. The client's approval is documented (email confirmation at minimum; signature on the SOW is preferred). After client approval, the SOW is baselined.

7.6 SOW Templates and Examples

The Capstone Design Program provides a SOW template on Canvas. Use it as a starting point but do not treat it as a fill-in-the-blank form. Your SOW should reflect the unique characteristics of your project, not generic placeholder text. Review SOW examples from previous capstone teams (available from your PA or on Canvas) to calibrate the expected level of detail.

> Write the SOW as if you are handing it to a new team that knows nothing about your project. Can they understand the scope, deliverables, and constraints from the SOW alone? If not, add detail. The SOW should be self-contained.

7.7 Negotiating Scope with Clients

Scope negotiation is one of the most important professional skills you practice in capstone. Clients often want more than is achievable in two semesters. Your job is not to say "no" but to present options with trade-offs.

The "Yes, and" technique: instead of rejecting a request, acknowledge it and present the cost. "Yes, we can add wireless connectivity, and the impact would be 3 additional weeks of development and approximately $150 in components. To stay on schedule, we would need to defer the color display to a future version. Which is more important to you?"

Minimum Viable Product (MVP): identify the smallest set of features that delivers value to the client. This becomes your "Must Have" scope. Everything else is "Should" or "Could." If the project goes perfectly, you deliver it all. If challenges arise (and they will), the MVP is protected.

Version planning: frame the two-semester capstone as "Version 1.0." Features that cannot fit in V1.0 are documented as "Future Work" in the SOW and the FDR report. This respects the client's vision while being honest about what is achievable. Clients who understand version planning are far more satisfied than clients who expect everything and receive a rushed partial implementation.

The 80/20 rule: in most projects, 80% of the value comes from 20% of the features. Identify the high-value features early and protect them. The remaining features; the ones that are nice-to-have but do not fundamentally change the system's value proposition; are the first candidates for deferral when schedule pressure mounts.

7.8 Building the Risk Register

The SOW includes a preliminary risk summary (top 3–5 risks). The full **risk register**, developed during Sprint 1–2 and maintained throughout the project, expands this into a living management tool.

Each risk entry contains: (1) **Risk ID** (R-001, R-002, etc.), (2) **Description** (specific, not vague: "Thermocouple accuracy degrades above 800 °C due to oxidation" not "temperature measurement may be inaccurate"), (3) **Likelihood** (1–5 scale), (4) **Impact** (1–5 scale), (5) **Risk Score** (Likelihood × Impact), (6) **Mitigation strategy** (what action reduces the likelihood or impact), (7) **Owner** (which team member is responsible for monitoring and executing the mitigation), and (8) **Status** (Open, Mitigated, Retired, Occurred).

Risk categories to consider:

- **Technical**: sensor accuracy, actuator performance, software bugs, EMI, thermal limits, structural failure.
- **Schedule**: procurement delays, fabrication overruns, testing reveals redesign need, team member availability (illness, other course demands, job interviews).
- **Budget**: component costs higher than estimated, shipping surcharges, PCB revision costs, travel costs exceed estimate.
- **External**: client changes priorities, regulatory requirement discovered late, facility access restricted (InnoHub closure, equipment breakdown).
- **Team**: unequal contribution, key member drops the course, personality conflicts escalate, knowledge concentrated in one person (bus factor = 1).

Review the risk register at every sprint retrospective. Update likelihood and status based on new information. A risk register that has not been updated since Sprint 1 is not a management tool; it is a decoration.

7.9 Documenting Assumptions and Constraints

The SOW should explicitly document what you are assuming to be true and what constraints bound the design:

Assumptions are things you believe to be true but have not verified. Examples: "The client will

provide access to the test site by Sprint 4," "The operating temperature will not exceed 50 °C," "Shop air supply provides 90 psi minimum." When an assumption proves false, it becomes a risk or a scope change.

Constraints are non-negotiable boundaries. Examples: "Total budget not to exceed $2,500," "System must fit within a 12"×12"×8" enclosure," "Must comply with IEC 61010 safety standard," "Must use ESP32 microcontroller per client specification."

Documenting assumptions protects the team: when the client says "I assumed you would also build a mobile app," you can point to the SOW assumptions section and initiate a scope change process rather than absorbing unplanned work.

7.10 The Stakeholder Communication Matrix

Different stakeholders need different information at different frequencies. Document this in a communication matrix:

Stakeholder	Information Need	Frequency	Method
Client	Progress, decisions, blockers	Biweekly	Status email + meetings
PA	Sprint progress, risks, grades	Weekly	Sprint review + email
Technical Advisor	Technical approach, analysis	As needed	Meeting + report sections
Capstone Purchasing	Orders, budget status	Per purchase	Forms + email
InnoHub Staff	Fabrication plans, training	Per session	In-person + email
Team Members	Tasks, blockers, decisions	Daily	Stand-up + task board

The Communication Lead owns this matrix and ensures that every stakeholder receives their information on schedule. When a stakeholder contacts the team asking "what's the status?"; that is a failure of proactive communication.

7.11 The SOW as a Living Document

While the SOW is baselined at Sprint 1, it is reviewed and potentially amended at each design review:

At PDR: confirm that the scope, deliverables, and schedule are still realistic given the design direction selected. If the selected concept requires capabilities not in the original scope (e.g., wireless communication was not planned but the selected concept requires it), amend the SOW with client and PA approval.

At CDR: confirm that the build plan, budget, and timeline are still achievable. If procurement delays or design changes have shifted the schedule, amend the SOW.

At FDR: the SOW is the benchmark against which your Project Goal Attainment is evaluated. Any features delivered beyond the SOW scope are "bonus"; any features not delivered require explanation (deferred to future work, design change, requirement waived).

Each SOW amendment is tracked with a revision number, date, description of change, and approver signatures.

7.12 Risk Mitigation Strategy Types

For each risk in the register, select one or more mitigation strategies [8]:

Avoidance: eliminate the risk by changing the design approach. If the risk is "motor controller IC may not be available due to supply chain shortage," avoid it by selecting a different, readily available IC. Avoidance is the strongest strategy but may require design compromises.

Reduction: decrease the likelihood or impact of the risk. If the risk is "PCB may have a layout error," reduce it by: having a second team member review the layout, running DRC (Design Rule Check), and ordering 5 boards instead of 2 (so a revision still leaves spares from Rev 1). Reduction does not eliminate the risk but makes it manageable.

Transfer: shift the risk to a party better equipped to handle it. If the risk is "CNC machining may exceed our skill level," transfer it by outsourcing to a professional machine shop (Xometry, Protolabs). The cost is higher, but the quality and schedule certainty improve.

Acceptance: acknowledge the risk and plan to deal with the consequences if it occurs. If the risk is "the client may change priorities after PDR," accept it by: building schedule buffer, documenting the current scope in the SOW, and planning the scope change process (Chapter 7). Acceptance is appropriate for low-probability risks where the cost of mitigation exceeds the expected cost of the risk.

Contingency planning: for high-impact risks that cannot be fully mitigated, develop a specific contingency plan: "If the custom motor does not arrive by Sprint 10, we will substitute [specific alternative motor] which meets [X% of the torque requirement] and can be procured locally within 2 days." The contingency plan converts an open-ended crisis into a pre-planned response.

7.13 What "Done" Looks Like: Defining Acceptance Criteria

For every deliverable listed in the SOW, define what "done" means in terms the client can evaluate:

Physical deliverables: "A functioning temperature control system that maintains the setpoint within $\pm 2\,^{\circ}\mathrm{C}$ over the operating range of 20–200 °C, demonstrated in the client's presence at FDR. Delivered with: assembled prototype, firmware source code, engineering drawings, BOM, O&M manual, and digital design archive."

Document deliverables: "A Final Design Report (FDRt) containing all sections specified in the Capstone Design Program Format Guide, formatted per program standards, submitted as PDF via Canvas and delivered to the client by email at least 3 days before the FDR presentation."

Presentation deliverables: "A 25-minute CDR presentation to the PA, client, and TAs, covering: design evolution since PDR, detailed design with engineering analysis, BOM and procurement status, manufacturing plan, VCRM status, and schedule/budget projections. Followed by 10 minutes of Q&A."

Vague acceptance criteria lead to disputes: the team thinks the work is done; the client expected something different. Specific criteria prevent this misalignment. When in doubt, ask: "If I delivered

exactly this, would you consider the requirement met?"

7.14 The SOW Signature and Approval Process

The SOW requires explicit approval from three parties:

Team: all members review and acknowledge the SOW, confirming that their roles, responsibilities, and the scope are accurately represented. Each member's agreement signifies their commitment to the project plan.

PA: reviews the SOW for completeness, feasibility, and alignment with capstone program requirements. The PA may request revisions before approving.

Client: reviews the SOW for alignment with their needs and expectations. The client's approval is documented (email confirmation or signature). After client approval, the SOW is **baselined**; changes require formal change control.

The approval process should be completed by the end of Sprint 1. Do not proceed to concept generation (Sprint 2) without an approved SOW. Building a design against an unapproved scope is building on sand.

7.15 Track-Specific SOW Considerations

While the SOW structure is the same for all tracks, the content emphasis differs:

Build Track: the SOW should emphasize: the physical deliverable description (what will the client receive?), the fabrication plan (where will it be built? what equipment is needed?), the procurement timeline (when must long-lead items be ordered?), and the testing strategy (where and how will the prototype be tested?). The schedule should show fabrication milestones alongside design review dates.

Paper Track: the SOW should emphasize: the analytical deliverable description (what models, simulations, or analyses will be produced?), the validation strategy (how will the results be verified against reference data or independent calculations?), the computational resources required (software licenses, computing time, data storage), and the documentation plan (the final deliverable is typically a detailed technical report, not a physical prototype). The schedule should show analysis milestones (model development, mesh convergence, validation, parametric studies) alongside design review dates.

Competition Track: the SOW should emphasize: the competition rules and compliance plan (which rules constrain the design? which are hardest to meet?), the registration and travel timeline (deadlines are typically non-negotiable), the budget allocation between build and travel, and the testing strategy (how will competition performance be validated before the event?). The schedule must account for the competition date as an absolute deadline that cannot be extended.

Hybrid projects: some projects combine elements of multiple tracks (e.g., a Build Track project with a significant analytical component, or a Competition Track project that also serves an external client). The SOW should clearly state which deliverables belong to each track element and how the team's effort is allocated.

7.16 Common SOW Negotiation Scenarios

These negotiations occur frequently during Sprint 1:

"The client wants more than is achievable." Present a phased plan: "In the two-semester capstone cycle, we can deliver [MVP scope]. If the project continues with a future team or through a sponsored research agreement, [Phase 2 scope] could be added." Most clients understand phasing once you present a concrete plan.

"The team disagrees on the scope." Use the MoSCoW framework [10] to reach consensus. If the team cannot agree on what is Must-Have vs. Should-Have, the disagreement reveals uncertainty about the client's priorities. Schedule a client meeting to clarify: "Which of these features is essential for the system to be useful to you?"

"The PA thinks the scope is too ambitious." Your PA has seen dozens of capstone projects and has a calibrated sense of what is achievable. If they say the scope is too large, take it seriously. Ask: "Which elements would you recommend deferring?" The PA's experience-based judgment is one of the most valuable inputs you will receive all semester.

"The budget does not cover the desired scope." Present the trade-off explicitly: "With the current budget of $1,500, we can deliver Options A and B. Adding Option C requires an additional $600. Would you prefer to request additional funding or defer Option C?" Never commit to scope that exceeds your budget without a documented plan for the funding gap.

7.17 SOW Lessons from Past Teams

"We should have defined 'done' more precisely." A team's SOW stated the deliverable as "a working prototype." At FDR, the client expected a prototype with a painted enclosure, labeled connectors, and a user manual. The team delivered a functional but unfinished-looking breadboard system. Both parties were right based on their interpretation. A precise deliverable description ("a functioning prototype in a 3D-printed enclosure with labeled connectors, mounted on a demonstration base, accompanied by a printed O&M manual") would have prevented the misunderstanding.

"We should have documented the client's verbal agreements." A client verbally approved a concept during a meeting. Three months later, at CDR, the client said "I don't remember agreeing to that approach." Without meeting minutes documenting the decision, the team had no evidence. The scope discussion had to be repeated, consuming a week of Sprint 8.

"We should have been more aggressive about Out of Scope." A team's SOW listed 8 In-Scope items and 0 Out-of-Scope items. By Sprint 4, the client had requested 3 additional features, each of which the team accepted because "it wasn't explicitly excluded." The project became unmanageable. A team that had listed those 3 features as Out of Scope from the beginning would have had a clean basis for the scope change conversation.

7.18 Practice Questions

1. Write the Scope section (In Scope and Out of Scope) for a project to automate grain silo monitoring for a brewery. Include at least 3 items in each category with rationale.

2. Your client requests a new feature in Sprint 4 that was listed as "Out of Scope" in the SOW. Draft the email that initiates the scope change process.
3. Create three SMART goals for the first semester of a Build Track solar irrigation controller project.
4. Explain why the "Out of Scope" section of the SOW is as important as the "In Scope" section.

Chapter 8

Preliminary Design Review (PDR)

8.1 Purpose and Scope

For the sprint-level PDR preparation timeline and deliverable checklist, see the **Student Guide, Sprint 5 section**.

PDR answers: *"Should this design work?"* You demonstrate that you understand the problem, have well-defined requirements, have generated and evaluated multiple concepts, have selected a feasible approach with engineering justification, and have a credible plan to develop it further. PDR is not a final design; it is a **commitment to a direction**, backed by analysis and evidence.

PDR serves as a design gate [35] and is your first opportunity to receive formal, structured feedback from the PA, client, and TAs on the entirety of your engineering approach. Teams that treat PDR as a checkbox exercise produce weak designs; teams that use PDR as a genuine review emerge with stronger designs and clearer paths forward.

8.2 Timeline

PDR occurs during SDI Sprint 5 (Weeks 11–13):

1. **Week 1**: finalize PDRt. Send front-end material to client in advance.
2. **Week 2**: conduct Mock PDR; practice presentation with PA or another team. Incorporate feedback.
3. **Week 3**: deliver formal PDRw to PA, client, and TAs.

8.3 PDR Report (PDRt)

Formal written document submitted to PA and shared with client before the presentation:

- Executive summary (1 page)
- Problem statement and stakeholder analysis
- Complete SDRD with IDs, SHALL/SHOULD, rationale, traceability, TAID [2], MoSCoW [10]
- Competitive benchmarking and prior art (product survey, patents, standards)
- Concept generation: $\geq$3 distinct, graphically communicated concepts
- Concept selection: Pugh screening [3] + weighted decision matrix + sensitivity analysis
- Preliminary architecture: functional decomposition, block diagram, allocation matrix, interfaces
- Risk register and initial FMEA

- Proof-of-Concept results with data or analysis
- Sustainability assessment (CLO 5 tool applied)
- Engineering standards identified and mapped to requirements
- VCRM (plan): all requirements with assigned methods and phases
- Schedule and budget update with projections through SDII

8.4 PDR Presentation (PDRw)

Structure as a narrative answering "should this work?":

Section	Content	Time
Problem	What are you solving, for whom, why does it matter?	2–3 min
Requirements	Key performance requirements, constraints, MoSCoW	3–4 min
Concepts & Selection	Concepts considered, decision matrix, sensitivity	5–7 min
Architecture	Block diagram, interfaces, critical subsystems	3–5 min
Risk & Feasibility	Top risks, mitigations, PoC results	2–3 min
Plan Forward	Schedule through SDII, procurement timeline	2–3 min
Q&A	Listen, take notes, commit to follow-up	5–10 min

The Mock PDR Is Not Optional

Teams that conduct a mock PDR consistently perform better. The mock reveals: dense slides, poor transitions, unsupported claims, timing problems, and questions you cannot answer. Schedule the mock $\geq$3 days before formal PDR to allow revision.

8.5 PDR Evaluation Criteria

Your PA evaluates PDR on:

1. **Problem definition**: is the problem clearly stated, with measurable gaps and identified stakeholders?
2. **Requirements quality**: are requirements testable, traceable, and appropriately prioritized?
3. **Concept generation breadth**: were multiple distinct concepts developed and graphically communicated?
4. **Selection rigor**: is the decision matrix defensible with sensitivity analysis?
5. **Feasibility evidence**: does the PoC support the selected concept?
6. **Risk assessment**: are risks comprehensive (technical, schedule, budget, team)?
7. **Sustainability**: is the CLO 5 tool applied meaningfully?
8. **Report quality**: organization, clarity, professionalism, proper citations?
9. **Presentation quality**: narrative flow, visual aids, timing, Q&A handling?

8.6 Common PDR Failure Patterns

1. Requirements that are not testable ("the system shall be reliable").
2. Only one concept generated (no evidence of divergent thinking).
3. Decision matrix with criteria that obviously bias toward a predetermined choice.
4. Risk register listing only technical risks.
5. No PoC evidence (just claims of feasibility).

6. Sustainability as an afterthought appendix.
7. Running over time or poorly coordinated team transitions.
8. Team members unable to answer questions about their own sections.

8.7 After PDR

1. Send **Next Steps Email** to client: outcome, feedback, plan for Sprint 6.
2. **Baseline the SDRD**: freeze the document. All subsequent changes tracked under configuration management with change rationale and PA approval.
3. Sprint 6 (Weeks 14–15): incorporate feedback, refine architecture, prepare SDII transition.
4. Submit **PDRt Update** and **Next Steps Email** documenting how feedback was addressed.

8.8 The Review Culture: Feedback Is a Gift

Design reviews exist to improve the design, not to judge the team. The best PDRs are the ones where tough questions lead to real improvements. A reviewer who asks "What happens if the sensor cable is severed during operation?" has identified a failure mode you missed. A reviewer who says "Your decision matrix weights seem arbitrary; how did you determine them?" is pushing you toward a more defensible selection.

Adopt this mindset: **every piece of feedback makes the design better**. Record every question and comment during the review. After the review, categorize feedback into: (1) actions to take immediately (clear deficiencies), (2) actions to investigate (questions that need analysis), and (3) acknowledged but deferred (valid concerns that are out of scope or low priority). Report back to the reviewer on all three categories.

Teams that respond defensively to feedback ("We already thought about that" or "That's not really an issue") miss opportunities to improve and alienate the people who are trying to help them succeed.

8.9 What Distinguishes a Strong PDR from a Weak One

Strong PDRs share these characteristics:

- Requirements are specific, testable, and traced to client needs.
- At least three genuinely different concepts were generated, graphically communicated, and systematically evaluated.
- The decision matrix includes sensitivity analysis and the team can articulate why the selected concept is best *and* what its weaknesses are.
- Proof-of-Concept evidence (data, photos, measurements) supports the feasibility claim.
- The risk register includes schedule, budget, and team risks; not just technical ones.
- Every team member can answer questions about the sections they are responsible for.
- The presentation has a clear narrative arc and respects the time limit.

Weak PDRs share these characteristics:

- Requirements are vague ("the system shall be reliable") or missing categories (no non-functional requirements).
- Only one concept was developed, with alternatives added retroactively.

- The decision matrix produces a "winner" that matches the team's initial preference, with no sensitivity analysis.
- No proof-of-concept evidence; feasibility is claimed but not demonstrated.
- Sustainability assessment is a one-paragraph afterthought.
- Team members cannot explain sections they did not write.
- The presentation runs over time and Q&A reveals unprepared areas.

8.10 Incorporating Feedback: The Sprint 6 Recovery

Sprint 6 (Weeks 14–15) exists specifically for incorporating PDR feedback and preparing the transition to SDII. This is not a "relax after PDR" sprint; it is a critical refinement period. Actions:

- **Categorize all PDR feedback** into action items with owners and deadlines.
- **Update the SDRD**: revise requirements flagged as vague, add missing non-functional requirements, update traceability.
- **Refine the architecture**: address any architectural concerns raised at PDR (interface ambiguities, single points of failure, unresolved technology selections).
- **Begin procurement planning**: identify long-lead-time components and initiate orders for SDII.
- **Prepare SDII transition**: document the current state of the design, open questions, and the plan for Sprint 7–14. This documentation ensures continuity if team dynamics shift over the break.
- **Complete Peer Evaluation 2** (SDI): provide honest feedback to teammates.
- **Submit End-of-Semester deliverables**: Final Course Survey, Individual Performance Evaluation, PDRt Update.

8.11 PDR Grading Criteria

PDR evaluation typically weighs the following dimensions (see Canvas for the specific rubric):

Problem definition and requirements (∼25%): Is the problem clearly stated? Are requirements testable, traceable, and prioritized? Are engineering standards identified?

Concept generation and selection (∼25%): Were multiple distinct concepts generated? Is the selection methodology rigorous (decision matrix with sensitivity analysis)? Is the recommendation justified?

Feasibility and risk (∼20%): Does the PoC evidence support the selected concept? Is the risk register comprehensive (technical, schedule, budget, team)? Are mitigations assigned?

Sustainability and broader impacts (∼10%): Is a CLO 5 tool applied meaningfully? Are sustainability requirements integrated into the SDRD?

Communication quality (∼20%): Is the report well-organized, clearly written, and professionally formatted? Is the presentation clear, on time, and well-coordinated? Can team members answer questions about their sections?

Teams that score poorly at PDR almost always recover in SDII if they incorporate the feedback aggressively during Sprint 6. Teams that ignore PDR feedback carry the same weaknesses into

CDR, where the consequences are more severe.

8.12 PDR Day Logistics

Before the presentation: arrive 15 minutes early. Test the projector, laptop connection, and any live demonstration equipment. Have a USB backup of your slides in PDF format. Assign one team member as the slide advancer and another as the scribe (to record reviewer feedback).

During the presentation: speak to the audience, not to the screen. Make eye contact. When a reviewer asks a question, listen completely before responding. If the question is for a specific team member, the Scrum Master directs it: "That is a sensor selection question; let me pass it to Jordan." Take notes on every question and comment; these become your Sprint 6 action items.

After the presentation: thank the reviewers. Send the Next Steps Email to the client within 48 hours. Hold a team debrief within 24 hours: what feedback did we receive? What are the action items? Who owns each? When will they be complete?

If the presentation does not go well: PDR is a learning experience, not a final exam. A tough PDR where reviewers identify significant weaknesses is more valuable than a smooth PDR where no one pushes back. The question is not whether you had a perfect PDR; the question is what you do with the feedback in Sprint 6. Teams that respond aggressively to tough feedback consistently produce the best FDR outcomes.

8.13 Writing the PDR Report: Tips from Reviewers

PA reviewers consistently identify these patterns in strong vs. weak PDR reports:

Strong executive summaries: state the problem, the selected concept, the key feasibility evidence, and the plan forward in one page. A reviewer who reads only the executive summary should understand the entire project.

Requirements that tell a story: organize requirements by functional area (sensing, processing, actuation, communication, power, safety), not by the order they were written. Each area should trace logically from a client need to a SHALL statement to a verification method. The requirements section should read as a coherent engineering argument, not a disconnected list.

Concept graphics that communicate: the three concepts should be presented with annotated sketches or block diagrams, not walls of text. Each concept graphic should be self-contained: a reviewer should understand how the concept works from the figure and caption alone. Label key components, show signal/energy/material flows, and annotate dimensions or performance targets.

The trade study narrative: do not just show the decision matrix; tell the story. "We evaluated five criteria derived from our MoSCoW requirements. Performance and safety received the highest weights because they correspond to Must-Have requirements. Concept A scored highest overall, with particular strength in manufacturability. However, the sensitivity analysis showed that if cost weight increases by 15%, Concept B becomes preferred. Because our budget has adequate margin, we recommend Concept A with cost managed through detailed BOM tracking."

Honest risk assessment: a risk register that lists only "sensor may not be accurate enough" and "schedule may slip" is too generic. Good risks are specific: "The ESP32 ADC has known nonlin-

earity at the extremes of its range, which may cause measurement error >1 °C at temperatures above 45 °C. Mitigation: use an external ADS1115 ADC with guaranteed linearity."

PoC evidence that is evidence: a photograph of a breadboard with the caption "our proof-of-concept" is not evidence. Evidence includes: measured data compared to expected values, a graph showing sensor response over a range, a video demonstrating a physical function, or a simulation result compared to an analytical solution. The PoC should retire the project's single largest technical risk.

8.14 Presenting Concepts Effectively at PDR

The concept generation and selection section is often the weakest part of student PDR presentations. Strengthen it:

Show the divergent thinking: present all concepts generated, not just the final three. A slide showing "we generated 15 concepts using brainstorming, SCAMPER [27], and morphological chart analysis [28], then screened to 3 using Pugh matrix [3]" demonstrates creative rigor. Show the morphological chart or the brainstorming results briefly.

Make each concept visually distinct: use annotated sketches or block diagrams, not paragraphs of text. Each concept should be immediately visually distinguishable from the others. Side-by-side comparison on a single slide is powerful.

Be honest about weaknesses: for the selected concept, state its weaknesses explicitly and your plan to mitigate them. "Concept A has the highest technical risk due to the unproven motor controller, but it also has the highest performance ceiling. We mitigate the motor risk with our Sprint 4 PoC, which demonstrated motor control at 80% of the target speed." This builds credibility.

Show the sensitivity analysis: a single slide showing "when we vary the weight of [criterion] by ±20%, the selection changes / does not change" tells the reviewer that your selection is robust (or identifies where it is fragile). If the selection is fragile, explain how you will manage the risk.

8.15 The Proof-of-Concept Presentation

The PoC is the evidence that your selected concept is feasible. Present it as evidence, not as a show-and-tell:

State the risk being retired: "The highest technical risk for our project is whether the thermocouple can measure accurately in the high-EMI environment near the motor. Our PoC specifically tested this."

Show the test setup: photograph or diagram of the PoC hardware configuration.

Present the data: measured values compared to expected/required values. "We measured ±1.8 °C accuracy with the motor running, which is within the ±2 °C requirement."

State the conclusion: "The PoC demonstrates that thermocouple measurement is feasible in our operating environment. The primary technical risk is retired."

A PoC that does not clearly state what risk it retires, what was measured, and what the result means has not served its purpose.

8.16 The Sustainability Assessment at PDR

CLO 5 requires a sustainability, equity, or societal impact assessment. At PDR, this assessment should be integrated into the design rationale, not relegated to an afterthought appendix:

Tool selection: choose the appropriate assessment framework based on your project type. LEED [17] (building-related projects), Envision [18] (infrastructure projects), Design for Environment (product design), or Engineering for Social Justice (community-focused projects). If none of these fits your project precisely, consult with your PA to select an appropriate framework or adapt one.

Integration with requirements: the sustainability assessment should produce at least 2–3 non-functional requirements in your SDRD. Examples: "The system shall consume $\leq$50 W during normal operation" (energy efficiency), "All materials shall be RoHS-compliant" (hazardous substance restriction), "The enclosure shall be designed for disassembly using standard hand tools" (end-of-life recyclability).

Presentation at PDR: dedicate 2–3 slides to the sustainability assessment. Show: which framework you selected, the key findings, how the findings influenced your design decisions, and the sustainability-related requirements added to the SDRD. Reviewers should see that sustainability was a design driver, not a compliance exercise.

Common sustainability assessment mistakes: (1) selecting a framework but not actually applying it ("We chose LEED" without any LEED analysis), (2) treating sustainability as separate from the design (the assessment lives in an appendix disconnected from the design narrative), (3) making generic claims ("our design is environmentally friendly") without specific, quantified evidence.

8.17 Mock PDR Best Practices

The mock PDR is a rehearsal that dramatically improves the formal PDR. Schedule it 3–5 days before the formal review:

Audience: invite your PA (ideal), another capstone team, or at minimum, run it with just your own team timing each section. Fresh eyes are the most valuable aspect.

Full simulation: run the mock exactly as you would the formal review: timed, with transitions, with Q&A. Do not stop and fix slides mid-presentation; note the problems and fix them afterward.

Feedback collection: the mock audience should focus on: (1) are the key messages clear? (2) are there claims without evidence? (3) are there sections that are confusing or too long? (4) what questions would a hostile reviewer ask? (5) is the time management on track?

Post-mock action items: prioritize fixes by impact. Slide reorganization and content gaps are high priority. Font size adjustments and animation tweaks are low priority. You have 3 days; spend them on substance, not polish.

Student Guide Cross-Reference

For the sprint-level PDR preparation timeline and deliverable checklist, see the **Student Guide, SDI Sprint 5** section.

8.18 SDRD Completeness Checklist for PDR

Before submitting the SDRD as part of the PDR package, verify:

- Every requirement has a unique ID (FR-001, PR-001, NF-001 format).
- Every requirement uses SHALL (Must-Have) or SHOULD (Should-Have) language.
- Every requirement is a single, independently verifiable statement (no compound requirements).
- Every requirement has quantitative acceptance criteria where applicable (numbers, units, tolerances).
- Every requirement traces upward to a specific client need (forward traceability).
- Every requirement has a TAID verification method assigned (Test, Analysis, Inspection, or Demonstration).
- Every requirement has a MoSCoW priority (Must, Should, Could, Won't).
- The SDRD includes at least: functional requirements, performance requirements, interface requirements, environmental requirements, safety requirements, and regulatory/standards requirements.
- Rationale is documented for each requirement (why this specific value, why this priority).
- No requirement specifies a design solution ("shall use Arduino") instead of a performance need ("shall provide $\geq$8 analog inputs").
- The total number of Must-Have requirements is achievable within two semesters (typically 15–25 for a capstone project).
- The SDRD has been reviewed by at least one team member who did not write it.

An SDRD that passes this checklist is ready for PDR. An SDRD that fails multiple items will receive significant feedback from reviewers; feedback that is much more expensive to incorporate after PDR than before.

8.19 Proof-of-Concept Best Practices by Track

The PoC demonstration at PDR must retire the project's highest technical risk. What constitutes an effective PoC differs by track:

Build Track: demonstrate at least one physical function with measured data. The PoC does not need to be pretty or complete; it needs to *work*. A thermocouple reading correctly on a breadboard, a motor spinning at the target speed with a bench power supply, or a 3D-printed mechanism that articulates as designed. Show: the test setup (photo), the measurement method, the expected result, the actual result, and the conclusion ("risk retired" or "additional investigation needed").

Paper Track: demonstrate computational goal attainment. A converging FEA mesh study, a validated sub-model that matches published benchmark data, or a simulation that produces physically reasonable results under boundary conditions. Show: the model setup, the validation case, the comparison to reference data, and the convergence metrics.

Competition Track: demonstrate compliance with the highest-risk competition rule. If the competition requires the vehicle to fit through a specific opening, demonstrate a mockup that fits. If the competition requires a specific sensor accuracy, demonstrate the sensor meeting that accuracy. The PoC should address the rule that is hardest to meet or most likely to cause a tech inspection failure.

The PoC Risk Retirement Question

Before presenting your PoC at PDR, answer this question: "What was the single biggest uncertainty about whether our concept would work, and does the PoC resolve that uncertainty?" If you cannot articulate the risk being retired, the PoC has not served its purpose. A PoC that demonstrates something easy ("the LED blinks") while leaving the hard question unanswered ("can the motor maintain position under vibration?") wastes Sprint 4 effort.

8.20 The Top 10 PDR Reviewer Questions

Prepare answers for these questions (they are asked at nearly every PDR):

1. "How did you determine the weights in your decision matrix?" (Trace each weight to a client priority or requirement category.)
2. "What happens if your selected concept does not work?" (Describe your fallback plan) is there a second-ranked concept you could pivot to?)
3. "How will you verify this requirement?" (Point to the TAID entry in the VCRM and describe the specific test or analysis.)
4. "What is your highest technical risk, and how are you mitigating it?" (Reference the risk register and the PoC results.)
5. "What standards apply to your design?" (Identify specific clauses, not just standard numbers.)
6. "How does your sustainability assessment influence the design?" (Point to specific requirements or design decisions driven by the assessment.)
7. "What is your procurement plan for long-lead items?" (Show the BOM items with lead times and when they will be ordered.)
8. "What did the client think of your concept selection?" (Describe the client approval meeting and any concerns raised.)
9. "How do you plan to manage scope if the project falls behind?" (Reference the MoSCoW priorities; which Could/Won't requirements would be deferred first?)
10. "What would you do differently if you started this project again today?" (Demonstrate self-awareness and learning from the Sprint 0–5 experience.)

8.21 The PDR-to-CDR Transition: What Happens Next

After PDR, the team has 3–4 months before CDR. Use this time deliberately:

Sprint 6 (SDI Weeks 14–15): incorporate PDR feedback, refine architecture, begin long-lead procurement. Submit the PDRt Update and Next Steps Email. Complete Peer Evaluation #2 and end-of-semester deliverables.

Semester break: maintain team communication. The Scrum Master sends a brief check-in email before SDII starts. Individual team members review PDR documentation to refresh their knowledge.

Sprint 7 (SDII Week 1): re-engage with client. Set SDII goals. Complete the Hazard Assessment and Safety Plan. Review and update the Product Backlog. This sprint sets the pace for the entire build semester.

Sprint 8 (SDII Weeks 2–5): complete the detailed design, Engineering Calculation Package,

and BOM. Order all remaining components. Complete the Manufacturing Review Meeting. Prepare and deliver CDR.

The teams that use this transition period productively (ordering components, completing training, refining the design) arrive at CDR with confidence. The teams that treat the semester break as a vacation arrive at CDR scrambling to catch up.

8.22 Concept Sketching for Engineers

You do not need to be an artist to communicate concepts visually. Effective concept sketches follow simple rules:

Use block diagrams for systems: rectangles connected by labeled arrows showing signal/energy/material flow. Every block has a clear label. Every arrow has a label describing what crosses the boundary. This is the universal language of systems engineering and is expected at PDR.

Use annotated sketches for physical concepts: draw the concept in 2D or isometric 3D. Label key components, dimensions, and materials. Use call-out lines to annotate features that are not obvious from the drawing. The sketch does not need to be to scale; it needs to communicate the concept clearly.

Use comparison layouts: when presenting 3 concepts for selection, show them side-by-side on a single slide or page with the same visual style, same level of detail, and same orientation. This enables instant visual comparison and prevents the bias that occurs when Concept A has a polished CAD rendering while Concept B has a rough hand sketch.

Tools: hand sketches (photographed or scanned) are perfectly acceptable at PDR. If you prefer digital tools: draw.io (free, web-based) for block diagrams, Inkscape (free) for technical illustrations, or PowerPoint/Keynote for quick annotated diagrams. CAD renderings are impressive but not required at the concept stage; spending 20 hours in SolidWorks on a concept that may be eliminated by the decision matrix is poor time management.

8.23 Practice Questions

1. Your reviewer says "All three concepts scored within 5% of each other." What does this indicate and what should you do?
2. Create a PDR preparation checklist: 10+ items organized as "1 week before," "3 days before," "day of."
3. Three requirements were flagged as untestable. Rewrite each: (a) "The system shall be efficient," (b) "The system shall be easy to use," (c) "The system shall be durable."
4. Your Mock PDR ran 8 minutes over time. Diagnose three causes and fix each.

Chapter 9

Critical Design Review (CDR)

9.1 Purpose and Scope

For the sprint-level CDR preparation timeline, see the **Student Guide, Sprint 8 section**.

CDR in 60 Seconds

CDR answers: "Will this design work?" By CDR, you must have: (1) a complete detailed design with CAD and drawings, (2) an Engineering Calculation Package proving feasibility, (3) a BOM with all components ordered or received, (4) a manufacturing plan with the InnoHub Manufacturing Review complete, and (5) an updated FMEA with mitigations. The design freezes after CDR.

CDR answers: *"Will this design work?"* The design must be complete enough to build or model in full fidelity. Every component specified with vendor part numbers. Every interface defined. Every critical analysis complete. Procurement in progress or complete. CDR is the **design freeze** [2], [35]; after CDR, architectural changes require formal change control.

CDR separates the conceptual work of SDI from the execution work of SDII. If the design is not solid at CDR, the build phase will be a series of crises. The question is not "is the design perfect?" but "is the design complete enough to build with confidence?"

9.2 Timeline

SDII Sprint 8 (Weeks 3–5):

1. Sprint 7 (Week 1): re-engage with client, review Product Backlog, set SDII goals, complete Hazard Assessment and Safety Plan.
2. Sprint 8 Weeks 1–2: finalize CDRt and Engineering Calculation Package. Send to client before CDRw.
3. Sprint 8 Weeks 2–3: CDRw presentations (may span two weeks).

Feedback Loop Peer Evaluation 1 occurs around this time.

9.3 CDR Report (CDRt)

Builds on updated PDR report and adds:

- **Complete detailed design**: CAD assembly with dimensions/tolerances, wiring diagrams with values and pin assignments, software architecture with state machines, piping layouts, structural drawings.
- **Bill of Materials**: every component with description, manufacturer, part number, vendor, unit cost, quantity, total cost, lead time, status (ordered/received/backordered).
- **Engineering Calculation Package**: formal analyses; hand calculations, FEA/CFD, circuit simulation, thermal analysis, structural analysis. Each calculation: purpose, assumptions, methodology, results with units, conclusion, and source citations. This is a narrative engineering document.
- **Updated FMEA** [15]: showing mitigations since PDR and reduced RPNs.
- **Manufacturing plan**: fabrication sequence, tools and facilities needed, procurement status. InnoHub Manufacturing Review must be completed.
- **Updated VCRM** [34]: analysis verifications complete. Some subsystem test results available.
- **Updated budget and schedule**: actuals vs. planned with projections.

9.4 The Engineering Calculation Package in Detail

The Calculation Package is the engineering backbone of CDR. It demonstrates that the team has performed the analyses necessary to confirm design feasibility *before* committing resources to fabrication.

Every calculation must include:

1. **Purpose**: which requirement or design question is being addressed (by ID).
2. **Assumptions**: stated explicitly (e.g., "ambient temperature assumed 25 °C," "material yield strength from vendor datasheet page 3").
3. **Methodology**: equations used, with textbook or standard references. For simulations: mesh settings, boundary conditions, convergence criteria.
4. **Results**: numerical results with appropriate units and significant figures. Include plots, stress contour maps, or time-domain responses as applicable.
5. **Conclusion**: "The safety factor under maximum load is 3.2, exceeding the required 2.0. The design is structurally adequate." Or: "The motor stall torque of 0.8 N·m is insufficient to meet the 1.2 N·m requirement. Recommend selecting a larger motor (see Section 4.3)."

9.5 Track-Specific CDR Expectations

Build Track: fabrication-ready drawings at manufacturing-quality detail. All critical components ordered or received. InnoHub Manufacturing Review Meeting completed and checkoff signed on purchasing form.

Paper Track: complete model/simulation architecture with preliminary results demonstrating convergence or feasibility. Validation plan with identified benchmark data or experimental datasets. Documentation sufficient for another engineer to reproduce the analysis without contacting the team.

Competition Track: competition compliance checklist showing how each rule is addressed. Registration confirmed. Travel plan (Chapter 16) with PA-approved budget.

Student Guide Cross-Reference

For the sprint-level CDR timeline and deliverable checklist, see the **Student Guide, SDII Sprint 8** section.

9.6 CDR Presentation (CDRw)

The CDR presentation narrative answers "will this design work?":

1. **Design evolution since PDR** (2–3 min): summarize how the design changed based on PDR feedback.
2. **Detailed design** (5–8 min): CAD assembly (exploded views help), wiring diagrams, software architecture, key subsystem details.
3. **Engineering analysis** (5–8 min): walk through the key calculations from the Engineering Calculation Package. Focus on the analyses that prove feasibility: structural safety factors, thermal margins, power budgets, control stability.
4. **BOM and procurement** (2–3 min): BOM summary with critical items highlighted. What is ordered, received, or backordered?
5. **Manufacturing plan** (2–3 min): facilities, tools, fabrication sequence, quality checkpoints. InnoHub Manufacturing Review complete?
6. **VCRM status** (2–3 min): which verifications are complete (analysis), which are planned (test), what is the test timeline?
7. **Schedule and budget** (1–2 min): actuals vs. planned, projections through FDR.
8. **Q&A** (5–10 min).

9.7 BOM Management

The BOM is the financial and procurement backbone. Every entry must include: item number, description, manufacturer, part number, vendor, unit cost, quantity, extended cost, lead time, and procurement status (Not Ordered / Ordered / Received / Backordered).

Common BOM deficiencies at CDR: missing fasteners (every bolt, nut, washer, standoff must be listed), missing wire, cable, and connectors, missing consumables (solder, heat shrink, adhesive, filament), and missing enclosure hardware. A BOM that lists only the "exciting" components but omits the "boring" ones is a budget bomb.

The BOM total must reconcile with the Budget Tracker. If they differ, investigate: the BOM reflects what you need; the tracker reflects what you have spent. The gap is your remaining procurement obligation.

9.8 The Build Phase Transition

CDR marks the transition from design to execution. Common pitfalls:

Continuing to redesign: the design is frozen after CDR. Teams that keep "optimizing" instead of building arrive at FDR with a beautiful CAD model and no working prototype.

Underestimating fabrication time: assembly, wiring, debugging, and rework always take longer than estimated. Budget 2× your initial fabrication time estimate.

Big-bang integration: assembling everything at once and hoping it works. Build and test one subsystem at a time, then integrate incrementally. This follows the **debugging pyramid**:

The Debugging Pyramid

Test in layers, from the bottom up:

1. **Component level**: does each individual component work on the bench? (sensor reads correctly, motor spins, relay clicks)
2. **Interface level**: do two components talk to each other? (sensor data reaches the microcontroller, PWM signal drives the motor controller)
3. **Subsystem level**: does each functional group perform its role? (the sensing subsystem acquires and filters data correctly under expected conditions)
4. **System level**: do all subsystems work together to meet requirements? (end-to-end functionality against the VCRM)

Problems found at Level 1 take minutes to fix. The same problem found at Level 4 can take days to isolate. Never skip levels.

Neglecting build documentation: photograph every assembly step, every test setup, and every intermediate result. At FDR, you will need these images and they are impossible to recreate after the fact.

9.9 After CDR

- **Next Steps Email (post-CDR)**: CDR outcome and build/test plan.
- Begin fabrication, assembly, and/or full-fidelity modeling in Sprint 9.
- Post-CDR changes: documented change control with rationale, impact, PA approval.

The Spring Break Shipping Trap

Spring Break falls after CDR. Campus facilities **closed for a full week**. Order replacement components *before* break. Plan critical procurement to arrive one week before break. This is the most common procurement failure in capstone.

9.10 CDR Grading Criteria

CDR evaluation typically weighs:

Design completeness (∼30%): Is the detailed design complete enough to build/model? Are all components specified with part numbers? Are interfaces defined? Are drawings at manufacturing quality?

Engineering analysis (∼25%): Is the Engineering Calculation Package rigorous? Are assumptions stated? Are conclusions supported by the analysis? Do safety factors meet requirements?

Procurement and manufacturing readiness (∼15%): Is the BOM complete with costs and lead times? Are critical items ordered? Is the Manufacturing Review Meeting complete? Is the fabrication plan realistic?

Risk and VCRM status (∼10%): Has the FMEA been updated with design mitigations? Are analysis-based VCRM entries complete? Is the test plan realistic for the remaining schedule?

Communication quality (∼20%): Report professionalism, presentation clarity, team coordination, Q&A preparedness.

9.11 The CDR "Gotcha" List

Items that are frequently missing at CDR and cause immediate reviewer concern:

1. **Incomplete BOM**: missing fasteners, wire, connectors, or consumables. If the reviewer can see components in the CAD model that are not in the BOM, credibility drops.
2. **No procurement status**: the BOM lists items but does not indicate whether they have been ordered or received. At CDR, all critical items should be ordered.
3. **Calculations without conclusions**: presenting a page of equations without stating "therefore, the safety factor is 2.3, which exceeds the requirement of 2.0." The reviewer should not have to interpret your analysis.
4. **Missing interface definitions**: the block diagram shows subsystems connected by arrows, but the arrows are not labeled with signal type, voltage level, protocol, or connector.
5. **No test plan timeline**: the VCRM shows 20 requirements to be verified by Test, but the schedule does not allocate specific weeks for testing. When will each test occur?
6. **Unrealistic schedule**: the Gantt chart shows fabrication, assembly, integration, and system testing compressed into 3 weeks. Reviewers know this is insufficient; plan realistically.

9.12 Engineering Calculation Package Best Practices

The Engineering Calculation Package is one of the most important CDR deliverables. It demonstrates that your design is based on engineering analysis, not guesswork. Best practices:

One calculation per section: each analysis should be a self-contained section with: (1) purpose (which requirement or design question it addresses), (2) given information (loads, dimensions, material properties with sources), (3) assumptions (stated explicitly; each assumption is a potential failure point), (4) methodology (equations with variable definitions and sources), (5) solution (showing the arithmetic or referencing the simulation), (6) result with units and significant figures, and (7) conclusion ("The safety factor is 2.3, exceeding the required 2.0. The bracket is structurally adequate under the specified load.").

Show your work: a single number without derivation is unverifiable. Even if you used MATLAB or Excel, show the equations and the input values so a reviewer can check your work.

Simulation documentation: for FEA, CFD, or circuit simulation, include: the model description (geometry, boundary conditions, material properties), the mesh details and convergence study (show that the result does not change significantly when the mesh is refined), the results (stress contours, flow patterns, voltage waveforms), and the validation (comparison to hand calculation, published data, or experimental measurement). Screenshots without context are not engineering documentation.

Units consistency: one of the most common calculation errors is mixing unit systems. State all inputs in consistent units (SI preferred). If converting between systems, show the conversion. Check dimensional analysis: the units on the left side of the equation must equal the units on the right.

Peer review: have a teammate check every critical calculation before CDR. A second pair of

eyes catches arithmetic errors, wrong unit conversions, and flawed assumptions that the original analyst has become blind to.

9.13 Design Freeze Discipline

CDR establishes the **design freeze**; the point at which the design is stable enough to build. After CDR, changes are permitted only through formal change control. This discipline is critical because:

Changes after CDR cascade: modifying one component often affects adjacent components, interfaces, the BOM, the manufacturing plan, the test procedures, and the schedule. A "small" change to the sensor mounting bracket may require: new mechanical drawings, a revised enclosure design, new 3D print files, updated wiring routes, and modified test procedures. What seemed like a 2-hour task becomes a 2-week task.

Procurement is committed: by CDR, critical components are ordered or received. Changing the design may mean components you already purchased are useless, and new components must be ordered with their own lead times.

Team paralysis: a design that keeps changing demoralizes the team. Members who built a subsystem to the CDR specification only to learn that the specification changed will lose confidence in the design and the process. Freeze the design, build to the frozen design, and address shortcomings through tested improvements; not through untested redesign.

When changes ARE justified: safety issues discovered after CDR are always justified. Client-directed scope changes with full impact assessment are justified. Test results that reveal a fundamental flaw (not just a marginal miss) are justified. In all cases, document the change through the change control process: what changed, why, what is the impact, and who approved.

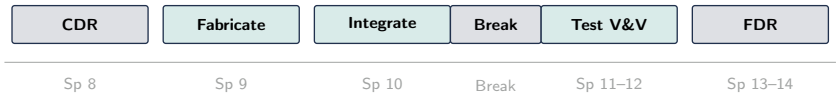

Figure 9.1: The CDR-to-FDR build phase. Fabrication and subsystem testing overlap; system integration occurs before Spring Break; V&V testing fills the remaining sprints. Start testing early to preserve redesign margin.

9.14 Starting Subsystem Testing Before Full Assembly

Do not wait until the entire system is assembled to start testing. Begin testing individual subsystems as soon as they are built:

Sprint 9: test individual subsystems in isolation. Does the sensor read correctly when connected to the amplifier and ADC? Does the motor spin at the correct speed with the driver? Does the firmware control the actuator correctly with a simulated sensor input?

Sprint 10: test subsystem pairs across interfaces. Does the sensor subsystem communicate correctly with the control subsystem? Does the power distribution subsystem maintain voltage regulation under the actuator load?

Sprint 11: system integration test. Connect all subsystems and verify end-to-end functionality. Test subsystems individually before connecting them.

Sprint 12–13: V&V testing against requirements. Execute formal test procedures from the VCRM. Record data. Disposition results.

This progressive testing approach (the debugging pyramid: components → interfaces → subsystems → full system) catches problems early when they are cheap to fix, rather than late when they are expensive and schedule-breaking.

9.15 The CDR-to-Build Transition Timeline

The weeks following CDR are the most schedule-critical period of the entire capstone. Figure 9.1 shows the overall timeline. A realistic breakdown for Build Track:

CDR Week (Sprint 8, Week 3–5): design freeze. All remaining component orders submitted. Manufacturing Review Meeting completed if not already done.

Sprint 9 (Week 6–7): fabrication begins. 3D print enclosures, machine brackets, assemble PCBs, begin firmware development. Test each subsystem individually as it is built (debugging pyramid Level 1–2).

Sprint 10 (Week 8–9): continue fabrication. Begin subsystem integration testing (Level 3). Order replacement parts for anything that failed during subsystem testing. This is typically when the first "this doesn't work as expected" moments occur.

Sprint 11 (Week 9–11): system integration (Level 4). All subsystems connected. Begin system-level V&V testing against requirements. Broader Impacts Essay and Peer Evaluation #2 also fall in this window.

Sprint 12 (Week 12–13): continue V&V testing. Address failed requirements (redesign, retest, or disposition). Begin writing FDR report sections.

Sprint 13 (Week 13–14): finalize FDR report. Submit poster for printing. Complete all VCRM entries. Write O&M manual. Prepare FDR presentation.

Sprint 14 (Week 15–16): FDR presentation. Design Showcase. Final deliverables. Checkout.

The Integration Squeeze

Most capstone teams underestimate the time between "all subsystems work individually" and "the complete system works." Integration typically takes 2–3 weeks for a moderately complex system. If your first integration attempt occurs in Sprint 12, you have less than one week to debug integration issues before FDR documentation must begin. Plan integration for Sprint 10–11.

9.16 The Engineering Calculation Package: What Reviewers Expect

The Eng Calc Package is often the weakest CDR deliverable. Reviewers expect:

Structural analysis: for any load-bearing component, show the free body diagram, the governing equation (bending stress, shear, buckling, fatigue), the material properties with source (datasheet, Matweb, textbook table number), the calculated stress/deflection, the allowable stress/deflection (from the requirement), and the safety factor. SF $= \sigma_{\text{allow}}/\sigma_{\text{actual}} \geq 2.0$ for most capstone applications.

Thermal analysis: for any system with heat generation or temperature sensitivity, show the heat source (power dissipation from datasheets), the thermal path (conduction through materials, convection to air, radiation if significant), the thermal resistance model ($R_{\text{total}} = R_1 + R_2 + \ldots$), the calculated temperature rise ($\Delta T = P \times R_{\text{total}}$), and the comparison to the maximum operating temperature of the most temperature-sensitive component.

Electrical analysis: for any power distribution design, show the voltage regulation chain (source $\rightarrow$ regulators $\rightarrow$ loads), the current budget (maximum current draw per rail, sum vs. regulator capacity), the power budget ($P = V \times I$ per rail, total system power vs. source capacity), and the battery life calculation (capacity in Ah / average current draw = runtime in hours).

Control analysis: for any feedback control system, show the plant model (transfer function or state-space representation), the controller design rationale (PID gains with tuning method), the predicted closed-loop response (settling time, overshoot, steady-state error vs. requirements), and the stability margin (gain margin >6 dB, phase margin >30°).

Each analysis section concludes with a clear statement: "The calculated [metric] is [value] with units, which [meets/does not meet] the requirement of [value]. The safety factor is [value]." A calculation without a conclusion is incomplete.

9.17 SDII Week 1: The Re-engagement Sprint

The first week of SDII (Sprint 7) sets the tone for the entire semester. Key actions:

Re-engagement email to client: this is a required deliverable. It must contain five elements: (1) reintroduce the team (the client may not remember names after the break), (2) summarize the SDI status (selected concept, PDR outcome, key open items), (3) propose SDII goals and the plan to achieve them, (4) request a meeting to align on priorities, and (5) attach the Group Reflection and Action Plan.

Review and update the Product Backlog: the client's priorities may have shifted during the break. Schedule a meeting within the first week to review the backlog, confirm SDII goals, and identify any new requirements or changes.

Set SDII semester goals: document SMART goals with PA and client approval. Upload to Canvas. These goals become the benchmark for your Project Goal Attainment assessment at FDR.

Complete the Hazard Assessment and Safety Plan: required before any fabrication begins. Use the template on Canvas. Identify all hazards, assess risks, and document controls. PA must approve.

Review team dynamics: SDII teams may have different membership or different dynamics than SDI. Hold a team discussion: what worked in SDI? What needs to change? Update the Team Contract if needed.

9.18 Subsystem Testing Before CDR

While most V&V testing occurs post-CDR, some subsystem testing should begin during PDR-to-CDR transition:

Analysis-based verification (A): these VCRM entries should be completed by CDR. If the structural analysis shows a safety factor of 2.3, mark that requirement as "Verified by Analysis" in the VCRM with a reference to the Engineering Calculation Package section.

Critical component testing: if there is a single component whose performance is uncertain (e.g., a sensor in a harsh environment, a motor under the expected load, a communication link at the required distance), test it before CDR. A CDR that claims "we will test this in Sprint 10" when the result could change the entire design direction is a CDR with unretired risk.

Interface testing: verify that key interfaces work before freezing the design. Can the MCU communicate with the sensor via I2C? Can the motor driver handle the expected current? Can the power supply maintain regulation under the expected load? Interface failures discovered after CDR require design changes that disrupt the build schedule.

Prototype demonstration: if your proof-of-concept from Sprint 4 (SDI) demonstrated partial functionality, show the updated version at CDR with improved performance. A CDR that shows working hardware (even partial) is dramatically more convincing than one that shows only CAD models and calculations.

The goal is not to complete V&V before CDR; it is to retire the highest-risk uncertainties so that the post-CDR build phase proceeds with confidence.

9.19 BOM Validation Before CDR Submission

The BOM is the financial and procurement backbone of the build phase. Validate it before CDR:

Completeness check: walk through the CAD assembly and verify that every component visible in the model has a corresponding BOM entry. Common omissions: fasteners (bolts, nuts, washers, standoffs (count them individually), wire and cable (estimate length, add 20% margin), connectors (male and female), heat shrink tubing, thermal paste, adhesives, and consumables (solder, flux, filament).

Price verification: check that BOM prices are current (vendor prices change). Verify: are these the unit prices or the total prices? Is shipping included? Is tax excluded (it should be) Mines is tax-exempt)?

Availability check: for every BOM item, verify that the vendor shows "In Stock" with sufficient quantity. A component that was in stock when you added it to the BOM may be out of stock when you order 4 weeks later. For critical items, check availability at multiple vendors.

Lead time verification: confirm that every item can arrive before the build start date. Items with lead times >2 weeks should already be ordered. Items with lead times >4 weeks should have been ordered in Sprint 6 of SDI.

Alternative identification: for every critical component (any item whose absence would stall the build), identify a backup source or substitute component in the BOM notes column. This dual-source strategy (Chapter 15) protects against stock-outs and shipping delays.

9.20 The CDR Readiness Self-Assessment

Before presenting CDR, the team should assess readiness against this checklist:

- ☐ Detailed design complete: CAD assembly with all components, drawings with dimensions and tolerances, wiring diagrams with pin assignments and component values.
- ☐ Engineering Calculation Package complete: every critical analysis documented with assumptions, methodology, results, and conclusions.
- ☐ BOM complete and validated: every component listed with vendor, cost, lead time, and procurement status.
- ☐ Critical components ordered or received.
- ☐ Manufacturing Review Meeting completed (for InnoHub builds).
- ☐ Hazard Assessment and Safety Plan approved by PA.
- ☐ FMEA updated with design mitigations.
- ☐ VCRM updated: analysis-based verifications complete, test plan documented.
- ☐ CDR Report reviewed by team (integration meeting completed).
- ☐ CDR Presentation rehearsed at least once.
- ☐ Client received CDR Report at least 3 days before presentation.

If more than 2 items are unchecked, the team may not be ready for CDR. Discuss with your PA whether to request additional preparation time or present with documented gaps.

9.21 Post-CDR Change Management in Practice

After CDR, the design is frozen; but reality intrudes. Here is how to handle common post-CDR changes:

Component substitution: the specified motor is out of stock. Process: (1) identify an alternative that meets the same specifications (torque, speed, voltage, mounting dimensions), (2) verify electrical compatibility (driver IC can handle the new motor's current), (3) verify mechanical compatibility (shaft diameter, mounting pattern), (4) update the BOM and wiring diagram, (5) document the change with rationale, (6) get PA approval. If the substitute has different performance characteristics, re-run the relevant analysis from the Engineering Calculation Package.

Design modification to fix a test failure: a subsystem test reveals that the amplifier saturates at high signal levels. Process: (1) diagnose the root cause (gain too high, input range exceeded), (2) propose a specific fix (reduce gain resistor from $100\,\mathrm{k\Omega}$ to $47\,\mathrm{k\Omega}$), (3) verify the fix does not create new problems (lower gain may reduce SNR below the requirement threshold), (4) implement, test, and document. If the fix requires a PCB revision, factor the fabrication lead time into the schedule.

Scope reduction: testing reveals that a Should-Have requirement cannot be met within the remaining schedule. Process: (1) present the evidence to the PA and client, (2) propose deferral to future work, (3) document the deferral in the change log and the VCRM, (4) adjust the Sprint Backlog to focus remaining effort on Must-Have requirements.

All post-CDR changes, no matter how minor, must be documented in the change log. At FDR, the change log tells the complete story of how the design evolved from CDR to the final as-built configuration.

9.22 Practice Questions

1. Create a CDR readiness checklist with 12+ items across design, analysis, procurement, and documentation.
2. Your reviewer says the BOM has two critical components showing "not yet ordered" with 3-week lead times. CDR is today. What is the impact and corrective action?
3. Write a 200-word Engineering Calculation narrative for a cantilever bracket with safety factor $\geq$2.0 under a 500 N tip load.
4. A teammate says "We should wait until after CDR to order components since the design might change." Explain why this approach is risky.

Chapter 10

Final Design Review (FDR) and Design Showcase

10.1 Purpose and Scope

For the sprint-level FDR preparation timeline and final deliverable checklist, see the **Student Guide, Sprints 12–14 sections**.

FDR in 60 Seconds

FDR answers: "How well did it work?" By FDR, you must have: (1) a complete VCRM with every requirement dispositioned, (2) test data and analysis demonstrating verification results, (3) an as-designed vs. as-built comparison, (4) an O&M manual, (5) lessons learned and future work recommendations, and (6) a Showcase poster submitted for printing.

FDR answers: *"How well did it work?"* This is the culmination of two semesters. You present the completed design, verification results, honest assessment of successes and failures, lessons learned, and recommendations for future work. FDR is not a sales pitch; it is a technical accounting.

The quality of FDR documentation (depth of engineering evidence, honesty of assessment, professionalism of presentation) is the primary determinant of the final project grade. A team whose prototype works perfectly but whose documentation is thin will score lower than a team that encountered failures but documented them honestly and dispositioned them systematically.

10.2 Timeline

SDII Sprint 13–14 (Weeks 13–15). FDRt submitted before FDRw. Showcase poster submitted for printing approximately one week before the event (Week 16).

10.3 FDR Report (FDRt)

The comprehensive final engineering document:

- **Executive summary**: project overview, key results, recommendations.
- **Final VCRM** [34]: every requirement dispositioned; Verified (with evidence reference), Failed (with disposition), Waived (with client approval), Deferred (with plan). The most important table in the report.

- **As-designed vs. as-built**: every deviation documented with reason, performance impact, and engineering justification.
- **Test data and analysis**: raw data, processed results, statistical analysis, plots, engineering interpretation. Each test mapped to a requirement.
- **O&M documentation**: operating procedures, maintenance schedule, troubleshooting guide, spare parts list, end-of-life instructions.
- **Sustainability results**: performance against sustainability requirements and CLO 5 evaluation.
- **Lessons learned**: specific, honest reflection.
- **Future work**: specific technical tasks beyond scope or time.
- **Final BOM**: actual vs. budgeted costs.

10.4 FDR Presentation (FDRw)

Structure around **results, not process**:

Section	Content	Time
Recap	Problem, requirements, design approach	1–2 min
Final System	Photos, video, live demo (Build); key results (Paper)	3–5 min
Verification	Walk through VCRM: passed, failed, dispositioned	5–8 min
Lessons Learned	Honest assessment, surprises, what you would change	2–3 min
Recommendations	Future work, next steps for the design	1–2 min
Q&A		5–10 min

10.5 Design Showcase

The public exhibition, typically during finals week:

Poster: follow the Program Format Guide on Canvas. Submit for printing one week before Showcase. Content: problem, approach, key results, significance; designed for a viewer standing 3 feet away. Large fonts ($\geq$24 pt body), clear graphics, minimal dense text. A good poster communicates the project in 2–3 minutes without the presenter speaking.

Live demonstration: Build Track teams demonstrate the prototype. Paper Track teams present results and visualizations. Prepare a 60-second elevator pitch.

Attendance: mandatory for all SDII students. SDI students send 2 members to attend SDII Showcase.

Capstone Awards Breakfast: recognition event during Showcase week.

10.6 The Operations and Maintenance Manual

The O&M manual is a professional deliverable that accompanies the system at handoff. It is not an academic exercise; it is the document that allows the client (or the client's staff) to operate, maintain, troubleshoot, and eventually decommission your system without contacting your team. After graduation, you will not be available for tech support.

Structure the O&M manual as follows:

1. **System overview**: what the system does, where it is installed, and what it connects to. Include a labeled photograph and a system block diagram.
2. **Startup procedure**: step-by-step instructions for powering on and initializing the system, from "plug in the power supply" to "verify the display shows the home screen."
3. **Normal operation**: how to use the system for its intended purpose. Describe the primary use case from the user's perspective.
4. **Shutdown procedure**: how to safely power down and store the system.
5. **Routine maintenance schedule**: what needs to be inspected, cleaned, calibrated, or replaced, and how often. Example: "Clean the sensor probe with isopropyl alcohol every 30 days. Replace the air filter every 90 days."
6. **Troubleshooting guide**: a table of common symptoms, probable causes, and corrective actions. Example: "Display shows NO SIGNAL → check sensor cable connection at J3 → if loose, reseat and restart."
7. **Spare parts list**: components that are likely to need replacement over the system's lifetime, with part numbers and suppliers.
8. **End-of-life instructions**: how to safely decommission the system, what materials require special disposal (batteries, chemicals), and where to recycle each component.
9. **Contact information**: Mines Capstone Design program contact for future inquiries.

Write the O&M Manual for a Non-Engineer

The person who operates your system may not be an engineer. Write clearly, use photographs and diagrams liberally, avoid jargon (or define it), and number every step. If the user must perform a sequence in a specific order, make the order explicit and explain why.

10.7 Poster Design for the Design Showcase

The Showcase poster is a visual communication piece, not a miniaturized version of your FDR report. Design for a viewer standing 3 feet away with 2–3 minutes of attention.

Layout: five sections, roughly equal space: (1) Problem and motivation (why this matters), (2) Approach and design (how you solved it), (3) Key results (what happened, with data visualizations), (4) Lessons learned or impact (what it means), (5) Team information and acknowledgments. Use large fonts (title $\geq$72pt, body $\geq$24pt), high-quality images (not screenshots of slides), and minimal text (target $<$400 words total on the poster).

Data visualization: use plots and charts, not tables of numbers. A well-designed bar chart showing "requirement vs. measured performance" communicates instantly what a 20-row VCRM table takes minutes to parse. Label all axes, include units, and use color consistently.

The poster is not the report: do not try to include everything. Select the 3–5 most important results and communicate them clearly. The poster invites conversation; the conversation provides the depth.

Submit the poster for printing approximately one week before the Showcase. Coordinate with CP for printing logistics and payment.

10.8 The SDII Sprint Flow: Build, Test, Document

The 14 weeks of SDII have a natural rhythm:

Sprints 7–8 (Weeks 1–5): re-engagement, CDR preparation, CDR execution. Design freeze.

Sprints 9–11 (Weeks 6–11): the build-and-test core. Fabricate, assemble, debug subsystems, integrate, begin system testing. The Broader Impacts Essay and Mid-Semester Course Survey also fall in this window. Peer Evaluation 2 occurs around Sprint 10.

Sprints 12–13 (Weeks 12–14): final testing, V&V completion, FDR preparation. This is where documentation debt catches up with teams that have not been writing as they go.

Sprint 14 (Weeks 15–16): FDR presentation, Design Showcase, final deliverables, and checkout.

The teams that succeed in SDII are the ones that **start testing early**. If your first test occurs in Sprint 12, you have left zero margin for redesign. If your first test occurs in Sprint 9, you have time to find problems, fix them, and retest.

10.9 Final Deliverables and Checkout

- **Project Goal Attainment**: evaluation of semester goals.
- **Digital Design Archive**: all files organized by category (CAD, code, simulation, test data, photos, videos, deliverables).
- **"What I Wish I Would Have Known" team video**: reflective video for future teams. What would you tell a team starting this project tomorrow?
- **Project Synopsis**: short summary for program records.
- **Next Steps Email (post-FDR)**: final client update.
- **Final Checkout**: closing meeting with PA; hand off materials, return equipment, close the project.

10.10 Self-Reflection

CLO 17 requires self-evaluation and improvement strategies. Specific reflection is valued: "I underestimated procurement lead times by three weeks, compressing our test schedule from four weeks to one. Next time I would order all critical components at CDR rather than waiting for post-CDR confirmation" demonstrates more engineering maturity than "I learned about teamwork."

The Three Questions of Capstone

PDR: "Should this design work?" **CDR**: "Will this design work?" **FDR**: "How well did it work?"
The quality of your answer (engineering evidence, honesty of assessment, professionalism of documentation) determines your grade and your growth.

10.11 FDR Grading Criteria

FDR evaluation focuses on:

Verification results (∼30%): Is the VCRM complete with all requirements dispositioned? Are test results presented with data, analysis, and acceptance criteria? Are failures honestly reported and dispositioned?

Documentation quality (∼25%): Is the FDR report comprehensive and professionally written? Is the as-designed vs. as-built comparison complete? Is the O&M manual usable by a non-expert?

Engineering depth (∼20%): Does the test data demonstrate rigorous engineering (multiple measurements, statistical treatment, environmental variations)? Are lessons learned specific and actionable?

Showcase quality (∼10%): Is the poster professional and communicative? Is the demonstration prepared and functional? Can all team members explain the project to visitors?

Individual contribution (∼15%): Does Feedback Loop evaluation and PA observation indicate equitable, professional contribution from each team member?

10.12 The Final Sprint Countdown

The last 4 weeks of SDII are the most intense period of capstone. A realistic timeline:

4 weeks before FDR: system integration complete. Begin system-level V&V testing. Start FDR report outline.

3 weeks before FDR: primary V&V testing complete. Begin writing FDR report sections. Start O&M manual.

2 weeks before FDR: FDR report first draft complete. Team peer review. VCRM disposition for all requirements. Submit poster for printing.

1 week before FDR: FDR report finalized and sent to client. Presentation rehearsed. Showcase demonstration tested. All final deliverables in progress.

FDR week: present FDRw. Submit all final deliverables. Attend Showcase. Final Checkout with PA.

Documentation Debt

The number one FDR crisis is documentation debt; teams that spent all their time building and testing but did not write anything until the last week. The FDR report is not a one-week deliverable; it is the culmination of two semesters of engineering documentation. Write as you go: update the SDRD when requirements change, update the VCRM when tests are completed, photograph every fabrication step, and draft report sections during the sprint in which the work was done.

10.13 Writing the FDR Report: A Chapter-by-Chapter Guide

The FDR report is the most comprehensive deliverable of the entire capstone experience. Plan your writing timeline: assign each section to a team member, set internal deadlines for drafts, and schedule a team review session at least one week before submission.

Chapter 1: Executive Summary (write last, 1 page). Problem, approach, key results, and recommendations. A reviewer who reads only this page should understand the project's outcome.

Chapter 2: Introduction and Background. Project context, client, problem statement, operating environment. Update from PDR/CDR versions to reflect the current understanding.

Chapter 3: Requirements and VCRM. The complete, final VCRM is the centerpiece. Present it as a table with clear pass/fail indicators. Discuss the overall verification results: how many requirements passed, failed, were waived, or deferred. For each failure or waiver, provide the disposition rationale.

Chapter 4: Design Description. The final design: CAD renderings, wiring diagrams, software architecture, piping layouts. Use the as-built configuration, not the as-designed (CDR) version.

Chapter 5: As-Designed vs. As-Built. Document every deviation from the CDR design. For each: what changed, why, what is the impact on performance. This section demonstrates engineering honesty and configuration awareness.

Chapter 6: Verification Results. Test data, analysis results, and inspection records organized by requirement. Show raw data, processed results, and engineering interpretation. Include plots, photos, and video references.

Chapter 7: Operations and Maintenance. The O&M manual (see earlier section). This may be a separate document appended to the report or integrated as a chapter.

Chapter 8: Sustainability Assessment. Results of the CLO 5 evaluation tool applied to the final design.

Chapter 9: Lessons Learned and Future Work. Honest, specific, actionable. See the framework below.

Chapter 10: References and Appendices. IEEE-format bibliography, detailed test data, engineering calculations, component datasheets, and supplementary materials.

10.14 The Lessons Learned Framework

Effective lessons learned are specific, causal, and actionable. Use this structure for each lesson:

What happened: describe the event factually. "The first PCB revision had a footprint error on the MOSFET driver IC."

Why it happened: identify the root cause. "The footprint was selected from a community library without verifying against the manufacturer datasheet. The library footprint had the drain and source pins swapped."

What was the impact: quantify the consequence. "The board required a second revision, adding 3 weeks to the schedule and $45 in fabrication costs. System testing was compressed from 4 weeks to 1 week."

What we recommend: state the actionable lesson. "Always verify PCB footprints against the manufacturer's recommended land pattern before submitting Gerber files. A 5-minute datasheet check would have prevented a 3-week delay."

Aim for 5–8 lessons learned, covering both technical and process topics. The best lessons learned sections are the ones that future teams can use directly to avoid the same mistakes.

10.15 The "What I Wish I Would Have Known" Video

Each SDII team produces a short reflective video (3–5 minutes) sharing advice for future capstone teams. This is one of the most valuable resources for incoming students. Structure your video around specific, actionable advice:

Strong content: "Start ordering long-lead-time components in Sprint 6 of SDI, not Sprint 8 of SDII. We lost 3 weeks waiting for a motor that could have been on our shelf before SDII started." "Schedule your InnoHub Manufacturing Review Meeting in the first week of SDII. We waited until Week 3 and the slots were full for two more weeks."

Weak content: "Communication is important." "Start early." "Work as a team." These are true but too generic to be useful.

Every team member should appear in the video. Be honest about what went wrong (future teams learn more from your mistakes than from your successes.

10.16 The Digital Design Archive

The Digital Design Archive is the permanent project record, submitted during Final Checkout. Organize it by category, not by date:

- `/01_Reports/`) SOW, PDRt, CDRt, FDRt (final versions)
- `/02_Presentations/` ; PDRw, CDRw, FDRw slide decks
- `/03_CAD/` ; SolidWorks/AutoCAD files, assembly and part files
- `/04_Drawings/` ; Engineering drawings (PDF), wiring diagrams
- `/05_Code/` ; Firmware, scripts, simulation files (with README)
- `/06_Analysis/` ; Engineering Calculation Package, FEA/CFD results
- `/07_Test_Data/` ; Raw data files, processed results, test photos/videos
- `/08_BOM_Budget/` ; Final BOM, Budget Tracker, purchase records
- `/09_Media/` ; Prototype photos, demo videos, Showcase poster
- `/10_Admin/` ; Meeting minutes, client correspondence, Team Contract

Include a top-level `README.txt` describing the project, team members, and archive organization. A future team or researcher should be able to understand and navigate the archive without contacting you.

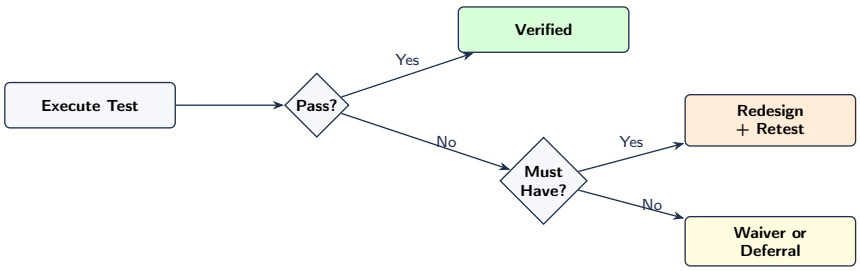

Figure 10.1: The requirement disposition flow at FDR. Every requirement must be dispositioned. Must-Have failures require redesign; Should-Have failures may be waived or deferred with client approval.

10.17 Dispositioning Failed Requirements

Not every requirement will be met. Professional engineering requires honest disposition of failures [2]. Figure 10.1 illustrates the decision flow for each requirement at FDR.

Redesign and retest: if time and budget permit, modify the design to meet the requirement and re-verify. This is the preferred disposition for Must-Have requirements. Document: what was changed, why, the retest results, and the impact on other requirements.

Waiver: the client agrees to accept the system despite the requirement not being met, because the shortfall is minor and the system is still useful. Document: the specific shortfall ("requirement: $\pm 1.0\,^{\circ}\mathrm{C}$; achieved: $\pm 1.3\,^{\circ}\mathrm{C}$"), the client's written approval, and the rationale ("the $0.3\,^{\circ}\mathrm{C}$ excess is within the measurement uncertainty of the reference instrument and does not affect the system's operational utility").

Deferral to future work: the requirement cannot be met within the current project timeline but can be addressed in a future revision. Document: the technical reason for deferral, the proposed approach for the future revision, and the client's acknowledgment. This is appropriate for Should-Have and Could-Have requirements.

Requirement removal: the requirement is no longer relevant (the client's needs changed, the operating environment changed, or the requirement was based on an incorrect assumption). Document: why the requirement is no longer applicable, the change control approval, and the impact on the overall system.

What is NOT acceptable: ignoring a failed requirement, reporting it as "in progress" at FDR, changing the requirement retroactively to match the achieved performance without documented change control, or blaming the failure on external factors without proposing a solution. Every requirement in the VCRM must be dispositioned at FDR. "Open" entries are not permitted.

10.18 The Showcase Preparation Checklist

Two weeks before the Showcase:

- Poster content finalized and reviewed by PA.
- Poster submitted for printing (coordinate with CP for printing logistics and payment).
- Demonstration script written and rehearsed. Every team member can explain the project in 60 seconds.
- Backup plan if the live demo fails (video of the demo, screenshots of key results, a prepared explanation of "what you would see if the demo were working").
- Display materials: table cover, team nameplate, project signage, business cards (optional but impressive).
- Dress code: business casual at minimum. Coordinate team attire.
- Each team member has prepared 3 talking points they can confidently discuss with any visitor.

10.19 The As-Designed vs. As-Built Comparison

Every capstone project deviates from the CDR design during fabrication. The as-designed vs. as-built comparison documents these deviations honestly:

What to document: every change from the CDR design, no matter how small. Examples: "Mounting hole moved 3 mm to clear the wire harness," "Resistor value changed from 10 kΩ to 12 kΩ to reduce amplifier gain by 17% (eliminated saturation observed during testing)," "MOSFET substituted from IRF540N to IRLZ44N due to stock-out (electrically compatible, lower $R_{DS(on)}$)."

How to document: create a table with columns: Item, CDR Design, As-Built, Reason for Change, Impact on Performance. Include the change control approval reference for each modification.

Why it matters: the as-built comparison demonstrates three things: (1) you maintained configuration awareness throughout the build phase, (2) you made engineering-justified changes (not random modifications), and (3) you can account for the differences between your CDR analysis (which was based on the CDR design) and your FDR test results (which are based on the as-built system).

A team that presents identical CDR and FDR designs is either: (a) extraordinarily disciplined and lucky (rare), or (b) not tracking changes (common). Reviewers know the difference.

10.20 The Design Showcase: Logistics and Setup

The Showcase is typically held during finals week in the Labriola Innovation Hub. Logistics:

Space allocation: each team receives a table or display area. Arrive early (at least 30 minutes before the event opens) to set up. Bring: poster (printed and delivered by CP), prototype, demonstration equipment, laptop for video backup, extension cord, table cover, and team nameplate.

Power and connectivity: verify that your display area has power outlets. Bring a power strip. If your demonstration requires internet connectivity, test the WiFi in the Showcase space beforehand; large crowds can overwhelm WiFi networks.

Demonstration preparation: test your demonstration setup in the Showcase space before visitors arrive. Murphy's Law applies: the demo that worked perfectly in the lab may fail in a new location due to different lighting, power quality, WiFi, or simply the stress of performing in front of 50 visitors. Have a backup plan: video of the demo, screenshots of key results, or a prepared explanation.

Team coordination at the Showcase: not all team members need to be at the display at all times. Rotate: 2–3 members at the display engaging visitors while others take breaks, visit other teams' displays, or attend the Capstone Awards Breakfast. Ensure at least one member who can explain the technical depth is always present.

Engaging visitors: stand in front of your display, not behind it. Make eye contact with approaching visitors. Offer the 60-second overview. If the visitor is interested, transition to the 3-minute guided tour of the poster. Let the visitor's questions guide the depth of discussion. Thank every visitor for their time.

10.21 The Project Synopsis

The Project Synopsis is a brief (1-page) summary document submitted during Final Checkout. It is used for program records, accreditation documentation, and future reference. Include: project title, team members, client name, problem statement (2–3 sentences), approach (2–3 sentences),

key results (2–3 sentences with quantified outcomes), and a representative photograph of the final system.

Write the synopsis for an audience that knows nothing about your project. It should be self-contained, professionally written, and suitable for publication in a program annual report or displayed on a department website.

Student Guide Cross-Reference

For the complete FDR/Showcase deliverable checklist and final checkout procedure, see the **Student Guide, SDII Sprints 12–14** sections.

10.22 The FDR Narrative Arc

Structure the FDR presentation around results and evidence, not chronology:

Opening (1–2 min): "We set out to [problem]. We designed and built [system]. Here are the results." The audience attended PDR and CDR; they know the background. Provide a brief reminder, not a re-presentation.

The system (3–5 min): show the final system with photos, video, or live demonstration. Paper Track: show key results and validated models. This is the moment the audience has been waiting for; show them the thing you built.

Verification results (7–10 min): walk through the VCRM systematically. Group requirements by category (functional, performance, safety). For each: state the requirement, show the test setup (photo), present the result (data plot or table), and state the disposition (pass/fail/waiver). Spend more time on failures and waivers than on passes; the audience learns more from how you handled problems than from things that worked perfectly.

Lessons learned (2–3 min): be specific, honest, and forward-looking. "Our biggest lesson was that integration testing should have started two weeks earlier; the three issues we discovered in Sprint 12 could have been found in Sprint 10 if we had followed the debugging pyramid more rigorously." Generic lessons ("communication is important") add no value.

Future work (1–2 min): specific recommendations for what a future team or the client should do next. "The system meets 20 of 22 requirements. The two unmet requirements (wireless range and battery life) could be addressed by upgrading to the ESP32-S3 with an external antenna and a higher-capacity LiPo cell."

Q&A (5–10 min): the most important part. Every team member should be prepared to answer questions about the areas they led.

10.23 Practice Questions

1. Your VCRM shows 18 Verified, 2 Failed, 1 Waived. One failure is "Should" (data logging rate); the other is "Must" (emergency shutoff 0.3 s over spec). How should each be dispositioned?

2. Design a Showcase poster layout: identify 5 sections and allocate space percentages. What is the maximum word count a viewer should encounter?

3. Write a 150-word lessons-learned paragraph about a specific capstone failure demonstrating engineering maturity.
4. Your O&M manual is blank two weeks before FDR. Create an O&M outline for a temperature-controlled enclosure with 8 section headings and 2–3 items per section.

Part III

Project Management and Operations

Chapter 11

Hybrid Project Management for Capstone

This chapter covers everything your team needs to manage the capstone project from Sprint 0 through Final Checkout. It starts with the framework (why hybrid, how Scrum adapts to capstone) and sprint mechanics, then moves into the tools and artifacts you will use day-to-day (backlog, Gantt chart, task tracking, velocity), and concludes with the processes that keep the project under control when things change (change control, escalation, retrospective tracking). Refer back to specific sections as they become relevant to your current sprint.

11.1 Why Hybrid, Not Pure Agile or Waterfall

Capstone requires both predictability (fixed milestones: PDR, CDR, FDR, Showcase) and adaptability (evolving requirements, unexpected technical challenges, shifting client priorities). Pure Waterfall is too rigid; a detailed plan created in Sprint 1 will be obsolete by Sprint 4 as technical realities emerge. Pure Agile lacks the milestone structure that external stakeholders (clients, PAs, CF) require.

The hybrid approach uses **Waterfall for the milestone backbone** [8] (fixed gates: SOW, PDR, CDR, FDR, Showcase with non-negotiable dates) and **Agile (Scrum) for sprint-level execution** [6] within each milestone window. You know *what* must be demonstrated at each gate; you use sprints to figure out *how* to get there.

11.2 The Scrum Framework for Capstone

In Scrum terminology [6], your capstone roles map as follows:

Product Owner = the Client. The client owns the **Product Backlog**; the prioritized list of everything the system should do, ordered by value. The client decides *what* is most important; they do not tell you *how* to build it.

Scrum Master = the student team lead. The Scrum Master facilitates the process: runs sprint planning, conducts stand-ups, removes impediments, and ensures commitments are met. The Scrum Master is *not* the boss; they are the process guardian.

Development Team = the project team. Everyone contributes to the technical work. The team is self-organizing: they decide how to accomplish sprint goals, who works on what, and how to coordinate.

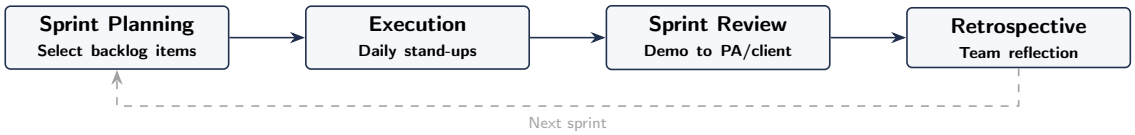

Figure 11.1: The sprint cycle. Each 2-week sprint follows the same cadence: planning, execution with daily stand-ups, review (demo completed work), and retrospective (team reflection and improvement).

11.3 The Sprint Cycle

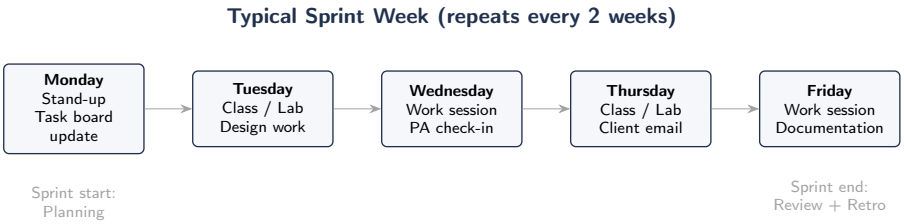

Figure 11.2: A typical sprint week. Monday: stand-up and task board update. Tuesday/Thursday: class and lab sessions. Wednesday/Friday: dedicated work sessions. The PA check-in (Wednesday) and client status email (Thursday) maintain communication cadence. Sprint Reviews and Retrospectives occur on the Friday of the sprint's second week.

The typical sprint week (Figure 11.2) provides a weekly rhythm that teams should adapt to their specific schedule. The key elements are non-negotiable: the stand-up (coordination), the PA check-in (accountability), and the client communication (professionalism). The work sessions between these touchpoints are where the engineering happens.

Capstone uses approximately 2-week sprint cycles. SDI contains Sprints 0–6 (Weeks 1–15); SDII contains Sprints 7–14 (Weeks 1–16). Each sprint follows the cadence shown in Figure 11.1.

Sprint Planning (beginning of sprint): the team selects items from the Product Backlog for this sprint based on priority and capacity. For each item, define what "done" means; this prevents ambiguity at the sprint review. Commit to a Sprint Backlog that is achievable in two weeks. Over-committing and under-delivering is worse than under-committing and over-delivering; adjust your capacity estimate based on the previous sprint's actual throughput.

Daily Stand-Ups [6] (throughout sprint): brief team check-ins (10–15 minutes, standing up to keep it short). Each member answers three questions: (1) What did I complete since the last meeting? (2) What will I work on next? (3) What is blocking me? The Scrum Master tracks blockers and works to remove them between meetings. Stand-ups are about *coordination*, not status reporting; the goal is to identify and resolve impediments quickly.

Sprint Review (end of sprint): demonstrate completed work to the PA (and client when possible). Show *working functionality*, not slide decks about functionality. "Here is the sensor reading on the screen" beats "here is a slide showing our plan to read the sensor." The sprint review is evidence-based. Several lectures are devoted to Sprint Reviews/Retrospectives; the Sprint Review and Retrospective assignments are graded.

Sprint Retrospective (end of sprint): the team reflects on the process itself, not the product.

Three questions: What went well this sprint? What should we improve? What specific actions will we take in the next sprint to improve? **Record the actions and assign owners**; a retrospective without follow-through is a waste of time. The Scrum Master checks retrospective action items at the next planning session.

The Velocity Diagnostic

Sprint velocity [6] is the number of backlog items (or story points) completed per sprint. Track it starting with Sprint 1. If velocity is declining over time, the team is losing momentum; investigate burnout, scope creep, or unresolved blockers. If velocity is increasing, the team is improving its process. Present velocity trends to your PA at sprint reviews.

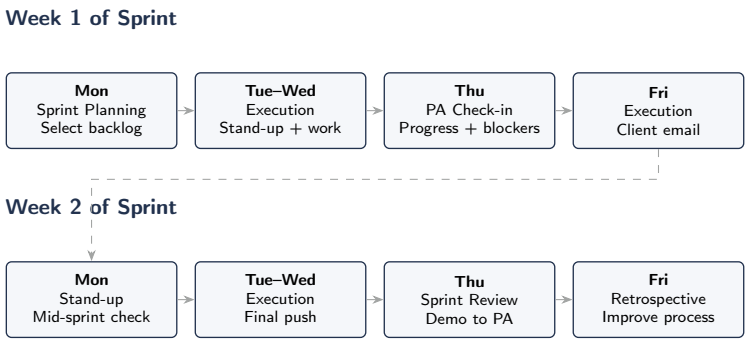

Figure 11.3: The Capstone Sprint: a typical two-week cycle. Week 1 focuses on planning and execution; Week 2 culminates in the Sprint Review (demo to PA) and Retrospective. Adapt the specific days to your team's schedule.

The weekly cadence in Figure 11.3 is a starting template, not a rigid prescription. Adapt it to your team's class schedule and PA availability. The non-negotiable elements are: sprint planning at the start, at least two stand-ups per week, the PA check-in, the sprint review (with a demo), and the retrospective. Teams that skip any of these elements consistently underperform teams that maintain the full cadence.

Student Guide Cross-Reference

For a quick, sprint-level checklist of deliverables and priorities at each phase, see the **Student Guide** sections for your current sprint. The Student Guide tells you *what to do this week*; this Course Text explains *why and how*.

11.4 Product Backlog and Sprint Backlog

The **Product Backlog** is the master list of all work the project requires, owned by the client and maintained by the Scrum Master. Each item has: a description, an estimate (story points, hours, or T-shirt sizes S/M/L/XL), a priority (from the client), and dependencies on other items. The backlog is a living document; items are added, refined, split, and reprioritized throughout the project.

The **Sprint Backlog** is the subset selected for the current sprint. The team commits to completing these items within the sprint. Items not completed roll back to the Product Backlog for the next sprint; they are not automatically carried forward. Consistently failing to complete the Sprint

Backlog indicates over-commitment; reduce scope in the next sprint and investigate root causes (task estimates too optimistic? external blockers? team availability issues?).

11.5 Work Breakdown Structure and Gantt Charts

While Scrum manages sprint-level execution, the **Work Breakdown Structure (WBS)** [8] decomposes the entire project into progressively smaller deliverables and tasks. The WBS ensures no major work element is overlooked. Decompose until each leaf task can be completed in one sprint or less by one person or a small sub-team. Typical capstone WBS has 3–4 levels of decomposition and 50–150 leaf tasks.

The **Gantt chart** [8] visualizes the WBS over time, showing task durations, dependencies (finish-to-start, start-to-start, etc.), and the **critical path**; the longest chain of dependent tasks that determines the minimum project duration. Tasks on the critical path have zero float (slack); any delay on the critical path delays the project end date. Non-critical tasks have float and can slip within that margin without affecting the overall schedule.

Identify Your Critical Path Early

Most capstone critical paths run through **procurement and fabrication**, not through design or analysis. A 3-week PCB fabrication lead time, a 2-week specialty sensor shipping delay, or a 1-week InnoHub Manufacturing Review scheduling wait puts those tasks on the critical path. Identify procurement-dependent tasks in Sprint 1 and order early. The critical path determines your schedule; everything else is secondary.

11.6 Task Tracking Tools

Use a visual task management tool that the entire team (and PA) can access: Trello boards, Microsoft Teams Planner, or GitHub Projects. Organize tasks in a Kanban-style board with columns: Backlog → To Do (this sprint) → In Progress → Review → Done. Update the board at every stand-up. A board that has not been updated in a week is a warning sign that the team has lost process discipline.

Each task card should include: a clear description, the assigned owner, the estimated effort, the deadline, and any blockers. Attach relevant files (drawings, datasheets, code commits) to the card for context.

11.7 The Team Charter and Team Contract

The **Team Charter** (SDI) establishes the team's mission, vision, goals, roles, communication norms (tools, response times, meeting schedule), and decision-making process (majority vote? consensus? technical lead decides for technical issues?).

The **Team Contract** (especially recommended for SDII) goes further, expressing minimum commitments that every member agrees to uphold:

- Presence at all team meetings, arriving on time.
- Minimum due dates for writing contributions (e.g., 48 hours before the team submission deadline).

- Minimum level of weekly contribution to the project.
- Expected professional behavior: communicating proactively when late or absent, responding to team messages within 24 hours, completing assigned tasks by the agreed date.
- Consequences for repeated violations: escalation to PIP process (Chapter 13).

Share the Team Contract with your PA. SDII teams should write or revise the contract at the start of the second semester.

11.8 Project Completion Plan

At each semester's start, establish explicit, measurable goals with your PA and client. Document them on Canvas. Goals should follow the SMART framework: Specific, Measurable, Achievable, Relevant, Time-bound. At semester's end, the **Project Goal Attainment** document evaluates performance against these goals. Teams not accomplishing their agreed-upon goals will not receive an A regardless of other scores.

11.9 Effective Meeting Management

Capstone teams attend many meetings: team stand-ups, PA meetings, client meetings, sprint reviews, retrospectives, design reviews, and ad-hoc technical discussions. Each meeting should be productive.

Before the meeting: prepare an agenda. For stand-ups, the agenda is fixed (three questions). For all other meetings, circulate the agenda at least 24 hours in advance. The agenda includes: purpose (what decision or outcome is needed), topics (numbered, with time estimates), and pre-reads (documents attendees should review beforehand).

During the meeting: start on time regardless of who is present. Assign a facilitator (keeps discussion on track) and a scribe (records decisions and action items). When a discussion goes off-topic, the facilitator parks it: "Good point; let us add that to the action items and discuss it at our next meeting." Limit meetings to the allocated time.

After the meeting: the scribe distributes meeting minutes within 48 hours. Minutes include: attendees, key decisions (with rationale), action items (owner, description, due date), and open questions. Copy the PA on all client meeting minutes.

Remote and hybrid meetings: in this era of remote work, some team members may need to join by Zoom or Teams. The expectation is face-to-face attendance for class and recitation. For team meetings outside of class, set guidelines with your team and PA. Remote participants should have video on, use a good microphone, and participate actively. A remote participant who is muted with camera off for the entire meeting is not present.

11.10 Common PM Anti-Patterns in Capstone

The Invisible Scrum Master: the Scrum Master creates a Trello board in Sprint 1 and never updates it again. Without visible task tracking, the team has no idea who is doing what, blockers go unresolved, and sprint reviews reveal that half the planned work was not completed. Fix: the Scrum Master updates the board at every stand-up and checks it before every sprint review.

The Perpetual Sprint 1: the team plans aggressively in Sprint 1, fails to complete the backlog, carries items forward, plans aggressively again in Sprint 2, fails again, and establishes a pattern of perpetual over-commitment. Fix: reduce Sprint 2 scope based on Sprint 1 actual velocity. Plan based on demonstrated capacity, not optimistic estimates.

The Retrospective Skip: "We don't have time for a retrospective; we need to build." This is like saying "I don't have time to sharpen the saw; I need to keep sawing." Retrospectives identify process inefficiencies that, when fixed, save more time than the retrospective consumes. Fix: retrospectives are non-negotiable. Keep them short (15–20 minutes) and focused on 2–3 specific improvements.

The Single Point of Failure: one team member owns a critical subsystem and nobody else understands it. When that person is sick, busy with other courses, or disengaged, the entire project stalls. Fix: pair programming and pair design. Every critical subsystem should have at least two team members who understand it. Document decisions in the project repository, not in one person's head.

The Phantom Stand-Up: stand-ups where everyone says "I'm still working on the same thing as last time" and no blockers are raised. This indicates either that the work items are too large (break them into smaller tasks) or that the team is not being honest about challenges. Fix: break backlog items into tasks completable in 1–2 days. If a task has been "in progress" for a week, it is either stuck (surface the blocker) or too large (decompose it).

11.11 PM Tools Comparison

Tool	Strengths	Limitations	Best For
Trello	Visual Kanban, simple, free tier	No Gantt, limited reporting	Sprint task tracking
Teams Planner	Integrated with Mines O365, assignments	Less flexible than Trello	Teams already using Teams
GitHub Projects	Integrated with code repos	Learning curve, developer-centric	Software-heavy projects
Excel/Sheets	Flexible, familiar, Gantt templates	Manual updates, no automation	WBS and Gantt charts

Most capstone teams benefit from using **two tools**: a Kanban board (Trello or Planner) for sprint task tracking, and a spreadsheet (Excel or Sheets) for the WBS, Gantt chart, and budget tracker. Do not over-invest in PM tooling; the tools should serve the process, not become the process.

11.12 The Complete Sprint Mapping

The following table provides the definitive sprint-to-milestone mapping for the two-semester capstone sequence. Use this table to plan your workload and anticipate deliverable deadlines.

Sprint	Semester	Focus	Gate
0	SDI	Project Expo, team formation, client kickoff	;
1	SDI	SOW, Project Plan, Scrum setup, product backlog	SOW
2	SDI	Concept generation, trade study, Career Day	;
3	SDI	Concept development, client approval, SDRD	;
4	SDI	Proof-of-Concept, ethics assignment, standards	;
5	SDI	PDR report, mock PDR, PDR presentation	**PDR**
6	SDI	Design refinement, SDII handoff, procurement start	;
7	SDII	Re-engagement, goal setting, hazard assessment	;
8	SDII	CDR report, eng. calc package, CDR presentation	**CDR**
9	SDII	Fabrication begins, subsystem testing	;
10	SDII	Integration, system testing begins	;
11	SDII	V&V execution, Broader Impacts essay	;
12	SDII	FDR preparation, final V&V, O&M manual	;
13	SDII	FDR report, poster design, final testing	**FDR**
14	SDII	FDR presentation, Showcase, final checkout	**Showcase**

11.13 Schedule Risk Management

Schedule risk is the probability that a task will take longer than planned. In capstone, schedule risk concentrates in three areas:

Procurement risk: components arrive late, are incorrect, or are out of stock. Mitigation: order early, identify backup sources, and have alternative components pre-selected in the BOM.

Fabrication risk: machining takes longer than estimated, 3D prints fail, PCB revisions are needed. Mitigation: add 50–100% margin to fabrication time estimates, prototype early, and start with the highest-risk fabrication tasks.

Integration risk: subsystems that work individually fail when connected. This is the most dangerous schedule risk because it occurs late in the project when margin is smallest. Mitigation: use the debugging pyramid approach (test components individually, then interfaces, then subsystems, then the full system), integrate incrementally (not all at once), and start integration testing as early as Sprint 10.

The schedule buffer strategy: instead of distributing slack evenly across all tasks, concentrate it in a **project buffer** at the end; a block of unallocated time before FDR that absorbs delays from any task. A 2-week project buffer (Sprint 12) that is consumed by integration debugging is far better than no buffer and a compressed FDR.

11.14 Communication Protocols by Sprint Phase

The communication cadence should intensify as milestones approach:

Normal sprints (Sprint 1–4, 7, 9–11): weekly team meetings (60–90 min), daily async stand-ups (Slack/Teams channel post), biweekly client status email, sprint review/retrospective at sprint end.

Pre-review sprints (Sprint 5, 8, 12–13): daily team meetings (30 min stand-up + work session), daily task board updates, client communication as needed for review preparation, mock review 3 days before formal review.

Crisis mode (integration failures, component failure, schedule slip): daily stand-ups become twice-daily check-ins, PA is notified immediately, client is notified within 24 hours with a recovery plan. Crisis mode is temporary; if it lasts more than one sprint, the project scope needs adjustment.

11.15 Velocity Tracking and Forecasting

Once your communication cadence is established, the next question is whether your team is making progress fast enough. Sprint velocity (the number of story points (or tasks) completed per sprint) is the most reliable predictor of future performance:

Tracking: at the end of each sprint, record: planned story points, completed story points, and the ratio (completion rate). After 3 sprints, you have enough data to forecast.

Forecasting: if your average velocity is 15 points/sprint and you have 45 points remaining in the backlog, you need ∼3 sprints to complete the work. If you have 2 sprints remaining, you must either reduce scope (remove 15 points from the backlog) or increase capacity (more hours, more team members on the critical path, or reduced task overhead).

The velocity plateau: most teams see velocity increase through Sprint 2–4 as they learn the process, then plateau. A velocity that is declining after Sprint 4 indicates: team burnout, unresolved blockers, increasing technical debt, or disengagement. Investigate at the retrospective.

Velocity is not comparable across teams: one team's "15 points" is not equivalent to another team's "15 points" because estimation calibration differs. Use velocity only for within-team forecasting.

11.16 Building an Effective Gantt Chart

Velocity tells you how fast you are moving; the Gantt chart tells you whether you are moving in the right direction and whether you will finish on time. It is the primary schedule visualization tool in capstone. Build it correctly:

Start with the WBS: every task in the Gantt must trace to a WBS element. If a task appears in the Gantt but not the WBS, either the WBS is incomplete or the task is unnecessary.

Define dependencies: for each task, identify what must be completed before it can start (predecessors). Common dependency types: Finish-to-Start (FS, most common: "PCB layout must finish before Gerber generation starts"), Start-to-Start (SS: "firmware development can start when the hardware design starts, using the pin assignment document").

Estimate durations realistically: the best estimate is based on experience. Since most capstone tasks are novel (you have not CNC-machined this specific bracket before), estimate conservatively. A useful heuristic: estimate the time, then add 50%. If you think the PCB layout will take 5 days, plan for 8.

Identify the critical path: the longest chain of dependent tasks. Color-code it in red on the Gantt. Every team member should know which tasks are on the critical path, because a delay on any of them delays the entire project. Tasks not on the critical path have float (schedule margin) and can slip within their float without affecting the project end date.

Add milestones: mark the fixed dates (PDR, CDR, FDR, Showcase, Spring Break) as diamond

milestones on the Gantt. These are non-negotiable. The schedule must work backward from these dates.

Update weekly: the Gantt is a living document. At each sprint retrospective, update task progress (percent complete, actual vs. planned dates), add new tasks discovered during the sprint, and recalculate the critical path. A Gantt that was created in Sprint 1 and never updated is not a management tool.

Tools: Microsoft Excel with conditional formatting works for simple Gantts. Microsoft Project provides automated critical path calculation but has a learning curve. Free alternatives include GanttProject and TeamGantt. For most capstone projects, an Excel-based Gantt is sufficient.

11.17 Change Control Process

After the SDRD is baselined at PDR and the design is frozen at CDR, changes must follow a formal change control process:

Step 1: Change request. Any team member can submit a change request. The request documents: what is being changed (specific section, requirement, or component), why the change is needed (test result, client request, safety concern, component unavailability), and the proposed new state.

Step 2: Impact assessment. The team evaluates the impact on: other requirements (does changing this requirement cascade?), the design (does the change require redesigning other components?), the schedule (how much time does the change add?), the budget (what does the change cost?), and the risk profile (does the change introduce new risks or retire existing ones?).

Step 3: Approval. The PA must approve all changes. If the change affects scope or performance, the client must also approve. For safety-critical changes, the CF may need to be notified.

Step 4: Implementation. The change is implemented in the design, the SDRD is updated, the BOM is updated, and the change is recorded in the change log with: change ID, date, description, rationale, impact, approver, and implementation status.

Step 5: Verification. After implementation, verify that the change achieves its intended effect and does not introduce new problems (regression testing).

Teams that skip change control arrive at FDR with a design that does not match their documentation. This mismatch makes the VCRM meaningless and undermines the credibility of the entire project.

11.18 When to Escalate Schedule Issues

Not all schedule problems require PA involvement. Use this escalation guide:

Handle within the team: a single task is 2–3 days late but can be recovered by reallocating effort or adjusting the sprint backlog. The Scrum Master adjusts the plan and reports the adjustment at the next sprint review.

Notify the PA: a task on the critical path is >1 week late, a procurement delay affects the build timeline, or the team's sprint velocity has declined for 2+ consecutive sprints. The PA should be aware and may offer advice or resources.

Escalate to PA and client: a milestone (PDR, CDR, FDR) is at risk of not being ready on time, a scope change is needed to maintain the schedule, or an external dependency (client access, facility availability) is blocking progress. Both the PA and client need to be involved in the recovery plan.

Escalate to CF: the project is fundamentally behind schedule with no viable recovery plan within the current scope, or the team has a systemic dysfunction (multiple members disengaged, leadership vacuum) that the PA cannot resolve alone.

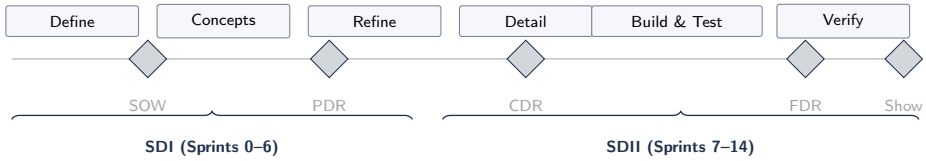

Figure 11.4: The capstone project timeline showing the five major milestones (SOW, PDR, CDR, FDR, Showcase) across the two-semester cycle. Phase widths are approximate; actual timing varies by sprint schedule.

The project timeline (Figure 11.4) shows the milestone backbone that all capstone projects follow. The phases between milestones have different characteristics: the Define and Concepts phases are intellectually exploratory; the Detail phase demands precision; the Build & Test phase demands execution discipline; and the Verify phase demands honesty. Successful teams adapt their working style to each phase rather than treating the entire project the same way.

11.19 The Retrospective Action Item Tracker

Retrospective action items that are not tracked are retrospective action items that are not completed. Maintain a simple tracker:

Sprint	ID	Action Item	Owner	Status
3	R3-1	Update task board daily, not weekly	All	Done
3	R3-2	Send client status email every Friday	Comm Lead	Done
4	R4-1	Start writing report sections during sprint	All	In Progress
4	R4-2	Order backup motor before Spring Break	CFO	Done

Review this tracker at the start of every sprint planning session. Close completed items. Carry forward open items. If an item has been open for 3+ sprints, either do it, delegate it, or delete it with an explanation of why it is no longer relevant.

11.20 Practice Questions

1. Your team completed only 60% of the Sprint 3 backlog. Diagnose three possible causes and propose a corrective action for each.
2. Draw a simplified WBS for a Build Track project involving a temperature-controlled enclosure. Identify the critical path assuming PCB fabrication takes 3 weeks and all other tasks take 1–2 weeks.
3. Your retrospective identified that "we need to communicate better." This is too vague to be actionable. Rewrite it as three specific, measurable action items with owners.

4. Explain the difference between a Product Backlog and a Sprint Backlog. Who owns each, and what happens to items not completed in a sprint?

Chapter 12

Fabrication Spaces, Resources, and Manufacturing

12.1 Labriola Innovation Hub Overview

The Labriola Innovation Hub provides dedicated spaces for capstone teams, both for project work during class time and for extended hours outside of class. You must complete required training before accessing any shop or workspace.

Thorson Capstone Design Center: primary workspace for all capstone teams, reserved for capstone students only. This open design space doubles as your build space and after-hours workspace. Blastercard access required. See the Labriola Canvas page for hours, Grad TA contacts, locker assignments, and storage policies. The Thorson Center is available during posted hours with TA supervision.

xWorks Project Bays: dedicated bays for teams working on vehicles or large-footprint projects (e.g., concrete canoe, Formula SAE car). Each qualifying team is assigned one bay for the semester. Review the expectations page for Blastercard enrollment.

Labriola Conference Rooms: second-floor conference rooms reserved for capstone students TTH 8:00–11:00 AM for Sprint Reviews and Design Reviews only. See the Labriola page for reservation procedures and limitations.

12.2 InnoHub Shops and Makerspace

The InnoHub includes specialized shops, each staffed and equipped for specific fabrication processes:

Metal Shop: manual mills, lathes, drill presses, band saws, grinders, MIG and TIG welders, sheet metal equipment. CNC mills and lathes available after completing manual machine training first.

Wood Shop: table saw, band saw, wood lathe, drill press, sanders, routers. CNC router available with training.

Electronics Shop: soldering stations, hot air rework station, oscilloscopes, bench power supplies, component storage, PCB assembly workspace.

Composites Shop: vacuum bagging, wet layup, curing oven. Specialized safety training required due to resin sensitization risks.

Makerspace: FDM and resin 3D printers, laser cutters, vinyl cutters, general hand tools, and fabrication workspace.

Training Is Mandatory; No Exceptions

Complete machine-specific safety training on the **Labriola InnoHub Canvas site** for every piece of equipment you plan to use. No drop-ins are accepted; you will be turned away. Trainings from other facilities (ME Machine Shop, physics lab, personal experience) do **not** substitute for InnoHub training. CNC training requires completion of the corresponding manual machine training first. Training prioritization: if a TA needs a machine or work area for safety training, you will be asked to return at a different time.

Extended machine use: if you plan to use manual mills/lathes, welders, wood lathe, or table saw for more than a **4-hour period**, get manager approval in advance.

12.3 Manufacturing Review Meeting

If your team plans to build your final project in *any* Labriola facility (Metal Shop, Wood Shop, Electronics Shop, Composites Shop, or Makerspace) you **must schedule a 30-minute Manufacturing Review Meeting** with one of the shop managers or the operations manager before ordering materials.

This meeting is a **required checkoff item on your purchasing form**. It must be completed before CP will process material orders for your build. Bring your design materials (CAD, drawings, BOM) to the meeting. The InnoHub team will:

- Provide feedback and guidance on your manufacturing procedures
- Suggest additional tools you may need
- Recommend alternative materials or approaches that may save time and money
- Identify which InnoHub machines and resources can improve your process

Schedule through the booking link on Canvas. The InnoHub team has blocked time specifically for these meetings. Do not wait until the last minute; scheduling fills up quickly around CDR.

12.4 3D Printing Policies

3D printing is the most commonly used fabrication method in capstone. InnoHub policies balance access across all teams:

- Each team may use **one printer at a time**.
- Teams receive **1,000 g of PLA Basic filament free**. Usage subject to standard time and gram limits.
- Individual prints must be $\leq$**250 g** without TA or manager approval.
- **Daytime prints**: up to 5 hours unless approved for exception.
- **Overnight prints**: up to 12 hours, prior TA approval required.
- After exhausting the free allotment: purchase filament rolls or pay per print. Purchased filament still subject to time limits.
- **Advanced materials**: TPU (flexible) and PETG (chemical resistant) available at cost, as are resin printers (SLA/DLP) for high-detail parts.

- **Payment**: all payments processed *before printing* via worktag transfer or credit card through the InnoHub payment portal.

12.5 Paint Booth and Surface Finishing

Spray painting must use the **designated paint booth** only. Complete the paint booth pre-quiz on Canvas before your first use. Sign up for work time through the booking system.

12.6 ME Machine Shop

The ME Machine Shop and Plasma Cutter in Brown Hall have **limited access** due to ME course usage. Use InnoHub as your primary fabrication resource.

12.7 General InnoHub Policies

- Shops available during posted hours with TA supervision only.
- **TA authority**: Labriola TAs have authority in all shops. If they provide guidance or request something, comply promptly. Contact Julia Roos (jroos@mines.edu), InnoHub Operations Manager, for policy concerns.
- **Tool Crib**: return all tools by closing time daily. Purchase fasteners from project budget after prototyping.
- **Cleanliness**: clean workspace before leaving. No overnight storage without approval.
- Some materials and services available for purchase from InnoHub directly.

12.8 Manufacturing Process Selection

Choosing the right manufacturing process for each component of your design is a key engineering decision. Consider:

3D Printing (FDM): best for complex geometries, rapid iteration, low-volume production. Lead time: hours to days. Limitations: anisotropic strength (weak between layers), limited material properties, surface finish requires post-processing. Use for: enclosures, brackets, jigs, fixtures, form-fit prototypes.

Laser cutting: best for flat parts from sheet material (acrylic, plywood, MDF, thin metal). Lead time: minutes to hours. Excellent dimensional accuracy in-plane ($\pm 0.1\,\mathrm{mm}$). Use for: panels, structural frames, gaskets, decorative elements.

CNC machining: best for metal parts requiring high strength, tight tolerances, and good surface finish. Lead time: hours to days (InnoHub) or weeks (outsourced). Use for: structural brackets, motor mounts, shaft couplings, bearing housings, heat sinks.

Hand fabrication: sheet metal bending, drilling, tapping, filing, soldering, adhesive bonding. Often the fastest approach for one-off parts. Lead time: minutes to hours. Use for: simple brackets, wire harnesses, electrical enclosures, quick modifications.

Outsourced fabrication: PCB manufacturing (JLCPCB, PCBWay), sheet metal fabrication (SendCutSend, OSH Cut), CNC machining (Xometry, Protolabs). Lead time: 1–4 weeks including

shipping. Use when the required process or material is not available in the InnoHub or when volume justifies the cost.

When choosing a manufacturing process, consider the **whole lifecycle**: not just the first prototype, but also replacement parts, maintenance, and potential future production. If the client may eventually manufacture your design at scale, design for the production process (injection molding, die casting) even if the prototype uses a different method (3D printing, machining).

12.9 Quality Control During Fabrication

Professional fabrication includes quality checkpoints:

Incoming inspection: when components arrive, verify: correct part number, correct quantity, no shipping damage, and visual quality. For precision parts, measure critical dimensions with calipers before assembly. A bracket that is 0.5 mm too thick will not fit the enclosure.

In-process checks: verify dimensions after each machining operation. Check 3D prints for warping, layer adhesion, and dimensional accuracy before assembling. Test electrical sub-assemblies before integration.

Final inspection: before the system is considered "built," inspect all fasteners for proper torque, all electrical connections for continuity, all fluid connections for leaks, and all moving parts for proper clearance and alignment.

Document quality issues and their resolution. At FDR, the as-designed vs. as-built comparison (Chapter 10) should include any fabrication deviations and their impact on performance.

Student Guide Cross-Reference

For the InnoHub training sequence and Manufacturing Review Meeting timeline, see the **Student Guide, SDI Sprint 4** and **SDII Sprint 8** sections.

12.10 Your First Week in the InnoHub

If this is your first time using the InnoHub shops, follow this sequence:

1. **Enroll** in the Labriola InnoHub Canvas site (link on the main Capstone Canvas page).
2. **Complete general safety training** on the Canvas site.
3. **Complete machine-specific training** for every piece of equipment you plan to use. Each machine has its own training module. Allow 1–2 weeks for training availability.
4. **Schedule your Manufacturing Review Meeting** with a shop manager (required before ordering build materials).
5. **Get your Blastercard activated** for Thorson Design Center access.
6. **Locate your team's locker/storage** in the Thorson Center.
7. **Introduce yourself to the Grad TAs**; they are your primary support for tools, safety questions, and fabrication guidance.

TA contact information: the current semester's Capstone Design Grad TAs are listed on the Labriola Spaces page on Canvas. These TAs support teams with hand and power tool needs, safety

questions, and storage. If TAs cannot assist you, contact Prof. Kristy Csavina (Capstone Design Director, kcsavina@mines.edu). For general InnoHub questions, contact innohub@mines.edu.

12.11 Fabrication Tips from Past Teams

- **Prototype early, prototype often**: your first 3D print will reveal design problems that CAD did not. Print a rough version early and iterate.
- **Test-fit before committing**: before machining the final aluminum bracket, make a cardboard or PLA test piece to verify dimensions, clearances, and mounting hole alignment.
- **Label everything**: label every wire, every connector, every tube, every PCB. During debugging at 11 PM, clear labels save hours.
- **Take photos during assembly**: document the build process with photos at every stage. These are essential for the FDR as-built documentation and the O&M manual.
- **Clean as you go**: a messy workspace leads to lost parts, damaged components, and safety hazards. Spend the last 10 minutes of every build session organizing your workspace.

12.12 Outsourced Fabrication Services

When InnoHub cannot provide the required process, or when the quality requirements exceed student fabrication capabilities, outsourced fabrication is an option:

PCB fabrication: JLCPCB (cheapest for prototypes: 5 boards for \$2–5 + shipping), PCBWay (more options, good for assembly), OSH Park (US-based, higher quality, \$5/in^2 for 3 boards). All accept Gerber files generated from KiCad, Altium, or EasyEDA.

Sheet metal: SendCutSend (laser cutting and bending from uploaded DXF files, 1–2 week turnaround), OSH Cut (similar), and local sheet metal shops for complex assemblies.

CNC machining: Xometry and Protolabs offer instant quoting from CAD files. More expensive than InnoHub but higher precision and faster turnaround. Use for parts that require tolerances tighter than ± 0.05 mm or materials not available in the InnoHub.

Waterjet and laser cutting: available through SendCutSend and local vendors. Excellent for thick metal plates, complex profiles, and materials that cannot be machined conventionally (composites, ceramics).

When outsourcing, add the vendor lead time to your procurement timeline and budget the fabrication cost explicitly. Order from outsourced fabrication vendors at CDR at the latest to have parts in hand for Sprint 10 assembly.

12.13 Working Safely with Power Tools

Even with completed InnoHub training, maintain constant vigilance with power tools:

The 5-second rule: before starting any power tool, pause for 5 seconds and mentally confirm: (1) workpiece is secure, (2) tool guards are in place, (3) PPE is on, (4) path of tool/workpiece is clear of hands and obstructions, (5) emergency stop is accessible. This mental checklist prevents the most common shop injuries.

Never reach across a running tool: turn off the tool and wait for it to stop completely before adjusting the workpiece, clearing chips, or measuring.

Fatigue awareness: tired people make mistakes. If you have been in the shop for more than 4 hours, take a break. If you are working late at night, consider whether the task can wait until morning when you are alert. The most serious shop injuries occur during the last hour of long sessions.

12.14 CNC Machining Tips for Capstone

If your project involves CNC-machined parts (aluminum brackets, motor mounts, bearing housings), these practices save time and money:

Design for 3-axis: unless you have specific 4- or 5-axis requirements, design parts that can be machined in a single setup on a 3-axis mill. This means: features accessible from one direction, no undercuts requiring repositioning, and flat reference surfaces for vise clamping.

Stock material selection: start with near-net-shape stock to minimize material removal. A bracket cut from 0.5" plate requires much less machining than one cut from a 2" block. Standard stock sizes: plate (various thicknesses), bar (round, square, hex), and angle/channel (structural shapes).

Coordinate with InnoHub staff: discuss your part with a shop manager before programming the CNC. They can suggest: optimal tool paths, appropriate feeds and speeds for the material, fixturing strategies, and whether the part is better made on the CNC or by manual machining (sometimes a simple bracket is faster on the manual mill).

Bring drawings: even though the CNC runs from G-code, bring a dimensioned drawing to the shop. The machinist (you or the TA) needs to verify the setup, check the first article, and inspect the finished part against the drawing.

12.15 Welding Considerations

Welding joins metal permanently with high strength. InnoHub offers MIG and TIG welding with training:

MIG welding: easiest to learn, fastest process. Good for steel and aluminum $\geq$1/8" thick. Produces adequate structural welds for most capstone applications. Not suitable for thin sheet metal (<1/16") or high-precision cosmetic applications.

TIG welding: slower, requires more skill, produces cleaner and more controlled welds. Necessary for thin material, stainless steel, and applications where weld appearance matters. Requires more training time than MIG.

Design for welding: provide adequate access for the welding torch (the nozzle is $\sim$1" diameter). Specify weld type (fillet, butt, corner) and size on your drawings. Design joints with adequate material thickness (minimum 1/8" for structural MIG welds). Account for heat distortion: weld assemblies may warp; design fixtures to hold parts in alignment during cooling.

Safety: welding produces intense UV radiation (arc eye), hot spatter (burns), and toxic fumes (metal oxide particles). Full PPE required: welding helmet with auto-darkening lens ($\geq$shade

10), welding gloves, long sleeves (leather jacket recommended), closed-toe boots, and adequate ventilation. Never weld near flammable materials, compressed gas cylinders, or battery packs.

12.16 3D Print Design Optimization for Capstone

Beyond the basic guidelines in Chapter 4, optimize your 3D prints for capstone-specific constraints:

Print orientation for function: orient parts so that: (1) the primary load is perpendicular to layer lines (strongest direction), (2) the most important surface faces up or away from the build plate (best surface finish), (3) support material is minimized (less post-processing, less waste of your 1,000 g free allotment).

Infill strategy: 20% infill is the default; adequate for most non-structural parts. For structural parts (brackets, mounts, enclosures under stress): use 50–100% infill with 4+ perimeter walls. For weight-sensitive parts: use 10% infill with gyroid pattern. For water-tight parts: use 100% infill or vapor-smooth the exterior.

Iteration strategy: print version 1 at 0.3 mm layer height (fast, rough) to check fit and form. Print version 2 at 0.2 mm (standard) after confirming dimensions. Print the final version at 0.15 mm (fine) only if surface quality matters for the client or showcase. Each quality step doubles print time; use the lowest quality that meets your need.

Heat-set inserts: for any part that will be assembled/disassembled more than once, use heat-set brass inserts (M3 are the most common in capstone). Press them into undersized holes using a soldering iron at 220 °C for PLA, 250 °C for PETG. The result is a metal thread in a plastic part that survives dozens of assembly cycles.

Combining 3D printing with other processes: print the complex geometry (organic shapes, internal channels, mounting features) and machine the precision interfaces (bearing seats, shaft bores, sealing surfaces). This hybrid approach leverages the strengths of each process.

12.17 Laser Cutting for Capstone

Laser cutters in the InnoHub Makerspace cut flat stock quickly and accurately:

Materials: acrylic (3–6 mm), plywood (3–6 mm), MDF, cardboard, felt, fabric. The laser cutter cannot cut metal (except thin aluminum foil on some machines) or polycarbonate (produces toxic fumes).

Design for laser: use 2D CAD (DXF or SVG format). All geometry must be in a single plane. Design with kerf compensation: the laser removes ∼0.1–0.2 mm of material width. For press-fit joints, offset the slot by half the kerf width.

Tab-and-slot construction: laser-cut panels can be assembled into 3D structures using tab-and-slot joints. Design tabs with a slight interference fit (0.05–0.1 mm). Glue with cyanoacrylate (instant bond) or acrylic cement (chemical weld for acrylic-to-acrylic joints). This technique produces rigid enclosures and frames in a fraction of the time of 3D printing.

Living hinges: thin cuts in plywood or acrylic create flexible hinges. Useful for enclosure lids and folding structures. Design hinge width based on material: 0.5 mm for 3 mm acrylic (fragile), 1 mm for 3 mm plywood (more durable).

12.18 Electronics Assembly in the InnoHub

The Electronics Shop in the InnoHub provides soldering stations, a hot air rework station, and component storage. Best practices for electronics assembly:

Soldering technique: use a temperature-controlled iron at 300–350 °C for leaded solder or 350–380 °C for lead-free. Apply flux to the joint, heat the pad and component lead simultaneously (not just one), then touch solder to the heated joint (not to the iron). A good joint is shiny (leaded) or slightly matte (lead-free), concave, and wets both the pad and the lead.

Common soldering defects: cold joint (dull, grainy (insufficient heat), solder bridge (excess solder connecting adjacent pads) use solder wick to remove), tombstoned component (one end lifted (reheat both pads), insufficient solder (pad visible through the solder) add more). Inspect every joint under 10× magnification before powering the board.

SMD soldering by hand: for SOT-23, SOIC, and 0805/0603 passives, use fine-tip solder and a magnifier. Tack one pad first, then solder the remaining pads. For QFP and TQFP packages (≤0.5 mm pitch), use drag soldering with flux: apply generous flux, tack one corner pin, then drag the solder-loaded iron tip across the row of pins. The flux prevents bridges; use solder wick to clean any that form.

Through-hole assembly: solder from the bottom of the board. Insert the component from the top, flip the board, solder, and trim the leads. For multi-pin connectors, tack one pin first to verify alignment before soldering the rest.

ESD protection: electrostatic discharge can damage CMOS ICs invisibly; the component appears functional but has reduced reliability. Use an ESD wrist strap connected to a grounded surface when handling bare ICs, PCBs, and sensitive modules. The InnoHub Electronics Shop provides ESD mats and wrist straps.

12.19 Material Sourcing and Stock Preparation

Before starting fabrication, ensure all materials are on hand and prepared:

Metal stock: order from McMaster-Carr or Online Metals. Allow 3–5 days for delivery. Verify dimensions with calipers upon receipt; raw stock can be slightly over or under nominal dimensions. Cut stock to rough size (leaving 3–5 mm excess per side) before machining to final dimensions.

3D printing filament: verify you have enough filament remaining in your free allotment (check with InnoHub TAs). If additional filament is needed, purchase through the InnoHub payment portal before printing. Different colors of the same material have identical mechanical properties; choose color for aesthetics, not performance.

Fastener kits: order assortment kits (M3, M4, M5 metric, or #4-40, #6-32, #8-32 imperial) rather than individual sizes. The per-unit cost is lower and you will inevitably need a size you did not anticipate. McMaster-Carr assortment kits are organized by size and stored in labeled compartments.

Wire and cable: for electronics projects, order hookup wire in at least 4 colors (red=VCC, black=GND, plus 2 signal colors), in both solid core (for breadboards) and stranded (for connections that flex). 22 AWG is the standard for signal wiring; use heavier gauge (18–14 AWG) for power distribution.

12.20 Workspace Organization and Waste Management

The clean workspace principle: a messy workspace is a slow workspace. You waste time searching for tools, risk losing small components, and increase the chance of assembly errors (wrong screw in the wrong hole because everything is jumbled together). Spend 10 minutes at the end of every build session organizing your workspace.

Component organization: use labeled bins, bags, or compartmentalized boxes for: received components (organized by vendor/order), installed components (keep a running installation checklist), spare components (clearly marked as spares), and rejected/damaged components (do not discard (they may be needed for failure analysis at FDR).

Waste disposal: dispose of materials properly. Metal chips and shavings: sweep into the designated metal waste bin (never vacuum) metal chips damage vacuum motors). 3D printing supports and failed prints: PLA is not recyclable through standard recycling; dispose in general waste unless your facility has a filament recycling program. Chemical waste (solvents, adhesives, resin): follow the SDS disposal instructions and use the designated chemical waste containers in the InnoHub.

Tool return: return all InnoHub tools to the Tool Crib before leaving each session. Check out tools through the designated system. If a tool is damaged during use, report it to the TA immediately; do not return it silently. The next team to use a damaged tool may be injured.

12.21 The Build Phase Daily Workflow

During active fabrication (Sprints 9–11), establish a daily workflow:

Start of session (10 min): review the task board. Identify today's build tasks and their dependencies. Gather required tools, materials, and drawings. Verify that all components for today's assembly are available.

Build activities (2–4 hours): execute the build tasks. Follow the assembly sequence documented in the manufacturing plan. Take photographs at each major assembly step. Record any deviations from the plan ("drilled hole 2 mm left of drawing dimension to avoid wire interference").

Test during build: after each subsystem is assembled, perform a basic functionality check before proceeding to the next subsystem. Catching a wiring error after assembling one subsystem is a 30-minute fix; catching it after the enclosure is sealed is a 3-hour disassembly.

End of session (15 min): update the task board. Log what was completed, what remains, and any blockers. Clean the workspace. Return tools. Store the project securely.

Documentation checkpoint (weekly): at the end of each week, update: the as-built documentation (photos, deviation notes), the BOM (mark installed components), and the test log (subsystem test results). This weekly checkpoint prevents documentation debt from accumulating to unmanageable levels before FDR.

12.22 Practice Questions

1. List the training sequence required before you can use the CNC mill in the InnoHub Metal Shop.

2. You have used 800 g of your free PLA allotment and need to print a 400 g enclosure. What are your options and required steps?
3. Your Manufacturing Review revealed that your planned sheet metal bending is not feasible on InnoHub equipment. Propose three alternative fabrication approaches.

Chapter 13

Team Dynamics, Leadership, and Professional Conduct

13.1 Team Formation and the Tuckman Model

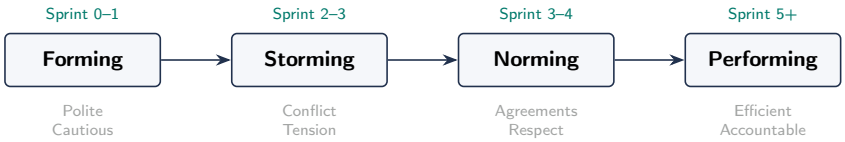

Figure 13.1: Tuckman's team development stages mapped to typical capstone sprint timing. Storming is normal and necessary; teams that suppress it stall.

Teams progress through predictable stages [11] (Figure 13.1): **Forming** (polite, cautious, uncertain), **Storming** (conflict emerges as work styles and expectations clash), **Norming** (working agreements established, mutual respect develops), and **Performing** (efficient operation with shared accountability). Most capstone teams reach Storming by Sprint 2–3 of SDI. Teams that suppress conflict stall permanently; teams that address it constructively reach Performing faster.

Storming Is Normal and Necessary

If your team has not had a disagreement by Sprint 3, either you are exceptionally well-aligned (rare) or conflicts are being suppressed (common and dangerous). Suppressed conflict does not disappear; it surfaces at CDR as passive-aggressive behavior, missed deadlines, and finger-pointing. Address disagreements early, directly, and professionally.

13.2 Team Roles and Responsibilities

All roles are captured in the SOW (Chapter 7). Core roles and their specific duties:

Scrum Master / Project Manager: facilitates the sprint process, maintains and updates the task board (Trello recommended), runs stand-ups asking the three questions, tracks attendance for all team members at all meetings. The Scrum Master is the first person the PA talks to about schedule and process. The team may select a new Scrum Master if the current lead wants to step down or if another member wants the experience.

Communication Lead: takes attendance at every team meeting (this documentation is **essential evidence for PIPs**. Records meeting minutes and distributes them within 48 hours. Manages

client communication cadence (Chapter 17). The Communication Lead's attendance records are the team's documentary evidence when performance issues arise; without dates and records, a PIP cannot proceed.

Chief Financial Officer (CFO): submits all purchase orders through Capstone Purchasing, maintains the Budget Tracker (updated and uploaded to Canvas monthly by the deadline), completes the CFO Form on Canvas, and coordinates all financial transactions. In SDII, the CFO role is **significantly busier**) purchasing accelerates, reimbursements multiply, and travel coordination may be added. Designate a separate Purchasing Lead if the workload is too high.

Technical Lead: makes architecture and integration decisions, ensures technical consistency across subsystems, oversees subsystem interfaces, and serves as primary contact for Technical Advisors.

Fabrication Lead / Documentation Lead: project-specific roles based on team skills and project needs.

13.3 The Team Contract

We encourage all teams (and **strongly recommend for SDII**) to create a Team Contract that expresses minimum expectations every member commits to:

- Presence at all team meetings, arriving on time
- Writing contributions due 48 hours before team submission deadlines
- Minimum weekly contribution level (quantified: hours, tasks, deliverables)
- Professional behavior: proactive communication when late/absent, 24-hour message response time
- Meeting time outside of class if the build schedule requires it
- Consequences for violations: informal warning → formal discussion → PIP

Share the contract with your PA.

13.4 Feedback Loop Peer Evaluations

Feedback Loop is used for formal peer evaluations. SDI: 2 evaluations. SDII: 3 evaluations. Results directly scale individual grades; if your peers report you did not contribute fairly, your individual scores are adjusted downward.

Every low evaluation must include **specific, documented evidence**: "Student X missed team meetings on [specific dates], did not complete their assigned CDR report section by the [specific date] deadline, and did not respond to team messages for [specific duration]." Vague claims without evidence cannot be acted upon.

Individual Score Thresholds

Students receiving less than **70% in SDI** or **60% in SDII** on the combined individual assignment scores (Professional Development, Individual Writing, and Individual Performance Evaluations) will receive a **failing grade**, regardless of team scores. You cannot rely on team scores to compensate for insufficient individual contribution.

13.5 Performance Improvement Plans (PIP)

When informal feedback fails to resolve participation issues, the formal PIP process provides structure:

1. Team documents specific concerns with **evidence**: missed meetings (dates from Communication Lead's log), incomplete work (what was assigned vs. delivered), communication failures (unanswered messages with timestamps).
2. Scrum Master or Communication Lead and PA jointly issue the PIP with a **two-week improvement timeline** and specific, measurable improvement goals.
3. Copy to Course Faculty.
4. After two weeks, team and PA evaluate improvement.
5. If insufficient: team may remove the member. The individual finishes all work products independently or is assigned a different project at CF discretion.

Communicate concerns *early* (first to the team member, then to the PA. Waiting until FDR to raise issues that existed since Sprint 2 is neither fair nor helpful.

13.6 Individual Performance Evaluations

PAs assess individual performance based on: Feedback Loop peer evaluation results and the PA's own observation of your technical and professional contributions at Sprint Reviews, Design Reviews, and other interactions. Both *what* you contributed and *how* you contributed are evaluated.

13.7 Creating a Productive Team Culture

The best capstone teams share a set of cultural attributes that can be deliberately cultivated:

Psychological safety [39]: team members feel safe to admit mistakes, ask questions, and raise concerns without fear of ridicule or punishment. When a teammate says "I think my wiring is wrong but I'm not sure how to fix it," the productive response is "let's look at it together," not "how did you mess that up?" Teams where mistakes are hidden (because admitting them is socially costly) find problems later, when they are harder and more expensive to fix.

Shared ownership: the project belongs to the team, not to the person who designed the subsystem. When a subsystem has a problem, the team rallies to fix it rather than waiting for the "owner" to handle it alone. Shared ownership also means shared accountability: every team member should understand the project well enough to explain the overall system to a reviewer, even if they did not design every subsystem.

Constructive disagreement: when team members disagree on a technical approach, the disagreement should be resolved with data and analysis (Chapter 4: decision matrix), not with volume or seniority. A team that cannot disagree productively will either suppress good ideas (the loudest person always wins) or fracture (unresolved disagreements turn personal).

Proactive communication: problems do not improve with silence. If you are behind on a task, tell the team *before* the deadline, not after. If you cannot attend a meeting, notify the team 24 hours in advance. If you disagree with a decision, say so in the meeting, not in a private conversation afterward.

Celebrating progress: acknowledge milestones and individual contributions. "The sensor subsystem tested perfectly today because Sam spent 8 hours debugging the ground loop last week" costs nothing to say and builds the kind of mutual respect that sustains teams through difficult sprints.

13.8 Managing Hybrid and Remote Team Members

Even in a synchronous course, circumstances occasionally require a team member to participate remotely: illness, family emergency, schedule conflicts with other courses, or cooperative education assignments. Managing this effectively:

- **Set expectations early**: the Team Contract should address remote participation) under what circumstances is it acceptable, how to participate effectively, and what "present" means remotely (camera on, microphone tested, actively participating).
- **Use shared digital tools**: maintain all project documents in the shared repository (Teams, Google Drive, or equivalent) so remote members have the same access as in-person members.
- **Assign remote-compatible tasks**: if a member must be remote for a week, assign documentation, analysis, code development, or research tasks rather than fabrication or hands-on testing.
- **Over-communicate**: remote members miss hallway conversations, whiteboard sessions, and casual updates. Compensate by posting key decisions, photos of progress, and meeting minutes promptly.

Being remote is not an excuse for disengagement. A remote team member who does not respond to messages, does not attend virtual meetings, and does not complete assigned tasks is an absent team member; and absence is documented toward PIP thresholds.

13.9 Conflict Resolution

Most team conflicts fall into three categories: (1) unequal workload distribution, (2) disagreement on technical approach, and (3) communication style clashes. For each:

Unequal workload: make contributions visible through the task board. Assign tasks with deadlines and track completion. If contributions are genuinely unequal, the data speaks for itself in Feedback Loop and PIP documentation.

Technical disagreement: use the decision matrix (Chapter 4). Frame technical disagreements as "which option best serves the requirements?" rather than "whose idea is better?" Let the data decide.

Communication clashes: establish norms in the Team Contract. If a team member's communication style is causing friction, address it directly and professionally: "When you [specific behavior], the impact is [specific consequence]. Going forward, I need [specific alternative]."

13.10 Self-Reflection and Professional Growth

CLO 17 requires self-evaluation and improvement strategies. Deliverables: end-of-semester reflection narrative, "What I Wish I Would Have Known" team video (SDII), and the Final Course

Survey. Connect capstone to your career: what technical skills did you develop, what professional skills did you practice, what would you do differently?

13.11 Leveraging Diverse Perspectives

Capstone teams are intentionally multidisciplinary: ME, EE, CE, EnvE, CS, and other majors working together on a single project. This diversity is a strength, not a complication. Different disciplines bring different problem-solving approaches, different technical vocabularies, and different design sensibilities.

Technical diversity: an EE sees the power distribution challenge; an ME sees the thermal management challenge; a CS student sees the data processing architecture. When these perspectives are combined early (in requirements and concept generation), the resulting design is more robust than any single-discipline solution.

Communication across disciplines: the biggest challenge of multidisciplinary teams is vocabulary. "Load" means something different to an EE (electrical current) than to an ME (mechanical force) than to a CE (structural demand). Establish a shared glossary in your project repository. When you catch yourself using jargon that your teammates do not understand, define it.

Inclusive team practices: ensure that all team members have the opportunity to contribute, not just the most vocal. In meetings, deliberately solicit input from quieter members: "Alex, you have experience with embedded systems; what do you think about this sensor interface approach?" Assign tasks based on skills and learning goals, not just on who volunteers first.

Respecting different work styles: some team members are planners who want everything defined before starting; others are experimenters who learn by building. Both approaches have value. The sprint structure accommodates both: the planning happens at sprint start, and the experimentation happens during execution.

13.12 Building a Productive PA Relationship

Your PA is your most important professional resource. They have seen dozens of capstone projects and can identify problems before they become crises. Maximize the relationship:

Come prepared to meetings: have an agenda, show progress (working demonstrations, not just slides), and ask specific questions. "How should we handle this trade-off between cost and accuracy?" is far more productive than "What should we do next?"

Ask for feedback early: do not wait until PDR to learn that your requirements are vague. Show draft requirements, concept sketches, and analysis approaches to your PA in Sprint 2–3. Early feedback is cheap; late feedback is expensive.

Report problems promptly: PAs would rather hear about a problem in Sprint 3 (when there is time to adjust) than in Sprint 13 (when there is not). Honest, early reporting of challenges builds trust; late revelation of hidden problems destroys it.

Accept feedback gracefully: your PA's job is to make you better engineers, which sometimes means telling you that your approach has a fundamental flaw. This is not personal criticism; it is professional mentorship. The appropriate response is "Thank you; how would you recommend we address this?" not "We already thought about that."

Leverage their network: PAs know faculty, industry contacts, and alumni who can serve as Technical Advisors for your project. Ask: "Do you know anyone with expertise in [specific domain]?" PAs are usually happy to make introductions.

13.13 Cross-Functional Collaboration with Other Teams

In some capstone configurations, multiple teams work on related subsystems of a larger project. Cross-team collaboration requires:

Interface agreements: define the physical, electrical, and data interfaces between subsystems early (Sprint 2–3). Document these interfaces in ICDs (Chapter 4). Changes to agreed interfaces require approval from all affected teams.

Regular coordination meetings: hold brief cross-team sync meetings (weekly, 15 minutes) to share progress, surface integration issues, and coordinate testing schedules.

Shared testing: when subsystems must work together, coordinate a joint integration test session where both teams are present. Do not throw your subsystem over the wall and expect the other team to integrate it without your support.

13.14 Team Health Indicators

Healthy teams and struggling teams exhibit observable patterns. Monitor these indicators throughout the semester:

Healthy team signs: task board updated regularly, sprint velocity stable or increasing, all members contributing at reviews, retrospective actions being completed, client receiving regular updates, PA hearing about problems early, team members helping each other across subsystem boundaries.

Warning signs: task board stale for >1 week, sprint velocity declining, one member doing most of the talking at reviews, retrospective actions not being tracked, client has not heard from the team in >2 weeks, PA learning about problems only at reviews, team members working in silos with no cross-pollination.

Crisis indicators: team member has stopped attending meetings (check Communication Lead's attendance log), Feedback Loop scores show ≤ 2 out of 5 for any member, deliverables being submitted incomplete or late, team conflict has become personal rather than professional, PA has had to intervene in team dynamics more than once.

If you observe warning signs, address them at the retrospective. If you observe crisis indicators, involve the PA immediately. The PIP process (described earlier) exists for a reason; use it before the situation becomes unrecoverable.

13.15 The Escalation Path for Team Issues

Not all team problems can be resolved internally. Here is the escalation path, in order:

1. **Direct conversation**: address the issue one-on-one with the person involved. Use "I" statements: "I notice that the firmware task has been in progress for two weeks without an update. Can we talk about what is blocking you?"

2. **Team discussion**: if the direct conversation does not resolve the issue, bring it to a team meeting. Frame it as a process problem, not a personal attack: "We have a pattern of tasks not being completed on time. How can we adjust our process?"
3. **PA involvement**: if team-level discussion does not resolve the issue, bring it to your PA. Bring documented evidence (attendance records, task board history, Feedback Loop scores). The PA may mediate or recommend a PIP.
4. **Formal PIP**: if PA-mediated discussion does not produce improvement, initiate the formal PIP process with a two-week timeline and specific, measurable improvement goals.
5. **Course Faculty**: if the PIP does not produce improvement, the PA escalates to the CF for potential team removal or course-level consequences.

Most issues are resolved at Step 1 or 2. Escalation is not failure; it is using the institutional support structure that exists specifically for these situations.

13.16 Making Feedback Loop Evaluations Effective

Feedback Loop evaluations are most useful when they are specific, evidence-based, and constructive:

For high-performing teammates: acknowledge specific contributions: "Alex consistently delivered firmware milestones on time and helped debug the sensor subsystem even though it was not their assigned area. Their initiative during Sprint 10 integration was critical to our on-time CDR." Specific praise reinforces the behaviors you want to see continue.

For underperforming teammates: document specific incidents with dates: "Jordan missed team meetings on 2/15, 2/22, and 3/1. The firmware task assigned on 2/10 (due 2/24) was not completed until 3/8, delaying integration testing by one week." Vague criticism ("Jordan doesn't pull their weight") is not actionable; specific documentation is.

For average teammates: identify both strengths to reinforce and areas for growth: "Sam consistently completes assigned tasks on time and contributes well to team discussions. An area for growth would be taking more initiative on unassigned tasks; during Sprint 9, several integration issues went unaddressed while Sam waited for them to be formally assigned."

Self-evaluation: be honest about your own contributions and shortcomings. Over-rating yourself while under-rating teammates is transparent to the PA and undermines your credibility. Acknowledging a specific area where you could improve ("I should have communicated my Sprint 11 delay earlier instead of waiting until the retrospective") demonstrates professional maturity.

The PA uses Feedback Loop data alongside their own observations. Consistent patterns across multiple evaluators carry more weight than a single outlier evaluation.

13.17 The SDI-to-SDII Team Transition

The semester break between SDI and SDII is a critical vulnerability for team dynamics. Momentum is lost, members may disengage, and the client relationship cools. Manage the transition deliberately:

Before the break: complete all Sprint 6 deliverables. Document the current state of the design comprehensively; assume that if a team member reads only the documentation (not their memory),

they can resume work in January. Create a "SDII Ready" checklist: What components are ordered? What is the build plan? What are the top 3 risks? What are the first actions for Sprint 7?

During the break: the Scrum Master sends a brief check-in email to the team one week before SDII starts: "We begin Sprint 7 on [date]. Our first priority is the re-engagement email to the client and the Hazard Assessment. Please review the Sprint 6 documentation before class."

First week of SDII: hold a team reset meeting. Review: what went well in SDI (reinforce), what needs to change (new norms), any role changes (new Scrum Master?), and the updated Team Contract. A team that starts SDII by addressing SDI problems proactively outperforms a team that hopes the problems went away during break.

Team membership changes: occasionally, a student drops the second semester, studies abroad, or is added to the team. When membership changes, redistribute roles and responsibilities explicitly. Update the SOW team section. Ensure the new member (or remaining members) understand the project context by reviewing the PDR report and Sprint 6 documentation.

13.18 Celebrating Milestones and Sustaining Morale

Capstone is a marathon, not a sprint. Sustaining team energy over 30 weeks requires deliberate attention to morale:

Acknowledge achievements: when the PCB works on the first try, when the client gives positive feedback, when a challenging analysis produces a good result (call it out. "The motor controller Jordan built works perfectly. That was a difficult design and it is exactly what we needed." Recognition costs nothing and builds the kind of mutual respect that carries teams through hard sprints.

Mark milestones: celebrate completing the SOW, passing PDR, receiving the first PCB, achieving the first successful system test. These do not need to be elaborate) a team lunch, a group photo with the prototype, or a brief acknowledgment at the sprint retrospective.

Manage the mid-semester slump: around Sprint 10–11 of SDII, many teams experience fatigue. The initial excitement of building has passed, testing is revealing problems, and FDR feels both distant and looming. This is when process discipline matters most. Stick to the sprint cadence, maintain the task board, and use the retrospective to identify and address morale issues.

13.19 Effective Meeting Facilitation for the Scrum Master

The Scrum Master's effectiveness is largely determined by how well they facilitate meetings:

The time-boxed stand-up: stand-ups must be $\leq$15 minutes. Enforce this by: standing (people talk less when standing), using a timer visible to all participants, and intervening when discussions go off-topic ("That sounds like a detailed technical discussion; let's take it offline after the stand-up. Jordan and Sam, can you stay for 5 minutes after?"). The stand-up surfaces blockers; it does not solve them.

The productive retrospective [40]: retrospectives become stale if they follow the same format every sprint. Vary the approach: the "Start/Stop/Continue" format (what should we start doing, stop doing, continue doing?), the "4Ls" format (Liked, Learned, Lacked, Longed for), or the

"sailboat" metaphor (wind = what propels us, anchor = what holds us back, rocks = risks ahead). The format matters less than the quality of the discussion and the commitment to action items.

Sprint planning discipline: sprint planning determines the next two weeks of work. The Scrum Master ensures: (1) items are selected from the Product Backlog in priority order (not cherry-picked by preference), (2) the team commits to a realistic volume based on previous sprint velocity (not an optimistic wish list), (3) every selected item has a clear definition of "done" that all team members agree on, and (4) every item has an assigned owner (even if multiple people will contribute, one person is accountable).

Facilitating disagreements: when two team members disagree in a meeting, the Scrum Master's role is not to choose sides but to structure the disagreement productively: "I hear that Jordan prefers Approach A for [reasons] and Sam prefers Approach B for [reasons]. Let's identify the criteria that matter and evaluate both approaches against them. Can we agree on the criteria first?" This redirects emotional disagreement into analytical evaluation.

13.20 Knowledge Sharing and the Bus Factor

The **bus factor** is the number of team members who would need to be "hit by a bus" (or, more realistically, get sick, have a family emergency, or be overwhelmed by another course) before the project stalls. A bus factor of 1 means the project depends entirely on a single person's knowledge.

Increase the bus factor through deliberate knowledge sharing:

- **Pair work**: for every critical subsystem, have at least two team members who understand it. The primary developer works alongside a "shadow" who observes, asks questions, and can take over if needed.
- **Documentation in the repository**: design decisions, wiring diagrams, firmware architecture notes, and calibration procedures must be documented in the shared repository; not in one person's head. If the knowledge exists only in conversation, it is lost when that person is unavailable.
- **Cross-training at sprint reviews**: during sprint reviews, the subsystem owner presents their work to the team with enough detail that another member could debug or modify it. This is not extra work; it is the sprint review's purpose.
- **Code comments and README files**: for firmware and software, comment liberally and maintain a README that explains: how to set up the development environment, how to build and upload, what each major module does, and how to run the test suite.

A team with a bus factor of 1 is one illness away from a project crisis. A team with a bus factor of 3+ can absorb any single disruption without missing a sprint.

Student Guide Cross-Reference

For the Feedback Loop evaluation schedule and tips on peer evaluations, see the **Student Guide**, Section: The Feedback Loop Survival Guide.

13.21 Practice Questions

1. Two members consistently arrive 15–20 minutes late. They complete technical work but miss stand-ups. What steps do you take, in what order?

2. Design a Team Contract for a 5-person Build Track team with at least 6 specific, measurable commitments.
3. A teammate received low Feedback Loop scores and says the feedback is "unfair." As Scrum Master, how do you handle this productively?
4. Describe a technical disagreement resolution process that uses evidence rather than opinion.

Chapter 14

Budgeting and Financial Management

14.1 Why Budgets Matter

You are spending real money; from Capstone Purchasing funds, client accounts, or Mines institutional funds. Your team is accountable for every dollar. Overspending without authorization may result in **personal financial liability**. Professional engineers manage budgets on every project; capstone is your first practice.

A budget serves three purposes: (1) it forces realistic planning, (2) it provides accountability to funders, and (3) it documents resource usage for the project record.

14.2 Creating a Project Budget

Develop an estimated draft budget *before* your first meeting with Capstone Purchasing:

- **Materials and components**: itemized list with vendor links, unit costs, quantities, total costs, and shipping estimates.
- **Fabrication services**: PCB fabrication, CNC machining, 3D printing beyond free allotment, laser cutting.
- **Test equipment/supplies**: calibration standards, consumables, specialized tools.
- **Travel** (if applicable): see Chapter 16.
- **Contingency**: **15–20%** of total for unexpected costs, replacements, and design changes. Projects budgeting zero contingency always exceed their budget.

14.3 The Budget Tracker

Maintain the Budget Tracker spreadsheet throughout the project. Upload to Canvas **monthly** by your PA's deadline. If late, PA and CLT Director are notified. The tracker shows: budgeted amount per category, committed costs (orders placed), actual costs (invoices received), remaining budget, and cumulative variance. Review at every retrospective and flag categories trending over budget.

14.4 Funding Sources

Capstone Purchasing (CP): primary source for most teams. Route all purchases through CP when possible. **External client funds**: some clients provide direct funding or authorize their

vendor accounts. **Mines faculty index**: for Mines-sponsored projects, funds may come from a faculty client's research account.

14.5 Cost Estimation Techniques

Bottom-up: itemize every component and sum. Most accurate; use for the BOM. **Analogous**: estimate from similar past projects. Ask your PA for precedent data. **Parametric**: cost-per-unit metrics (cost/board, cost/machined part). These estimation techniques are standard project management practice [8].

14.5.1 Sample Budget Structure

Table 14.1 shows a typical capstone budget structure. Your actual categories will vary by project.

Table 14.1: Sample budget structure for a Build Track electronics project.

Category	Budgeted	Actual	Variance
Electronic components (sensors, ICs, passives)	$350	;	;
PCB fabrication and assembly	$120	;	;
Mechanical components (fasteners, bearings, brackets)	$150	;	;
3D printing (beyond free allotment)	$50	;	;
Enclosure and hardware	$80	;	;
Wiring, connectors, cables	$60	;	;
Power supply and battery	$100	;	;
Shipping (estimated 5 orders × $12 avg)	$60	;	;
Miscellaneous/consumables	$30	;	;
Subtotal	$1,000	;	;
Contingency (20%)	$200	;	;
Total	$1,200	;	;

14.5.2 Tracking Actuals vs. Budget

At every sprint retrospective, review the budget tracker and compare actuals to the plan. Key questions: Are any categories trending over budget? If so, what can be reduced or deferred? Are there unspent categories that can absorb overruns? Has the contingency been used, and if so, for what?

When actuals exceed the budget in a category by more than 15%, flag it to your PA immediately. Do not wait until the end of the semester to discover that you are $300 over budget with no path to recovery.

14.5.3 End-of-Project Reconciliation

At FDR, the final BOM should show actual costs alongside budgeted costs for every line item. The total variance (actual minus budgeted) should be explained: which items cost more than expected (and why), which cost less, and whether the contingency was used appropriately. This reconciliation is part of the FDR report and demonstrates financial management competence.

14.6 Tax-Exempt Status

Colorado School of Mines is tax-exempt. CRS 39-26-114 (State, RTD, Use Taxes), CRS 29-2-105 (Local Taxes). Certificate #98-02565 (State), Federal #84-730123K, F.E.I.N. #84-6000551. **Taxes will not be reimbursed.** Provide the exemption certificate to vendors. If the vendor cannot process tax-exempt, route through CP. The certificate is available on Canvas.

Student Guide Cross-Reference

For the CFO-specific purchasing workflow and form submission checklist, see the **Student Guide, CFO and Purchasing Survival Guide** section.

14.7 Budget Planning Timeline

- **Sprint 1 (SDI)**: initial budget estimate in the SOW. Identify major cost categories.
- **Sprint 3–4 (SDI)**: refined budget as the design takes shape. Update component costs with actual vendor quotes.
- **Sprint 5 (PDR)**: budget presented with procurement timeline. Contingency allocated.
- **Sprint 8 (CDR)**: BOM-based budget with actual costs for ordered items. Track committed vs. actual.
- **Monthly (SDII)**: Budget Tracker uploaded to Canvas. Variance analysis at each sprint retrospective.
- **Sprint 14 (FDR)**: final budget reconciliation. Actual vs. planned comparison in FDR report.

14.8 Variance Analysis and Corrective Action

At each budget review, calculate the variance for each category: Variance = Actual − Budgeted. Positive variance means over budget; negative means under budget.

If a category is trending over budget, investigate: was the estimate wrong (update the estimate and adjust other categories), did the design change (document the change and its cost impact), or was there waste (identify and eliminate the source)?

Corrective actions: substitute a lower-cost component, reduce the scope of a non-essential feature, negotiate with the client to defer a capability, or request additional funding from the client or CP. All corrective actions require PA awareness and, if scope-affecting, client approval.

Do Not Hide Budget Problems

If you are trending over budget, tell your PA and CFO immediately. Budget problems get worse with time, not better. A team that reports a $200 overrun in Sprint 10 can adjust; a team that reports a $800 overrun in Sprint 14 is in crisis.

14.9 End-of-Project Financial Reconciliation

At FDR, the CFO presents the final budget reconciliation: budgeted amount per category, actual amount spent, variance, and explanation for significant variances. This is included in the FDR report and demonstrates financial accountability; a professional skill that employers value highly.

14.10 Typical Capstone Budget Ranges

Budget allocations vary widely by project type, but typical ranges for Mines capstone projects:

Build Track (electronics-focused): $500–$2,000. Major costs: PCB fabrication ($50–200 for 2 revisions), electronic components ($100–500), sensors and actuators ($100–400), enclosure and mechanical parts ($50–200), miscellaneous (wire, connectors, fasteners, consumables: $50–150), contingency (15–20%).

Build Track (mechanical-focused): $1,000–$4,000. Major costs: raw materials (metal stock, sheet metal, composites: $200–1,000), machining services if outsourced ($100–500), 3D printing beyond free allotment ($50–200), fasteners and hardware ($100–300), electronics and controls ($200–500), contingency.

Paper Track: $100–$500. Major costs: software licenses (often available through Mines), reference materials, printing, travel to site for surveys or data collection.

Competition Track: $2,000–$10,000+. Major costs: registration fees ($200–1,000), travel ($1,000–5,000), materials and components ($500–3,000), shipping ($100–500).

These are estimates only. Your actual budget depends on your specific project, client, and funding source. Develop your budget from the bottom up (itemized BOM) rather than from the top down ("we have $2,000 to spend").

14.11 Hidden Costs That Break Budgets

1. **Shipping**: $10–50 per order adds up fast when you are ordering from 8–10 vendors. Batch orders from the same vendor to reduce shipping charges.
2. **PCB revisions**: if the first PCB has a design error (common), a second revision doubles the PCB cost and adds 2–3 weeks. Budget for at least one revision.
3. **Replacement components**: a motor that burns out during testing, a sensor that is damaged during assembly, or a connector that is the wrong gender. Budget 10% of component cost for replacements.
4. **Consumables**: solder, flux, heat shrink, wire, zip ties, adhesive, sandpaper, tape, cleaning supplies. Individually cheap; collectively significant.
5. **3D printing overruns**: the free 1,000 g allotment goes fast when you are iterating enclosure designs. Each revision burns 100–300 g.
6. **Tax on personal purchases**: if you buy something personally and forget to use the tax exemption certificate, the tax is not reimbursable.

14.12 Working with Your PA on Budget Decisions

Your PA is your primary advisor for budget decisions that involve trade-offs between cost and technical performance. When facing a budget constraint:

Present options, not problems: instead of "We can't afford the good sensor," say "We have two sensor options: Option A meets the accuracy requirement and costs $85; Option B exceeds the requirement by 2× and costs $350. Given our remaining budget of $400, we recommend Option A, which meets the requirement with adequate margin."

Track the cost of design changes: when the team decides to change a component, update the BOM and Budget Tracker immediately. A "small" change ($20 different connector) multiplied by 10 design changes adds up to a significant budget impact.

Client-funded vs. CP-funded: understand who is paying for what. Some clients provide materials directly; others fund through Mines accounts. The funding source affects the purchasing workflow. Clarify this in Sprint 1.

When you are over budget: do not panic and do not hide it. Present the situation to your PA: how much over, what caused it, and what your options are (reduce scope, substitute cheaper components, request additional funding from client). Most budget issues can be resolved if caught early.

14.13 Presenting Budget Information at Design Reviews

Each design review should include a budget summary:

At PDR: present the estimated budget by category, the funding source, and the contingency allocation. The reviewer wants to know: is the budget realistic for this scope? Are there any high-cost items that represent procurement risk? Is the contingency adequate?

At CDR: present the BOM-based budget with actual costs for items already ordered. Show: budgeted vs. committed vs. received. Highlight any categories that are trending over budget and your corrective plan. The reviewer wants to know: is procurement on track? Are there surprises?

At FDR: present the final budget reconciliation: budgeted, actual, and variance for each category. Explain significant variances. The reviewer wants to know: did the team manage its resources responsibly? What was the total project cost?

14.14 Budget Documentation for the Project Record

The budget is a project deliverable. Include in your FDR report:

- Final BOM with actual unit costs and extended costs.
- Budget Tracker summary showing monthly actuals vs. budget.
- Variance analysis for categories that deviated >15% from the original estimate.
- Lessons learned: what would you estimate differently next time?

This documentation serves two purposes: it demonstrates financial accountability for this project, and it provides cost data for future capstone teams working on similar projects.

14.15 Real-Time Cost Tracking

The CFO should update the Budget Tracker not just monthly but **at every purchase event**: when an order is submitted (committed cost), when the item is received (verify actual cost matches estimate), and when a reimbursement is processed. This real-time tracking prevents the common scenario where the team discovers at FDR that they overspent by $400 because nobody tracked the last six orders.

Committed vs. actual: a committed cost is an order that has been placed but not yet invoiced. An actual cost is a verified, received expenditure. The Budget Tracker should distinguish between

these: committed costs are your best estimate of what will be spent; actual costs are confirmed. The difference matters when reconciling at FDR.

Budget review at retrospectives: at each sprint retrospective, the CFO presents a 2-minute budget update: total budgeted, total committed, total actual, remaining budget, and any categories trending over budget. This takes minimal time but prevents budget surprises.

14.16 When Things Cost More Than Expected

Budget overruns happen. The professional response:

Immediate notification: as soon as you realize a cost will exceed the budget, inform your PA and the team. Do not wait until the monthly Budget Tracker upload to reveal a problem.

Root cause analysis: why did the cost exceed the estimate? Was the initial estimate unrealistic? Did the design change? Did a component fail and need replacement? Was shipping more expensive than budgeted? Understanding the cause prevents recurrence.

Corrective options: (1) reallocate budget from an under-spending category (e.g., travel came in under budget, redirect to materials), (2) substitute a lower-cost component that still meets requirements, (3) defer a non-essential feature to reduce material cost, (4) request additional funding from the client or CP (with justification).

14.17 Budget Allocation by Project Phase

A well-structured budget allocates spending by phase, not just by category:

SDI phase (∼15–25% of total budget): proof-of-concept materials, development boards, basic electronic components, initial 3D printing. Keep SDI spending low; the design is not yet finalized, and components ordered for a concept that changes at PDR are wasted money.

SDII pre-CDR (∼10–15%): long-lead-time components ordered in Sprint 6–8, custom PCB fabrication (Rev 1), and specialized materials. This is "investment spending" based on the PDR-approved design direction.

SDII post-CDR (∼50–60%): the bulk of materials, components, and fabrication services. The BOM drives this spending. Order as much as possible immediately after CDR to maximize build time.

Contingency (∼15–20%): held in reserve for PCB revisions, component replacements, shipping surcharges, and design modifications discovered during testing. Do not plan to spend the contingency; plan to have it available when (not if) something goes wrong.

Late-stage spending (∼5–10%): consumables during the build phase (wire, solder, filament, fasteners, adhesive), finishing materials (paint, labels, anti-slip feet), and any last-minute replacement parts.

14.18 Negotiating Budget with Clients and CP

If your project requires more funding than the standard CP allocation:

Prepare a business case: document what additional funding is needed, why it is needed (specific components or services with costs), and what the impact would be of not receiving it (reduced performance, deferred features, inability to meet a requirement).

Present options: "With the standard $1,500 allocation, we can deliver a system that meets 80% of requirements. With an additional $800 from the client, we can meet 100% of requirements by adding [specific capability]. Which option do you prefer?"

Explore alternatives: can the client provide materials or equipment directly? Can you borrow equipment from another lab? Can a faculty member provide access to a research account? Are there alternative components that meet the requirement at lower cost?

Document the agreement: whatever funding arrangement is reached, document it in the SOW budget section with the amount, source, and any conditions ("Client will purchase the motor directly and ship to campus").

14.19 Defending Budget Decisions at Design Reviews

At every design review, reviewers may challenge budget decisions. Prepare to defend them:

"Why didn't you choose the cheaper option?": present the cost-benefit analysis. "The cheaper sensor ($15) does not meet the accuracy requirement; the selected sensor ($45) does. The $30 difference avoids a redesign that would cost $200+ in additional components and 3 weeks of schedule."

"Your budget seems high for this project.": show the BOM line-by-line. Most capstone budgets are dominated by 5–10 high-cost items. Identify these and justify each: "The motor ($120) and driver ($35) together represent 40% of our materials budget. We evaluated three motor options (see trade study in Chapter 4) and selected this one because it is the only option that meets the torque requirement within the weight constraint."

"What happens if you go over budget?": present the contingency plan. "Our contingency reserve is $300 (18% of total). The highest-risk cost items are the PCB revision ($50 if needed) and potential motor replacement ($120). Even if both occur, we remain within the contingency reserve."

"Can the client provide additional funding?": present the funding conversation status. "We discussed additional funding with the client in Sprint 8. The client offered to purchase the motor directly from their vendor account, reducing our CP expenditure by $120."

14.20 Cost-Benefit Analysis for Design Decisions

When choosing between design options with different costs, perform a simple cost-benefit analysis:

Step 1: identify the options and their costs (purchase price + shipping + installation time + testing time).

Step 2: identify the benefits of each option in terms that matter: which requirements does it meet? What risks does it reduce? How much schedule margin does it provide?

Step 3: calculate the cost per requirement met, or the cost per risk point reduced. The option with the lowest cost per benefit unit is the best engineering value; not necessarily the cheapest

option.

Step 4: present the analysis to the PA and client. "Option A costs $80 and meets 4 of 5 performance requirements. Option B costs $140 and meets all 5. The incremental cost of the 5th requirement is $60. Given that this is a Must-Have requirement, we recommend Option B."

This structured approach to cost decisions builds engineering judgment and demonstrates to reviewers that your budget is not arbitrary.

14.21 Financial Lessons Learned from Past Capstone Projects

These financial lessons recur every semester:

The "just one more order" trap: a team places 12 separate small orders from DigiKey over the semester. Each order has a $4–8 shipping charge. Total shipping: $60–96 that could have been saved by batching orders into 3–4 larger purchases. *Lesson: batch orders by vendor. Place one large DigiKey order per sprint, not one per component.*

The "it's only $10" accumulation: individual small purchases seem insignificant, but 25 "small" purchases of $10–20 each consume $250–500 of the budget without any single item triggering budget concern. *Lesson: track every purchase in the Budget Tracker immediately, no matter how small. Review the cumulative total at every retrospective.*

The "free" 3D printing illusion: teams use their 1,000 g free PLA allotment iterating enclosure designs; printing V1 (200 g), V2 (250 g), V3 (300 g), V4 (280 g). By the time the design is finalized, the free allotment is exhausted and the final production print must be purchased. *Lesson: print rough test-fit versions at low quality and low infill. Save the free allotment for the final production print.*

The untracked client purchase: a client offers to buy the motor directly from their vendor account. The team does not track this in the Budget Tracker because "it's not our money." At FDR, the total project cost is unclear. *Lesson: track all expenditures in the Budget Tracker, regardless of funding source. Separate the "CP-funded" and "client-funded" columns.*

14.22 Multi-Semester Financial Planning

The capstone budget spans two semesters with very different spending patterns:

SDI spending is typically low: development boards ($20–50), basic electronic components ($30–80), 3D printing filament within the free allotment, and miscellaneous prototyping supplies ($20–50). Total SDI expenditure: $75–250 for most projects.

SDII spending is front-loaded to CDR: the bulk of BOM items are ordered in Sprint 7–8. Custom PCB fabrication ($50–200 for 2 revisions), motors and actuators ($50–300), sensors ($30–200), structural materials ($50–300), and fasteners/hardware ($50–100) all arrive in a 3–4 week window. Total SDII expenditure: $400–3,000+ depending on the project.

Cash flow planning: because most spending occurs in a short window around CDR, ensure your CP allocation is available when you need it. Submit purchase requests early in Sprint 8; CP processes orders in the order received, and the CDR window generates a surge of requests from all capstone teams simultaneously. Teams that submit early get their orders processed first.

Tracking across semesters: maintain a single Budget Tracker that spans both semesters. The SDI ending balance becomes the SDII opening balance. This continuity ensures no expenditure is lost in the transition and the FDR budget reconciliation covers the entire project lifecycle.

14.23 The CFO Handoff

If the CFO role changes between SDI and SDII (or within a semester), execute a formal handoff:

1. The outgoing CFO walks the incoming CFO through: the Budget Tracker (every column and formula), all pending orders and their status, the relationship with CP (contact names, preferred communication method), the tax exemption process, and any financial issues in progress.
2. The incoming CFO emails CP to introduce themselves and confirm they are the new financial point of contact for the team.
3. The incoming CFO presents the current budget status at the next sprint retrospective to demonstrate understanding.

A handoff that consists of "here's the spreadsheet" without context is a handoff that will produce budget errors.

14.24 Budget Risk Scenarios and Responses

Anticipate these common budget risks and plan responses:

Scenario: PCB revision required. Impact: +$50–200 and +2–3 weeks. Response: budget for at least one revision in the contingency. Order Rev 1 as early as possible to maximize time for Rev 2 if needed.

Scenario: Motor burns out during testing. Impact: +$50–300 and +1–3 weeks (lead time for replacement). Response: if the motor is critical-path, order a spare with the initial purchase. If budget is tight, identify a lower-cost alternative that meets minimum requirements as a backup.

Scenario: Shipping costs higher than estimated. Impact: +$20–100 per order. Response: batch orders by vendor to amortize shipping. Use standard shipping (not expedited) for non-critical items. For international orders, budget explicitly for customs duties.

Scenario: Client requests additional scope. Impact: +$100–500+ depending on scope. Response: use the scope change process (Chapter 7). Present the cost impact and trade-offs. If the client provides additional funding, great. If not, identify what existing scope can be deferred to accommodate the new request within the current budget.

Scenario: InnoHub material costs exceed estimate. Impact: variable. Response: check InnoHub pricing before CDR (prices are posted on the InnoHub materials page). For 3D printing beyond the free allotment, estimate filament usage based on slicer estimates (grams per part) and current InnoHub pricing per gram.

14.25 The End-of-Project Financial Summary

The FDR report includes a financial summary chapter demonstrating responsible resource stewardship:

Budget vs. actual table: present a side-by-side comparison for each category (materials, fabrication, travel, consumables, contingency) showing: original budget, final actual, variance, and percent variance. Highlight any category that exceeded the budget by >15% with an explanation.

Cost per requirement: calculate the total project cost divided by the number of verified requirements. This metric helps future teams estimate budgets for similar projects and demonstrates cost-consciousness.

Return on contingency: report how much of the contingency reserve was consumed and by what events. A team that budgeted 20% contingency and used 12% managed risk well. A team that budgeted 0% contingency and overspent by 25% learned a lesson worth documenting.

Recommendations for future teams: based on your experience, recommend specific budget adjustments. "We underestimated shipping costs by 40%; future electronics-heavy projects should budget $8–12 per vendor order for shipping rather than our initial estimate of $5." These recommendations have direct practical value for the next cohort.

Figure 14.1: The budget lifecycle. The budget evolves from a rough estimate (SOW) through progressive refinement (design development) to a committed BOM-based budget (CDR), then is tracked monthly and reconciled at FDR.

The budget lifecycle (Figure 14.1) illustrates that budgeting is not a one-time activity. The estimate accuracy improves at each stage: ±50% at the SOW stage (order-of-magnitude), ±25% at PDR (refined with vendor research), ±10% at CDR (BOM-based with actual quotes), and actual costs at FDR. Professional cost estimators call this the "cone of uncertainty" [37]; early estimates are inherently imprecise, and this is expected. What matters is that the estimate converges to reality as the design matures.

14.26 Practice Questions

1. Create a preliminary budget for a Build Track project with an ESP32, 4 sensors, custom PCB, 3D-printed enclosure, and solar panel. Include all categories and contingency.
2. Your tracker shows 75% of materials budget spent at 40% through the build. What corrective actions?
3. Give three specific examples of unexpected costs that justify a 15–20% contingency reserve.

Chapter 15

Purchasing Procedures

15.1 The Purchasing Workflow

All purchases follow: (1) team identifies need and selects item/vendor, (2) **PA approves** the purchase, (3) team submits purchase form to CP, (4) CP processes the order, (5) delivery to campus (Labriola 204), (6) team picks up. **PA approval is required for all purchases.** Failure to obtain prior approval may result in personal financial liability.

15.2 Capstone Purchasing Team

Kirsten Kelly, Finance and Operations Manager (kkelly@mines.edu). Graduate Purchasing Assistants (capstonepurchasing@mines.edu). Office: Labriola 204. Schedule meetings via the booking link on Canvas.

When emailing CP, always include: team name/number, submission date, vendor name, and form used.

15.3 Purchase Forms

Three forms on Canvas:

- **Purchase Form**: standard orders from approved vendors.
- **Shopping List**: multiple items, same vendor.
- **Reimbursement Form**: student-made purchases (very limited; requires PA pre-approval, itemized receipt, no tax).

15.4 Approved Vendor List

Only order from approved vendors. Unapproved vendors require a review process and may be denied. Check the list before selecting components. If you need an unapproved vendor, discuss with CP *early*; do not discover this issue the week before CDR.

15.5 Lead Time Planning

Order early. Common lead times: Amazon/DigiKey standard (3–7 days), specialty sensors/actuators (2–4 weeks), custom PCBs from JLCPCB/PCBWay (1–2 weeks fab + 1–2 weeks

shipping), machined parts from external vendors (2–4 weeks). Build a procurement timeline during Sprint 1 and update at every sprint.

Figure 15.1 shows a typical procurement timeline for a Build Track project.

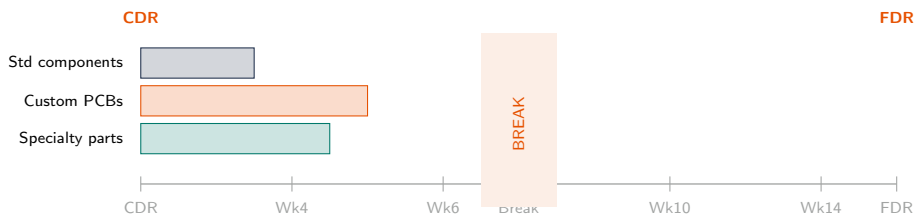

Figure 15.1: Procurement timeline for a Build Track project. Standard components (3–7 days) should be ordered at CDR. Custom PCBs (3–4 weeks) and specialty parts (2–4 weeks) must be ordered before Spring Break to avoid schedule-breaking delays.

The Spring Break Shipping Trap

Spring Break means campus shipping/receiving, InnoHub, and all lab facilities are **closed for an entire week**. If a critical component fails during testing and you need a replacement, order it *before* break (not after. Plan all critical procurement to arrive at least one week before break begins. This is the single most common procurement failure in capstone) avoid it.

15.6 Manufacturing Review Checkoff

If building in any Labriola facility, the Manufacturing Review Meeting is a **required checkoff on the purchasing form**. CP will not process material orders for your build until this checkoff is signed. See Chapter 12.

15.7 Receiving and Inventory

Packages arrive at Labriola 204. Pick up promptly. Maintain an inventory list of all received components; this feeds the BOM and budget tracker. Report shipping errors, damaged items, or missing components immediately.

15.8 Common Purchasing Mistakes

1. Ordering without PA approval → personal liability.
2. Waiting until after CDR to order critical components → schedule crisis.
3. Forgetting the Manufacturing Review checkoff → order rejected by CP.
4. Not checking the approved vendor list → weeks of delay for vendor review.
5. Paying sales tax when Mines is tax-exempt → non-reimbursable loss.
6. Not tracking inventory → cannot reconcile budget at FDR.

15.9 Purchasing Timeline for Build Track Projects

The following timeline assumes a Spring semester. Adjust dates for Fall semester accordingly.

Sprint 1 (Weeks 3–4): identify long-lead-time components. Begin researching vendors and prices. No orders yet (design not stable enough).

Sprint 3–4 (Weeks 7–10): order components needed for proof-of-concept prototyping. These are typically development boards, breakout sensors, basic electronic components, and 3D printing filament (if beyond free allotment).

Sprint 5–6 (Weeks 11–15, SDI): order long-lead-time components for SDII build: custom PCBs, specialty sensors, motors, mechanical stock. **This is critical**; items ordered in Sprint 6 of SDI arrive in time for Sprint 8 of SDII. Items ordered in Sprint 8 of SDII may not arrive until Sprint 10.

Sprint 8 (SDII Weeks 2–5): order all remaining components from the BOM. Manufacturing Review Meeting must be complete before ordering build materials. All critical components should be ordered by the end of Sprint 8.

Sprint 9 (SDII Weeks 6–7): verify all orders received. Order replacements for any missing, damaged, or incorrect items. This is your last comfortable procurement window before Spring Break.

After Spring Break: emergency orders only. Every day spent waiting for parts is a day lost from testing.

15.10 Working with Capstone Purchasing (CP)

CP processes hundreds of orders per semester across all capstone teams. Help them help you:

- **Be organized**: fill out forms completely and accurately. Missing information means CP has to email you for clarification, adding days to the process.
- **Batch orders**: combine multiple items from the same vendor into one Shopping List rather than submitting five separate Purchase Forms. This reduces CP workload and shipping costs.
- **Plan ahead**: CP cannot expedite orders that were submitted late. A "rush" order submitted on Friday afternoon will not be processed faster than a normal order submitted on Monday morning.
- **Communicate changes**: if you need to cancel or modify an order, contact CP immediately. Once an order is placed with the vendor, cancellation may not be possible.
- **Be respectful of CP staff time**: schedule meetings during posted hours, respond to their emails promptly, and pick up your packages when notified.

15.11 The Reimbursement Process

Sometimes a student must make a purchase directly (e.g., buying materials at a local hardware store for an urgent repair). Reimbursement is possible but limited:

1. **Get PA pre-approval** before making the purchase. Document the approval (email is sufficient).
2. **Use the Mines tax exemption**: show the certificate to the vendor. If the vendor does not accept it (some local stores will not), you will pay tax; and tax is **not reimbursable**.
3. **Get an itemized receipt**: the receipt must show each item, its price, and the total. A credit card charge slip alone is not sufficient.

4. **Submit the Reimbursement Form**: fill out the form on Canvas, attach the receipt and PA approval email, and submit to CP.
5. **Wait**: reimbursement processing takes 2–4 weeks. Do not rely on reimbursement for urgent cash-flow situations.

Route Through CP Whenever Possible

Reimbursement is the fallback, not the default. CP has established vendor accounts, tax-exempt purchasing authority, and institutional purchasing processes. Route orders through CP to avoid tax, get better pricing, maintain proper documentation, and protect yourself from personal financial exposure.

15.12 Vendor Selection Tips

- **DigiKey / Mouser**: electronic components. Huge selection, reliable datasheets, fast shipping. Use for ICs, passives, connectors, wire.
- **McMaster-Carr**: mechanical components. Extensive catalog with CAD models for every product. Premium pricing but guaranteed quality and fast shipping.
- **Amazon**: general supplies. Fast shipping, wide selection, but verify: (1) the seller is reputable, (2) the product is genuine (not counterfeit), and (3) the product matches the datasheet specs.
- **JLCPCB / PCBWay**: PCB fabrication. Upload Gerber files, select options, order. 5–10 boards for $2–20 + shipping. JLCPCB also offers SMD assembly.
- **McMaster / Online Metals / Metals Depot**: raw materials (aluminum, steel, plastic sheet/rod/tube).
- **Adafruit / SparkFun**: development boards, breakout modules, and tutorials. Higher cost than raw components but pre-assembled and well-documented. Good for prototyping; switch to raw components for production.

15.13 International Shipping and Customs

Some components are only available from international vendors (especially specialty sensors, motors, and materials). International shipping adds complexity:

Lead time: international shipping typically adds 1–3 weeks beyond domestic. Use expedited shipping if the component is on the critical path. Budget $30–80 for expedited international shipping.

Customs and duties: Mines' tax-exempt status covers US state and federal taxes but does not exempt you from import duties on international shipments. Budget 5–15% of item cost for potential customs charges.

Vendor reliability: verify the international vendor is reputable. Check reviews, shipping history, and whether they accept returns. For Chinese vendors (AliExpress, Banggood), quality varies widely; order early enough to return and reorder if the product does not match the listing.

CP handling: CP has experience with international orders. Discuss international procurement with CP early in the process; they can advise on the best approach and any institutional restrictions.

15.14 Emergency Procurement

When a component fails during testing and you need an immediate replacement:

1. Check if the same component is available locally (Home Depot, Ace Hardware, local electronics shop, another team's spare parts).
2. If not local: order from Amazon with 1–2 day shipping (verify the item is Prime-eligible and ships from a US warehouse).
3. If not Amazon: order from DigiKey or Mouser with overnight shipping ($20–40 premium, worth it if you are testing next week).
4. Document the emergency purchase and submit the reimbursement form with a note explaining the urgency.
5. Update the Budget Tracker immediately.

For emergency purchases, get PA approval by phone or text if email response will take too long. Follow up with an email confirmation.

15.15 The Purchasing Decision Tree

When you need to buy something, follow this decision tree:

1. **Is it available from the InnoHub?** Some materials and consumables are available from InnoHub inventory at cost. Check the InnoHub materials page first.
2. **Is the vendor on the approved list?** If yes, submit a Purchase Form to CP. If no, contact CP to request vendor approval (allow 1–2 weeks for review).
3. **Is this a build material order?** If yes, verify the Manufacturing Review Meeting checkoff is complete. CP will not process the order without it.
4. **Is the cost within your remaining budget?** If yes, proceed with PA approval. If no, discuss with PA and CFO before committing.
5. **Is this urgent?** If a component has failed during testing and you need a replacement immediately, consider local sources or Amazon Prime before the standard CP workflow (which takes 3–5 business days to process).
6. **Does the vendor accept tax exemption?** If yes, provide the certificate. If no, route through CP or accept that tax is non-reimbursable.

15.16 Inventory Management

Maintain an inventory list (spreadsheet) of all components received. For each item: date received, description, quantity, location (which drawer/bin/shelf in your workspace), and status (unused, installed, spare, consumed). Benefits:

- Prevents duplicate orders ("I didn't know we already had those resistors").
- Enables quick substitution ("The motor failed; check inventory for the backup we ordered").
- Supports the FDR BOM reconciliation (actual quantities used vs. ordered).
- Prevents loss (components scattered across multiple workspaces without tracking are components that disappear).

15.17 Tracking Orders from Submission to Receipt

Maintain an order tracking log (spreadsheet or Trello board) with one row per order:

Order #	Vendor	Items	Date Submitted	Expected Arrival	Status
PO-001	DigiKey	ICs, passives	2/15	2/22	Received
PO-002	JLCPCB	PCB Rev 1	2/18	3/4	In transit
PO-003	McMaster	Bearings, bolts	2/20	2/25	Processing

The CFO or Purchasing Lead updates this log when: the order is submitted to CP, CP confirms the order is placed, a tracking number is received, and the package is picked up from Labriola 204.

Chasing delayed orders: if an order has not arrived by the expected date, contact CP. CP can check with the vendor for tracking information. Do not assume delayed orders will "show up eventually"; proactive follow-up prevents schedule surprises.

Receiving inspection: when packages arrive, immediately verify: correct items received (compare to the order), correct quantities, no shipping damage, and components match the datasheet specifications (correct package size, correct part number markings). Report discrepancies to CP within 24 hours.

15.18 Procurement Risk Management

Procurement is on the critical path for most Build Track projects. Manage it proactively:

Identify long-lead items in Sprint 1: during the first SOW development, scan the preliminary BOM for items with lead times >2 weeks. These include: custom PCBs, specialty sensors, motors and gearboxes, structural materials, and anything shipping from overseas.

Dual-source critical components: for components where a stock-out or shipping delay would stall the project, identify a backup source (different vendor, compatible alternative part). Document the backup in the BOM: "Primary: Adafruit #4871 ($14.95, 3-day). Backup: SparkFun #SEN-13770 ($15.95, 3-day). Pin-compatible."

Order in waves: Wave 1 (Sprint 6/SDI or Sprint 7/SDII): long-lead-time and critical-path items. Wave 2 (Sprint 8): all remaining BOM items after CDR design freeze. Wave 3 (Sprint 9+): replacement parts and additional consumables as needed. This wave structure prevents both premature ordering (before the design is stable) and late ordering (after the schedule margin is consumed).

Track order status actively: the CFO or Purchasing Lead checks order status weekly during the build phase. If an order has not shipped within the expected timeframe, contact CP immediately. Do not assume "it will show up eventually"; proactive follow-up prevents schedule surprises.

The "Will Call" option: for components available locally (Home Depot, Ace Hardware, local electronics shops), consider buying locally for urgent needs. This avoids the 3–5 day CP processing time. Remember: PA pre-approval is still required, and sales tax must be avoided using the exemption certificate.

15.19 Evaluating Vendor Reliability

Not all vendors are equal. Evaluate potential vendors before committing budget:

Delivery reliability: does the vendor ship on time consistently? Check reviews on Reddit, EEVblog forums, and other engineering communities. For Chinese PCB vendors (JLCPCB, PCB-Way), standard shipping is 7–14 days; express (DHL/FedEx) is 3–5 days. Factor the actual delivery time into your schedule, not the vendor's optimistic estimate.

Product quality: for electronic components, buy from authorized distributors (DigiKey, Mouser, Newark, Arrow) whenever possible. Amazon and eBay sellers frequently sell counterfeit or sub-standard components; an "Arduino Nano" from an unknown seller may use a clone chip that is unreliable or incompatible with your libraries.

Return and warranty policies: before ordering, understand the return policy. DigiKey and Mouser accept returns within 30 days for unopened items. Amazon return policies vary by seller. JLCPCB will re-fabricate boards with manufacturing defects but will not refund design errors (and should not; the design is your responsibility).

Technical support: some vendors (Adafruit, SparkFun, Pololu) provide excellent documentation, tutorials, and community forums. This support can save hours of debugging. The component may cost $5 more than a bare-bones alternative, but the documentation makes the $5 investment pay for itself many times over.

15.20 Handling Incorrect or Damaged Shipments

When a shipment arrives with problems:

1. **Document immediately**: photograph the packaging and contents before disturbing anything. Note the order number, shipping tracking number, and a description of the problem.
2. **Contact CP**: report the issue to capstonepurchasing@mines.edu with: team name, order number, description of the problem, and photos. CP will initiate the return or replacement process with the vendor.
3. **Do not discard anything**: keep all packaging, packing slips, and the incorrect/damaged items until CP confirms the resolution. The vendor may require the item to be returned.
4. **Order the replacement immediately**: do not wait for the return to be processed before ordering the correct item. The schedule impact of waiting for a return cycle (1–2 weeks) plus a new order (1–2 weeks) is unacceptable on most capstone timelines.

15.21 Vendor-Specific Ordering Tips

DigiKey/Mouser: search by parametric filters (voltage, current, package, temperature range) rather than browsing categories. Use the "In Stock" filter. Check minimum order quantities (MOQ); some components require ordering 100+ units. For passive components (resistors, capacitors), order from cut tape rather than reels to get small quantities.

JLCPCB: use their online Gerber viewer to verify your board before ordering. Select "HASL with lead" for the cheapest option or "ENIG" for better surface finish (recommended for fine-pitch SMD). Standard green soldermask is cheapest; other colors cost $1–3 more. Their SMD assembly

service (PCBA) can populate common components from their parts library, saving hours of hand soldering.

McMaster-Carr: their 3D CAD models are excellent; download the model for every purchased component and integrate it into your assembly to verify fit before ordering. McMaster ships same-day for in-stock items, making them ideal for urgent needs. Their prices are premium but the reliability and convenience justify the cost for critical items.

Amazon: verify seller ratings and reviews. For electronic components, prefer Amazon-fulfilled listings over third-party sellers. **Beware of counterfeits**: Arduino boards, OLED displays, and sensor modules from unknown sellers may use clone chips with reliability or compatibility issues. If in doubt, buy from Adafruit or SparkFun for guaranteed quality.

Local hardware stores (Home Depot, Ace, Harbor Freight): useful for hardware (bolts, nuts, washers), raw materials (aluminum bar, angle, sheet), tools, paint, adhesives, and electrical supplies. Bring the tax exemption certificate. Local purchases avoid shipping delays and costs.

15.22 Component Selection Best Practices

Selecting the right component is an engineering decision that balances performance, availability, cost, and risk:

Start with the requirement: what does the component need to do? Translate the system-level requirement into component-level specifications. A "temperature measurement accuracy of $\pm 1\,^{\circ}\mathrm{C}$" translates into sensor specifications: accuracy, range, response time, output type, and operating environment.

Search parametrically: use DigiKey or Mouser parametric search to filter components by the critical specifications. This narrows thousands of options to 10–20 candidates. Sort by: "In Stock" first, then by price or relevance.

Evaluate the datasheet: for each candidate, read the datasheet; not just the first page, but the electrical characteristics, absolute maximum ratings, application circuit, and recommended PCB layout. The datasheet is the manufacturer's contract with you about what the component will do.

Check availability and lead time: a perfect component that is backordered for 12 weeks is useless for your timeline. Verify in-stock quantity and expected lead time before finalizing the selection. Check multiple vendors; a component out of stock at DigiKey may be available at Mouser or Newark.

Prefer components with application notes: manufacturers publish application notes (app notes) that show how to use the component in common circuits, with design equations, recommended values, and PCB layout guidance. A component with a 20-page app note saves hours of design time compared to one with only a basic datasheet.

Prefer components used by the maker community: components with Arduino/ESP32 libraries, SparkFun or Adafruit breakout boards, and YouTube tutorials are dramatically easier to integrate. The community has already found and documented the common pitfalls. For capstone, this is a significant advantage.

15.23 When Components Are Discontinued or Unavailable

Supply chain disruptions affect capstone teams regularly. When your selected component becomes unavailable:

1. **Check other vendors**: the component may be available from a different distributor (Mouser vs. DigiKey vs. Newark vs. Arrow).
2. **Check the manufacturer's cross-reference**: most manufacturers publish a list of pin-compatible, spec-compatible replacements for discontinued parts.
3. **Search for functional equivalents**: use parametric search with the critical specifications to find alternatives. Verify that the alternative meets *all* critical specs, not just the primary one.
4. **Use a breakout module**: if the bare IC is unavailable, a development module (Adafruit, SparkFun, Pololu) that uses the same IC may be in stock. The form factor is larger but the functionality is identical.
5. **Redesign the circuit**: if no drop-in replacement exists, redesign the circuit around an available component. This is expensive (schedule and engineering time) but sometimes necessary.

Document the substitution: original component, replacement component, reason for change, and verification that the replacement meets all affected requirements. This goes in the CDR change log.

15.24 Buying from International Vendors: Practical Guidance

Many capstone components are cheapest from Chinese vendors (AliExpress, Banggood, Taobao via agents). Practical considerations:

Quality variation: Chinese marketplace vendors range from excellent to terrible. Components may be: genuine branded parts at a discount, compatible clones that work but have lower specifications, or outright counterfeits with falsified markings. For critical components (ICs, voltage regulators, power transistors), buy from authorized distributors only. For non-critical components (LEDs, generic passives, enclosures), marketplace sources are usually acceptable.

Shipping time: standard free shipping from China takes 2–6 weeks. ePacket takes 1–3 weeks. Express (DHL, FedEx) takes 3–7 days at $15–50 per shipment. Factor the *actual* delivery time into your procurement timeline, not the vendor's optimistic estimate. Track every international shipment.

Customs and duties: shipments valued >$800 may incur customs duties (5–15% of declared value). Shipments valued <$800 are typically duty-free under the US de minimis threshold. The vendor's declared value may differ from the actual value; this is the vendor's practice, not something you control.

Returns: returning items to China is usually impractical (return shipping costs more than the item). Most Chinese vendors offer refunds without returns for defective items if you provide photo evidence. Document any defects immediately with photos and contact the vendor within the dispute window (typically 15–60 days after delivery).

CP handling: Capstone Purchasing may have restrictions on international vendors. Discuss international procurement with CP before placing orders. They may require the order to go through an approved US-based distributor instead.

15.25 End-of-Project Purchasing Closeout

At the end of the project, the CFO completes the purchasing closeout:

Final receipt reconciliation: verify that every purchase has a corresponding receipt or purchase order in the Budget Tracker. Reconcile the CP records with your team's records. Flag any discrepancies.

Return unused items: if the project purchased spare components that were not used, discuss with the PA and CP whether items can be returned for credit or stored for future capstone teams.

Return borrowed equipment: if the team borrowed equipment from the InnoHub, ME labs, or other facilities, return everything during Final Checkout. Document what was returned and to whom.

Asset disposition: some project deliverables (the prototype, test equipment, remaining materials) must be transferred to the client. Others remain with the program. The SOW should specify the disposition of all physical assets. During Final Checkout, verify that assets are transferred as agreed.

Final Budget Tracker upload: the final version of the Budget Tracker, showing all actual costs reconciled against the budget, is submitted to Canvas as part of the FDR deliverables. This is the financial record of the project and serves as a reference for future capstone budget planning.

Student Guide Cross-Reference

For a quick-reference purchasing workflow and common CFO mistakes, see the **Student Guide**, Section: CFO and Purchasing Survival Guide.

15.26 Practice Questions

1. Map a procurement timeline for a project needing custom PCBs, specialty motors, and standard electronics. CDR is SDII Week 4, Spring Break is Week 10.
2. Your PA is unavailable when you urgently need a replacement motor. What do you do?
3. A teammate wants a component from an unapproved vendor at 30% lower cost. Walk through the decision.

Chapter 16

Travel Planning and Policies

16.1 When Travel Is Required

Competition teams (SAE, ASCE, AIAA, Formula SAE, EcoCAR), field survey projects, and client site visits may require travel. All travel must be planned, budgeted, and approved well in advance.

16.2 The Go/No-Go Decision

Before any travel spending, the PA must approve a **Go/No-Go decision** with the Capstone Leadership Team (CLT). The Go/No-Go considers: is the trip essential? Is the budget available? Is the team prepared? The PA determines the **travel roster**; the minimum number of travelers needed for the mission.

16.3 PA and Student Responsibilities

PA: confirm whether staff must travel (competition rules), determine roster, review and approve budget, support team logistics, secure OneCard for airfare/rental bookings.

Student Travel Lead: create and manage the travel budget, follow all Mines and Capstone travel policies, serve as primary logistics contact, manage confirmations and check-in details, coordinate flight selection with FROSCH, verify traveler names match government IDs, ensure on-time arrivals.

16.4 Budget Planning for Travel

Prepare a draft budget *before* meeting with CP:

- Multiple lodging options with links and cost comparisons
- Airfare per traveler (include FROSCH service fee if applicable)
- Ground transport: rental car, mileage for personal vehicles, fuel estimates (note: fuel not reimbursable for personal vehicles)
- Per diem for all travelers for all days
- Registration and competition fees
- Shipping for the project/equipment

Use the Budget Template on Canvas. Review the "Example of a Great Team Budget."

16.5 Key Travel Policies

- **Carpooling required**. Multiple personal vehicles for the same trip will not be reimbursed.
- **Mileage**: Mines rate ($0.60/mile as of 2/26/24 (confirm current rate). Calculate from 1500 Illinois St., Golden CO 80401. Save the Google Maps route as PDF. **Gas not reimbursable** for personal vehicles.
- **Meals**: itemized receipts required. **No alcohol.** Receipts containing alcohol rejected *in full*) no partial reimbursement.
- **Post-travel receipts**: submit within **one month** of trip completion.
- **Airfare**: use FROSCH (university travel agency) when required. Coordinate with Travel Lead.

16.6 Competition-Specific Considerations

Registration deadlines (often months ahead), equipment shipping logistics and costs, team shirt production (InnoHub, competition teams only), and the competition schedule's impact on the academic calendar. Plan early; competition travel budgets are among the largest expenditures in capstone.

16.7 Travel Budget Template

A well-organized travel budget includes these categories with line-item detail:

Category	Items	Est. Cost
Airfare	Round-trip per person × N travelers	$XXX
Lodging	Hotel rate × nights × rooms (2 per room)	$XXX
Ground Transport	Rental car + fuel, OR mileage (personal vehicle)	$XXX
Meals	Per diem × days × travelers	$XXX
Registration	Competition/event registration fees	$XXX
Shipping	Project equipment shipping (outbound + return)	$XXX
Miscellaneous	Parking, tolls, public transit	$XXX
Total		**$XXX**

Get actual quotes (not guesses) for airfare, lodging, and registration. Use university-contracted rates where available. Include links to the specific hotels and flight options you are proposing.

16.8 The Post-Trip Checklist

After returning from travel, complete these tasks within one week:

1. Collect all receipts from all travelers. Ensure every receipt is itemized and includes the date, vendor name, and items purchased.
2. Verify no receipts contain alcohol (check carefully; some restaurants list it under cryptic names).
3. Complete the mileage reimbursement form with Google Maps PDF documentation.
4. Complete the reimbursement forms for all out-of-pocket expenses.
5. Submit all forms and receipts to CP within one month.
6. Update the Budget Tracker with actual travel costs.
7. Debrief with the team: what went well, what logistics should improve for next time?

The Alcohol Receipt Rule Is Absolute

If *any* receipt includes alcohol (even one drink on a $150 team dinner receipt) the **entire receipt is rejected**. There is no partial reimbursement and no exception to this policy. Teams learn this the hard way every semester. The solution: pay for alcohol separately on a personal card, or do not order alcohol on team travel.

16.9 Travel Safety and Logistics

Emergency contacts: before departing, share the team's travel itinerary with your PA and ensure every traveler has the PA's phone number. For competition travel, identify the nearest hospital to the competition venue.

Equipment transport: if shipping the prototype or competition vehicle, plan the logistics well in advance. Measure and weigh the shipment. Get shipping quotes from multiple carriers (FedEx, UPS, freight carriers for large items). Ship early enough to arrive before you do; having your prototype arrive after the competition is a preventable disaster.

Personal vehicle travel: when driving personal vehicles, follow the Mines driving policy (available from EHS). Get adequate sleep before long drives. Switch drivers every 2–3 hours. Do not drive more than 10 hours in a day. Ensure vehicle insurance is current and covers the trip.

Hotel logistics: book rooms with two occupants per room to control costs. Reserve rooms with a free cancellation policy in case plans change. Verify the hotel is within reasonable distance of the competition venue or client site. Request a late checkout on the last day if your event runs late.

Documentation during travel: photograph and video-record everything relevant; competition performance, client site conditions, field survey data, team activities. This content is invaluable for reports, presentations, and the Showcase poster.

16.10 Competition-Specific Travel Considerations

Competition travel has unique requirements beyond standard capstone travel:

- **Registration deadlines**: competition registration often closes months before the event. The Travel Lead must track deadlines and submit registration as soon as the Go/No-Go is approved.
- **Tech inspection preparation**: many competitions require pre-inspection paperwork (safety forms, technical declarations, driver certifications). Complete these before traveling.
- **Pit area setup**: plan what tools, spare parts, and equipment you need at the competition. Create a packing list and check it before departing.
- **Team coordination at the event**: designate roles for the competition day; who operates the prototype, who handles documentation, who interfaces with judges, who manages logistics.
- **Post-competition debrief**: before leaving the venue, conduct a hot debrief while memories are fresh. What went well? What failed? What should be improved for next year?

16.11 International Travel Considerations

Some competitions and client projects involve international travel. Additional requirements:

Passport: verify all travelers have valid passports with at least 6 months remaining before expiration. Passport processing takes 6–8 weeks (standard) or 2–3 weeks (expedited). Apply early.

Mines international travel registration: Mines requires all university-affiliated international travel to be registered through the Office of Global Education. Complete the registration at least 4 weeks before departure.

Travel insurance: Mines provides institutional travel insurance for registered university travel. Verify coverage details with the Office of Global Education.

Equipment export: if transporting technical equipment or prototypes internationally, verify that no export control restrictions apply. The US Department of Commerce Export Administration Regulations (EAR) and the International Traffic in Arms Regulations (ITAR) restrict export of certain technologies. When in doubt, consult the Mines Office of Research Administration.

16.12 What Happens When Plans Change

Travel plans rarely survive first contact with reality. Common disruptions and responses:

Flight cancellation: rebook through the airline or FROSCH immediately. Document any additional costs (hotel night, meals) for reimbursement. Notify your PA and the competition organizer.

Team member illness: if a team member cannot travel, determine whether the mission can be completed with fewer people. Adjust roles accordingly. The PA may need to approve a modified travel roster.

Equipment damage during transport: this is why you ship the prototype separately and insure it. If damage occurs, assess whether repair is possible on-site. Bring a basic tool kit and spare parts for the most failure-prone components.

Weather delays: build a buffer day into the travel schedule when possible. Arriving the day before the competition starts allows recovery from a weather delay without missing the event.

16.13 The Pre-Trip Checklist

One week before departure, verify:

- All travelers have confirmed flight/hotel reservations (names match IDs).
- Rental car reservation confirmed (driver has valid license and is $\geq$21 years old).
- Competition registration confirmed and all pre-event paperwork submitted.
- Prototype packed and shipping confirmed (if shipping separately).
- Tool kit packed: basic hand tools, multimeter, soldering iron, spare components, zip ties, tape, USB cables.
- Printed copies of: travel itinerary, hotel confirmation, rental car confirmation, competition schedule, campus map, emergency contacts.
- All team members have PA phone number and competition emergency contact.
- Sufficient cash or card for meals and incidentals (receipts will be needed for reimbursement).
- Camera/phone charged for documentation.
- Team shirts packed (competition teams).

16.14 Travel Contingency Planning

Anticipate problems and have backup plans:

If the prototype does not work at the competition: bring a basic repair toolkit and spare parts for the most failure-prone components. Identify a workspace near the venue (hotel room, parking lot with a table) where repairs can be attempted.

If a team member cannot travel: redistribute roles before departure. Ensure at least two people understand every critical subsystem so the team can function with one member absent.

If weather delays travel: build a buffer day into the schedule. For competitions with fixed start times, this buffer is essential. An extra hotel night ($50–100) is cheap insurance against missing the event entirely.

If costs exceed the budget: track spending in real time during the trip. If you are trending over budget on meals, switch to grocery store food for the remaining days. If an unexpected cost arises (parking fees, toll roads, equipment replacement), document it immediately for the reimbursement process.

16.15 Rental Car and Personal Vehicle Policies

Rental cars: the PA or a designated student with a OneCard can book a rental car through the university's contracted rental agency. All drivers must be listed on the rental agreement. Most rental companies require drivers to be $\geq$21 years old; some charge a surcharge for drivers under 25. Verify the fuel policy (return full, prepay, etc.) and follow it precisely (rental fuel surcharges are expensive.

Personal vehicles: Mines reimburses mileage at the official rate (confirm with CP before the trip). Mileage is calculated from the Mines campus (1500 Illinois Street, Golden, CO 80401) to the destination. **Save the Google Maps route as a PDF before departure**) this is required documentation for reimbursement. Gas purchases are **not reimbursable** for personal vehicles (the mileage rate covers fuel). Carpooling is required; multiple personal vehicles for the same trip will not be reimbursed.

Parking and tolls: reimbursable with receipts. Use hotel parking if available (often included in the room rate). Airport parking receipts must show entry and exit dates matching the travel dates.

16.16 Documenting the Trip for Academic and Project Credit

Travel is not just logistics; it is an opportunity to gather project data and professional experience:

Competition teams: photograph and video-record every event run, tech inspection interaction, and scoring session. Document equipment performance under competition conditions. Record failure modes and repair procedures. This content feeds directly into the FDR report.

Client site visits: photograph the operating environment, existing equipment, user workflows, and any conditions that affect your design requirements. Take notes during every client conversation. These observations often reveal requirements that were not captured during remote meetings.

Field survey teams: follow your data collection protocol precisely. Document GPS coordinates, weather conditions, equipment calibration, and any deviations from the planned procedure. Raw field data that lacks metadata (where, when, how, by whom) cannot be used in analysis.

16.17 The Travel Planning Timeline

Travel planning should begin as soon as the project involves a potential trip:

8–12 weeks before: identify the travel need (competition, client site visit, field survey). Discuss with PA. Initiate the Go/No-Go conversation with CLT.

6–8 weeks before: designate the Travel Lead. Begin budget development. Research lodging and airfare options. Register for the event if applicable.

4–6 weeks before: submit the travel budget to PA and CP for approval. Book lodging. Begin airfare booking (through FROSCH if required). Confirm the travel roster with PA.

2–4 weeks before: finalize all bookings. Prepare the equipment packing list. Ensure all travelers have required documents (ID, registration confirmation, competition paperwork). Ship equipment if required.

1 week before: final pre-trip meeting. Distribute the itinerary to all travelers and PA. Review the pre-trip checklist. Confirm all reservations.

During the trip: document everything. Track spending. Follow all Mines travel policies.

Within 1 month after: submit all receipts and reimbursement forms. Update the Budget Tracker. Debrief with the team.

Teams that wait until 2 weeks before the trip to start planning consistently encounter: sold-out hotels, expensive last-minute airfare, missed registration deadlines, and avoidable logistical crises.

16.18 Delegating Travel Duties

For multi-day trips, distribute logistics across the team to prevent the Travel Lead from being overwhelmed:

Travel Lead: overall coordination, budget management, flight booking.

Lodging Coordinator: hotel research, booking, check-in/check-out.

Ground Transport Lead: rental car booking, driving schedule, fuel stops.

Equipment Manager: packing list, shipping, setup at venue, teardown.

Documentation Lead: photos, videos, competition results, travel receipts collection.

Each role has specific responsibilities, but all roles share the responsibility of being on time, communicating proactively, and following Mines travel policies.

16.19 Lessons from Past Capstone Travel

These lessons come from actual capstone team experiences at Mines:

The forgotten adapter: a team traveled to a competition with their laptop charger but forgot the USB-C cable needed to program the ESP32. They spent 2 hours finding a Best Buy in an unfamiliar city. *Lesson: create a tool/cable checklist specific to your project and verify it the day before departure.*

The unviewable video: a team recorded excellent competition footage on a phone that ran out of storage mid-day. They deleted "unnecessary" videos to free space, including the one showing their best performance run. *Lesson: bring a dedicated camera or ensure ample phone storage. Back up footage to a laptop each evening.*

The tax receipt: a team submitted a $187 dinner receipt for reimbursement. The receipt included two glasses of wine totaling $24. The entire $187 was rejected; not just the $24. The team absorbed the loss personally. *Lesson: the alcohol policy is absolute. Pay for alcohol separately on a personal card, every time.*

The missed flight: a team member's driver's license was expired. TSA would not let them through security. They missed the flight, reboked at their own expense ($380), and arrived a day late. *Lesson: the Travel Lead must verify that all traveler IDs are valid and match the name on the booking, at least 1 week before departure.*

The shipping delay: a team shipped their prototype via ground shipping to arrive 3 days before the competition. A weather delay added 2 days. The prototype arrived the morning of the competition with no time for setup testing. *Lesson: ship equipment at least 5 business days before the event, or use expedited shipping with tracking.*

16.20 The Equipment Packing List

For competition and field trips, prepare a project-specific packing list:

Essential for all trips: laptop + charger, USB cables for all project hardware, multimeter, basic hand tools (screwdrivers, pliers, wire strippers), soldering iron + solder + flux (if electrical work may be needed), electrical tape + zip ties, spare batteries/power supplies, printed copies of all documentation (schematics, firmware, O&M procedures), camera for documentation.

Competition-specific: spare components for failure-prone items (fuses, connectors, motors), team shirts, competition rulebook (printed), tech inspection paperwork, safety equipment required by competition rules, display materials (poster, banner, table cover).

Field survey-specific: GPS unit or phone with offline maps, field notebook (waterproof), measurement instruments with calibration certificates, sample containers and labels, weather-appropriate clothing, first aid kit, water and snacks, emergency communication device.

16.21 Post-Competition and Post-Trip Documentation

The work does not end when you return to campus:

Financial closeout: collect all receipts from all travelers within 3 days of return. The CFO organizes receipts by category, verifies no alcohol receipts, completes all reimbursement forms, and submits to CP within 2 weeks (well within the 1-month deadline). Update the Budget Tracker with actual travel costs.

Technical debrief: within 1 week of return, hold a structured team debrief: (1) what went well (specific successes to replicate), (2) what failed (specific failures to analyze and prevent), (3) what we would do differently (actionable improvements for next year's team or for your own future professional practice). Document the debrief in your project repository.

Client communication: if the trip involved a client site visit, send a follow-up email within 48 hours summarizing: observations, data collected, decisions made, and next steps. Attach relevant photos and documentation.

Competition results documentation: for competition teams, document: overall placement, scores in each category, judge feedback (often provided informally; write it down immediately), photos of the prototype in competition, and any equipment issues or damage. This documentation feeds into the FDR report and provides valuable data for future teams.

"Lessons for Next Year" document: competition teams should create a brief (1–2 page) document capturing specific advice for the next year's team. Include: what to pack, what to order early, which competition rules caused problems, hotel and restaurant recommendations, and any logistical tips specific to the venue. Store this in the Digital Design Archive so future teams can find it.

16.22 Per Diem and Meal Management

Per diem is the daily allowance for meals and incidental expenses during travel. Mines per diem rates follow federal GSA guidelines (rates vary by city):

How per diem works: the per diem rate is a maximum, not an entitlement. You are reimbursed for actual meal costs up to the per diem rate, with itemized receipts. If the per diem for your destination is $59/day and you spend $40, you are reimbursed $40. If you spend $70, you are reimbursed $59.

Meal receipt requirements: every meal receipt must be: (1) itemized (showing each item ordered, not just a total), (2) free of alcohol, (3) dated within the travel period, and (4) from a restaurant or food vendor (grocery store receipts for meal supplies are acceptable). A credit card slip showing only a total amount is not an itemized receipt (request the detailed bill from the server.

Group meals: when the team eats together, one receipt is acceptable with a note listing all team members present. However, the per diem allocation is per person) a $180 dinner for 4 people at $45/person is within most per diem rates, but a $300 dinner for 4 people may exceed the per diem even though the total seems reasonable.

The alcohol rule (again): this rule is worth repeating because teams violate it every semester. If any item on the receipt is an alcoholic beverage, the **entire receipt** is rejected for reimbursement. Not just the alcohol; the entire meal. The solution: (1) do not order alcohol on team travel, or (2) if you choose to drink, pay for the alcoholic beverages on a separate, personal transaction with a separate receipt. Ask the server for a split check *before* ordering.

16.23 Coordinating Large Group Travel

For teams of 5+ or multi-team travel (e.g., EcoCAR teams sending 8–12 students):

Central booking: designate one person (Travel Lead or PA with OneCard) to make all hotel and flight reservations. This ensures consistent pricing, group rates, and organized documentation. Having 6 people make individual hotel bookings produces 6 different hotels, 6 different rates, and 6 different cancellation policies.

Shared transportation: for driving trips, arrange carpooling with the minimum number of vehicles. Each vehicle needs: a designated driver, a backup driver, a phone with navigation, a charged phone for emergencies, and knowledge of the route. Plan rest stops every 2–3 hours.

Communication during travel: create a group text/chat for real-time coordination. Designate one person as the "home base" contact (typically the PA) who has the full itinerary and can coordinate if a vehicle breaks down, a flight is cancelled, or a team member needs assistance.

Emergency protocols: every traveler should have: the PA's phone number, the campus emergency number, the hotel address and phone number, and basic insurance information. For competition travel, add the competition organizer's emergency contact.

16.24 Domestic vs. International Travel Differences

Most capstone travel is domestic (within the US), but some competitions and client projects involve international travel. Key differences:

Documentation: domestic travel requires a valid government-issued ID (driver's license or state ID). International travel requires a valid passport with at least 6 months remaining before expiration. Some countries require visas (check requirements at least 3 months before departure.

Institutional requirements: Mines requires all international travel to be registered through the Office of Global Education. Complete the registration at least 4 weeks before departure. This registration activates institutional travel insurance coverage.

Customs considerations: if transporting a prototype or technical equipment internationally, verify that no export control restrictions apply (ITAR, EAR). Carry a letter from the Capstone Program Director describing the equipment and its educational purpose) this can expedite customs clearance.

Communication: international cell phone service is expensive. Activate an international plan before departure or purchase a local SIM card at your destination. Ensure at least one team member has reliable communication at all times.

16.25 Travel as a Professional Development Opportunity

Travel in capstone is not just logistics; it is a professional development experience that prepares you for business travel throughout your career:

Professional behavior while traveling: you represent Colorado School of Mines and the Capstone Design Program at every interaction during travel; at the airport, the hotel, the competition venue, and the restaurant. Dress professionally when interacting with competition officials, judges, and other teams. Be punctual. Be courteous to service staff. Your behavior reflects on the program and on future Mines teams.

Networking at competitions and conferences: competitions bring together students from dozens of universities. This is a networking opportunity: introduce yourself to other teams, ask about their design approaches, and exchange contact information. The student you meet at an SAE competition may be your colleague at an automotive company in 5 years.

Observing other teams: at competitions, observe how top-performing teams organize their pit area, present to judges, and recover from technical failures. These observations provide lessons that improve your own team's performance. Document what you observe; specific, actionable observations are more valuable than vague impressions.

Post-travel reflection: after returning, reflect (individually and as a team) on: what professional skills did you practice during travel? What would you do differently? How does the competition or site visit experience change your perspective on the project? This reflection feeds into your end-of-semester Individual Performance Evaluation narrative.

16.26 Practice Questions

1. Create a travel budget for a 4-person team attending a 3-day SAE competition in Detroit. Include all categories with realistic estimates.
2. A dinner receipt includes two beers. What happens to the entire receipt?
3. You forgot to save the Google Maps PDF before a client site visit. How do you document mileage?

Part IV

Professional Skills

Chapter 17

Client Engagement and Management

17.1 The Professional Client Relationship

Your client is a professional with a real problem who has partnered with Mines to find a solution. This is not an instructor-student relationship; it is a professional services engagement. The client does not grade you (your PA and CF do), but their satisfaction with your work, communication, and professionalism directly affects your PA's assessment and, ultimately, your grade.

Treat every client interaction as you would treat a meeting with a paying customer in your first engineering job. Respond to emails within 24 hours. Show up on time, prepared, and professionally dressed (business casual minimum). Deliver what you promise by the date you promise it. If you cannot meet a commitment, communicate the delay *before* the deadline, not after.

17.2 Setting Expectations

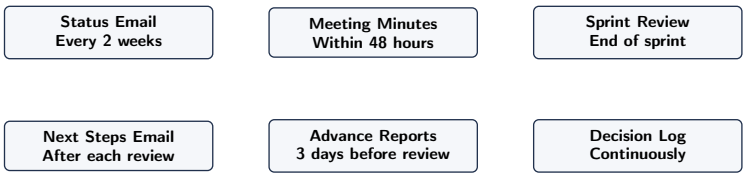

Figure 17.1: The client communication cadence. Six communication touchpoints ensure the client is never surprised. The Communication Lead owns the cadence; every team member contributes content.

At the kickoff meeting (Chapter 2), establish mutual expectations explicitly. Document them in the SOW.

What the client can expect from you: regular status updates at an agreed frequency (biweekly is typical), meeting minutes sent within 48 hours, honest assessment of progress and problems (no sugar-coating), professional communication, and timely delivery of reports and presentations in advance of design reviews.

What you can expect from the client: timely responses to questions (agree on a response window; 48 hours is reasonable), access to the operating environment or data you need, feedback on deliverables within agreed timelines, attendance (or delegation) at design reviews, and availability for key decisions that block your progress.

The Client Communication Cadence

Figure 17.1 shows the six communication touchpoints that keep the client informed throughout the project. At minimum, the client should hear from you: (1) a brief status email before each sprint review summarizing progress, blockers, and upcoming milestones, (2) meeting minutes within 48 hours of every client interaction, (3) advance delivery of reports at least 3 days before presentations so the client can review, and (4) Next Steps Emails after each design review. If the client has not heard from you in two weeks, they are worried; or worse, they have concluded you are not making progress.

17.3 Managing Scope Changes

Client needs evolve as the project progresses and as the client sees your work. New ideas emerge; priorities shift; external constraints change. This is normal. The professional response is not to simply agree or refuse, but to evaluate the impact systematically:

1. **Document the request**: write down exactly what the client is asking for. Confirm your understanding with the client ("Let me make sure I understand: you would like us to add wireless data logging to the system. Is that correct?").
2. **Assess the impact**: how does this change affect the schedule? The budget? Other requirements? The risk profile? Be specific: "Adding wireless logging requires an ESP32 module ($8), antenna design (1 week), firmware development (2 weeks), and additional testing (1 week). Total impact: 4 weeks and $50."
3. **Present the trade-off**: show the client what the change costs and what must be deferred or dropped to accommodate it. "We can add wireless logging if we defer the color LCD display to a future version. Or we can add both, but FDR would slip by 3 weeks past the deadline."
4. **Get explicit approval**: the client and PA must agree before the team implements the change. Document the decision in the SOW under change control.

Small changes accumulate. Without this discipline, you will arrive at FDR having built something substantially different from what was planned, with no documentation of why the scope shifted or who approved the changes.

17.4 Delivering Bad News

Delays happen. Components fail. Tests produce unexpected results. Analyses reveal fundamental problems. The professional response is **immediate, honest communication**: what happened, what the impact is, and what your plan is to address it. Structure the conversation:

1. State the problem clearly and specifically: "Our motor driver failed testing on Tuesday. The H-bridge IC overheats at the required current load."
2. State the impact: "This means we cannot demonstrate motor control at CDR. The CDR presentation will show the analysis and the revised design, but not the physical demonstration."
3. State your plan: "We have identified an alternative driver IC rated for higher current. It arrives Wednesday. We expect to have a working motor subsystem by Sprint 10."
4. Ask for input: "Do you have any concerns about this approach, or suggestions for alternatives?"

Never hide a problem hoping it will resolve itself. The worst client meeting is the one where they learn about a problem you have known about for three weeks. Early disclosure builds trust; late

disclosure destroys it.

17.5 The Client as a Resource

Your client has domain expertise that your team lacks. They understand the operating environment, the end users, the regulatory landscape, and the practical constraints in ways that no amount of internet research can replicate. Leverage this expertise:

- Ask the client to review your ConOps and requirements for accuracy (Chapter 2–3).
- Invite the client to suggest testing scenarios based on their operational experience.
- Ask for introductions to other domain experts (potential Technical Advisors).
- Request access to the operating site for contextual inquiry and validation testing.

17.6 Intellectual Property and Confidentiality

Some clients require Non-Disclosure Agreements (NDAs) before sharing proprietary information. If your client provides confidential data, you are **legally and ethically obligated** to protect it. Do not share proprietary information with other teams, post it publicly (including in GitHub public repositories), or include it in materials shared beyond the project team, PA, and CF without explicit written permission. If uncertain about what you can share, ask your PA.

17.7 When the Client Is Wrong

Clients are domain experts but not always engineering experts. When a client requests something technically infeasible, unsafe, or contradictory to good engineering practice, it is your **professional obligation** to push back; respectfully, with evidence. Present data: cost estimates, risk analysis, simulation results, or test data that demonstrate why an alternative approach is preferable. The client may still choose their original direction (it is their project), but they should do so with full information about the technical consequences.

17.8 Tips for Productive Client Relationships

- **Mirror their communication style**: some clients prefer formal emails; others prefer quick Slack messages. Adapt.
- **Come with options, not open-ended questions**: "We have identified three approaches and recommend Option B because [reasons]. Do you agree, or would you prefer Option A or C?" is far more productive than "What should we do?"
- **Underpromise, overdeliver**: commit to what you are confident you can achieve by the deadline. Surprising a client with early delivery builds trust. Surprising them with a missed deadline erodes it.
- **Acknowledge feedback explicitly**: when the client gives feedback at a design review, write it down in front of them, confirm you understood it, and report back on how you addressed it at the next meeting.
- **Send a thank-you at the end**: a professional thank-you email after the project closes is a small gesture that maintains the relationship for future networking.

17.9 Handling Difficult Client Situations

Not all client relationships are smooth. Common challenging scenarios and professional responses:

The unresponsive client: the client does not return emails or attend meetings. Response: (1) try alternative communication channels (phone call, PA-mediated introduction to a different client contact), (2) document every contact attempt with dates and methods, (3) escalate to your PA if two weeks pass without response, (4) continue working based on the last documented agreement while awaiting response.

The scope-expanding client: the client adds features at every meeting. Response: use the scope change process (Chapter 7). Acknowledge the request, assess the impact, present the trade-off, and get explicit approval before implementing. If the client insists on adding scope without removing anything, involve the PA in the conversation.

The technically directive client: the client dictates specific implementation details ("Use this exact sensor, use this programming language, use this vendor") that may not be the best engineering choice. Response: implement the client's direction unless it compromises safety, exceeds budget, or violates engineering best practice. If it does, present your analysis: "We evaluated your recommended sensor and found that it does not meet the IP67 requirement. Here is an alternative that does. May we proceed with the alternative?"

The absent client: the client attends the kickoff meeting and then disappears until FDR. Response: maintain the communication cadence (status emails, meeting minutes) even if the client does not respond. At FDR, you can demonstrate that you communicated consistently; the client cannot claim they were uninformed.

Conflicting client stakeholders: two client contacts give you contradictory direction. Response: document both positions, present the conflict to the PA, and request a single point of contact with decision authority. Do not try to satisfy contradictory requirements simultaneously.

17.10 Working Across Organizations and Cultures

Some capstone clients are large organizations with complex internal structures. Navigating these requires cultural awareness:

Large companies: your primary client contact may not have full decision authority. Decisions may require approval from their manager, legal department, or procurement. Allow extra time in the schedule for internal approvals. Do not be frustrated by slow responses; organizational processes take time.

Government agencies: subject to procurement regulations, security requirements, and bureaucratic processes. Documentation requirements may be more formal. Access to government sites may require background checks or escort. Plan for these requirements in Sprint 0.

Nonprofits and community organizations: may have limited technical sophistication but deep domain knowledge. Translate engineering terminology into plain language. Be sensitive to budget constraints; what seems like a small expenditure to you may be significant to a nonprofit.

International clients or partners: be aware of time zone differences, communication style differences, and cultural norms. In some cultures, direct disagreement is considered rude; in others,

silence does not mean agreement. When in doubt, confirm understanding in writing: "Based on our discussion, my understanding is [summary]. Please confirm or correct."

Academic clients (Mines faculty): understand that faculty operate on academic timelines (semester cycles, grant deadlines, conference schedules). They may be less available during grant proposal season or conference travel. Communicate proactively about scheduling constraints.

17.11 Email Templates for Common Client Communications

Professional email communication follows consistent patterns. Here are templates for the most common client emails:

Meeting Request Email: Subject: "[Team XX] Meeting Request – [Topic]." Body: greeting, purpose of meeting, proposed dates/times (offer 3 options), expected duration, agenda items, and any pre-reads. Close with "Please let me know which time works best for you."

Meeting Follow-Up Email: Subject: "[Team XX] Meeting Summary – [Date]." Body: thank the client for their time, list key decisions made (numbered), list action items with owners and deadlines, note any open questions, and state the next scheduled touchpoint. Send within 48 hours. Copy your PA.

Status Update Email: Subject: "[Team XX] Sprint [N] Status Update." Body: brief summary of progress since last update (3–5 bullet points), any blockers or risks that need client input, upcoming milestones, and a note inviting questions or feedback.

Next Steps Email (post-review): Subject: "[Team XX] [PDR/CDR/FDR] Summary and Next Steps." Body: thank reviewers, summarize key feedback received (categorized as: will address, will investigate, acknowledged and deferred), state the plan for the next sprint cycle, and attach the final version of the review report.

Bad News Email: Subject: "[Team XX] Update on [Specific Issue]." Body: state the problem directly (do not bury it), describe the impact on schedule/budget/performance, present your plan to resolve it (with timeline), and request any client input needed. Never use a bad-news email as a substitute for a phone call or meeting for serious issues (call first, then follow up in writing).

> Every client email should be: (1) addressed to the client by name, (2) professional in tone, (3) specific about action items and deadlines, (4) proofread for grammar and spelling, and (5) copied to your PA. Treat every email as if it will be forwarded to the client's boss—because it might be.

17.12 Preparing for Client Meetings

Every client meeting should have a purpose, an agenda, and a desired outcome. The Communication Lead prepares:

Before the meeting:

- Circulate an agenda 24–48 hours in advance. Include: purpose, topics with time estimates, any documents the client should review, and the desired outcome for each topic.
- Assign roles: facilitator (keeps discussion on track), scribe (records decisions and actions), presenter (for any prepared material), and timekeeper.

- Review the decision log and open action items from the previous meeting. Be prepared to report on closed items.
- Prepare visual aids if presenting design work (slides, CAD renderings, test data plots). Do not walk into a client meeting with only words.

During the meeting:

- Start on time. End on time or earlier.
- Take notes on every decision made and every action item assigned. Read back decisions: "So we are agreeing that the system will use wireless data transmission rather than wired. Is that correct?" Confirmation prevents misunderstandings.
- When disagreements arise, note them explicitly: "The client prefers Option A; the team recommends Option B for [reasons]. We will present a trade-off analysis at the next meeting for a final decision."

After the meeting:

- Send meeting minutes within 48 hours. Include: attendees, date, key discussion points, all decisions made (numbered), all action items (owner, description, due date), and next meeting date.
- Copy your PA on all client correspondence.
- Update the decision log with any new client decisions.
- Add action items to the task board with owners and deadlines.

17.13 Building Long-Term Client Trust

Client trust is built through consistent, reliable behavior over two semesters. The compounding effect:

Sprint 0–2: trust is fragile. The client is evaluating whether your team is competent and professional. Every missed email, every late deliverable, and every unprepared meeting erodes trust at this stage. Invest heavily in professionalism: respond to emails within 24 hours, arrive on time, deliver meeting minutes promptly, and come prepared with questions and visual aids.

Sprint 3–5: trust is developing. If you have been consistent through the first sprints, the client begins to rely on your team and share more openly. They may reveal constraints or preferences they did not mention initially. This is a sign that the relationship is working.

Sprint 5 (PDR): trust is tested. PDR is the first time the client sees the full scope of your engineering work. A strong PDR solidifies trust; a weak PDR raises concerns. Even if the PDR has weaknesses (it usually does), a team that responds to feedback aggressively in Sprint 6 restores confidence.

Sprint 7–14 (SDII): trust is deepened. Regular communication during the build phase (especially honest communication about challenges) builds the kind of professional trust that leads to job offers, references, and long-term career relationships. Clients who trust your team will advocate for you to their colleagues.

Post-FDR: trust becomes a career asset. A client who had a positive capstone experience is a professional contact for life. They will provide references, make introductions, and potentially offer employment opportunities. A brief thank-you email after the project closes is a small investment that maintains this relationship.

17.14 When to Involve Your PA in Client Communications

Most client communication should be managed by the team directly. Involve your PA when:

- The client is requesting a scope change that significantly affects the project (Chapter 7).
- The client is unresponsive for more than 2 weeks despite multiple contact attempts.
- A disagreement has arisen between the team and client that cannot be resolved at the team level.
- The client requests something that conflicts with course requirements or academic integrity policies.
- Confidential or sensitive information (NDA, proprietary data) needs to be discussed.
- The team needs to deliver significant bad news (major delay, fundamental design flaw, budget overrun >25%).

In all cases, the team should attempt direct resolution first. When escalating to the PA, bring: a summary of the situation, what the team has already tried, specific documentation (emails, meeting minutes), and a proposed resolution.

17.15 Professional Boundaries with Clients

While client relationships should be warm and productive, maintain professional boundaries:

Communication channels: use professional channels (email, scheduled meetings, Teams) for all project communication. Avoid personal social media connections during the project. If the client contacts you via text or personal channels, respond professionally and suggest moving to email for documentation purposes.

Scope of the relationship: the client is a professional partner, not a friend, employer, or professor. You owe them professional deliverables and honest communication; they owe you timely feedback and access to information. The power dynamic is different from a student-professor relationship: you are providing a service, and the client depends on your expertise.

Confidentiality: some clients share proprietary information under NDA. Treat all client information as confidential unless explicitly told otherwise. Do not discuss client-specific details with other capstone teams, post project details on social media, or store client data on personal devices. When in doubt, ask: "May I share this information with [specific audience] for [specific purpose]?"

Managing expectations about future availability: after FDR and Showcase, the project is complete. You are not obligated to provide ongoing support, debugging, or maintenance after graduation. However, a brief "transition document" included in the O&M manual (listing known issues, recommended improvements, and suggested next steps) is a professional courtesy that most clients appreciate.

When things go wrong: if a client behaves unprofessionally (inappropriate comments, unreasonable demands outside the SOW, pressure to cut safety corners), document the behavior and involve your PA immediately. You have the right to a professional working environment, and the PA and capstone program exist to protect that.

17.16 The Client Satisfaction Metric

Your PA evaluates the project partly based on client satisfaction. While client satisfaction is not a formal grade component, it influences the PA's assessment of your professionalism and communication.

Client satisfaction correlates most strongly with three factors: (1) **predictable communication**—the client always knows what is happening and when the next update will arrive, (2) **honest reporting**—the client trusts that what the team says is accurate, even when the news is bad, and (3) **delivered promises**—the team does what it committed to do in the SOW, on the agreed schedule.

Note what is *not* on this list: a perfect prototype. Clients understand that student projects have limitations. What they do not accept is being surprised, misled, or ignored. A team that delivers a flawed prototype but communicated honestly throughout the project will have a more satisfied client than a team that delivers a beautiful prototype but went dark for 6 weeks.

17.17 Managing Virtual Client Relationships

Many capstone clients are not local to Golden, CO. Virtual client relationships require additional effort:

Technology setup: agree on a video conferencing platform (Zoom, Teams, Google Meet) during Sprint 0 and verify that all parties can connect before the first substantive meeting. Test: video, audio, screen sharing, and recording (if the client consents to recording for the team's reference).

Time zone management: if the client is in a different time zone, propose meeting times that are reasonable for both parties. A 9:00 AM Mountain Time meeting is 11:00 AM Eastern (reasonable) but 6:00 AM Pacific (unreasonable). Use a shared scheduling tool (Doodle, When2Meet) for scheduling.

Compensating for reduced bandwidth: virtual communication has less bandwidth than in-person. Compensate by: sending agendas and pre-reads farther in advance (48 hours instead of 24), using screen sharing to show visual aids during every discussion (never rely on verbal description alone), and sending more detailed follow-up emails (virtual meetings generate more misunderstandings than in-person meetings).

Building rapport remotely: small talk at the beginning of virtual meetings builds the relationship. Ask about the client's week, mention relevant news from their industry, or share a brief project update before diving into the agenda. These 2–3 minutes of human connection pay dividends in client engagement throughout the semester.

Site visits: for virtual clients, plan at least one in-person site visit if geography and budget permit. A single day at the client's facility provides more context than months of virtual meetings. If a visit is not possible, request a video tour of the operating environment.

17.18 The Client Thank-You and Project Closeout

After FDR and Showcase, close the client relationship professionally:

The thank-you email: send within one week of FDR. Thank the client for their partnership, summarize the project outcome, reference specific contributions the client made (access to their facility, technical guidance, prompt feedback), and provide the location of all deliverables (Canvas links, shared drive, physical handoff).

The handoff meeting: if the client will continue using or developing the system, schedule a brief handoff meeting. Walk them through: how to operate the system (O&M manual), where all documentation is stored (Digital Design Archive), known limitations and recommended future work, and who to contact at Mines if questions arise after your graduation.

LinkedIn connections: after the project is complete (not during), connect with the client on LinkedIn with a personalized message: "Thank you for the opportunity to work on [project]. I learned a great deal about [domain] and appreciated your guidance throughout the semester." This maintains the professional relationship for future networking.

The reference conversation: if you plan to list the client as a professional reference, ask permission explicitly: "Would you be comfortable serving as a professional reference for me? If so, I'll send you my resume and a summary of the positions I'm applying for so you can speak to the most relevant aspects of our work together." Never assume permission; always ask.

17.19 Practice Questions

1. Your client asks you to add a feature that was explicitly listed as "Won't Have" in the SOW. Walk through the scope change process, including the specific conversation you would have with the client.
2. Your team has discovered a critical design flaw three weeks before CDR. The fix requires redesigning the power subsystem. Draft the key points of your communication to the client.
3. Your client has not responded to your last two emails (sent 5 and 12 days ago). What steps do you take?

Chapter 18

Professional Communication ; Written

18.1 Engineering Report Structure

Every major deliverable (SOW, PDR report, CDR report, FDR report) follows a similar structure: **executive summary** (the most-read and least-written section; it should stand alone as a complete summary—write it last), **introduction** (context, scope, purpose), **body sections** organized by topic (not by team member), **conclusions and recommendations**, **references** (IEEE format), and **appendices** (detailed data, drawings, datasheets).

The executive summary deserves special attention. A busy client or reviewer may read only the executive summary. It must convey: what the project is, what was accomplished, what the key results are, and what the recommendations are—in one page or less. If the executive summary is unclear, the report has failed regardless of what follows.

18.2 Technical Writing Best Practices

Active voice: "The team measured the output voltage" (clear subject and action) not "The output voltage was measured" (who measured it?). Active voice is shorter, clearer, and more accountable.

Specific language: replace vague words with numbers and units. "The temperature rose by 12.3 °C over 45 seconds" not "The temperature rose significantly." "Significantly" conveys no engineering information.

Figures and tables: every figure and table must be **numbered**, **captioned** (the caption should make the figure self-contained (a reader should understand it without reading the surrounding text), and **referenced in the text** ("as shown in Figure 3"). Never include a figure without discussing it. Never discuss a result without showing the data.

Document organization: organize by topic, not by team member. The reader does not care who wrote which section; they care about the logical flow of the engineering argument. Use consistent formatting, heading hierarchy, and terminology throughout.

18.3 Citation and Reference Practices

Use **IEEE citation format** [24]. Cite all sources. Attribute all images, data, diagrams, and content not produced by the team. "Hallucinated" AI citations—references that do not actually

exist—are an **academic integrity violation**. Every reference must be real and verified by clicking the link or locating the source.

18.4 Document Version Control

Engineering documents evolve throughout the project. Without version control, you end up with five different versions of the SDRD in five different folders and nobody knows which is current. Implement a simple version control system from Sprint 1:

File naming: use a consistent convention. Example:

`TeamXX_SDRD_v2.1_2026-03-15.docx`

where XX is your team number, v2.1 is the version (major.minor), and the date is ISO format. The major version increments at each design review baseline (v1 = PDR baseline, v2 = CDR baseline, v3 = FDR baseline); the minor version increments for internal revisions.

Change tracking: use track changes (Word) or revision history (Google Docs) when editing team documents. Before submitting, review all tracked changes and accept or reject them as a team. The final submitted version should not contain visible track changes.

Single source of truth: designate one location (Teams folder, Google Drive folder) as the authoritative repository. All other copies are working drafts. When in doubt about which version is current, check the repository. The Documentation Lead is responsible for keeping the repository organized and current.

18.5 Common Writing Errors in Capstone Reports

These patterns appear in nearly every semester of capstone deliverables:

Passive voice overuse: "The temperature was measured" (who measured it?). "The analysis was performed" (by whom?). Active voice is clearer, shorter, and more accountable: "The team measured the temperature" or "The structural analysis showed a safety factor of 2.3." Use active voice as the default; passive voice only when the actor is truly unknown or irrelevant.

Imprecise language: "The system performed well" (by what metric?). "The results were close to the requirement" (how close?). Replace subjective assessments with numbers: "The system measured temperature with $\pm 0.7\,^{\circ}\mathrm{C}$ accuracy, meeting the $\pm 1.0\,^{\circ}\mathrm{C}$ requirement."

Unsupported claims: "This is the best approach" (according to whom?). "Industry standard practice is…" (cite the standard). Every factual claim must be supported by either your own data (test results, analysis) or an external source (textbook, standard, peer-reviewed paper, manufacturer datasheet).

Missing transitions: sections that jump from topic to topic without connecting paragraphs. The reader should understand how each section relates to the next. A one-sentence transition ("Having established the requirements, we now describe the concept generation process") maintains narrative flow.

Orphaned figures: a figure appears in the document but is never referenced in the text. Every figure must be numbered, captioned, and discussed in the body text ("As shown in Figure 3, the output voltage increases linearly with temperature over the calibrated range").

Disorganized appendices: critical data buried in appendices that are not referenced from the main text. Appendices supplement the main text; they do not replace it. Reference each appendix from the section that relies on it ("Complete test data is provided in Appendix B").

Inconsistent terminology: calling the same component "the sensor module," "the measurement unit," and "the sensing element" in different sections. Pick one term and use it throughout the document. Define it in the glossary if it is technical.

18.6 The Report Revision Process

Writing a professional engineering report is an iterative process, just like the design process itself. Plan for multiple revision cycles:

First draft (individual contributions): each team member writes their assigned sections. Focus on content completeness, not polish. Get the engineering substance on paper.

Integration and peer review (team activity): assemble all sections into a single document. Read the entire report as a team. Check for: consistent terminology, logical flow between sections, complete cross-references (figures, tables, sections), and gaps in content. Assign one person as the integration lead who reviews the entire document for consistency.

Technical review: ask your PA (or a teammate who did not write a particular section) to read critically. Can they understand the engineering argument without asking clarifying questions? If not, the section needs revision.

Final polish: proofread for grammar, spelling, formatting, and compliance with the Program Format Guide. Read the document aloud; errors that are invisible when reading silently become obvious when read aloud.

Timeline: the first draft should be complete 7–10 days before the submission deadline. Integration and peer review: 3–5 days before. Final polish: 1–2 days before. A report written in the final 48 hours will read like a report written in 48 hours.

18.7 Report Template and Organization

The Capstone Design Program Format Guide on Canvas specifies formatting requirements. Beyond the format, organize your report for readability:

Section numbering: use hierarchical numbering (1, 1.1, 1.1.1). Do not go deeper than three levels; if you need a fourth level, the section is too granular and should be restructured.

Executive summary: write this last, not first. It should stand alone as a complete summary (a reader who reads only the executive summary should understand: what the project is, what was accomplished, what the key results are, and what the recommendations are. One page maximum.

Introduction: set the context. What is the project? Who is the client? What problem does the system solve? This is not a copy of the executive summary; it provides the detailed background that the executive summary condenses.

Body sections: organize by topic, not by team member. The reader does not care who wrote which section; they care about the logical flow of the engineering argument. Within each section,

follow the pattern: introduce the topic, present the methodology, show the results, and draw conclusions.

Appendices: use appendices for detailed data that supports the main text but would interrupt the narrative flow if included inline. Always reference appendices from the main text ("Complete test data is provided in Appendix B").

18.8 Visual Communication in Reports

Engineering reports communicate through visuals as much as text. Best practices:

Figures: use figures to show what words cannot efficiently describe—system architecture, test setups, data trends, CAD renderings, fabrication photos. Every figure must be numbered, captioned, and referenced in the text. The caption should make the figure self-contained.

Tables: use tables for structured data comparisons; requirements tables, BOM summaries, test result summaries, material properties. Every table must be numbered and captioned.

Data visualization: choose the right plot type. Line charts for trends over time or continuous variables. Bar charts for categorical comparisons. Scatter plots for correlation between two variables. Box plots for distribution comparisons. Never use 3D charts, pie charts with many slices, or gratuitous color when a simple plot would suffice.

Consistent formatting: use the same font, line weight, color scheme, and labeling convention throughout the document. Inconsistent formatting signals a document assembled hastily from disparate sources.

18.9 The Ethics Assignment: Structure and Common Mistakes

The Ethics Assignment (SDI Sprint 4) is a **1000–1500 word argumentative essay**; not an informative report, not a book summary, and not a list of ethical principles. You must:

1. **Take a controversial stance** on an ethical issue directly connected to your specific project's hardware, software, or system. "Engineers should be ethical" is not controversial. "Autonomous irrigation systems should be prohibited in drought-restricted areas even when they reduce total water consumption" is controversial and specific.
2. **Defend that stance with evidence**: cite data, case studies, regulations, or expert opinions that support your position. At least 3 credible sources.
3. **Address opposing viewpoints honestly**: present the strongest argument against your position and explain why your position is still preferable. This demonstrates critical thinking, not just advocacy.
4. **Write with a distinct voice**: the essay should sound like you, not like a generic AI-generated summary of ethical principles. Personal engagement with the specific ethical tension in your project is what distinguishes excellent essays.

The 7 most common mistakes (avoid all of them):

1. Writing an informative essay instead of an argumentative one (no thesis, no position taken).
2. Taking a position so obvious that no reasonable person would disagree ("engineers should not build unsafe products").
3. Discussing ethics in general instead of connecting to your specific project.

4. Failing to address counterarguments.
5. Exceeding the word limit (1500 words maximum; concision is a skill).
6. Using AI to generate the essay without genuine personal engagement.
7. Modifying the provided template formatting (use the template exactly as provided).

18.10 Figure and Table Formatting Standards

Every figure and table in your reports must comply with these standards:

Figures: numbered sequentially (Figure 1, Figure 2, ...). Caption below the figure. Caption is a complete sentence describing what the figure shows and why it matters. All axes labeled with units. Legend if multiple data series. Referenced in the body text before the figure appears ("As shown in Figure 3, the output voltage increases linearly with temperature").

Tables: numbered sequentially (Table 1, Table 2, ...). Caption above the table. Column headers with units. Consistent decimal places within each column. Referenced in the body text. Use horizontal rules (top, header, bottom) per the booktabs convention; avoid vertical rules and excessive gridlines.

Equations: numbered if referenced later. Variables defined immediately after the equation ("where V is voltage in volts, I is current in amperes, and R is resistance in ohms"). Units stated explicitly.

18.11 Collaborative Writing Workflows

Capstone reports are written by teams, not individuals. This creates coordination challenges that must be managed:

The outline-first approach: before anyone writes prose, the team agrees on a complete outline: section titles, subsection titles, key content points for each section, and the responsible author. This prevents overlap (two people writing about the same topic) and gaps (no one writing about a critical topic).

The integration meeting: after first drafts are complete, hold a 2-hour team meeting where the integration lead reads the entire document aloud. The team listens for: inconsistent terminology (the sensor is called "the thermocouple module" in one section and "the temperature sensor" in another), gaps in logic (a result is referenced but never presented), redundant content (the same information appears in two sections), and tone inconsistencies (one section is formal, another is casual).

The single-voice edit: after integration, one team member (preferably the strongest writer) does a final "voice pass" through the entire document to make it read as if written by a single author. This pass fixes: inconsistent formatting, varying heading hierarchies, different citation styles, and tone shifts between sections.

Deadline discipline: the team writing schedule should work backward from the submission deadline. Example for a PDR report due Friday:

- Monday (5 days before): individual section drafts due to shared repository.
- Tuesday: integration meeting; assemble sections, identify gaps and conflicts.
- Wednesday: gap-filling and revision of flagged sections.

- Thursday: single-voice edit, formatting check, PA review (if time permits).
- Friday morning: final proofread, PDF generation, submission.

A team that starts writing on Thursday night for a Friday submission will produce a document that reads like it was written Thursday night.

18.12 The Capstone Design Program Format Guide

All deliverables should follow the Format Guide available on Canvas unless superseded by specific client requirements. The guide specifies: file format, heading styles, margin widths, citation format, and deliverable naming conventions. Comply with your PA and client preferences for any additional formatting requirements.

Key formatting rules that are frequently violated:

- **Page numbers**: every page numbered. Front matter uses roman numerals; body uses arabic numerals.
- **Heading hierarchy**: consistent throughout (Title Case or Sentence case, not both).
- **Figure/table numbering**: sequential within the document. Every figure and table referenced in the text.
- **Citations**: IEEE format. Numbered in order of first appearance, not alphabetical.
- **File naming**: use the convention specified by your PA. Typical format:

  `TeamXX_[Deliverable]_[Version]_[Date].pdf`

18.13 The Writing Center

The Writing Center at Colorado School of Mines (https://writing.mines.edu) provides free support for technical reports, essays, and presentations. Make an appointment at https://mines.mywconline.com. This is one of the most underutilized and most valuable campus resources. The Writing Center staff are trained to help engineers communicate more effectively; they understand that engineering writing has different conventions than humanities writing.

18.14 The Final Proofread: A Systematic Approach

The difference between a B+ report and an A report is often the final proofread. Use this systematic approach:

Pass 1: Structure check: read only the headings and topic sentences. Does the document have a logical flow? Does each section follow naturally from the previous one? Are there gaps (topics mentioned but never explained) or redundancies (the same content in two places)?

Pass 2: Technical accuracy: read every number, unit, equation, and figure caption. Does every number have appropriate significant figures and units? Does every equation define its variables? Does every figure caption describe what the figure shows? Do the numbers in the text match the numbers in the figures and tables?

Pass 3: Citation check: does every factual claim have a citation? Is every citation formatted correctly (IEEE numbered format)? Does every reference in the bibliography actually exist (no AI-hallucinated citations)? Is every figure and table cross-referenced in the text?

Pass 4: Grammar and style: read the document aloud (or use text-to-speech). Awkward sentences, run-on sentences, and missing words become obvious when heard. Check for: subject-verb agreement, consistent tense (past tense for completed work, present tense for ongoing truths), and parallel construction in lists.

Pass 5: Format compliance: check against the Capstone Design Program Format Guide. Page numbers present? Margins correct? Heading styles consistent? File named per the convention? All required sections present?

This five-pass approach takes 2–3 hours for a 30-page report. It is the highest-ROI activity you can perform before submission.

18.15 Common Team Writing Pitfalls

These patterns recur in capstone team reports every semester:

The "assembled, not integrated" report: each team member writes their section independently, the sections are concatenated into a single document, and nobody reads the result. The report has: three different names for the same component ("the sensor module" in Chapter 2, "the measurement unit" in Chapter 4, "the sensing subsystem" in Chapter 6), two sections that explain the same concept differently, an introduction that promises content the body does not deliver, and formatting that changes between sections. Fix: the integration meeting and single-voice edit (described earlier) prevent this.

The "requirements rewrite" report: the team copies the SDRD into the report as Chapter 3, then rewrites the same requirements in different words throughout the design, analysis, and testing chapters. The result is a document where the reader encounters the same requirement 4–5 times in slightly different phrasing, creating confusion about which version is authoritative. Fix: present requirements once (in the SDRD chapter), then reference them by ID in subsequent chapters ("As specified in PR-004, the system shall maintain $\pm 1\,^{\circ}\mathrm{C}$ accuracy").

The "screenshot dump" appendix: the team includes 40 pages of unprocessed screenshots (oscilloscope captures without annotations, CAD models without dimensions, code output without interpretation. Appendix content must be organized, labeled, and referenced from the main text. Every figure in the appendix should have a caption explaining what it shows and why it matters. An appendix that nobody can navigate is an appendix that adds volume but not value.

The "written at midnight" conclusion: the final chapter is visibly rushed) shorter than preceding chapters, full of vague statements ("the project was successful"), and lacking the specificity of the rest of the document. The conclusion and recommendations chapter is the last thing the reviewer reads and strongly influences their overall impression. Write it with the same rigor as the analysis chapter.

The missing "so what": the report presents data but does not interpret it. "Figure 7 shows the temperature response over 60 seconds" tells the reader what the figure contains. "Figure 7 shows that the system reaches the setpoint within 12 seconds with 4% overshoot, meeting the PR-003 requirement of $\leq$15 seconds settling time and $\leq$10% overshoot" tells the reader what the data *means*. Every data presentation must include engineering interpretation.

18.16 IEEE Citation Format: Quick Reference

Capstone reports use IEEE citation format [24]. Key conventions:

In-text citations: numbered in square brackets in order of first appearance: "The Nyquist theorem requires sampling at twice the signal frequency [1]." Multiple citations: "Several studies confirm this approach [2], [5], [7]." Citation ranges: "[3]–[6]."

Reference list format (common types):

Journal article: [1] A. B. Author, "Title of article," *Journal Name*, vol. XX, no. YY, pp. ZZ–ZZ, Month Year.

Conference paper: [2] A. B. Author, "Title of paper," in *Proc. Conference Name*, City, Country, Year, pp. ZZ–ZZ.

Book: [3] A. B. Author, *Title of Book*, Edition. City: Publisher, Year.

Datasheet: [4] Manufacturer, "Part Number Datasheet," Document Number, Year. [Online]. Available: URL.

Website: [5] Organization, "Page Title." URL (accessed Month Day, Year).

Common citation mistakes:

- Citing a URL without a title, author, or access date.
- Citing Wikipedia as a primary source (cite the original sources that Wikipedia references).
- Including references that are never cited in the text (or citing references not in the list).
- AI-hallucinated citations that do not exist; **verify every reference by locating the actual source**.
- Inconsistent formatting (some references have periods, others do not; some have volume numbers, others do not).

18.17 Writing for Different Deliverable Types

Each capstone deliverable has a distinct purpose and audience:

The SOW: primarily a *contractual* document. Language is precise, commitments are explicit, and ambiguity is the enemy. Use SHALL/SHOULD for obligations. Define terms that could be interpreted differently by different readers. The SOW is not a design document; it defines *what* will be delivered, not *how*.

The PDR/CDR/FDR reports: primarily *technical* documents. Language balances precision with readability. The narrative should flow logically: problem → approach → analysis → results → conclusions. Support every claim with evidence (data, calculations, references). The audience is technically sophisticated (PA, TAs, client engineers) and will challenge unsupported assertions.

The Ethics and Broader Impacts essays: primarily *argumentative* documents. Language is persuasive but evidence-based. You are not reporting facts; you are defending a position. Use first person where appropriate ("I argue that..."), acknowledge counterarguments honestly, and demonstrate personal engagement with the ethical tension.

The O&M manual: primarily an *instructional* document. Language is simple, direct, and action-oriented. Use imperative mood ("Press the START button. Wait for the green LED. Verify

the display shows HOME.") with numbered steps. The audience may not be an engineer; assume no technical background unless the client specifies otherwise.

Meeting minutes: primarily a *record* document. Language is factual and concise. Document decisions (not discussions), action items (with owners and deadlines), and attendance. Meeting minutes are legal-weight documents in professional practice; in capstone, they are the evidence base for PIPs and grade disputes.

18.18 Practice Questions

1. Rewrite this sentence in active voice with specific data: "It was found that the system performed adequately under the tested conditions."
2. Write a one-paragraph executive summary for a hypothetical CDR report for your project.
3. A figure in your PDR report shows test data but has no caption, no axis labels, and no reference in the text. List everything that needs to be fixed to meet professional standards.

Chapter 19

Professional Communication ; Oral

19.1 Design Review Presentations

Structure every design review around a **narrative arc**: problem → approach → results → next steps. The audience should feel like they are following a logical story, not watching a random sequence of slides. Time your presentation and rehearse (going over time signals poor preparation; going significantly under time signals insufficient depth.

Coordinate team transitions: each speaker introduces the next ("Now Sarah will walk you through our concept selection process"), and all team members use a consistent slide template and visual style. A presentation where each section looks like it was made by a different team in a different course undermines credibility.

19.2 Slide Design Best Practices

- **One message per slide**. If you need more than 30 seconds to explain a slide, it has too much content.
- **Use visuals, not text walls**. Show the system architecture diagram, the decision matrix, the test data plot. Do not put the full text of your requirements on a slide—show a summary table.
- **Data visualization**: choose the right chart type. Bar charts for comparisons. Line charts for trends. Scatter plots for correlations. Tables for exact values. Label all axes with units.
- **Font size**: minimum 18pt for body text, 24pt for titles. If you have to say "I know this is hard to read," the slide has failed.

19.3 Handling Q&A

Q&A is not a threat; it is the most valuable part of the review. Tough questions from your PA, client, or TA improve the design. Listen to the entire question before responding. If you do not understand, ask for clarification ("Could you clarify what you mean by 'thermal margin'?"). If you do not know the answer, say so honestly and commit to following up: "That is an excellent question. I do not have the data at hand, but I will have an answer by our next meeting." Never make up an answer.

19.4 The Elevator Pitch

Every team member should be able to explain the project in **60 seconds** to a non-technical audience. Structure: what problem you are solving (one sentence), for whom (one sentence), how your approach works (one sentence), and why it matters (one sentence). Practice until it is natural. You will use this at Career Day, at the Showcase, and in every job interview for the next decade.

19.5 Sprint Review Presentations

Sprint reviews (Chapter 11) are less formal than design reviews but still require preparation. Show *working functionality*: a sensor reading on the screen, a motor spinning, a simulation result that matches the benchmark. "We are still working on it" is not a sprint review; it is an excuse.

19.6 Video Demonstrations

For subsystem demos (SDI) and system demos (SDII), you will record narrated video demonstrations. Narrate what you are doing, what the expected result is, and what the actual result is. A silent video of a blinking LED tells the viewer nothing. Verify your video link is accessible by opening it in an incognito/private browser window. Unviewable links may receive a zero.

Good demonstration video practices: (1) plan the demo script before recording; what will you show, in what order, and what will you say at each step, (2) start with a 10-second introduction stating your name, team, and what you are demonstrating, (3) show the test setup with a wide shot, then zoom in on the relevant component, (4) narrate the expected behavior before triggering it ("When I apply 5 V to the input, we expect to see approximately 3.3 V at the output"), (5) show the actual result clearly (point to the multimeter reading, show the oscilloscope trace), (6) state whether the result matches expectation and what it means.

Recording tips: use a phone or webcam with a tripod (not handheld; shaky video is unwatchable), ensure adequate lighting, minimize background noise, and keep videos under 3 minutes per subsystem. Longer is not better; concise and clear is.

19.7 Preparing for Design Reviews: The Rehearsal

Every design review presentation should be rehearsed at least twice:

First rehearsal (internal, 4–5 days before): run through the full presentation with the team. Time each section. Identify: slides that are too dense (split them), sections that run long (cut or condense), transitions that are awkward (add bridging sentences), and questions you cannot answer (research before the formal review).

Second rehearsal / Mock Review (with PA or another team, 2–3 days before): present to fresh eyes. Note what questions they ask; these are the same questions your formal reviewers will ask. Practice answering concisely: a 30-second answer is almost always better than a 3-minute answer for a review question.

Day-of preparation: arrive 15 minutes early. Test the projector, screen, and any live demonstration equipment. Have backup slides (PDF on a USB drive) in case the primary laptop fails.

Assign one team member to advance slides so the presenter can focus on speaking.

19.8 Managing Presentation Nervousness

Nervousness before a presentation is universal and physiologically normal (elevated heart rate, sweaty palms, dry mouth). It is not a sign of inadequacy; it is a sign that you care about the outcome. Strategies:

Preparation is the best antidote: the single most effective way to reduce nervousness is thorough preparation. If you know your material deeply and have rehearsed twice, your confidence is grounded in competence. If you are reading from slides for the first time, your nervousness is grounded in justified uncertainty.

Structured breathing: before stepping up to present, take 4–5 slow, deep breaths (4 seconds in, 4 seconds hold, 6 seconds out). This activates the parasympathetic nervous system and reduces the fight-or-flight response.

Focus outward, not inward: instead of worrying about how you look or sound, focus on the audience. Are they understanding the content? Are they engaged? Do they have questions? Outward focus redirects attention from self-consciousness to communication.

Accept imperfection: you will stumble on a word, lose your place, or forget a detail. This is normal. Pause, take a breath, and continue. The audience cares about the engineering content, not about flawless delivery.

19.9 Coordinating Team Presentations

A design review presentation with 4–5 speakers requires coordination:

Transitions: each speaker introduces the next ("Now Sarah will walk through our concept selection process"). Never end a section with silence and a confused handoff.

Visual consistency: all slides should use the same template, font, color scheme, and labeling conventions. A presentation where each section looks like it was designed by a different person undermines the impression of a cohesive team.

Q&A protocol: agree in advance who answers questions about which topics. If a question is directed to the wrong person, the Scrum Master redirects: "That's a great question about the power subsystem; let me pass it to Jordan, who led that design." Never have two people start answering the same question simultaneously.

Time management: assign a timekeeper (not the current speaker). At the 2-minute warning, the timekeeper signals. At time, the speaker wraps up the current slide and transitions. Going over time cuts into Q&A, which is the most valuable part of the review.

19.10 Synthesizing Feedback

After every design review, consolidate all feedback (from PA, client, TAs, and peers) into an **action item list**. For each item: what was the feedback, what action is required, who is responsible, and when will it be completed. Report back on closed action items at the next sprint review. Feedback that is received but not acted upon is wasted effort by everyone involved.

19.11 Speaking to Different Audiences

The same project must be communicated differently to different audiences:

Technical audience (PA, TAs, fellow engineers): emphasize methodology, analysis rigor, quantitative results, and engineering trade-offs. Use technical terminology. Show equations, data plots, and detailed block diagrams. Expect probing questions about assumptions and limitations.

Client audience (especially non-technical clients): emphasize the problem being solved, the user experience, the key performance outcomes, and the value delivered. Minimize jargon. Show photos, demonstrations, and high-level block diagrams. Translate technical metrics into client-meaningful terms: "the sensor accuracy is $\pm 0.5\,^{\circ}\mathrm{C}$" becomes "the system will detect temperature changes as small as one degree, which is sufficient to prevent crop damage in your greenhouse."

General audience (Showcase visitors, prospective students, administrators): emphasize the "so what"; why does this project matter? What real-world problem does it address? What makes the solution interesting or innovative? Keep it accessible. A visitor should understand and appreciate your project in 2 minutes.

Adapting in real time: at design reviews and the Showcase, you will encounter all three audiences in rapid succession. Develop the ability to shift your communication register based on the listener's questions. If they ask "What's your control bandwidth?" (they are technical; go deep. If they ask "What does it do?") they are general; go broad.

19.12 Using Visual Aids Effectively

Visual aids (slides, posters, demos, videos) should support your message, not be your message. If you are reading your slides verbatim, the slides have replaced you as the presenter. Best practices:

The 6-word test: each slide should have a clear takeaway that can be stated in 6 words or fewer. If you cannot summarize the slide's message concisely, the slide is doing too much.

Build animations: for complex diagrams or data, reveal information progressively rather than showing everything at once. A block diagram that builds from left to right mirrors your verbal explanation.

Live demonstrations: nothing is more compelling than showing the system working in real time. But live demos can fail. Always have a backup video of the demonstration in case the hardware misbehaves. Test the demo in the presentation room before the review.

Pointing: when presenting data or diagrams, physically point to the relevant area of the slide. This directs the audience's attention and prevents them from trying to parse the entire figure while you are talking about one specific element.

19.13 The Showcase: Speaking to a General Audience

The Design Showcase is your public exhibition. Visitors include: other students, faculty, industry representatives, prospective students and their parents, university administrators, and media. Your communication must be accessible to all of them.

The 60-second version: have a memorized, natural-sounding overview that any visitor can understand. Structure: "We built [what] for [whom] because [why]. It works by [how, one sentence]. The key result is [what you achieved]." Practice until it flows naturally without sounding rehearsed.

The 3-minute version: for visitors who are interested and want depth. Walk them through the poster: problem (top left), approach (center), results (right), and impact (bottom). Point to the relevant section as you speak. Show the prototype if possible.

The technical deep-dive: for PAs, faculty, or industry engineers who ask detailed questions. Be prepared to discuss: your analysis methodology, specific design decisions and trade-offs, test results with quantitative detail, and lessons learned. Not every team member needs to be prepared for deep-dive questions on every topic; but every subsystem should have at least one team member who can go deep.

Handling "dumb" questions: no question is dumb at the Showcase. A visitor who asks "What's an Arduino?" is giving you the opportunity to explain your project to a non-engineer; a skill you will use throughout your career. Answer respectfully and use it as a chance to practice accessible communication.

19.14 Panel Presentations and Judged Events

Some capstone projects involve panel presentations (competition judging, client review boards, or external advisory panels). Panel communication differs from standard presentations:

Time discipline is critical: panels see many presentations in one session. Going over time is disrespectful to the panel and to the teams waiting after you. Practice to fit within the allocated time with 1–2 minutes of margin.

Address the entire panel: make eye contact with different panel members throughout the presentation, not just the one who asked the last question. Panels evaluate how well you communicate, not just what you communicate.

Anticipate panel-specific questions: if the panel includes a safety officer, expect safety questions. If it includes a budget manager, expect cost questions. Research the panel composition (if available) and prepare accordingly.

Defer gracefully: if a panel member asks a question outside your expertise, say "That is outside my area; let me pass it to [teammate] who led that subsystem." This is professional, not weak.

19.15 Advanced Q&A Strategies

Q&A is where the most valuable learning happens at design reviews. Prepare for it:

Anticipate questions: before every review, brainstorm 10–15 questions that reviewers are likely to ask. For each, prepare a concise answer (30–60 seconds). Common question categories: "Why did you choose X over Y?" (design decisions), "What happens if X fails?" (risk/failure modes), "How do you know X works?" (verification evidence), "What is the accuracy/uncertainty of X?" (measurement confidence), and "What would you do differently?" (self-awareness).

The structured answer: use the PREP framework for answering technical questions: **P**oint (state your answer in one sentence), **R**eason (explain why), **E**vidence (cite data, analysis, or test

results), **P**oint (restate the answer). This structure prevents rambling and ensures the reviewer gets a complete answer.

When you do not know: "That is a great question. I do not have the data to answer it right now, but I will investigate and have an answer by [specific date]." Record the question on the feedback log. Follow through. A reviewer who asks a question and receives a follow-up email with the answer within a week remembers a professional, competent team.

When the question challenges your approach: do not become defensive. Listen completely. Acknowledge the concern: "You raise a valid point about thermal expansion." Then either (a) explain the analysis you have already done that addresses the concern, or (b) acknowledge that you have not considered it and commit to investigating: "We have not analyzed the thermal expansion effect. We will add it to our Sprint 12 analysis plan and report back at FDR." Both responses are professional; defensiveness is not.

When a teammate gives a wrong answer: do not contradict them publicly. After the review, discuss privately. If the incorrect information could affect a decision, send a correction in the post-review Next Steps Email: "Upon further review, we want to clarify that the motor current draw is 2.4 A, not 1.4 A as stated during Q&A. This does not change our power supply selection, which is rated for 3 A."

19.16 Remote and Hybrid Presentations

When presenting remotely (Zoom, Teams) or to a hybrid audience:

Technology check: test your setup 30 minutes before the presentation. Verify: camera, microphone (use a headset, not laptop speakers), screen sharing, and internet connection. Have a backup plan: if screenshare fails, email the PDF to the meeting organizer and they share their screen.

Engagement: look at the camera, not the screen (this creates the impression of eye contact). Speak slightly slower and more clearly than in person; audio compression and latency make fast speech harder to follow. Pause after key points to allow questions.

Slide design for remote: increase font sizes by 20% compared to in-person slides. Remote audiences view on smaller screens (laptops, tablets) and cannot lean forward to read fine print. Use high-contrast colors; low-contrast slides that look fine on a projector may be unreadable on a compressed video stream.

Managing Q&A remotely: designate one team member to monitor the chat for questions while another presents. At the end of each section, the chat monitor relays questions: "We have a question from Dr. Smith in the chat about our sensor placement." This ensures remote reviewers are heard.

19.17 Turning Nervousness into Energy

The physiological symptoms of nervousness (elevated heart rate, adrenaline, heightened alertness) are identical to the symptoms of excitement. The difference is interpretation. Research shows that reframing anxiety as excitement ("I am excited to present our work") improves performance more than trying to calm down ("I need to relax") [41].

Practical techniques:

- **Power posing**: stand in an expansive, confident posture for 2 minutes before presenting. Research on this is debated, but many presenters report feeling more confident.
- **Warm-up**: say your first 30 seconds aloud in the hallway before entering the room. The hardest part of any presentation is the first sentence; once you are moving, momentum carries you.
- **Memorize the opening**: know your first 2–3 sentences by heart. This eliminates the "blank mind" moment that novice presenters fear. Once past the opening, your preparation and knowledge of the material take over.
- **Focus on the message, not on yourself**: you are not performing; you are communicating engineering work that matters. The audience wants to understand your project, not evaluate your presentation style. Shift your mental focus from "how do I look?" to "did the audience understand the key point?"

19.18 The Team Presentation Rehearsal Protocol

For design review presentations with 4–5 speakers:

Rehearsal 1 (5 days before, 90 min): full run-through with timing. No interruptions. Record video if possible. After: note time per section, identify over-long sections, flag missing content, and list slides that need redesign.

Rehearsal 2 (3 days before, 60 min): revised slides, focused on: transitions between speakers (smooth, with introductions), Q&A practice (team members ask each other tough questions), and time management. If still over time, cut content from the least essential sections.

Rehearsal 3 (day before, 30 min, optional): final dry run in the actual presentation room if possible. Test technology (projector, demo equipment). Walk through the first 2 minutes of each section to verify timing.

The investment of 3 hours in rehearsal consistently produces: a 10–15% improvement in reviewer-assessed presentation quality, better time management, smoother transitions, and more confident Q&A responses. This is the highest-ROI activity in design review preparation.

19.19 Data Visualization for Design Review Slides

Engineering presentations live and die on the quality of data visualization:

The "so what" title: every data slide should have a title that states the conclusion, not just the topic. Bad: "Temperature Test Results." Good: "System Maintains $\pm 0.7\,^{\circ}$C Accuracy, Meeting the $\pm 1\,^{\circ}$C Requirement." The title tells the reviewer what to conclude from the data before they even look at the plot.

Comparison plots: when presenting test results against requirements, show both on the same plot. A horizontal dashed line at the requirement threshold, with data points above or below, instantly communicates pass/fail. Color-code: green for passing, red for failing.

Before/after comparisons: when presenting design improvements ("we reduced noise by implementing star grounding"), show the before and after data side-by-side on the same scale. The visual contrast communicates the improvement more effectively than a percentage.

Simplify for the audience: a detailed MATLAB plot with 6 data series, 3 axis labels, and a legend in 8-point font may be scientifically complete but is unreadable on a projected slide. Simplify: show only the most relevant data series, increase font sizes, use distinct colors (not shades of the same color), and remove gridlines that do not aid interpretation.

Animation for complex data: for time-series data, sensor comparison, or process flow, consider animating the data (reveal data points sequentially) rather than showing everything at once. This guides the audience's attention through the narrative: "Here is the setpoint change at $t = 30\,\mathrm{s}$... now watch how the system responds... settling time is 8 seconds."

Tables vs. plots: use tables for exact values that the audience needs to reference (BOM summary, requirement compliance matrix). Use plots for trends, comparisons, and patterns. Never put a 50-row table on a slide and say "the key number is in the third column, eighth row."

19.20 Common Presentation Mistakes and Fixes

These patterns appear at nearly every design review cycle:

The text wall: slides with 8+ bullet points of full sentences. Fix: convert text to visuals. A requirements slide should show a summary table, not paragraphs. A design slide should show a block diagram, not a description. The presenter provides the narrative; the slides provide the visual evidence.

The speed run: the team practices once, discovers they are 5 minutes over, and decides to "just talk faster" instead of cutting content. Fix: remove slides. Identify the 3–5 slides that contribute least to the narrative and eliminate them. A concise 20-minute presentation is always better than a rushed 25-minute presentation.

The disconnected Q&A: the team gives a polished presentation, then falls apart during Q&A because members cannot answer questions about sections they did not write. Fix: every team member reviews the entire presentation and can speak to the key points of every section, even if they did not author it.

The invisible transition: the team changes speakers without any verbal bridge, leaving the audience confused about who is speaking and what the new topic is. Fix: every transition includes: "Thank you, [previous speaker]. Now [next speaker] will walk us through [topic]." This takes 5 seconds and dramatically improves flow.

The apologetic opener: "Sorry this slide is hard to read" or "We didn't have time to finish this analysis." Fix: if the slide is hard to read, redesign it before the review. If the analysis is incomplete, present what you have and state the plan to complete it; without apologizing. Confidence in what you *have* done is more professional than apology for what you have not.

19.21 Practice Questions

1. Write a 60-second elevator pitch for your capstone project (or a hypothetical one).
2. A reviewer at PDR says "I don't think your sensor has adequate range for the operating conditions." You are not sure whether their concern is valid. How do you respond in the moment, and what follow-up do you take?

3. Your team's CDR presentation runs 8 minutes over the allotted time. Diagnose three likely causes and propose fixes for the FDR presentation.

Chapter 20

Resumes, Interviews, and Career Preparation

20.1 Your Capstone Project as a Career Asset

Capstone is the richest source of resume content you have as a graduating senior. Unlike coursework, capstone involves real clients, real budgets, real deadlines, interdisciplinary teamwork, and technical problem-solving under uncertainty. Employers recognize this. When they ask about "relevant project experience" in an interview, your capstone project is the answer.

20.2 Writing a Capstone-Informed Resume

Quantify your contributions wherever possible. Numbers are more compelling than adjectives:

- "Managed a $3,500 project budget with procurement of 47 components across 8 vendors, delivering under budget by 12%."
- "Designed and tested a PID temperature controller achieving $\pm 0.3\,^{\circ}\mathrm{C}$ accuracy against a $\pm 1.0\,^{\circ}\mathrm{C}$ requirement."
- "Led a team of 5 engineers through PDR, CDR, and FDR for an external industry client over two semesters."
- "Designed a 4-layer PCB with mixed-signal layout, ordered from JLCPCB, and completed board bring-up with zero rework."

List technical skills demonstrated: specific software (SolidWorks, MATLAB, KiCad, LabVIEW, Python, ANSYS), fabrication methods (CNC machining, 3D printing, PCB assembly, soldering), analysis techniques (FEA, CFD, thermal analysis, statistical testing), and professional skills (design reviews, client presentations, technical writing).

20.3 Career Day Preparation

Career Day falls during SDI (no class that day). This is a prime networking opportunity. Prepare: update your resume (emphasize capstone contributions even if the project is early-stage), practice your elevator pitch, dress professionally (business casual minimum), and bring 10–15 printed resumes.

Use your capstone project as a conversation starter. Employers are interested in your *process* (how you approached the problem, how you made decisions, how you handled setbacks) as much as the *outcome*. Saying "I am working on a capstone project designing a solar-powered irrigation

controller for a community garden, and I am responsible for the sensor integration and PID control" immediately communicates initiative, technical skill, and teamwork.

20.4 Behavioral Interviews and the STAR Method

Behavioral interview questions ask you to describe past experiences as evidence of your skills and character. Common capstone-relevant questions:

- "Tell me about a time you had a conflict with a teammate. How did you resolve it?"
- "Describe a project where something went significantly wrong. How did you handle it?"
- "Give an example of when you had to make a technical decision with incomplete information."
- "Tell me about a time you had to deliver bad news to a stakeholder."
- "Describe a situation where you had to learn a new skill quickly to complete a task."

Use the **STAR method**: **S**ituation (set the context (your project, the phase, the stakeholders), **T**ask (what was your specific responsibility), **A**ction (what did you specifically do) not what the team did), **R**esult (what was the outcome, and what did you learn). Capstone provides abundant material for STAR responses. Prepare 5–7 STAR stories before interview season and practice telling each in 90 seconds.

20.5 Technical Interviews

Some employers conduct technical interviews where they assess your engineering depth. Your capstone project is excellent material. Be prepared to explain: your system architecture (block diagram level), the key engineering decisions (why did you choose this sensor over that one?), the analysis you performed (show you understand the physics), and the test results (including what did not work and what you learned). Interviewers are assessing whether you think like an engineer, not whether your project was perfect.

20.6 Networking Through Capstone

Your PA, client, and TAs are professional contacts who may be valuable throughout your career. After the project ends, they may serve as references, provide job leads, or connect you with their professional networks. Maintain the relationship:

- Send a thank-you email after the project closes.
- Connect on LinkedIn with a personalized note (not the default "I'd like to add you to my network").
- Provide occasional career updates (new job, graduate school acceptance, published paper).
- **Always ask permission before listing someone as a professional reference.**

20.7 LinkedIn Profile Optimization

Your LinkedIn profile is often the first thing a recruiter sees. Capstone provides excellent material:

Headline: go beyond "Student at Colorado School of Mines." Use: "Mechanical Engineering Senior | Capstone: Solar-Powered Irrigation Control System | Seeking Full-Time Roles in Mechatronics."

Experience section: add your capstone project as a position. Title: "Capstone Design Engineer" or "Senior Design Team Lead." Organization: "Colorado School of Mines; EDNS 491/492." Description: 3–5 bullet points quantifying your contributions, just like your resume.

Projects section: add the capstone project with a detailed description, your specific role, the technical skills used, and the outcome. Link to any public documentation (poster PDF, team video, GitHub repository).

Skills section: add the specific technical skills demonstrated in capstone: SolidWorks, KiCad, MATLAB, Python, FEA, PCB Design, PID Control, Project Management, Technical Writing, etc.

Featured section: upload your Showcase poster, a photo of your prototype, or a short video clip of the demonstration. Visual content dramatically increases profile engagement.

20.8 Professional References

When listing professional references on a job application:

- **Ask permission first**: never list someone as a reference without their explicit consent. Contact them before you submit the application, tell them about the position, and ask if they are willing to serve as a reference.
- **Prepare your references**: send them your resume, a description of the position, and 2–3 key points you would like them to emphasize (technical skills, leadership, problem-solving ability). A reference who knows what to highlight is far more effective than one caught off guard.
- **Best capstone references**: your PA (who observed your engineering and professional growth over two semesters), your client (who can speak to your professionalism and technical contributions), and your TA (who can speak to domain expertise). A reference from a capstone PA carries significant weight because PAs observe students in a professional context, not just an academic one.
- **Thank your references**: after the application process, send a brief thank-you note regardless of the outcome. If you got the job, let them know.

20.9 Technical Interviews and Capstone

Many engineering employers conduct technical interviews that assess your depth of understanding. Your capstone project is excellent material for these conversations. Be prepared to discuss:

System architecture: draw a block diagram of your system on a whiteboard. Explain each subsystem, the interfaces between them, and why the architecture was organized this way. This demonstrates systems thinking.

Key engineering decisions: why did you choose this sensor over alternatives? What trade-offs did you make? What was the sensitivity analysis? This demonstrates engineering judgment, not just execution.

Analysis methodology: walk through a calculation from the Engineering Calculation Package. State the assumptions, show the approach, and explain the conclusion. The interviewer is assessing whether you understand the physics, not whether you memorized the formula.

Test results and failures: what did you test? What passed? What failed? How did you diagnose and resolve failures? Employers value engineers who can troubleshoot as much as engineers who can design. Discussing a failure you resolved intelligently is more impressive than claiming everything worked perfectly.

Lessons learned: what would you do differently? This question assesses self-awareness and growth mindset. A thoughtful answer ("I would have ordered the PCB two weeks earlier to leave margin for a board revision") demonstrates maturity.

20.10 Portfolio Development

Create a visual portfolio documenting your capstone design process. Include: the problem statement, concept sketches, CAD renderings, fabrication photos, test setups, data visualizations, and the final product. A well-curated portfolio communicates your engineering process more effectively than a resume alone. Host it on a personal website, a GitHub repository with a README, or a LinkedIn featured section.

20.11 Translating Capstone Experience to Different Industries

Your capstone project is in a specific domain, but the skills transfer broadly:

Project management: "Managed a 5-person interdisciplinary team through a 2-semester development cycle with 3 formal design reviews." This applies to any industry.

Client communication: "Presented technical recommendations to a non-technical external client and incorporated feedback into 3 design revisions." Consulting, product management, and sales engineering all value this.

Technical problem-solving: "Diagnosed a ground-loop-induced sensor error and implemented a star-grounding solution that reduced measurement noise by 85%." This demonstrates systematic debugging; applicable to any technical role.

Budget management: "Managed a $2,500 project budget with 15% contingency, completing the project 8% under budget." Finance awareness is valued in every engineering role.

Documentation: "Authored a 45-page Final Design Report including engineering analysis, test procedures, and an operations manual." Technical writing skills are consistently rated as the most important non-technical skill for new engineers.

20.12 The Career Day Strategy

Career Day is not a job fair where you passively hand out resumes. It is a networking event where you actively seek connections:

Before Career Day: research the companies attending. Identify 5–8 that match your interests. Review their job postings. Prepare specific questions for each ("I saw your posting for an embedded systems engineer; my capstone project involves ESP32-based control systems; could you tell me about the team structure for that role?").

At Career Day: lead with your elevator pitch. Make eye contact. Hand over your resume while describing your capstone project. Ask about the company's engineering challenges, not just their open positions. Get business cards or LinkedIn contact information.

After Career Day: within 48 hours, send a brief follow-up email or LinkedIn connection request to every contact you made. Reference something specific from your conversation: "It was great talking with you about your thermal management challenges. I would love to learn more about the team." Most students skip this step; the ones who follow up stand out.

20.13 The Job Search Timeline for Graduating Seniors

The job search does not begin at graduation; it begins in SDI:

SDI (Fall semester): update resume with capstone project description (even early-stage). Attend Career Day. Apply to summer internships if applicable. Begin LinkedIn profile optimization.

SDII (Spring semester): apply to full-time positions starting in January–February. Use the capstone elevator pitch in interviews. Request references from PA and client. Update resume with CDR and testing accomplishments. Attend spring career events.

Post-Showcase: update resume and LinkedIn with final project outcomes, quantified achievements, and Showcase photos. Send thank-you notes to references. Follow up with contacts from Career Day and networking events.

The compound advantage: students who start the job search in SDI (attending Career Day, networking, updating LinkedIn) have a significant advantage over students who start in Sprint 13 of SDII. Employers fill positions on rolling timelines; the best opportunities may close before graduation if you are not already in the pipeline.

20.14 Common Resume Mistakes for Engineering Seniors

1. **Listing duties instead of achievements**: "Responsible for PCB design" tells the employer nothing about your competence. "Designed a 4-layer mixed-signal PCB with integrated power management, fabricated through JLCPCB, and completed board bring-up with zero rework" tells a story of competence.
2. **No quantification**: "Improved system performance" vs. "Reduced measurement error from $\pm 3.2\,^{\circ}\mathrm{C}$ to $\pm 0.7\,^{\circ}\mathrm{C}$, a 78% improvement." Numbers are more memorable and more credible than adjectives.
3. **Generic skills list**: "Microsoft Office, teamwork, communication" adds nothing. List specific, differentiating skills: "SolidWorks (500+ hours), KiCad (PCB layout), Python (data analysis and automation), ANSYS (FEA/thermal), PID control system design and tuning."
4. **Outdated formatting**: one page for undergraduates (two pages only if you have extensive research or work experience). Clean, consistent formatting. No photos, no decorative borders, no colored backgrounds. Let the content speak.
5. **Spelling and grammar errors**: these immediately signal carelessness. Proofread. Then have someone else proofread. Then proofread again.

20.15 The FE Exam: Your First Step to Licensure

The Fundamentals of Engineering (FE) exam is a 5-hour-and-20-minute, computer-based exam administered by NCEES [20]. Passing it makes you an Engineer Intern (EI) or Engineer-in-Training (EIT); the first step toward PE licensure (Chapter 22).

When to take it: during your senior year or within 1–2 years of graduation. The material is freshest while you are still in school. Many states allow you to register before graduation.

What it covers: mathematics, probability/statistics, engineering economics, ethics, statics, dynamics, thermodynamics, fluid mechanics, heat transfer, materials, and your discipline-specific topics (ME, CE, EE, etc.). The exam is open-reference (NCEES provides a digital reference handbook), but speed matters; you have ∼3 minutes per question.

How to prepare: use the NCEES Practice Exam ($50, official questions), review the NCEES FE Reference Handbook (free download), and study weak areas 2–3 months before the exam. Campus review sessions may be available through your department.

Why it matters for your career: PE licensure is legally required for offering engineering services to the public, signing engineering documents, and holding certain government and consulting positions. Starting the licensure path now (by passing the FE) demonstrates professionalism and forward planning to employers.

20.16 Building Your Professional Identity Beyond Capstone

Capstone is a launching pad, not a destination. Continue developing your professional identity after graduation:

Join a professional society: ASME, IEEE, ASCE, AIChE, SAE, or your discipline's equivalent. Student membership is free or discounted. Professional membership provides: conferences, publications, networking, and professional development credits needed for PE maintenance.

Seek mentorship: identify 2–3 professionals (former PAs, clients, managers, senior engineers) who can advise your career decisions. A mentor who has navigated the path you are starting can save you years of trial and error.

Document your work: maintain a portfolio of your best engineering work (capstone project, research, personal projects). Update it with each significant project. A well-maintained portfolio differentiates you from candidates with identical resumes.

Continue learning: the half-life of engineering knowledge is commonly cited as ∼5 years. Technologies, standards, and best practices evolve continuously. Read technical publications, attend conferences, and pursue relevant certifications (PMP, Six Sigma, AWS, specific software certifications).

20.17 Capstone as Interview Material: Specific Scenarios

Prepare to discuss these specific capstone experiences in interviews:

Technical decision under uncertainty: "We had to select between two sensor technologies before completing all our analysis. I gathered the available data, presented a comparison matrix

to the team, and recommended Option B based on its superior noise performance and lower integration risk. We validated this choice during testing, and the sensor met requirements with margin."

Conflict resolution: "Two team members disagreed on the motor controller architecture. I facilitated a structured discussion using our decision matrix framework, ensuring both perspectives were evaluated against the requirements. We reached consensus on a hybrid approach that incorporated the strengths of both proposals."

Schedule recovery: "Our PCB had a layout error that required a second revision, adding three weeks to our timeline. I reorganized the sprint backlog to parallelize firmware development with the board re-fabrication, so we recovered two of the three lost weeks and still completed testing on schedule."

Client management: "Our client requested a feature mid-project that would have required redesigning the power subsystem. I prepared a scope impact analysis showing the 4-week schedule extension and $200 cost increase, and presented three options: add the feature and extend the schedule, defer it to future work, or implement a simplified version within the current timeline. The client chose the simplified version."

Technical failure: "Our first integration test failed because the motor's EMI was corrupting the sensor readings. I systematically diagnosed the issue using the signal walk technique; measuring at each stage of the signal chain until I localized the noise to the shared ground path. We implemented star grounding, which reduced the noise by 90% and brought sensor accuracy within specification."

Each of these follows the STAR framework (Chapter 20) and demonstrates a specific professional competency. Prepare 5–7 such stories before interview season.

20.18 Networking Beyond Capstone: The Long Game

Your professional network grows throughout your career. Capstone provides your initial nodes:

Your PA: a professor or industry professional who knows your engineering capabilities and work ethic from 8 months of close observation. This is one of the strongest references you will ever have. Maintain the relationship.

Your client: an industry professional who worked with you on a real project. Client references carry enormous weight with employers because they demonstrate that you can operate in a professional (not just academic) context.

Your teammates: your future professional network includes your classmates. The teammate who struggles through a tough CDR with you may be the hiring manager at your dream company in 10 years. Treat every professional relationship as long-term.

Guest speakers and TAs: capstone may include guest lectures from industry professionals. Introduce yourself, ask a thoughtful question, and follow up with a LinkedIn connection. These are low-effort, high-value networking opportunities.

Alumni network: Mines alumni are disproportionately supportive of fellow graduates. Leverage the Mines alumni network (LinkedIn, alumni events, department connections) when job searching. A cold outreach to a Mines alum is substantially more likely to get a response than a cold outreach to a stranger.

20.19 Cover Letters That Reference Capstone

When applying for positions where a cover letter is requested, capstone provides excellent material for the body paragraphs:

Opening paragraph: state the position you are applying for and how you learned about it. If a capstone connection (PA, client, TA, Career Day contact) referred you, mention them by name: "Dr. Smith, who served as my capstone project advisor, recommended I apply for this position."

Body paragraph 1 (technical alignment): connect your capstone technical skills to the job requirements. "During my capstone project, I designed and implemented a PID-based temperature control system using an ESP32 microcontroller, which directly aligns with your team's work in embedded control for industrial automation."

Body paragraph 2 (professional skills): highlight soft skills demonstrated in capstone. "As Scrum Master for a 5-person interdisciplinary team, I facilitated sprint planning, tracked deliverables, and managed communication with our external industry client across two semesters. This experience prepared me for the collaborative, client-facing work described in your posting."

Closing paragraph: express enthusiasm and request an interview. Reference a specific aspect of the company that genuinely interests you (not generic flattery).

20.20 Following Up After Applications and Interviews

After submitting an application: if you have a contact at the company (from Career Day, LinkedIn, or a PA introduction), send a brief note: "I recently applied for [position]. I wanted to let you know because of our conversation about [topic] at Career Day. I believe my capstone experience in [domain] aligns well with the role."

After an interview: send a thank-you email within 24 hours to every interviewer. Reference a specific topic from the conversation: "I enjoyed discussing your team's approach to thermal management; it reminded me of the challenges we solved in our capstone project's enclosure design." Keep it brief (3–5 sentences).

After a rejection: respond graciously. "Thank you for letting me know. I appreciated the opportunity to learn about [company] and I would welcome the chance to be considered for future openings." Professional graciousness is memorable; bitterness is also memorable, but in the wrong way.

After an offer: before accepting, discuss the offer with a trusted advisor (PA, mentor, parent, career counselor). Negotiate respectfully if appropriate (salary, start date, relocation assistance). Once accepted, notify all other companies where you have active applications.

20.21 Salary Negotiation for New Graduates

When you receive a job offer, negotiation is expected and appropriate:

Research market rates: use Glassdoor, Levels.fyi, BLS Occupational Outlook Handbook, and Mines Career Center salary data to understand the range for your role, location, and experience level. Know the 25th, 50th, and 75th percentile salaries.

Consider total compensation: salary is one component. Also evaluate: signing bonus, annual bonus structure, stock/equity (for tech companies), retirement matching (401k match), health insurance quality and cost, paid time off, remote work flexibility, professional development budget (conference attendance, certifications), and relocation assistance. A $75k offer with $5k signing bonus, 6% 401k match, and full remote flexibility may be more valuable than an $82k offer with no bonus, 3% match, and required 5-day in-office.

How to negotiate: express enthusiasm for the role first ("I'm very excited about this opportunity"). Then present your case with data: "Based on my research, the market rate for this role in [city] is $X–$Y. Given my capstone experience in [relevant domain] and my [specific skill], I would like to discuss a salary of $Z." Be specific, not vague. Be prepared for the employer to counter-offer.

What is negotiable for new graduates: salary (usually 5–10% above the initial offer is achievable), start date, signing bonus, relocation assistance, and professional development budget. What is usually not negotiable: vacation days (often set by company policy for all new hires), job title (set by the HR classification system), and work location (set by the team's needs).

When not to negotiate: if the offer is already at or above market rate and includes strong benefits, aggressive negotiation may signal unrealistic expectations. Read the situation: a startup offering $70k may have no room to move; a Fortune 500 offering $70k almost certainly does.

Figure 20.1: The path to Professional Engineer (PE) licensure. Each step builds on the previous one. The FE exam is best taken during or shortly after your senior year while course material is fresh.

The licensure pathway (Figure 20.1) [20] is straightforward but time-bounded. The FE exam covers material from your undergraduate courses; every semester you wait after graduation, the material fades. Students who pass the FE during their senior year or within 6 months of graduation have the highest pass rates. The 4-year experience requirement begins accumulating as soon as you start working under a licensed PE's supervision; your capstone PA may qualify if they hold a PE license.

20.22 Your First 90 Days on the Job

The transition from capstone to your first engineering position involves a mindset shift:

You are no longer graded: performance is evaluated through project outcomes, peer feedback, and manager observation over months and years, not through rubrics on individual assignments. Consistency matters more than occasional brilliance.

You do not know the codebase/system/process: every new engineer spends 3–6 months learning the organization's tools, processes, standards, and culture. Ask questions. Take notes. The colleagues who seem to know everything were in your position 2–5 years ago.

Capstone skills transfer directly: the sprint cadence, the stand-up meetings, the design reviews, the client communication, the documentation discipline; these are standard industry practices. You are better prepared than you think.

20.23 Practice Questions

1. Write three resume bullet points describing your capstone contributions (real or hypothetical). Each must include a quantified achievement.
2. Prepare a STAR response for: "Tell me about a time you had to make a difficult technical decision under time pressure." Use a capstone scenario.
3. An interviewer asks "What was the biggest failure in your capstone project?" Write a response that demonstrates engineering maturity rather than defensiveness.

Part V

Safety, Compliance, and Professional Practice

Chapter 21

Environmental Health and Safety (EHS)

21.1 Mines EHS Framework

Colorado School of Mines Environmental Health and Safety (EHS) provides policies, training, oversight, and resources for all activities conducted on campus and at field sites. EHS is not a bureaucratic obstacle; it is the institutional mechanism that protects you, your teammates, your clients, and the public from foreseeable hazards. Professional engineers are personally responsible for the safety of their designs and their work practices. Developing strong safety habits now will serve you for the entirety of your career.

All capstone students must complete the required **EHS Safety Training module on Canvas** at the start of **each semester** (both SDI and SDII). This is not a one-time requirement; the SDII training includes content specific to the build and test phase. Completion is tracked by the CF and PA.

21.2 Hazard Assessment and Safety Plan

In SDII (Week 2), each team must complete a project-specific **Hazard Assessment and Safety Plan**. This is a formal document (not a checkbox exercise) that demonstrates your team has identified, evaluated, and planned controls for every significant hazard associated with your project's fabrication, testing, and demonstration activities.

The safety plan must include:

1. **Hazard identification**: systematic inventory of all hazards. Walk through every step of your build and test plan and ask: what could go wrong at each step? What energy sources are present (electrical, mechanical, thermal, chemical, pneumatic, potential)?
2. **Risk assessment**: for each hazard, evaluate the **likelihood** (how often will the team be exposed to this hazard?) and **severity** (what is the worst credible outcome?). Use the 5×5 risk matrix (Figure 21.1): likelihood (1=rare to 5=frequent) vs. severity (1=negligible to 5=catastrophic). Risks in the upper-right quadrant (high likelihood, high severity) require active engineering controls.
3. **Engineering controls**: design-level safety features. Examples: fuses on every power rail, current-limited bench supplies, thermal cutoffs on heaters, mechanical guards on rotating equipment, interlocks that disconnect power when enclosures are opened, ventilation for soldering and painting. Engineering controls are preferred because they do not depend on human behavior.

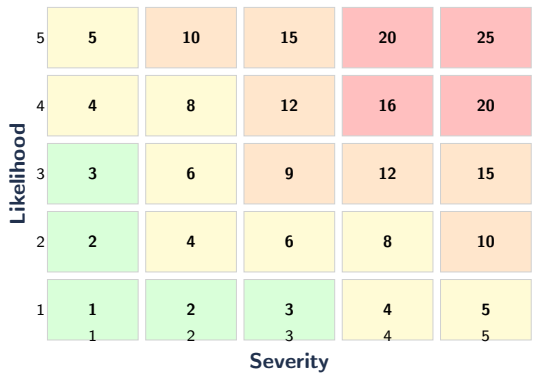

Figure 21.1: The 5×5 risk matrix. Risk score = Likelihood × Severity. Green (1–3): acceptable. Yellow (4–8): monitor. Orange (9–15): mitigate. Red (16–25): mitigate immediately or redesign.

4. **Administrative controls**: procedures, training, signage, and work practices. Examples: the buddy system for shop work (never operate power tools alone), mandatory safety briefing before each build session, posted hazard warnings at the workstation, lockout/tagout procedures for equipment maintenance. Administrative controls supplement engineering controls.
5. **Required PPE**: personal protective equipment matched to each hazard. Safety glasses (all shop work), hearing protection (>85 dBA [43]), closed-toe shoes (all shop spaces), nitrile gloves (chemical handling, resin printing; **not** for rotating machinery, which can catch gloves and pull your hand in), face shield (grinding, cutting), welding helmet and gloves (welding), dust mask (sanding, cutting composites).
6. **Emergency procedures**: specific to your project's hazards. Lithium battery thermal runaway: evacuate, place in metal container with sand, call 911. Chemical spill: evacuate, consult SDS, contact EHS. Electrical shock: de-energize before touching the victim, call 911. Injury: first aid, incident report, EHS notification.

The PA must **review and approve** the safety plan before fabrication begins. The plan is a living document; update it as the project evolves and new hazards are identified. If your design changes significantly after CDR (new materials, new processes, higher voltages, larger forces), update the safety plan.

21.3 Electrical Safety

Electrical shock severity depends on current through the body, which is determined by voltage and body resistance (approximately 100 kΩ for dry, intact skin but as low as 1 kΩ hand-to-hand with wet or broken skin) [42]. **No voltage should be treated as unconditionally safe.** A source as low as 24 VDC can deliver dangerous current under the right conditions (wet skin, cuts, conductive jewelry, medical implants). IEC 61010 [21] designates voltages below 30 VDC / 25 VAC RMS as "extra-low voltage," but this classification assumes dry-skin contact with intact insulation, not that the source is harmless. **The safest practice is always to de-energize before working.** Above 48 V, including all AC mains: **de-energize and lock out before touching any conductor**.

Design-level controls: fuse every power rail (fuse rating = 150% of maximum expected current), use current-limited bench supplies during development, include an emergency shutoff switch accessible without reaching into the enclosure, use connectors that cannot be reversed (keyed headers, Ander-

son Powerpole), and **never design your own AC-DC converter**. Always use a UL/CE-listed wall adapter.

Lockout/tagout (LOTO) [22], [23]: when working on energized equipment, disconnect the power source and apply a physical lock to prevent re-energization. In capstone, this means disconnecting the battery or unplugging the power supply before modifying wiring, not just turning off the switch.

21.4 Mechanical Safety

Rotating equipment (lathes, mills, drill presses, grinders) presents entanglement, pinch, and projectile hazards. Before operating any rotating machinery: tie back long hair, remove dangling jewelry, roll up loose sleeves, and remove gloves (counterintuitively, gloves increase the danger by providing a surface for the machine to grab). Wear safety glasses; a chip from a lathe can cause permanent eye damage.

Machine guards: never remove or bypass a machine guard. If a guard prevents you from performing an operation, the operation requires a different approach, not a removed guard.

Clamping: always clamp workpieces securely. A loose workpiece on a drill press becomes a spinning projectile. Use the correct chuck key and remove it before starting the machine.

21.5 Chemical Safety

Many capstone projects involve adhesives, solvents, paints, battery electrolyte, flux, and resin. Before using any chemical: read the **Safety Data Sheet (SDS)**. The SDS tells you: hazards, PPE requirements, first aid measures, storage requirements, and disposal procedures. SDS sheets are available online for every commercial chemical product.

Spray painting: must use the designated paint booth in the InnoHub. Complete the paint booth pre-quiz on Canvas before your first use. Spray paint contains flammable propellants and toxic solvents; using it in open lab areas creates fire and inhalation hazards for everyone in the building.

Resin handling (SLA/DLP 3D printing): uncured resin is a skin sensitizer (repeated exposure can cause permanent allergic reactions). Wear nitrile gloves when handling uncured prints. Do not pour resin down the drain; it is toxic to aquatic life. Use the designated wash and cure stations.

21.6 Lithium Battery Safety

Lithium-ion and lithium-polymer cells are energy-dense and present fire and explosion risks if mishandled. The energy stored in a fully charged 3S LiPo battery pack is sufficient to start a fire that can destroy a room.

Never: short-circuit, overcharge, over-discharge, puncture, crush, or overheat cells. **Always**: use a Battery Management System (BMS) for charging and discharge protection, charge in a fireproof LiPo safety bag, never charge unattended, use a charger designed for the specific chemistry (LiPo/Li-ion), and inspect cells before each use (swelling, hissing, or heat are warning signs of imminent failure).

If a cell swells, hisses, smokes, or begins to heat uncontrollably: **clear the area immediately**. If safe to do so, move the battery outdoors to a metal container (e.g., ammo can or metal bucket with sand) using insulated tools. Call 911 if fire occurs. For lithium-ion or LiPo battery fires: use water in large quantities to cool the cells and prevent thermal runaway propagation. Do **not** use a Class D extinguisher (those are designed for combustible metals such as magnesium or sodium, not lithium-ion cells). For small fires, an ABC dry chemical extinguisher may also be used.

21.7 Compressed Gas and Pneumatic Safety

Compressed gas cylinders contain enormous stored energy. A cylinder that falls and breaks its valve becomes an unguided missile. **Always**: chain cylinders upright to a wall, cart, or bench; use the correct regulator for the gas type; close the valve when not in use; and transport with the protective cap installed.

Pneumatic systems store energy in compressed air lines. Before disconnecting any fitting, **relieve all pressure** by closing the supply valve and exhausting the downstream line. Never direct compressed air at a person—even shop air at 90 psi can cause serious injury.

21.8 Noise Exposure

Power tools, CNC machines, and compressed air tools can produce noise levels above 85 dBA [43], which causes permanent hearing damage with prolonged exposure. Wear hearing protection (earplugs or earmuffs) when working near loud equipment. If you must shout to be heard at arm's length, the noise exceeds 85 dBA.

21.9 Emergency Procedures

Before beginning any work in any facility, identify the location of: the nearest fire extinguisher (CO_2 or dry chemical for electrical fires (**never** use water on electrical equipment), the first aid kit, the nearest eyewash station (for chemical splashes; flush for 15 minutes minimum), the nearest emergency exit, and the emergency phone number (911 from any campus phone).

Incident reporting: report all incidents, injuries, and near-misses to your PA, the InnoHub Operations Manager, and Mines EHS. Near-miss reporting is not a sign of weakness; it is a sign of professional safety culture. A near-miss report today prevents a real injury tomorrow.

21.10 InnoHub-Specific Safety

Labriola TAs have **authority in all InnoHub shops and workspaces**. If a TA asks you to stop an activity, comply immediately. TAs can require additional PPE, modify your approach, or deny access to equipment if safety concerns exist. If you disagree with a TA's decision, discuss it after complying; do not argue while standing next to running machinery.

The InnoHub shops are available **only during posted hours with TA supervision**. No unsupervised shop access is granted, regardless of training level. This policy exists because even experienced machinists have accidents, and a second person present can call for help.

21.11 Field Work and Site Visit Safety

Some capstone projects require work at client sites, field locations, or outdoor environments. Additional safety considerations:

Site assessment: before visiting a client site or field location, assess the hazards specific to that environment. Construction sites have fall hazards and heavy equipment. Wastewater treatment plants have confined space and chemical hazards. Agricultural sites have machinery and biological hazards. Ask the client about required PPE and safety orientation before arriving.

Travel safety: follow the guidelines in Chapter 16 for vehicle travel. For remote field sites, ensure: (1) someone knows your location and expected return time, (2) you have cell phone coverage or a satellite communicator, (3) you have a first aid kit, (4) weather conditions are checked before departure.

Environmental hazards: Colorado-specific considerations include: altitude sickness (stay hydrated, watch for headache and nausea at elevations >8,000 ft), lightning (common in afternoon thunderstorms May–September; seek shelter if thunder is audible), hypothermia (mountain temperatures can drop rapidly), and wildlife (mountain lions, bears, rattlesnakes (know the appropriate response for each).

Buddy system: never work alone at a field site or in the InnoHub shops. Always have at least one other person present who can call for help if something goes wrong. This is not a suggestion; it is a safety requirement.

21.12 PPE Selection Guide

Personal Protective Equipment (PPE) is the last line of defense—use it in conjunction with engineering and administrative controls, not as a substitute. Select PPE based on the specific hazard:

Hazard	Required PPE	Notes
Flying chips/sparks	Safety glasses (ANSI Z87.1)	All shop work, no exceptions
Grinding/cutting	Face shield + safety glasses	Shield alone is not sufficient
Chemical splash	Splash goggles + nitrile gloves	Check glove compatibility with chemical
Resin (SLA/DLP)	Nitrile gloves + safety glasses	Resin is a skin sensitizer
Soldering	Safety glasses	Flux fumes: ventilate workspace
Welding	Welding helmet + jacket + gloves	Shade ≥10 for arc welding
Loud equipment	Earplugs or earmuffs (≥25 dB NRR)	If shouting to be heard at arm's length
Heavy lifting	Steel-toe boots	Required for loads >50 lbs

PPE that is NOT appropriate: do **not** wear gloves when operating rotating machinery (lathes, mills, drill presses, grinders). The machine can catch the glove and pull your hand into the rotating component. Do **not** wear loose clothing, dangling jewelry, lanyards, or untied long hair near rotating equipment.

21.13 Incident Reporting and Near-Miss Culture

All incidents; injuries, property damage, chemical spills, and near-misses; must be reported to: (1) your PA, (2) the InnoHub Operations Manager, and (3) Mines EHS. The purpose of reporting

is not to assign blame; it is to prevent recurrence.

Near-miss reporting is particularly important. A near-miss is an event that could have caused injury but did not; a dropped tool that barely missed a foot, a circuit that sparked and tripped the breaker before causing a fire, a chemical spill that was contained before anyone was exposed. Near-misses are free lessons: they reveal hazards without anyone getting hurt. Report them so the hazard can be controlled before a real injury occurs.

The incident report should include: date, time, and location; who was involved; what happened (factual description); what injury or damage occurred (if any); what immediate action was taken; and what corrective action is recommended to prevent recurrence. Your PA and EHS will follow up on corrective actions.

A team that reports near-misses and minor incidents is not a "unsafe" team; it is a team with a strong safety culture. The teams that never report anything are the ones that should concern us.

21.14 Colorado-Specific Environmental Hazards

For projects involving outdoor work in Colorado:

Altitude: Golden is at 5,675 ft. Field sites in the mountains may be at 8,000–14,000 ft. Altitude reduces oxygen availability, causing headache, nausea, fatigue, and impaired judgment. Hydrate aggressively, ascend gradually, and do not perform strenuous or precision work until acclimated (24–48 hours at elevation).

Lightning: Colorado consistently ranks among the states with the highest lightning fatality rates per capita (NOAA Storm Data). Afternoon thunderstorms are common May–September. If you hear thunder, seek shelter immediately. The 30/30 rule: if the time between lightning flash and thunder is <30 seconds, seek shelter and stay sheltered for 30 minutes after the last thunder.

Temperature extremes: Colorado temperatures can range from $-20\,^{\circ}\mathrm{C}$ in winter to $+38\,^{\circ}\mathrm{C}$ in summer, sometimes with $20\,^{\circ}\mathrm{C}$ swings in a single day. Dress in layers. Carry water. Monitor team members for heat exhaustion and hypothermia symptoms.

Wildlife: mountain lions (back away slowly, do not run, make yourself appear large), black bears (store food properly, make noise while hiking, do not approach), and rattlesnakes (watch where you step, do not reach into rock crevices). Encounters are rare on campus but possible at mountain field sites.

21.15 Lab-Specific Safety Protocols

Different project types have different dominant hazards. Identify your project's hazard profile and prioritize accordingly:

Electronics projects: primary hazards are electrical shock (especially at voltages $>48\,\mathrm{V}$), burns from soldering irons and hot components, chemical exposure from flux and solvents, and eye strain from fine-pitch work. Mitigation: current-limited bench supplies, proper ventilation for soldering, safety glasses, and adequate lighting.

Mechanical fabrication projects: primary hazards are entanglement with rotating machinery, flying debris from cutting and grinding, noise exposure, and cuts from sharp edges and burrs.

Mitigation: machine guards, PPE (glasses, hearing protection, closed-toe shoes), clamping all workpieces, and deburring all edges.

Fluid system projects: primary hazards are pressure release (burst fittings), chemical exposure (process fluids, hydraulic oil), slip hazards (spilled fluids), and burns (hot fluids, steam). Mitigation: pressure testing before operation, secondary containment, spill response materials, and labeling all lines with contents and pressure.

Field projects: primary hazards are environmental (altitude, weather, wildlife), transportation (vehicle accidents), and site-specific (construction hazards, confined spaces, chemical exposure). Mitigation: site assessment before every visit, buddy system, weather monitoring, and site-specific PPE.

Software/modeling projects: primary hazards are ergonomic (repetitive strain, eye strain from extended screen time) and, if involving hardware interfaces, electrical hazards. Mitigation: proper workstation ergonomics, regular breaks (20-20-20 rule: every 20 minutes, look at something 20 feet away for 20 seconds), and standard electrical safety practices for any hardware interaction.

21.16 Safety Documentation for FDR

Your FDR report should include a safety section documenting:

- The Hazard Assessment and Safety Plan (updated from the Sprint 7 version to reflect any new hazards discovered during the build and test phase).
- All safety-related design features (fuses, current limiters, emergency stops, guards, interlocks, ventilation, labeling).
- Any incidents or near-misses that occurred during the project, with the corrective actions taken.
- Safety considerations for the end user documented in the O&M manual: operating hazards, required PPE, emergency procedures, and safe decommissioning instructions.
- Compliance with applicable safety standards (IEC 61010 [21], OSHA requirements [22], NFPA 70E [23], competition rules, site-specific requirements).

This documentation demonstrates that safety was integrated into the design process, not treated as an afterthought. Employers and PE boards view safety consciousness as a core engineering competency.

21.17 Lithium Battery Safety: A Detailed Protocol

Lithium-ion and lithium-polymer cells are common in capstone projects (portable devices, vehicles, drones) and present significant fire and explosion risks:

Charging safety: always use a charger designed for the specific battery chemistry (LiPo, Li-ion, LiFePO4). Never charge cells unattended. Charge in a fireproof LiPo safety bag or on a non-flammable surface (metal tray, ceramic tile). Set the charger to the correct cell count (1S, 2S, 3S, etc.); overcharging a 3S pack with a 4S charger setting will cause thermal runaway.

Storage safety: store batteries at 50–60% charge (storage voltage: 3.7–3.85 V per cell). Do not store fully charged or fully depleted cells for extended periods; both accelerate degradation. Store in a fireproof container away from flammable materials.

Inspection before every use: check for: swelling (puffy cells are damaged and must be retired), physical damage (dents, tears in the pouch), unusual odor (sweet chemical smell indicates electrolyte leakage), and hot spots (feel for localized heat). Any of these signs requires immediate retirement of the cell.

Battery Management System (BMS): every multi-cell battery pack must include a BMS that provides: overcharge protection (disconnects at $\sim$4.2 V/cell), over-discharge protection (disconnects at $\sim$3.0 V/cell), overcurrent protection (disconnects at the maximum safe discharge rate), and cell balancing (equalizes voltage across cells in series). Do not operate a multi-cell pack without a BMS (an unbalanced pack will overcharge the weakest cell, leading to thermal runaway).

Emergency response: if a cell swells, hisses, smokes, or begins to heat uncontrollably: (1) clear the area immediately, (2) if safe to do so, move the battery to a metal container with sand using insulated tools, (3) call 911 if fire occurs, (4) for lithium-ion or LiPo fires, use water in large quantities to cool the cells and prevent thermal runaway propagation; do **not** use a Class D extinguisher (those are for combustible metals, not Li-ion cells); for small fires, an ABC dry chemical extinguisher may also be used, (5) ventilate the area (lithium battery fires produce toxic hydrogen fluoride gas).

Disposal: damaged or end-of-life lithium cells must not be placed in regular trash. Use the campus battery recycling program or a certified electronics recycler. Tape the terminals of used cells before disposal to prevent short circuits in the recycling bin.

21.18 First Aid Essentials for the Lab

Every team working in the InnoHub should know the location of the nearest first aid kit, eyewash station, fire extinguisher, and AED (Automated External Defibrillator). Key first aid actions:

Minor cuts and abrasions: clean with soap and water, apply antibiotic ointment, and cover with a bandage. Report to the TA. Most shop cuts are minor but they must still be documented.

Burns: for thermal burns (soldering iron, hot metal, heater contact): cool under running cold water for 10–20 minutes. Do not apply ice directly (causes frostbite on damaged tissue). Do not apply butter, toothpaste, or other home remedies. For chemical burns: flush with water for 15+ minutes while consulting the SDS for specific neutralization instructions. Seek medical attention for burns larger than a quarter or on the face, hands, or joints.

Chemical splash to eyes: use the eyewash station immediately. Flush both eyes for at least 15 minutes, holding the eyelids open. Remove contact lenses during flushing if possible. Seek medical attention even if the irritation subsides; some chemical damage is delayed.

Electrical shock: do not touch the victim if they are still in contact with the electrical source. De-energize the circuit first (disconnect the power supply, pull the plug, trip the breaker). Once the victim is separated from the source: check responsiveness and breathing. Call 911 if the victim is unresponsive or was exposed to >48 V. Even if the victim feels fine, recommend medical evaluation; cardiac effects from electrical shock can be delayed.

Severe bleeding: apply direct pressure with a clean cloth. Elevate the injured limb if possible. Call 911 for bleeding that does not stop with direct pressure or for any arterial bleeding (bright red, pulsing). Maintain pressure until emergency responders arrive.

21.19 Emergency Contact Numbers

Post these numbers at your workstation:

Emergency	911 (from any phone)
Mines Police (non-emergency)	303-273-3333
Mines EHS	303-273-3316
InnoHub Operations	Julia Roos, jroos@mines.edu
Poison Control	1-800-222-1222
Capstone Program Director	Dr. Kristy Csavina, kcsavina@mines.edu

In any emergency: (1) ensure your own safety first, (2) call 911 if there is any injury or immediate danger, (3) administer first aid if trained and safe to do so, (4) notify your PA and the InnoHub Operations Manager within 24 hours, and (5) complete an incident report.

21.20 The Safety Mindset for Professional Engineers

Safety is not a checklist to complete; it is a mindset to internalize:

Anticipation: before starting any task, mentally walk through the steps and identify what could go wrong. This mental rehearsal takes 30 seconds and prevents the most common injuries.

Questioning assumptions: "It was fine last time" is not a safety justification. Conditions change. Check every time.

Stopping work: if something does not feel right (an unusual sound, a smell, a vibration) stop immediately and investigate. You have the authority and obligation to stop work when safety is in question.

Reporting: near-misses are learning opportunities. Every unreported near-miss is a missed opportunity to prevent a future injury. Teams with strong safety cultures report more near-misses because reporting is treated as a positive contribution, not a sign of failure.

These principles apply equally to capstone and to your entire engineering career.

Student Guide Cross-Reference

For the sprint-level Hazard Assessment deadline and safety plan approval requirements, see the **Student Guide**, SDII Sprint 7 section.

21.21 Practice Questions

1. Your team's project involves a 24 V, 10 A motor controller, a 3S LiPo battery, and CNC-machined aluminum parts. Create a hazard assessment table with at least 6 hazards, their risk ratings, and the controls you would implement for each.
2. A teammate proposes soldering a wire directly to a LiPo cell terminal instead of using a connector. Explain why this is dangerous and propose an alternative.
3. Your safety plan was approved at CDR. Since then, you have added a pneumatic cylinder (80 psi) to the design. What must you update, and who must approve the changes?

Chapter 22

Professional Ethics, Licensure, and Course Policies

22.1 Professional Engineering Licensure

CLO 6 requires you to define the process for attaining professional engineering licensure and explain why it is important. The SDI Professional Development modules on licensure and liability (Week 2) cover this in detail; this section provides the reference framework.

22.1.1 The Path to PE

1. **Accredited degree**: earn an ABET-accredited engineering degree [25]. You are completing this requirement now.
2. **FE exam**: pass the Fundamentals of Engineering exam, administered by NCEES [20]. The FE is typically taken during your senior year or shortly after graduation. It is a 5-hour-and-20-minute, 110-question computer-based exam covering mathematics, sciences, and engineering fundamentals. Passing the FE makes you an **Engineer Intern (EI)** or **Engineer-in-Training (EIT)**.
3. **Supervised experience**: accumulate 4 years (in most states) of progressive engineering experience under the supervision of a licensed PE. The experience must be verified by your supervisors.
4. **PE exam**: pass the Principles and Practice of Engineering exam in your specific discipline (Mechanical, Civil, Electrical, Environmental, etc.). Passing makes you a **Professional Engineer (PE)** licensed in that state.

22.1.2 Why Licensure Matters

A PE license is **legally required** to: offer engineering services directly to the public, sign and seal engineering documents (drawings, plans, specifications), and hold certain positions in government and consulting. Beyond legal requirements, licensure signals competence, ethical commitment, and personal accountability to the profession and the public. Many employers in consulting, construction, and government require or prefer PE licensure. The licensure process ensures that engineers offering services to the public have demonstrated a minimum level of competency and ethical commitment.

22.1.3 Involuntary vs. Voluntary Liability

The SDI Week 2 lecture introduces these legal concepts:

Involuntary liability (tort law): you are liable for harm caused by your engineering work even without a contract. If a product you designed injures someone, the injured party can sue for negligence. The standard is: would a reasonably competent engineer have foreseen and prevented this hazard?

Voluntary liability (contract law): you accept specific obligations when you sign a contract to provide engineering services. Failure to meet contractual obligations exposes you to breach-of-contract claims.

Understanding liability is not about being scared; it is about understanding why professional standards, thorough documentation, and safety margins exist. Every engineering decision you make in capstone (from material selection to safety factor to test procedure) reflects the same judgment that licensed engineers exercise daily.

22.2 NCEES Rules of Professional Conduct

The NCEES Model Rules [20] define the ethical obligations of licensed engineers. These rules apply to you as aspiring professionals, even before licensure. The key principles are summarized below; this is not the complete text of the Model Rules. For the full, authoritative document, consult the NCEES *Model Rules of Professional Conduct* at https://ncees.org/licensure/laws-and-rules/.

22.2.1 Section A: Obligations to the Public

Your first and highest obligation is to the **health, safety, and welfare of the public** [19], [20]. This is not a platitude; it is a legally enforceable professional duty. If your professional judgment is overruled and public safety is endangered, you have an obligation to report the situation to the appropriate authorities. You shall include all relevant information in professional reports and statements and shall not knowingly permit the publication of misleading information.

22.2.2 Section B: Obligations to Employers and Clients

Undertake only assignments for which you are **qualified by education or experience**. If a project requires skills you do not have, inform your client or employer. Do not reveal confidential information without the client's consent (NDA compliance; see Chapter 17). Disclose all **conflicts of interest**; situations where your personal interests could bias your professional judgment.

22.2.3 Section C: Obligations to Other Licensees

Do not **misrepresent your qualifications**. Do not injure the reputation of other licensees. Report known violations of the rules of professional conduct and material errors that may impact public safety.

22.3 Engineering Ethics in Practice

Ethics in engineering is not abstract philosophy. It is the daily practice of making decisions that affect real people, often under time pressure, incomplete information, and competing demands from stakeholders with different priorities.

22.3.1 The Ethics Assignment

The Ethics Assignment (SDI, Sprint 4) is a **1000–1500 word argumentative essay**. You must take a controversial stance on an ethical issue directly connected to your project's hardware, software, or system, defend it with evidence, and address opposing viewpoints. See Chapter 5 for essay structure and Chapter 18 for writing guidance.

The assignment requires you to move beyond generic ethical principles ("safety is important") to specific, nuanced dilemmas found in your actual project. Because AI tools can easily generate generic ethical statements (e.g., "Safety is important"), a top-tier essay must demonstrate **specific, nuanced critical thinking** that connects directly to the unique constraints of your Senior Design project.

22.3.2 Case Studies

Challenger (1986): Morton Thiokol engineers warned that O-ring seals would fail at low temperatures. Management overruled the recommendation. Seven crew members died (Presidential Commission on the Space Shuttle Challenger Accident, 1986). Lesson: the obligation to public safety overrides schedule pressure and management hierarchy.

Flint Water Crisis (2014–): cost-cutting decisions to change water sources without adequate corrosion control exposed a predominantly low-income, majority-Black community to lead contamination. Engineers who raised concerns were overruled or ignored (Michigan Civil Rights Commission, 2017). Lesson: equity and environmental justice are engineering responsibilities, not just political considerations.

Boeing 737 MAX (2018–2019): design decisions to minimize pilot training requirements for a new flight control system (MCAS) contributed to two crashes and 346 deaths (Joint Authorities Technical Review, 2019). Lesson: when safety-critical systems are designed to minimize cost rather than maximize safety, the consequences can be catastrophic.

Hyatt Regency Walkway Collapse (1981): a structural connection detail was changed during construction without adequate engineering review. 114 people died (NBS, 1982). Lesson: every change to a safety-critical design must be reviewed by a qualified engineer, no matter how minor it appears.

These case studies share a common thread: in each case, the technical failure was preceded by a *process* failure; suppressed dissent, inadequate review, conflicted incentives, or organizational pressure to prioritize cost and schedule over safety. The ethical engineer's job is to recognize and resist these process failures.

22.4 Generative AI Policy

Students must represent all work submitted for evaluation as their own intellectual product. The course AI policy:

Permitted uses: studying, brainstorming, proofreading, exploring AI capabilities, generating initial research directions, checking grammar and clarity.

Prohibited uses: submitting AI-generated text, code, images, or solutions as your own work without explicit instructor permission and proper attribution. "Hallucinated" AI citations (refer-

ences that do not actually exist) constitute an **academic integrity violation** and will be treated as such.

The purpose of this policy is not to prevent you from using AI tools; it is to ensure that the work you submit reflects *your* engineering judgment, *your* analysis, and *your* understanding. An AI can generate a requirements document; but only *you* can verify that those requirements accurately capture your client's needs and your project's constraints.

22.5 Attendance Policy

Attendance at lectures, recitation, and all team meetings is mandatory. Capstone Design is a synchronous course by design; your PA, client, and teammates depend on your presence.

Unexcused Absences	Consequence
1–3	No grade deduction, but grade of 0 for any missed in-class work
4	Final average lowered by 2 percentage points; formal PIP may be initiated
5–6	Final grade lowered by 5 percentage points
7 or more	Final grade of 0 assigned; withdrawal is recommended

The Scrum Master is encouraged to track attendance at all team meetings. Absences must be documented with dates, times, and context to support any PIP action (Chapter 13). It is not enough to say "student X missed several meetings"; you need dates and the record from the Communication Lead's attendance log.

22.6 Late Work and Grading

Late individual assignments: **10% penalty per 24 hours** late, at PA discretion. No individual assignments accepted after 5 calendar days. Team assignments: discuss extensions with your PA before the deadline; retroactive extensions are not granted.

Grades returned by PAs are preliminary; the CF may apply correction factors for consistency across sections. Grade scale: A (93.34–100%), A− (90–93.33%), B+ (86.67–89.99%), B (83.34–86.66%), B− (80–83.33%), C+ (76.67–79.99%), C (73.34–76.66%), C− (70–73.33%), D (60–69.99%), F (<60%).

Functional prototype/solution required: teams not accomplishing their agreed-upon prototypes or solutions will not receive an A, regardless of other assignment scores. This reflects the reality that engineering is ultimately about producing results, not just following a process.

22.7 Course Concerns Chain of Command

If you have concerns about grading, team dynamics, PA interactions, or course policies, follow this escalation path:

1. **Project Advisor (PA)**: your first point of contact for all project-related concerns.
2. **Course Lead Departmental Faculty**: if the concern cannot be resolved with your PA.
3. **Capstone Program Director**: Dr. Kristy Csavina (kcsavina@mines.edu). For program-level concerns.
4. **Dean**: for concerns that cannot be resolved at the program level.

Most concerns are resolved at Step 1. If you skip steps, you will be directed back to the appropriate level.

22.8 Engineering Judgment and Professional Responsibility

Engineering judgment is the ability to make sound decisions when the textbook does not provide a clear answer. It develops through experience, mentorship, and reflection; and capstone is where you begin developing it.

When codes and standards do not cover your situation: engineering standards cannot anticipate every design scenario. When your specific configuration is not addressed by the applicable standard, you must apply the standard's *intent* (protect public safety, ensure reliability, maintain performance) using your engineering knowledge and conservative assumptions. Document your reasoning: "No standard directly addresses this configuration. We applied the intent of IEC 61010 Section 6.3 by [specific approach], with a safety factor of 2.0 to account for the uncertainty introduced by the non-standard application."

When stakeholder interests conflict: the client wants the cheapest solution; the end user needs the safest solution; the schedule demands the fastest solution. Engineering judgment means recognizing these conflicts explicitly, evaluating the trade-offs with data, and recommending a course of action that satisfies the most important constraint (safety) while honestly communicating the trade-offs on the others.

When you do not know enough: the ethical response to uncertainty is not to guess confidently; it is to acknowledge the limitation and take appropriate action; research, consult an expert, add safety margin, or design a test to resolve the uncertainty. "I don't know, but I will find out" is always preferable to "I think it's fine" without evidence.

22.9 Generative AI: Detailed Guidance for Capstone

AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) are powerful engineering aids when used appropriately. The course policy is not anti-AI; it is pro-integrity.

Appropriate uses that make you more effective:

- **Research acceleration**: "What are the common failure modes of MOSFET H-bridges?" AI provides a starting point; verify with datasheets and application notes.
- **Brainstorming**: "Generate 10 concepts for a soil moisture sensor mounting mechanism." AI augments your ideation; evaluate each concept using your engineering judgment.
- **Code generation**: "Write an ESP32 function to read an I2C temperature sensor." AI produces draft code; review, test, and modify to match your specific hardware configuration.
- **Proofreading**: "Check this paragraph for grammar and clarity." AI catches errors you miss after staring at the document for hours.
- **Learning**: "Explain the Ziegler-Nichols PID tuning method with an example." AI provides a tutorial you can compare against the MEGN 300 reference.

Inappropriate uses that violate academic integrity:

- Submitting AI-generated text as your own writing in reports, essays, or deliverables without attribution.

- Using AI to generate "analysis" that you do not understand and cannot verify. If you cannot explain the calculation to your PA, you did not do the analysis.
- Generating citations that do not exist ("hallucinated references"). AI frequently invents plausible-looking citations that refer to papers, books, or websites that do not exist. **Every citation must be verified by locating the actual source.**
- Generating ethics essay content that is generic and disconnected from your specific project. The ethics assignment specifically requires engagement with *your* project's unique ethical dimensions; something AI cannot provide without deep project knowledge.

The attribution rule: when AI-generated content contributes substantially to a deliverable, disclose it: "AI assistance was used for [specific purpose, e.g., initial literature search, code template generation, grammar checking] and the output was reviewed and modified by the team." Transparency is always acceptable; concealment is always a violation.

The verification rule: AI output is a draft, not a final answer. Every AI-generated claim must be verified against authoritative sources. Every AI-generated calculation must be checked by hand or independent tool. Every AI-generated code snippet must be tested on your hardware. You are responsible for everything in your deliverables, regardless of its origin.

22.10 The Broader Impacts Essay

Broader Impacts Essay: Quick Reference

This section provides the complete guide for the SDII Broader Impacts Essay. For the SDI Ethics Assignment (a related but distinct deliverable), see the guidance in Chapter 18 (Professional Communication; Written).

The Broader Impacts Essay (SDII, Sprint 11) is a **1000–1500 word argumentative essay** on a societal impact connected to your project. It follows the same structure as the Ethics Assignment (Chapter 18) but focuses on broader societal effects rather than narrow ethical dilemmas:

Scope: how does your project's technology, application, or domain affect society beyond the immediate client? Consider: environmental impact, equity and access, labor displacement, privacy and surveillance, public safety, community resilience, economic effects on marginalized populations, and unintended consequences.

Structure: take a position ("Automated agricultural monitoring systems will exacerbate economic inequality among small farmers unless [specific policy intervention]"), defend it with evidence (cite economic data, case studies, policy analyses), address counterarguments honestly, and conclude with a recommendation.

Connection to your project: the essay must be grounded in your specific project's technology and application domain, not a generic discussion of "technology and society." The reader should understand exactly how your project relates to the broader impact you are analyzing.

Common mistakes: (1) writing an informative essay instead of an argumentative one (no thesis, no position), (2) discussing impacts so broad they are disconnected from the project ("AI will change everything"), (3) taking a position so bland that no one would disagree ("access to clean water is important"), (4) using AI to generate the essay without genuine engagement with the topic.

22.11 Academic Integrity in Capstone

Academic integrity in capstone extends beyond the standard prohibition on plagiarism. Capstone-specific integrity issues:

Data fabrication: presenting test results that were not actually obtained, or modifying data to make results appear more favorable. This is an automatic failing offense and potential expulsion. If a test failed, report the failure honestly and present your analysis of why it failed and how you would address it. Honest failure is infinitely preferable to fabricated success.

Contribution misrepresentation: claiming credit for work you did not perform. Feedback Loop evaluations and PA observations are designed to identify contribution imbalances. If you did not write a section of the report, do not claim authorship. If a teammate completed the analysis, credit them in the document.

Unauthorized collaboration: while capstone is a team project, individual assignments (Ethics Assignment, Broader Impacts Essay, Individual Performance Evaluations) must be your own work. Using a teammate's essay as a template, submitting substantially similar individual assignments, or dividing individual work among team members are all violations.

Source attribution: cite every source. This includes: datasheets, application notes, textbooks, online tutorials, StackOverflow answers, AI-generated content, and prior capstone team reports. When in doubt, cite it. Over-citation is never penalized; under-citation can be an integrity violation.

Consequences: academic integrity violations are adjudicated through the Mines Student Code of Conduct. Consequences range from a zero on the assignment to failure in the course to expulsion from the university, depending on the severity and whether it is a first or repeated offense. Your PA and CF take integrity seriously; do not test the system.

22.12 Professional Development Modules

Each semester includes Professional Development assignments completed on Canvas. These modules build skills that complement the project work:

SDI modules (typical selection): Introduction to Scrum, Creating Team Culture, Intellectual Property, Professional Licensure and Liability, Engineering Ethics (pre-essay preparation), and discipline-specific modules selected by the team.

SDII modules (typical selection): Advanced Scrum, Project Retrospective Techniques, Resume and Interview Preparation, and discipline-specific modules (e.g., CE construction management, EE PCB design best practices, ME FEA validation).

The team's **Professional Development Plan** identifies which modules the team will complete each semester. The plan is submitted at the start of each semester and reviewed with the PA. Some modules are required for all students; others are selected based on project needs and individual development goals.

Individual Professional Development: CLO 15 requires you to demonstrate functioning effectively as a member of a team. CLO 16 requires applying project management methodologies. CLO 17 requires self-evaluation and self-improvement strategies. These CLOs are assessed

through: your Professional Development module completions, your Feedback Loop self-evaluations, your PA's observation of your contributions, and your end-of-semester reflection narrative.

22.13 ABET Student Outcome Alignment

Capstone Design is the primary assessment vehicle for several ABET Student Outcomes [25]. Your work in capstone directly demonstrates your competence in these areas:

SO 1: an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics (demonstrated through: requirements engineering, engineering analysis, design calculations, test methodology).

SO 2: an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, welfare, global, cultural, social, environmental, and economic factors (demonstrated through: the complete design cycle from problem definition through V&V, sustainability assessment, ethics analysis).

SO 3: an ability to communicate effectively with a range of audiences (demonstrated through: PDR/CDR/FDR reports and presentations, client communication, Showcase poster and demonstration).

SO 5: an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives (demonstrated through: team charter, sprint management, Feedback Loop evaluations, PA observation of team dynamics).

SO 7: an ability to acquire and apply new knowledge as needed, using appropriate learning strategies (demonstrated through: self-directed learning of domain-specific technologies, professional development modules, end-of-semester reflection).

Your capstone deliverables are the evidence that you have achieved these outcomes. Strong capstone performance is a direct indicator of your readiness for professional engineering practice.

22.14 The Final Course Survey

At the end of each semester, complete the Final Course Survey on Canvas. This survey provides feedback to the capstone program on: PA effectiveness, course structure, resource availability, and suggestions for improvement. The survey is anonymous and does not affect your grade.

Your feedback matters: the capstone program has evolved significantly based on student survey responses. Specific, constructive feedback ("The CDR timeline was too compressed; starting CDR one week earlier would allow more time for the Engineering Calculation Package") is far more useful than generic comments ("The course was hard").

22.15 Ethics Beyond the Classroom

The ethical principles you practice in capstone apply throughout your career:

Whistleblowing: if you discover that your employer is knowingly producing unsafe products, falsifying test data, or violating environmental regulations, you have a professional and legal obligation to report the violation. The path: (1) raise the concern internally through your chain

of command, (2) if internal channels fail or retaliate, report to the appropriate regulatory agency (OSHA, EPA, SEC, state engineering board). Whistleblower protection laws exist, but the path is difficult. Studying the ethics case studies (Challenger, Flint, Boeing) prepares you to recognize the warning signs before a crisis occurs.

Conflicts of interest: as a PE or engineering employee, you must disclose any financial, personal, or professional interest that could bias your engineering judgment. If your company is evaluating a product made by your spouse's company, you must disclose the relationship and recuse yourself from the evaluation. Undisclosed conflicts of interest can result in loss of PE license, termination, and legal liability.

Continuing competence: the NCEES Rules require that you undertake only assignments for which you are qualified. As technology evolves, the boundary of your competence shifts. An engineer qualified to design analog circuits in 2026 may not be qualified to design AI-based control systems in 2036 without additional training. Continuing education (required for PE license maintenance in most states) ensures that your qualifications keep pace with technology.

Pro bono and public service: many engineering professional societies encourage members to donate engineering services to community organizations, disaster relief, and underserved populations. Engineers Without Borders, Bridges to Prosperity, and similar organizations provide opportunities to apply engineering skills for public benefit. Capstone projects with community clients provide your first experience with this form of professional service.

WE Engineering Judgment: The Safety Factor Decision

Your structural analysis shows that a mounting bracket has a safety factor of 1.8 under the maximum expected load. The requirement specifies SF $\geq$ 2.0. The bracket is machined from 6061-T6 aluminum using conservative material properties from the datasheet.

Option A: redesign the bracket with thicker material to achieve SF = 2.0. Impact: +\$15 material cost, +4 hours machining time, bracket weight increases by 30%.

Option B: accept SF = 1.8 and document the engineering rationale. The 1.8 value used minimum datasheet properties; typical 6061-T6 yield strength is 10–15% higher than minimum, which would bring the effective SF above 2.0. Additionally, the "maximum expected load" includes a 1.5× application factor that already accounts for dynamic loading uncertainty.

The judgment call: a strictly by-the-numbers approach says Option A (the calculated SF is below the requirement). An engineering judgment approach considers: the conservatism already embedded in the analysis (minimum material properties, 1.5× load factor), the consequence of failure (the bracket supports a sensor module, not a load-bearing structural element), and the schedule impact of redesign (4 hours during a schedule-critical sprint).

Professional practice: document both options. Present the analysis to your PA. If the PA and team agree that the conservatism in the analysis provides adequate margin, Option B may be acceptable with a documented rationale and client awareness. If the bracket supports a safety-critical function, Option A is required regardless of schedule impact.

This is engineering judgment: making defensible decisions when the numbers alone do not give a clear answer.

22.16 Practice Questions

1. Describe the four-step path to PE licensure. At what point in your career should you take the FE exam, and why is it advantageous to do so while still in school?
2. Your client asks you to omit a safety guard from the final design because "it gets in the way of operation." Using the NCEES Rules, explain your professional obligation and how you would handle the conversation.
3. A teammate used ChatGPT to generate a section of the CDR report and submitted it as their own work. The section includes two citations that do not exist. What policies has this violated, and what should happen?
4. Select one of the four case studies (Challenger, Flint, Boeing, Hyatt Regency) and identify the specific process failure that enabled the technical failure. How could a capstone team experience a similar process failure, and what safeguards would prevent it?
5. You receive a grade you believe is unfair. Walk through the chain of command and describe what information you would bring to each conversation.

Glossary of Key Terms

BOM (Bill of Materials)
Complete list of every component in the design with descriptions, part numbers, vendors, costs, quantities, and procurement status. The financial backbone of the build phase.

Build Track
Project track requiring a physical prototype that demonstrates functional performance.

CDR (Critical Design Review)
Design gate answering "Will this design work?" Occurs in SDII Sprint 8. Design freezes after CDR.

CF (Course Faculty)
The faculty member(s) responsible for the capstone course. Sets policies, resolves escalated issues, and assigns final grades.

CLO (Course Learning Outcome)
A specific, measurable learning objective that the course is designed to achieve. Capstone has 17 CLOs.

CLT (Capstone Leadership Team)
Program leadership that approves Go/No-Go decisions for travel and resolves program-level issues.

Competition Track
Project track organized around an external competition (SAE, ASCE, AIAA, etc.) with rules functioning as requirements.

ConOps (Concept of Operations)
Narrative description of how the system will be used in its operating environment.

CP (Capstone Purchasing)
The administrative team that processes all purchase orders, manages vendor relationships, and handles reimbursements. Office: Labriola 204.

DFE (Design for Environment)
Sustainability assessment framework for product design, evaluating lifecycle environmental impacts.

Digital Design Archive
Complete, organized submission of all project files at Final Checkout.

Engineering Calculation Package
Collection of formal, documented analyses (hand calculations, FEA, CFD, circuit analysis) supporting the CDR design.

ENVISION
Sustainability assessment framework for infrastructure projects.

ESJ (Engineering for Social Justice)
Framework for assessing equity, access, and social impact in engineering design.

FDR (Final Design Review)
Design gate answering "How well did it work?" Occurs in SDII Sprint 13–14.

FE Exam (Fundamentals of Engineering)
A 5-hour-and-20-minute, computer-based exam administered by NCEES. Passing the FE is the first step toward PE licensure and grants the title Engineer Intern (EI).

Feedback Loop
Peer evaluation tool used throughout capstone. Results scale individual grades.

FMEA (Failure Mode and Effects Analysis)
Systematic technique for identifying potential failures and prioritizing mitigation by Risk Priority Number (RPN $= S \times O \times D$).

FRL (Filter-Regulator-Lubricator)
Air preparation unit for pneumatic systems.

ICD (Interface Control Document)
Specification defining what crosses a subsystem boundary: signals, power, data, format, and constraints.

InnoHub
Labriola Innovation Hub (the primary fabrication facility for capstone teams, including Metal Shop, Wood Shop, Electronics Shop, Composites Shop, and Makerspace).

LEED
Leadership in Energy and Environmental Design (LEED), a sustainability assessment framework for building-related projects.

MoSCoW
Requirements prioritization method: Must Have, Should Have, Could Have, Won't Have (this time).

NCEES (National Council of Examiners for Engineering and Surveying)
Organization that develops and administers the FE and PE examinations and publishes the Model Rules of Professional Conduct.

MVP (Minimum Viable Product)
The smallest set of features that delivers value to the client.

O&M Manual
Operations and Maintenance manual delivered at FDR, enabling the client to operate, maintain, and troubleshoot the system.

PA (Project Advisor)
The faculty member or industry professional who mentors your team throughout both semesters. Primary point of contact for project-related concerns.

Paper Track
Project track requiring a comprehensive analytical deliverable (simulation, model, technical report) rather than a physical prototype.

PDR (Preliminary Design Review)
Design gate answering "Should this design work?" Occurs in SDI Sprint 5. SDRD is baselined after PDR.

PE (Professional Engineer)
Licensed engineering credential requiring an accredited degree, FE exam, 4+ years of progressive engineering experience under a licensed PE, and passage of the PE exam. See Chapter 22.

PIP (Performance Improvement Plan)
Formal process for addressing team member non-participation, with a two-week improvement timeline.

PoC (Proof of Concept)
Demonstration that retires the project's highest technical risk. Presented at PDR.

Product Backlog
Master list of all work the project requires, prioritized by the client.

Scrum Master
Team member who facilitates the sprint process, tracks tasks, runs stand-ups, and removes impediments.

SDI / SDII
Senior Design I (EDNS 491) and Senior Design II (EDNS 492).

SDRD (System Design Requirements Document)
The formal requirements document containing all SHALL/SHOULD statements with IDs, traceability, TAID assignments, and MoSCoW priorities.

SMART Goals
Specific, Measurable, Achievable, Relevant, Time-bound goals set at the start of each semester.

SOW (Statement of Work)
Foundational agreement defining scope, deliverables, schedule, budget, roles, assumptions, and risks.

Sprint Backlog
Subset of Product Backlog items selected for the current sprint.

Sprint Retrospective
End-of-sprint team reflection: what went well, what to improve, what actions to take.

Sprint Review
End-of-sprint demonstration of completed work to the PA and client.

STAR Method
Behavioral interview framework: Situation, Task, Action, Result.

TAID
Verification method categories: Test, Analysis, Inspection, or Demonstration.

Team Contract
Document expressing minimum commitments (attendance, deadlines, contribution, behavior) that all team members agree to uphold.

VCRM (Verification Cross-Reference Matrix)
Table mapping every requirement to its verification method, planned phase, result, and disposition.

WBS (Work Breakdown Structure)
Hierarchical decomposition of the project into progressively smaller tasks.

Why Capstone Projects Fail: Twelve Common Patterns

These patterns have been observed across hundreds of capstone projects at Mines and other institutions. Each pattern includes the warning signs and the prevention strategy. Reading this appendix at the start of SDI may save your project.

Pattern 1: Solution Before Problem

What happens: the team falls in love with a technology ("let's use a Raspberry Pi!") before understanding the problem. The design becomes a technology showcase rather than a client solution. **Warning signs**: concept selection in Sprint 1 before requirements are written; the team cannot articulate the client's need in one sentence. **Prevention**: complete the Five-Element Problem Statement and SDRD before generating concepts. The requirements define the solution space; the solution does not define the requirements.

Pattern 2: Scope Creep

What happens: the client suggests "one more feature" at every meeting. The team agrees to each individually reasonable request. By CDR, the project is 50% larger than the SOW and the schedule is impossible. **Warning signs**: the SOW has no "Out of Scope" section; client requests are accepted without impact assessment; the backlog grows every sprint but nothing is removed. **Prevention**: baseline the SOW at Sprint 1. Use the scope change process (Chapter 7) for every addition. Present the trade-off: what is deferred to accommodate the new feature?

Pattern 3: The Invisible Team Member

What happens: one team member stops attending meetings, does not complete assigned tasks, and does not respond to messages. The team compensates by doing the work themselves rather than addressing the issue. **Warning signs**: the Communication Lead's attendance log shows repeated absences; the task board shows tasks assigned to one person with no progress; Feedback Loop scores show a clear outlier. **Prevention**: address the behavior directly at the first occurrence (Chapter 13). Escalate to the PA if direct conversation does not resolve the issue. Initiate a PIP within 2 weeks if the pattern continues.

Pattern 4: Big-Bang Integration

What happens: the team builds all subsystems independently, then connects everything for the first time in Sprint 12. Nothing works. The interactions between subsystems (ground loops, timing conflicts, mechanical interference) produce failures that take weeks to debug. **Warning**

signs: no subsystem testing before Sprint 11; the task board shows "integration" as a single task rather than a phased process; no test results presented at sprint reviews. **Prevention**: follow the debugging pyramid (Chapter 6). Test at every level: component, interface, subsystem, system. Start integration in Sprint 10.

Pattern 5: Procurement Paralysis

What happens: the team waits until after CDR to order components. Lead times push critical deliveries past Spring Break. The build phase is compressed from 6 weeks to 2 weeks. **Warning signs**: the BOM shows "Not Ordered" for critical items at CDR; no components ordered during Sprint 6 of SDI; the procurement timeline does not account for Spring Break closure. **Prevention**: order long-lead items in Sprint 6 of SDI. Order all remaining BOM items within one week of CDR. Build the procurement timeline backward from the test start date.

Pattern 6: Documentation Debt

What happens: the team spends all of SDII building and testing but writes nothing until the last week. The FDR report is a rushed, thin document that does not reflect the quality of the engineering work. **Warning signs**: no report sections drafted before Sprint 12; the team says "we'll write it all at the end"; sprint retrospectives do not include writing tasks. **Prevention**: write as you go. Update the SDRD when requirements change. Draft report sections during the sprint in which the work occurs. The FDR report should be 80% written before the final sprint.

Pattern 7: Ignoring Advisor Feedback

What happens: the PA raises concerns at Sprint 2 ("your requirements are vague"), Sprint 4 ("your PoC does not retire the key risk"), and Sprint 5 (PDR feedback). The team dismisses each concern. At CDR, the same weaknesses remain. **Warning signs**: PDR feedback items are not tracked or addressed; the team responds to PA feedback with "we already thought about that" without demonstrating the analysis; the same issues recur at each review. **Prevention**: treat PA feedback as action items. Track them on the task board. Report back on how each item was addressed. The PA's experience-based judgment is one of your most valuable resources.

Pattern 8: The Overengineered Design

What happens: the team designs a system far more complex than the client needs. A simple monitoring task becomes a machine-learning-powered, cloud-connected, multi-sensor fusion platform. The complexity exceeds the team's ability to build and test it in the available time. **Warning signs**: the BOM exceeds $3,000 for a monitoring project; the system architecture has more subsystems than team members; the schedule shows no margin for testing. **Prevention**: design to the requirements, not to your ambitions. The MVP that meets all Must-Have requirements on time is superior to the ambitious design that meets 60% of requirements because the team ran out of time.

Pattern 9: The Absent Client

What happens: the client attends the kickoff meeting and then stops responding. The team makes assumptions about requirements, design decisions, and priorities without client input. At FDR, the client is disappointed because the system does not match their expectations. **Warning signs**: no client response to emails for >2 weeks; team making design decisions without client approval; no client attendance at sprint reviews. **Prevention**: maintain the communication cadence even if the client does not respond (Chapter 17). Escalate to the PA after 2 weeks of non-response. Document every contact attempt.

Pattern 10: Testing Theater

What happens: the team "tests" the system by demonstrating it once under ideal conditions and declaring all requirements met. No repeated measurements, no edge cases, no statistical analysis, no documentation of the test setup. **Warning signs**: the VCRM shows all requirements "Verified" but no supporting test evidence is attached; test "procedures" are one-sentence descriptions; no measurement uncertainty is reported. **Prevention**: write formal test procedures with pass/fail criteria before testing. Repeat measurements at least 3 times. Test at operating range extremes. Report mean, standard deviation, and comparison to the requirement threshold.

Pattern 11: The Personality Conflict Spiral

What happens: a personal conflict between two team members escalates over the semester. The conflict poisons team meetings, splits the team into factions, and eventually dominates the team dynamic. Engineering work suffers because the team cannot collaborate. **Warning signs**: two members do not speak to each other; meetings become tense; other team members start "choosing sides"; the PA hears complaints from multiple members about the same individuals. **Prevention**: address conflicts early and directly (Chapter 13). Use the escalation path. Involve the PA before the conflict becomes entrenched. Remember that you do not need to be friends; you need to be professional colleagues.

Pattern 12: The Last-Week Miracle

What happens: the team is behind schedule at Sprint 11 but believes they can "make it all up" in the final 2 weeks with an intense push. They pull all-nighters, skip documentation, cut corners on testing, and arrive at FDR exhausted with a prototype that sort of works. **Warning signs**: sprint velocity declining since Sprint 9; critical-path tasks consistently behind schedule; the team's plan requires 80-hour weeks in the final sprint; "we just need one more week" is said repeatedly. **Prevention**: the Gantt chart does not lie. If you are behind at Sprint 11, reduce scope; do not compress time. Deliver a complete, well-documented system that meets 80% of requirements rather than a rushed, poorly documented system that theoretically meets 100%.

Project Health Check: Team Self-Assessment

Use this form at the end of each semester (or before each design review) to honestly assess your team's risk profile. For each pattern, rate your team Green (no concern), Yellow (early warning

signs present), or Red (active problem). Discuss the results as a team and with your PA.

#	Failure Pattern	G / Y / R	Evidence / Notes
1	Solution Before Problem	☐G ☐Y ☐R	
2	Scope Creep	☐G ☐Y ☐R	
3	Invisible Team Member	☐G ☐Y ☐R	
4	Big-Bang Integration	☐G ☐Y ☐R	
5	Procurement Paralysis	☐G ☐Y ☐R	
6	Documentation Debt	☐G ☐Y ☐R	
7	Ignoring Advisor Feedback	☐G ☐Y ☐R	
8	Overengineered Design	☐G ☐Y ☐R	
9	Absent Client	☐G ☐Y ☐R	
10	Testing Theater	☐G ☐Y ☐R	
11	Personality Conflict Spiral	☐G ☐Y ☐R	
12	Last-Week Miracle	☐G ☐Y ☐R	

How to use: any Red rating should be discussed with your PA immediately with a specific action plan. Two or more Yellow ratings indicate systemic process issues that should be addressed at the next retrospective. A team with all Green ratings is either exceptionally well-managed or not being honest with themselves; review the evidence column to verify.

When to use: complete the Health Check at PDR (SDI), at CDR (SDII), and at FDR preparation (SDII). Share the results with your PA. The patterns you catch early are the patterns that do not derail your project.

Quick Reference Checklists

Sprint 0 Kickoff Checklist

☐ Team formed, roles assigned (Scrum Master, CFO, Comm Lead, Tech Lead)
☐ Team repository created (Teams group, shared file structure)
☐ Client contact information confirmed
☐ Kickoff meeting prepared (agenda, discovery questions, slide deck)
☐ Kickoff meeting conducted
☐ Follow-up email sent within 24 hours
☐ EHS Safety Training completed (Canvas module)
☐ Individual Design Log started

SOW Submission Checklist

☐ Project title and team information
☐ Track classification (Build / Paper / Competition) with justification
☐ Five-Element Problem Statement
☐ In-Scope and Out-of-Scope sections
☐ Deliverables list (SDI and SDII)
☐ Schedule with milestones and Gantt chart
☐ Budget estimate with contingency
☐ Team roles and responsibilities
☐ Assumptions and constraints
☐ Risk summary (top 3–5 risks)
☐ PA reviewed and approved
☐ Client reviewed and approved

PDR Readiness Checklist

☐ Complete SDRD (all requirements with IDs, SHALL/SHOULD, TAID, MoSCoW)
☐ $\geq$3 distinct concepts generated with graphics
☐ Pugh screening and weighted decision matrix with sensitivity analysis
☐ Proof-of-concept evidence (data, not just claims)
☐ Sustainability assessment with CLO 5 tool applied
☐ Engineering standards identified
☐ Risk register with mitigations
☐ VCRM plan (all entries Open with TAID assigned)
☐ PDR Report reviewed by team
☐ Mock PDR conducted $\geq$3 days before formal review
☐ PDR Report sent to client
☐ Presentation rehearsed, timed, within allotted time

CDR Readiness Checklist

☐ Detailed design complete (CAD, wiring, software architecture)
☐ Engineering Calculation Package (all critical analyses)
☐ BOM complete and validated (prices, availability, lead times verified)
☐ Critical components ordered or received
☐ InnoHub Manufacturing Review Meeting completed
☐ Hazard Assessment and Safety Plan approved by PA
☐ FMEA updated with design mitigations
☐ VCRM updated (analysis verifications complete)
☐ CDR Report reviewed by team
☐ CDR Report sent to client
☐ Presentation rehearsed

FDR Readiness Checklist

☐ All V&V testing complete
☐ VCRM: all requirements dispositioned (Verified / Failed / Waived / Deferred)
☐ As-designed vs. as-built comparison documented
☐ O&M manual written
☐ Lessons learned section complete
☐ Future work recommendations documented
☐ FDR Report reviewed by team
☐ FDR Report sent to client
☐ Showcase poster submitted for printing
☐ Demo tested in Showcase space
☐ Digital Design Archive organized
☐ All final deliverables in progress

Final Checkout Checklist

☐ FDR presented
☐ Showcase attended
☐ Digital Design Archive submitted
☐ "What I Wish I Would Have Known" video submitted
☐ Project Synopsis submitted
☐ Next Steps Email (post-FDR) sent to client
☐ Project Goal Attainment submitted
☐ Feedback Loop #3 completed
☐ Individual Performance Evaluation #2 completed
☐ Final Course Survey completed
☐ All borrowed equipment returned
☐ Workspace cleaned and locker emptied
☐ Budget Tracker final version uploaded
☐ Checkout meeting with PA completed

References and Further Resources

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Mines Resources

- **EHS**: https://www.mines.edu/ehs/
- **Labriola Innovation Hub**: https://labriola.mines.edu/
- **Writing Center**: https://writing.mines.edu/
- **Mines Online**: https://online.mines.edu/
- **Capstone Purchasing**: capstonepurchasing@mines.edu
- **NCEES FE Reference Handbook**: https://ncees.org/exams/study-materials/
- **AskNature (biomimicry database)**: https://asknature.org/
- **Google Patents**: https://patents.google.com/